

**SCOTT–VOGELIUS–NITSCHKE ON A POLYGONALLY APPROXIMATED
BOUNDARY: CONVERGENCE, THE MISSING PRESSURE TRACTION,
AND THE PENALTY THRESHOLD**

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ABSTRACT. We analyse the Scott–Vogelius–Nitsche method of Gjerde and Scott for two-dimensional Stokes flow whose no-slip curve is replaced by an inscribed polygon: continuous P_k velocities that are exactly divergence-free, with Dirichlet data imposed weakly on the polygon by Nitsche’s method. This answers Scott’s zero-gradient prize question, which asks whether the error behaves like $h_\Gamma^{3/2} + h_\Omega^k$ and, if not, why not.

For the method as printed, the answer is no as soon as the pressure is not constant on the boundary. The printed Nitsche form omits the traction $-pn$. We prove that the method converges, at the sharp rate h/μ in the H^1 seminorm and at the exact rate $(h/\mu)^{1/2}$ in the mesh-dependent energy norm. To leading order the error is an explicit field: the discrete velocity leaks through the wall with normal velocity $(h/\mu)(p - \bar{p})$, with constant exactly 1 in the slip and energy norms. The H^1 rate is 1 and not 1/2 only because incompressibility turns the normal part of the missing traction into a tangential derivative. For fixed γ and bounded ρ , (μ/h) times the error converges in H^1 to the Stokes flow driven by the leak as Dirichlet data; the resulting constant depends on the domain (2.7565 for our test, tending to $3/\sqrt{7}$ as the outer box grows). A penalty growing like $\mu \gtrsim h h_\Gamma^{-3/2}$ restores the H^1 rate 3/2 of the printed method, but no power-law scaling of the penalty lifts its energy rate above 1. When the wall pressure varies, its pressure converges in L^2 only at rate 1/2, because of an explicit boundary layer in the first row of triangles, and its drag error is, to leading order, $-(h/\mu)$ times an explicit reciprocity constant.

Adding the plain pressure term makes the square system singular. Subtracting the wall mean of the discrete pressure gives a well-posed, exactly divergence-free method; this mean-free correction is not new, since its velocity coincides with that of the zero-net-flux formulation of Frachon, Nilsson and Zahedi, and what we contribute is its analysis. We prove that it attains $h_\Gamma^{3/2} + h_\Omega^k$ for general Stokes data, for every fixed $k \geq 4$ on meshes satisfying (M0)–(M2) and every $k \geq 2$ on Clough–Tocher refinements. With flux-corrected outer data the H^1 bound holds uniformly in the penalty and with no relation between h_Γ and h_Ω ; its pressure then converges at the same rate and its drag at rate 2. Whenever the wall shear is not identically zero, we prove a matching lower bound $ch_\Gamma^{3/2}$ for the H^1 error, so the $h_\Gamma^{3/2}$ term of Scott’s rate is sharp. The lower bound is unconditional for $k \geq 7$; for $4 \leq k \leq 6$ it is conditional on a local condition on the boundary stars, which we verified numerically on all our meshes but have not proved.

The Nitsche penalty must scale like $\rho = h_\Omega / \min_e |e|$ on the meshes we can certify. Sufficiency is proved on every mesh; necessity is certified in exact rational arithmetic for every mesh whose shortest boundary edge carries a triangle with apex angle at least 35° and base angles at least 5° , and is open in general. The convergence theory extends to general smooth curved boundaries in two dimensions (nonconvex and multiply connected, with a nonempty polygonal component carrying strongly imposed data), to steady Navier–Stokes under a linearised-stability assumption that we prove for small data, and to three dimensions for a spherical obstacle, under a uniform inf-sup assumption. That assumption is known on Alfeld splits for $k \geq 3$, modulo the uniformity of the published constants over the perturbed domains, which we have not re-audited.

A computational study up to $N = 256$ boundary chords, for $k = 4, 5, 6$ and for Clough–Tocher refinements with $k = 2, 3$, confirms the rates and constants. It includes a pure-pressure test that isolates the traction response exactly, pressure errors, a penalty study in which the threshold is linear in h_Ω/h_Γ , with $\mu^* h_\Gamma/h_\Omega \approx 20$ for $k = 4$, and a comparison with strong imposition: in our computations the $h^{1/2}$ locking of strongly imposed no-slip appears on a mesh with wall vertices in only two triangles, but not on our production meshes.

1. INTRODUCTION

Background. *Exactly divergence-free elements.* Scott and Vogelius [40] showed that continuous piecewise- P_k velocities with discontinuous piecewise- P_{k-1} pressures satisfy $\operatorname{div} V_h = \Pi_h$ for $k \geq 4$ on meshes without singular vertices; the inf-sup constant was later made explicit in terms of the local mesh geometry [27]. The discrete velocity is then divergence-free pointwise, not only weakly. This gives exact mass conservation. On fitted meshes with strongly imposed no-slip data it also gives pressure robustness: the velocity error does not depend on the pressure, which matters when

pressure gradients are large or the viscosity is small [28]. Split meshes give similar pairs of lower degree [26]; see [36] for a review.

Curved walls. Exact incompressibility has a cost at a curved wall. A divergence-free field that is smooth at a polygon vertex and vanishes on both adjacent edges has zero gradient there, and piecewise-polynomial divergence-free fields inherit such gradient constraints at the vertices of a polygonal wall [39, 24]. If the wall is replaced by an inscribed polygon and no-slip is imposed strongly, the computed gradient can therefore be off by $O(1)$ at the polygon vertices, and the H^1 error of Scott–Vogelius elements then decays only like $h^{1/2}$ [39, 24], compared with $h^{3/2}$ for scalar problems [5, 42]. How strongly this locking acts depends on the mesh at the wall, at least in our computations (Section 14.8). For a conforming piecewise-polynomial field the constraints at a wall vertex depend on its star: with two triangles they force a zero gradient, with three they leave a one-parameter family. On a mesh in which alternate polygon vertices lie in only two triangles we reproduce the rate $h^{1/2}$ with an $O(1)$ vertex-gradient error. On the meshes used in the rest of this paper, where every polygon vertex lies in three triangles, strong imposition shows no locking in the range computed (H^1 rates decreasing from 1.93 to 1.61 at $N = 128$). We have not analysed strong imposition, and we do not claim that three triangles per wall vertex suffice in general; the observation suggests that the $h^{1/2}$ phenomenon needs wall vertices whose stars are singular or nearly singular.

One remedy is to fit the geometry, with isoparametric or Piola-mapped elements [37, 17]. Gjerde and Scott [24] keep the polygon and impose the wall condition weakly by Nitsche’s method [38], with penalty μ/h and no pressure term, and report errors orders of magnitude below those of strong imposition. Scott observed H^1 errors of order $h^{3/2}$ for this Scott–Vogelius–Nitsche method on a benchmark with constant pressure [39]. His zero-gradient prize [39] asks for a published proof that Scott–Vogelius–Nitsche converges, including a computational study of the influence of the Nitsche penalty. It states the expected error $h_\Gamma^{3/2} + h_\Omega^k$ and adds: *if this does not hold, why not?* We treat the question for general Stokes data.

Answer. For the method as printed in [24, (27)–(28)] and bounded γ , the expected estimate holds if and only if the pressure is constant on the wall. The printed Nitsche form contains no traction term $-pn$. When p varies along Γ , the penalty, not a traction, balances the pressure. The discrete velocity then leaks through the wall with normal velocity $(h/\mu)(p - \bar{p})$, and the error is dominated by this leak (Theorems A–C, Corollary B’ and Theorem B’'). The mean-free correction CNS* (cf. Frachon, Nilsson and Zahedi [20]), well posed for $\gamma \geq \gamma_2$, restores the expected rate (Theorem D), and its $h_\Gamma^{3/2}$ term cannot be improved whenever the wall shear is not identically zero (Theorem D’). On every mesh whose shortest boundary edge carries a triangle with apex angle at least 35° and base angles at least 5° , the penalty threshold must grow with the ratio of bulk to boundary mesh size (Proposition E).

Main results. The notation is fixed in Section 2. Throughout, $h = h_\Omega$ is the bulk mesh size used in Scott’s penalty μ/h , h_Γ is the length of the longest boundary edge, $\rho = h / \min_e |e|$, $\gamma = \mu h_\Gamma / h$ and $G = \|p - \bar{p}\|_{L^2(\Gamma)}$. Constants do not depend on h, h_Γ, μ, γ or ρ .

Theorem A (Theorem 6.1). *The printed method GS converges, and its error in the mesh-dependent energy norm is*

$$\|\tilde{u} - u_h\| \asymp (h/\mu)^{1/2} G + O((1 + \gamma^{1/2})h_\Gamma^{3/2} + \varepsilon_h),$$

with matching upper and lower bounds.

Theorem C (Theorem 6.4). $\|\nabla(\tilde{u} - u_h)\| \leq C_p h/\mu + C((1 + \gamma^{1/2})h_\Gamma^{3/2} + \varepsilon_h)$. *The rate 1 (rather than $1/2$) holds because $\operatorname{div} v = 0$ turns the normal load into a tangential derivative (Lemma 6.3).*

Theorem B (Theorem 6.6). *In the regime where the pressure term dominates, the slip satisfies $\|\tilde{u} - u_h\|_{L^2(\Gamma_h)} \asymp (h/\mu)G$ and $\|\nabla(\tilde{u} - u_h)\| \asymp h/\mu$. So Theorem C is sharp.*

Corollary B' (Corollary 6.7). *$\tilde{u} - u_h = (h/\mu)v_\varphi + o((h/\mu)^{1/2})$ in the energy norm, where v_φ is an explicit divergence-free field with $v_\varphi \approx -(p - \bar{p})n$ on Γ_h . In particular*

$$\frac{\|\tilde{u} - u_h\|_{L^2(\Gamma_h)}}{(h/\mu)G} \rightarrow 1, \quad \frac{\|\tilde{u} - u_h\|}{(h/\mu)^{1/2}G} \rightarrow 1.$$

Theorem B'' (Theorem 6.10, Corollary 6.11). *Let U be the Stokes flow in Ω with Dirichlet data $-(p - \bar{p})n$ on Γ and 0 on ∂R . For fixed γ and bounded ρ , $(\mu/h)(\tilde{u} - u_h) \rightarrow U$ in H^1 , so*

$$\frac{\|\nabla(\tilde{u} - u_h)\|}{(h/\mu)G} \rightarrow K := \frac{\|\nabla U\|_{L^2(\Omega)}}{G}.$$

The constant K is not universal: it depends on the domain. For the pressure of our test P it is $K = 2.7565157$ (series computation, inside a rigorous bracket $[2.0746, 3.0526]$), and $K \rightarrow 3/\sqrt{7}$ as the outer box grows. Convergence in the regime $\gamma h_\Gamma \rightarrow \infty$ is observed but not proved.

Theorem D (Theorems 7.2, 7.19 and 7.22). *The naive consistent form, which adds $\langle p_h, v \cdot n \rangle$, is singular; its L_0^2 variant is square but only gives $\operatorname{div} u_h \equiv c$, with $c \neq 0$ in general (Proposition 7.1). The mean-free correction CNS^* , which adds $\langle p_h - \bar{p}_\Gamma(p_h), v \cdot n \rangle$ to the momentum row only, is well posed for $\gamma \geq \gamma_2$ and keeps u_h exactly divergence-free; its velocity coincides with that of the zero-net-flux formulation of [20] (Remark 7.3). With the flux-corrected outer data $\tilde{g}_I$ of Definition 2.2, for general Stokes data, every fixed $k \geq 4$ and every $\gamma \geq \gamma_2$,*

$$\|\tilde{u} - u_h\|_{H^1(\Omega_h)} \leq C(h_\Gamma^{3/2} + h_\Omega^k),$$

with C independent of γ and with no relation between h_Γ and h_Ω . This is Scott's expected rate. With the plain interpolant g_I the same holds under the proviso that the boundary edges are not too short relative to the bulk ($h_\Gamma \geq h_\Omega^4$ for even k , $h_\Gamma \geq h_\Omega^2$ for odd k); without it a single lower-order term, due to the net flux of the interpolated outer data, remains. On Clough-Tocher refinements the estimate of Theorem 7.2 holds for every $k \geq 2$ (Theorem F).

Theorem D' (Theorem 7.13, Corollary 7.14). *Suppose the wall shear $W = \partial_n u \cdot \tau$ is not identically zero. For CNS^* , and for GS with any G ,*

$$\|\nabla(\tilde{u} - u_h)\| \geq c \|W\|_{L^2(\Gamma)} h_\Gamma^{3/2} - C h_\Gamma^2,$$

with c independent of γ, μ, ρ and the pressure. So, whenever $\varepsilon_h \leq C h_\Gamma^{3/2}$ and γ is bounded (or the data are flux-corrected and $h_\Omega^k \leq h_\Gamma^{3/2}$), the rate $h_\Gamma^{3/2}$ of Theorem D (and of Theorem A at $G = 0$) is exact; it is the $h_\Gamma^{3/2}$ term, not the h_Ω^k term, that is shown to be sharp. This is unconditional for $k \geq 7$. For $4 \leq k \leq 6$ it is conditional on a local, scale-invariant condition (M3) on the boundary stars, which we have not proved; we verified it numerically at every boundary vertex of our meshes, for N up to 1024.

Proposition E (Proposition 9.1, Theorem 9.2, Corollary 9.4). *The condition $\mu \geq 4\kappa C_I^2 \rho$, $\rho = h / \min_{e \in \mathcal{E}_h^\Gamma} |e|$, is sufficient for coercivity on every mesh. On every mesh, coercivity requires $\mu \geq \lambda_T h / |e|$ for each triangle T with an edge e on Γ_h (and not touching ∂R), with an explicit λ_T depending only on the shape of T . In exact rational arithmetic, $\lambda_T > 1$ whenever the apex angle is at least 35° and $\lambda_T > 9.3$ when it is at least 60° (base angles at least 5°). So $\mu \gtrsim \rho$ is necessary on every mesh whose shortest boundary edge carries a triangle with apex angle at least 35° and base angles at least 5° ; for tall boundary triangles $\lambda_T < 0$, and a necessity statement under (M0)–(M2) alone is open. Necessity is also certified on stars close to two reference stars.*

Theorem A with $G = 0$ gives $\|\tilde{u} - u_h\| \leq C((1 + \gamma^{1/2})h_\Gamma^{3/2} + \varepsilon_h)$. This recovers, for the printed method, the constant-pressure benchmark studied numerically in [24, 39] and analysed in [13], where the bound $\|u - u_h\|_{H^1} \leq C(h_\Omega^k + h_\Gamma^{3/2})$ is proved for that benchmark, for fixed $k \geq 4$ and also for $k \geq 2$ on Clough–Tocher splits [13, abstract and Cor. 5.3]. Our proof does not use [13]. Theorems A–D' hold for every fixed $k \geq 4$ on meshes satisfying (M0)–(M2).

Further results.

Theorem R (Theorems 7.19, 7.24, Corollaries 7.23, 7.25, Proposition 7.26). *The H^1 error of the printed method is bounded uniformly in the penalty: $\|\nabla(\tilde{u} - u_h)\| \leq C_p h_\Gamma/\gamma + C(h_\Gamma^{3/2} + h^k + h^{\sigma_k} h_\Gamma^{-1/2})$ for every $\gamma \geq \gamma_0$. So a growing penalty $\mu \gtrsim h h_\Gamma^{-3/2}$ ($\mu \gtrsim h^{-1/2}$ when $h \asymp h_\Gamma$) restores the H^1 rate $\frac{3}{2}$ of GS for general data. It cannot restore the energy norm: if $W \not\equiv 0$ the energy error is at least $c\|W\|\gamma^{1/2}h_\Gamma^{3/2} - C(h_\Gamma/\gamma)^{1/2}$ for large γ , so under a power-law penalty $\gamma \asymp h_\Gamma^{-\alpha}$ the energy rate of GS is $\min(\frac{1+\alpha}{2}, \frac{3-\alpha}{2}) \leq 1$ when $G > 0$ and $W \not\equiv 0$. The condition number of the velocity block grows by the factor $1 + \gamma$, that of the pressure Schur complement by the factor $1 + \mu/h$.*

Theorem P (Section 8). *Let $p^\circ = \tilde{p} - \bar{p}_\Gamma(\tilde{p})$. For CNS* with flux-corrected data and $\gamma \in [\gamma_2, \gamma_{\max}]$, $\|p^\circ - p_h\|_{L^2(\Omega_h)} \leq C(h_\Gamma^{3/2} + h_\Omega^k)$, including the constant, with C depending on $\gamma_{\max}$ (Theorem 8.7). For GS the pressure is uniquely determined, but carries an explicit first-layer field ξ_b produced by the symmetric Nitsche term acting on the leak; for fixed γ , bounded ρ and $G > 0$ the L^2 pressure error converges at the exact rate $\frac{1}{2}$, and not at all pointwise in the first layer (Theorem 8.11). The variational drag, lift and torque coincide with the integral of the Nitsche flux. For GS with fixed γ and bounded ρ their error is $-(h/\mu) \int_\Gamma (p - \bar{p})(R_\psi - \bar{R}_\psi) + O(h^{3/2})$, with R_ψ the pressure of a dual Stokes problem, so the rate is 1 (Theorem 8.17); for CNS* with $\gamma \geq \max(1, \gamma_2)$ and $\gamma h_\Gamma \leq 1$ it is $O(h_\Gamma^2 + \varepsilon_h)$ (Theorem 8.19).*

Related work and what is new. For Stokes, the consistent Nitsche form uses the full traction $\sigma(u, p)n$. It therefore contains the pressure term $\langle p, v \cdot n \rangle$ [21, 43, 4, 9], and this term is recognised as essential for consistency [11]. The Gjerde–Scott form omits it. On the geometric side, boundary-value correction [7] has been combined with Nitsche’s method for cut elements [10, 11] and for polygonal approximations of scalar problems [16]; the shifted boundary method [33] is a related approach. None of these works uses exactly divergence-free spaces.

With such spaces, the placement of the pressure term decides whether exact incompressibility survives. If the term also appears in the continuity equation, $\operatorname{div} u_h$ vanishes only away from an $O(h)$ boundary strip [31]. If it appears only in the momentum equation, the constant pressure mode must be handled separately. Frachon, Nilsson and Zahedi [20] show that fixing the pressure mean by a Lagrange multiplier makes $\operatorname{div} u_h$ equal to that multiplier, and avoid this by testing the momentum equation only with velocities of zero net flux. Our CNS* is this device in square form: its velocity coincides with theirs (Remark 7.3), and its wall-mean subtraction fixes the pressure constant automatically. The work closest to ours is that of Liu, Neilan and Otus [32], for Scott–Vogelius elements on Clough–Tocher splits of a mesh inside a curved domain. They add a mean-free boundary multiplier that approximates the wall pressure up to a constant, together with a zero-flux constraint and a Taylor boundary correction. Their velocity is exactly divergence-free and converges at the optimal order h^k ; without the constraint, they note, the method is ill posed [32, Remark 3.2]. Related multiplier formulations for cut elements are given in [12]. Exactly divergence-free methods also require discrete Dirichlet data with zero net flux [18]; our flux-corrected $\tilde{g}_I$ is a simple instance.

Our subject is the method as Gjerde and Scott printed it, and its minimal repair: no extra unknowns, no boundary correction, and the pressure itself on the wall. We do not claim the mean-free device as new. As far as we know, what is new is the analysis:

- the two-sided leak analysis of the printed method (Theorems A–C, Corollary B'), including the identification of its H^1 constant (Theorem B'') and of its pressure and drag defects (Theorem P);
- the proof of Scott's conjectured rate $h_\Gamma^{3/2} + h_\Omega^k$ for the mean-free correction on an inscribed polygon (Theorem D), with the penalty-uniform H^1 bound, and the proof that its $h_\Gamma^{3/2}$ term is sharp (Theorem D');
- the penalty threshold $\propto \rho$: sufficiency is a standard consequence of the trace-inverse inequality [43, 29], while the explicit necessity certificates of Proposition E appear to be new;
- the extensions: Clough–Tocher refinements, general smooth domains, steady Navier–Stokes and three dimensions (Theorems F–I), and the growing-penalty analysis (Theorem R).

That the pressure term is needed for consistency is not new [11]; the mechanism of the leak is the classical consistency error of boundary penalty methods [3], here acting only on the pressure and with vanishing penalty parameter h/μ , which is why the method converges at all.

Scope and related variants. We analyse the symmetric Nitsche form of [24]. The leak comes from the missing pressure term, not from the symmetry of the viscous terms, so we expect it in nonsymmetric variants as well; we have not analysed these, nor penalty-free variants [21, 8, 6]. Nitsche's method with Robin-type data is treated in [29]. Multiplier and boundary-correction methods [32, 12, 7, 16] reach higher order than $h_\Gamma^{3/2}$, at the price of extra unknowns or Taylor corrections. A common alternative restricts the pressure to L_0^2 and adds the plain term $\langle p_h, v \cdot n \rangle$, testing the continuity equation only with mean-zero pressures. Because $\operatorname{div} V_h = \Pi_h$, this gives only $\operatorname{div} u_h \equiv c$ with $c|\Omega_h| = \int_{\partial\Omega_h} u_h \cdot n$ (Proposition 7.1; compare [20, (5.9)]). The constant vanishes exactly when the naive system is solvable, equivalently when the CNS* pressure has zero mean on Γ_h . With weak imposition nothing forces the flux through Γ_h to vanish, even for flux-corrected data, and in our computations $c \neq 0$. Our proofs use Scott–Vogelius elements with $k \geq 4$, and $k \geq 2$ on Clough–Tocher splits, in two dimensions (with a three-dimensional extension, Theorem I). Other divergence-free families, and cut or unfitted meshes, are not treated. Gjerde and Scott also treat slip conditions by Nitsche's method [23]; we have not examined that method.

Extensions.

Theorem F (Theorem 10.8, Proposition 10.10). *On the Clough–Tocher refinement of any shape-regular mesh whose boundary edges have comparable lengths, Theorems A–D, Corollary B' and the sufficiency part of Proposition E hold for every $k \geq 2$, with no angle or singular-vertex condition. For example, CNS* satisfies $\|\tilde{u} - u_h\|_{H^1} \leq C(h_\Gamma^{3/2} + h^2)$ for $k = 2$ if $h_\Gamma \geq h^4$. The penalty necessity is recertified exactly on split stars.*

Theorem G (Section 11). *In two dimensions, Theorems A–D and Corollary B' extend to a bounded connected domain whose curved boundary consists of finitely many closed smooth curves, convex or not, with a nonempty polygonal component Σ on which the data are imposed strongly. The geometric lemmas use only C^3 curves; the results assume the regularity (R1) of the solution, which for generic f requires $\Gamma \in C^{k+1,\alpha}$, and the results that use the leak test field assume $\Gamma \in C^{k+3}$. With several curved components, the correct pressure constant is the global boundary mean. CNS* needs one global correction, and a per-component correction is inconsistent.*

Theorem H (Section 12). *For steady Navier–Stokes, assume the discrete linearisation at the exact solution is uniformly stable; we prove this under a small-data condition. Then GS converges*

at H^1 rate h/μ with the same leak (slip and energy constants 1), and CNS^* attains $h_\Gamma^{3/2} + h_\Omega^k$. No inflow/outflow correction of the convective term is needed for these statements.

Theorem I (Theorem 13.13). *In three dimensions, with an inscribed triangulated polyhedron, Theorems A–D and Corollary B' hold with the same rates, assuming a uniform discrete inf-sup constant. That assumption is known for Alfeld (barycentric) splits with $k \geq 3$, by the published split-mesh theory combined with a uniform Bogovskiĭ bound that we prove; we have not re-audited the published proofs for hidden domain dependence. Only a spherical obstacle is treated. Penalty necessity, sharpness and unsplit meshes remain open in 3D.*

Computations. Section 14 reports computations with up to $N = 256$ boundary chords, mostly for $k = 4$, with $k = 5, 6$ and Clough–Tocher $k = 2, 3$ in Section 14.11. They show the following.

- The CNS^* H^1 rate decreases monotonically to $3/2$, independently of the pressure, for $k = 4, 5, 6$.
- The printed method has rate 1 whenever $G > 0$.
- On a pure-pressure test that isolates the traction response exactly, the slip and energy constants of Corollary B' approach 1 for every element tested. For $k = 4$ the H^1 constants approach the K of Theorem B''; that they do so also for $k = 5, 6$ and on Clough–Tocher refinements is observed, not proved.
- In the penalty study, the coercivity threshold μ^* is linear in h/h_Γ over more than a decade, with $\gamma^* = \mu^* h_\Gamma / h \approx 18\text{--}21$ for $k = 4$. It is larger for higher k and on Clough–Tocher refinements.
- Strongly imposed no-slip shows the $h^{1/2}$ locking on a mesh with two-triangle wall vertices but not on our production meshes, while GS and CNS^* are insensitive to the change of mesh (Section 14.8).
- The pressure errors follow Theorem P: rate $\rightarrow \frac{3}{2}$ for CNS^* , and rate $\rightarrow \frac{1}{2}$ for GS when $G > 0$, carried by the first layer (Section 14.9).
- A growing penalty behaves as Theorem R predicts, and flux-corrected data remove the flux floor exactly (Section 14.10).

All code, raw outputs and logs are in the accompanying repository (Appendix A).

Organisation. Section 2 fixes the setting. Sections 3–5 collect geometric and discrete tools and the error equation. Section 6 treats the printed method and identifies its H^1 constant. Section 7 treats the pressure-consistent method, proves the lower bound, and treats flux-corrected data and a growing penalty. Section 8 treats the pressure, the boundary stress and drag. Section 9 treats the penalty threshold. Sections 10–13 contain the extensions: Clough–Tocher refinements, general domains, Navier–Stokes and three dimensions. Section 14 reports the computations.

2. SETTING, METHODS AND ASSUMPTIONS

Domain. Let $R = (-L, L)^2$ with $L > 1$, let D be the open unit disk, $\Omega = R \setminus \overline{D}$ and $\Gamma = \partial D$. Let P_h be a convex polygon with vertices on Γ , $\Gamma_h = \partial P_h$ and $\Omega_h = R \setminus P_h$. Since $P_h \subset D$, we have $\Omega \subset \Omega_h$. The strip $S_h = \Omega_h \setminus \overline{\Omega}$ lies between each chord and its arc.

Mesh. $\mathcal{T}_h$ is a conforming triangulation of Ω_h , and $\mathcal{E}_h^\Gamma$ is its set of edges on Γ_h . We use the parameters

$$h = h_\Omega := \max_{T \in \mathcal{T}_h} \text{diam } T, \quad h_\Gamma := \max_{e \in \mathcal{E}_h^\Gamma} |e|, \quad \rho := \frac{h}{\min_{e \in \mathcal{E}_h^\Gamma} |e|}, \quad \gamma := \frac{\mu h_\Gamma}{h}.$$

Here h is the h in Scott's penalty μ/h , and γ is the effective boundary penalty, so that $\mu/h = \gamma/h_\Gamma$.

Assumption 2.1.

- (M0) Shape regularity, and $c_0 h_\Gamma \leq |e| \leq h_\Gamma$ for all $e \in \mathcal{E}_h^\Gamma$. Hence $\rho/h \leq 1/(c_0 h_\Gamma)$. No quasi-uniformity is assumed; $h_\Gamma \ll h$ is allowed.
- (M1) No vertex of $\mathcal{T}_h$ is singular. A vertex is singular if it is an interior vertex whose incident edges lie on two lines, a boundary vertex with one incident triangle, or a boundary vertex whose incident edges lie on two lines.
- (M1') Every vertex of Γ_h has at least three incident triangles. The angle of Ω_h at a vertex of Γ_h is $\pi + O(h_\Gamma)$, so with two triangles $\Theta(z) = O(h_\Gamma) \rightarrow 0$, contradicting (M2).
- (M2) $\Theta(z) := \max_j |\sin(\theta_j + \theta_{j+1})| \geq \Theta_0 > 0$ at every vertex z , in the notation of Guzmán–Scott [27]. At boundary vertices the wrap-around term is omitted.
- (D) $\tilde{f}$ is smooth and bounded on a fixed neighbourhood of $\bar{\Omega}$, $\tilde{f} = f$ on Ω , and $h \leq h_*$ is small (hence so is $h_\Gamma \leq h$).

Spaces. Fix $k \geq 4$ (Section 10 treats $k \geq 2$ on Clough–Tocher refinements). V_h is the space of continuous piecewise- P_k vector fields, $V_h^R := \{v \in V_h : v|_{\partial R} = 0\}$ and $V_h^0 := \{v \in V_h : v|_{\partial \Omega_h} = 0\}$. Π_h is the space of discontinuous piecewise- P_{k-1} functions and $Z_h := \{v \in V_h^R : \operatorname{div} v = 0\}$. $W_h := \{z \in V_h : z|_{\partial R} = g_I, \operatorname{div} z = 0\}$, where g_I is the P_k Lagrange interpolant of g .

Definition 2.2 (Flux-corrected outer data). Let ν be the outward unit normal of R , $m_R := \int_{\partial R} g_I \cdot \nu$, and

$$\tilde{g}_I := g_I - \frac{m_R}{8L^2} x \quad \text{on } \partial R,$$

where x is the position field. A method *with flux-corrected data* is the same method with g_I replaced by $\tilde{g}_I$, and $\tilde{W}_h := \{z \in V_h : z|_{\partial R} = \tilde{g}_I, \operatorname{div} z = 0\}$.

Since x is linear, $\tilde{g}_I$ is again continuous and piecewise P_k on ∂R , and $\int_{\partial R} x \cdot \nu = \int_R \operatorname{div} x = 8L^2$, so $\int_{\partial R} \tilde{g}_I \cdot \nu = 0$ exactly. This is a simple instance of the flux-corrected Dirichlet interpolants that exactly divergence-free methods require [18]. We recommend it for CNS*: it removes the only term in Theorem 7.2 that needs a relation between h_Γ and h_Ω (Theorem 7.22). Gjerde and Scott use $k = 4$, as do the computations of Section 14 except those of Section 14.11 ($k = 5, 6$, and $k = 2, 3$ on Clough–Tocher refinements).

Forms. n is the outward unit normal of Ω_h on Γ_h (pointing into P_h), and $\langle \cdot, \cdot \rangle$ is the $L^2(\Gamma_h)$ pairing. Let

$$\begin{aligned} a_h(u, v) &= \frac{1}{2}(Du, Dv), \quad Du = \nabla u + (\nabla u)^\top, \\ N_h(u, v) &= a_h(u, v) - \langle \partial_n u, v \rangle - \langle u, \partial_n v \rangle + \frac{\mu}{h} \langle u, v \rangle, \quad \partial_n u := (\nabla u)n. \end{aligned}$$

Definition 2.3 (Method GS [24, (28)], as printed). Find $u_h \in V_h$ with $u_h|_{\partial R} = g_I$ and $p_h \in \Pi_h$ such that

$$N_h(u_h, v) - (p_h, \operatorname{div} v) = (\tilde{f}, v) \quad \forall v \in V_h^R, \quad (q, \operatorname{div} u_h) = 0 \quad \forall q \in \Pi_h.$$

Two details of the transcription. Gjerde and Scott write $b(v, p) = \int p \operatorname{div} v$ with a plus sign; our sign is the substitution $p_h \mapsto -p_h$. They also define V_h earlier as vanishing on Γ_h ; read literally, the Nitsche terms would then drop out of (28), so the evident intended test space is our V_h^R , which vanishes only on ∂R .

Definition 2.4 (Method CNS*: the mean-free correction). As GS, with momentum row

$$N_h(u_h, v) - (p_h, \operatorname{div} v) + \langle p_h - \bar{p}_\Gamma(p_h), v \cdot n \rangle = (\tilde{f}, v), \quad \bar{p}_\Gamma(q) := |\Gamma_h|^{-1} \int_{\Gamma_h} q.$$

The added term is the missing pressure traction with the wall mean of the discrete pressure removed. It appears in the momentum row only, so the continuity row, and with it the pointwise

incompressibility of u_h , is unchanged. This device is not new: its velocity is that of the zero-net-flux formulation of Frachon, Nilsson and Zahedi [20], transcribed to our setting (Remark 7.3). What we add is its analysis for Scott–Vogelius elements on an inscribed polygon.

Exact solution and extensions. (u, p) is smooth with $-\Delta u + \nabla p = f$ and $\operatorname{div} u = 0$ in Ω , $u|_\Gamma = 0$ and $u|_{\partial R} = g$; note $-\operatorname{div} D(u) = -\Delta u$. The stream function ψ with $u = \operatorname{curl} \psi = (\partial_y \psi, -\partial_x \psi)$ is single-valued because $u \cdot n = 0$ on Γ . Let $\tilde{\psi}$ be a smooth extension of ψ to a neighbourhood of $\bar{\Omega}$ and $\tilde{u} := \operatorname{curl} \tilde{\psi}$, so $\operatorname{div} \tilde{u} = 0$. Let $\tilde{p}$ be a smooth extension of p . Then $F := \tilde{f} + \Delta \tilde{u} - \nabla \tilde{p}$ vanishes on Ω and is supported in S_h .

Norms and constants. For v with traces on Γ_h ,

$$\|v\|^2 := \|\nabla v\|_{\Omega_h}^2 + \frac{\mu}{h} \|v\|_{\Gamma_h}^2 + \sum_{e \in \mathcal{E}_h^\Gamma} |e| \|\partial_n v\|_{L^2(e)}^2.$$

This is used for discrete fields and for errors $\tilde{u} - z$; since $\tilde{u}$ is smooth, every trace exists. We write $G := \|p - \bar{p}\|_{L^2(\Gamma)}$, with $\bar{p}$ the mean of p over Γ .

C and c depend on (M0)–(M2), k , L and norms of $(u, p, \tilde{\psi})$. They are independent of $h, h_\Gamma, \mu, \gamma, \rho$ unless displayed. C_p is a constant that may also depend on $\|p - \bar{p}\|_{W^{1,\infty}(\Gamma)}$. Where G is tracked explicitly, “ δ small” means $\delta \leq \delta_0(C_p)$.

3. GEOMETRY

Lemma 3.1 (Chords). *Let e be a chord of length ℓ and $x \in e$. Then $\operatorname{dist}(x, \Gamma) \leq \ell^2/8 + O(\ell^4) \leq \ell^2/4$. Let s be the signed arclength from the chord midpoint m_e , x^* the nearest point of Γ , and τ_e the unit tangent of e . Then, up to a fixed orientation sign,*

$$n(x^*) - n_e = -s \tau_e + O(\ell^2).$$

The arclength Jacobian of the radial projection $\Gamma_h \rightarrow \Gamma$ is $1 + O(\ell^2)$.

Proof. The sagitta is $1 - \sqrt{1 - \ell^2/4} = \ell^2/8 + \ell^4/128 + \dots$. The normal at x^* is the radial unit vector. Its angle differs from that of n_e by the central angle to x^* , which is $s + O(s^3)$. $\square$

Lemma 3.2 (Uniform trace, Friedrichs and Korn). *There are $C_{\text{tr}}, C_F, \kappa$, independent of h , such that every $v \in H^1(\Omega_h)^2$ with $v|_{\partial R} = 0$ satisfies*

$$\|v\|_{L^2(\Gamma_h)} \leq C_{\text{tr}} \|\nabla v\|, \quad \|v\| \leq C_F \|\nabla v\|, \quad \|\nabla v\|^2 \leq \kappa a_h(v, v).$$

Proof. (1) Write Γ_h in polar form as $r = \rho_h(\theta)$. On a chord with endpoint angles $\theta_m \pm \alpha$ we have $\rho_h = \cos \alpha / \cos(\theta - \theta_m)$. Hence $\|\rho_h - 1\|_\infty \leq Ch_\Gamma^2$ and $\|\rho'_h\|_\infty \leq Ch_\Gamma$.

(2) Let χ be smooth, with $\chi(1) = 1$ and $\chi \equiv 0$ outside a fixed collar in Ω . Define $\Phi_h(r, \theta) = (r + \chi(r)(\rho_h(\theta) - 1), \theta)$.

(3) Φ_h maps Ω onto Ω_h and Γ onto Γ_h , is the identity near ∂R , and is bi-Lipschitz with $\|D\Phi_h - I\|_\infty \leq Ch_\Gamma$ almost everywhere.

(4) For $w = v \circ \Phi_h$ we have $D(w) = D(v) \circ \Phi_h + E$ with $|E| \leq Ch_\Gamma |\nabla v \circ \Phi_h|$.

(5) On the fixed domain Ω , w vanishes on ∂R , a set of positive measure. So the first Korn inequality, the trace inequality and the Friedrichs inequality hold.

(6) Changing variables back gives $\|\nabla v\| \leq C(\|D(v)\| + h_\Gamma \|\nabla v\|)$; absorb the last term for $h_\Gamma \leq h_*$. The other two inequalities transfer the same way. $\square$

Lemma 3.3 (Extension). $\|\tilde{u}\|_{L^\infty(\Gamma_h)} \leq Ch_\Gamma^2$, so $\|\tilde{u}\|_{L^2(\Gamma_h)} \leq Ch_\Gamma^2$. For $v \in V_h^R$, $|(F, v)| \leq Ch_\Gamma^2 \|\nabla v\|$.

Proof. The first bound follows from Lemma 3.1 and $\tilde{u}|_\Gamma = 0$. For the second, apply a one-dimensional Poincaré inequality across strips of width at most Ch_Γ^2 , and use $|S_h| \leq Ch_\Gamma^2$, $\|F\|_\infty \leq C$ (by (D)) and Lemma 3.2. $\square$

Lemma 3.4 (Wall structure). *On Γ : $\nabla u = \partial_n u \otimes n$, $\partial_n u \cdot n = 0$ and $(\nabla u)^\top n = 0$.*

Proof. The tangential derivatives vanish, and $\operatorname{div} u = n \cdot \partial_n u$. $\square$

Lemma 3.5 (Transposed-gradient term). *For $v \in V_h^R$,*

$$| \langle (\nabla \tilde{u})^\top n, v \rangle | \leq C \sum_{e \in \mathcal{E}_h^\Gamma} (|e|^2 \|v\|_{L^1(e)} + |e|^2 \|\partial_\tau v\|_{L^1(e)}) \leq Ch_\Gamma^{3/2} \|\nabla v\|.$$

Pointwise the transposed term is only $O(h_\Gamma)$ on Γ_h , not $O(h_\Gamma^2)$: although $\tilde{u} = O(h_\Gamma^2)$ there, the chord normal n_e is tilted by $O(h_\Gamma)$ against $n(x^*)$ (step 1 below). The rate $h_\Gamma^{3/2}$ comes from the odd symmetry of the leading term in s (step 2), not from the size of $\tilde{u}$. In the energy dual it can also be obtained from the penalty alone, $| \langle (\nabla \tilde{u})^\top n, v \rangle | \leq Ch_\Gamma \|v\|_{\Gamma_h} \leq C\gamma^{-1/2} h_\Gamma^{3/2} \|v\|$ (Lemma 8.4); in the plain H^1 dual, without odd symmetry, it would be only $O(h_\Gamma)$.

The rate $h_\Gamma^{3/2}$ is sharp on V_h^R : a first-layer field with odd trace on each chord attains it, and the dual-norm test of Section 14 gives 1.56–1.57. For the method itself, which is tested on Z_h , Theorem 7.13 proves the matching lower bound $ch_\Gamma^{3/2}$ for the H^1 error, by a different route (unconditionally for $k \geq 7$, and under a local condition on the boundary stars for $4 \leq k \leq 6$). The CNS* H^1 rates of Table 4 decrease monotonically toward 3/2. Lemma 3.5 is a sharper dual-norm statement than the upper bounds need: by Remark 13.7, Lemma 5.2 and Theorems 6.4 and 7.2 also follow from the cruder energy-dual bound that holds without the odd-in- s cancellation.

Proof. (1) Taylor-expand about x^* and apply Lemmas 3.1 and 3.4:

$$(\nabla \tilde{u}(x))^\top n_e = n(x^*) (\partial_n u(x^*) \cdot (n_e - n(x^*))) + O(h_\Gamma^2) = s a_e + O(h_\Gamma^2),$$

where $a_e := n(m_e^*) (\partial_n u(m_e^*) \cdot \tau_e)$ is constant on e . The $O(h_\Gamma^2)$ remainder collects $|x - x^*|$, the $O(\ell^2)$ term of Lemma 3.1, and $s \cdot O(s)$.

(2) Since $\int_e s ds = 0$,

$$\left| \int_e s a_e \cdot v \right| = \left| \int_e s a_e \cdot (v - v(m_e)) \right| \leq \frac{|e|^2}{4} \|\partial_\tau v\|_{L^1(e)}.$$

(3) Use $\|\cdot\|_{L^1(e)} \leq |e|^{1/2} \|\cdot\|_{L^2(e)}$ and the inverse trace inequality $|e| \|\partial_\tau v\|_{L^2(e)}^2 \leq C_I^2 \|\nabla v\|_{T_e}^2$.

(4) Summing, $\sum_e |e|^{5/2} \|\partial_\tau v\|_{L^2(e)} \leq Ch_\Gamma^2 (\#\mathcal{E}_h^\Gamma)^{1/2} \|\nabla v\| \leq Ch_\Gamma^{3/2} \|\nabla v\|$. The zeroth-order term is handled the same way, finishing with Lemma 3.2. $\square$

4. DISCRETE TOOLS

Lemma 4.1 (Inverse traces). *For $v \in V_h$,*

$$\sum_{e \in \mathcal{E}_h^\Gamma} |e| \|\partial_n v\|_{L^2(e)}^2 \leq C_I^2 \|\nabla v\|^2, \quad \text{hence} \quad \|\partial_n v\|_{\Gamma_h} \leq C_I (\rho/h)^{1/2} \|\nabla v\|.$$

For $q \in \Pi_h$,

$$\|q\|_{\Gamma_h}^2 \leq C_I^2 \sum_{T \in \mathcal{T}_h^\Gamma} |e_T|^{-1} \|q\|_T^2 \leq C_I^2 (\rho/h) \|q\|^2.$$

Lemma 4.2 (Coercivity and continuity). *Let $\mu_0 := 4\kappa\rho C_I^2$. The condition $\gamma \geq \gamma_0 := 4\kappa C_I^2/c_0$ is sufficient for $\mu \geq \mu_0$. If $\mu \geq \mu_0$, then*

$$N_h(v, v) \geq c_1 \|v\|^2 \quad \forall v \in V_h^R, \quad |N_h(w, v)| \leq M \|w\| \|v\|,$$

with $c_1 = 1/\max(2\kappa(1 + C_I^2), 2)$ and $M = 3 + (\kappa C_I^2)^{-1/2}$. Both constants are independent of h , μ and ρ .

Proof. (1) By Lemma 4.1 and Young's inequality with $\varepsilon = 1/(2\kappa)$,

$$2|\langle \partial_n v, v \rangle| \leq 2C_I \|\nabla v\| (\rho/h)^{1/2} \|v\|_{\Gamma_h} \leq \frac{1}{2} a_h(v, v) + 2\kappa\rho C_I^2 h^{-1} \|v\|_{\Gamma_h}^2.$$

(2) Hence $N_h(v, v) \geq \frac{1}{2} a_h(v, v) + (\mu - 2\kappa\rho C_I^2) h^{-1} \|v\|^2$. For $\mu \geq \mu_0$ this is at least $(2\kappa)^{-1} \|\nabla v\|^2 + (\mu/2h) \|v\|^2$.

(3) Adding $\sum_e |e| \|\partial_n v\|^2 \leq C_I^2 \|\nabla v\|^2$ gives the coercivity constant.

(4) For continuity, $a_h(w, v) \leq 2 \|\nabla w\| \|\nabla v\|$. Each ∂_n -term is at most $(\rho/\mu)^{1/2} \|w\| \|v\| \leq (4\kappa C_I^2)^{-1/2} \|w\| \|v\|$. $\square$

Lemma 4.3 (Divergence range). *Under (M1), $\operatorname{div} V_h^R = \Pi_h$ and $W_h \neq \emptyset$. The velocities of GS and CNS* are pointwise divergence-free.*

Proof. (1) $\operatorname{div} V_h^0 = \Pi_h \cap L_0^2$ by the inf-sup condition of Lemma 4.5 below (Guzmán–Scott [27], extending Scott–Vogelius [40]). This holds on the doubly connected domain Ω_h ; there is no circularity, since Lemma 4.5 does not use the present lemma.

(2) The edge bubble $b = \beta_e n_e$ on one $e \subset \Gamma_h$ lies in V_h^R and has $\int \operatorname{div} b = \int_{\Gamma_h} b \cdot n = |e|/6 \neq 0$. This adds the constants, so $\operatorname{div} V_h^R = \Pi_h$.

(3) For any $\mathcal{G} \in V_h$ with trace g_I , we have $\operatorname{div} \mathcal{G} \in \Pi_h = \operatorname{div} V_h^R$. Hence $\mathcal{G} - v \in W_h$ for a suitable $v \in V_h^R$.

(4) For the discrete velocity, $u_h - \mathcal{G} \in V_h^R$, and the continuity row forces $\operatorname{div} u_h = 0$. $\square$

Remark 4.4. If a vertex of ∂R is singular (for example a corner with one triangle), then $A_z(\operatorname{div} \mathcal{G}) = 0$ is a condition on g_I that generic data violate, and $W_h = \emptyset$.

Lemma 4.5 (Uniform inf-sup constant). *There is $\beta > 0$, depending only on k , Θ_0 , shape regularity and L , such that for every $q \in \Pi_h \cap L_0^2(\Omega_h)$ there exists $v \in V_h^0$ with $\operatorname{div} v = q$ and $\|\nabla v\| \leq \beta^{-1} \|q\|$.*

Proof. Guzmán and Scott [27, Theorem 1] prove the inf-sup condition for $(V_h^0, \Pi_h \cap L_0^2)$ for $k \geq 4$ under (M1)–(M2). Their proof needs neither quasi-uniformity, nor small h , nor an interior vertex in each triangle. The discrete Fortin operator is built locally from vertex and edge patches, using shape regularity and (M2) only. The domain enters only through the continuous inf-sup constant (via the P_2 – P_0 Bernardi–Raugel pair) and a Poincaré constant. The Poincaré constant is uniform by Lemma 3.2, so it remains to show that the continuous inf-sup constant is uniform over the domains Ω_h .

- Cover Ω_h by the h -dependent pieces $W_j \cap \Omega_h$, where the W_j are fixed angular wedges of half-angle $\alpha < \pi/2$ about the origin. Each $W_j \cap R$ is convex, and $W_j \cap \Omega_h = (W_j \cap R) \setminus P_h$.
- Each $W_j \cap \Omega_h$ is star-shaped with respect to a ball $B(c_j, r_B) \subset W_j \cap R$ with $|c_j| = r_b$, provided every chord line meeting W_j leaves the ball on its outer side. This holds uniformly when $(r_b - r_B) \cos(\alpha + \arcsin(r_B/r_b) + h_\Gamma) > 1$, which can be arranged with fixed α, r_b, r_B because $P_h \subset D$.
- The pieces have diameters, ball radii and pairwise overlap measures bounded above and below independently of h .

Galdi's Lemma III.3.4 [22] then gives a Bogovskii constant, and hence a continuous inf-sup constant, independent of h . The equivalence of the inf-sup condition with a bounded right inverse of div gives the statement. $\square$

Lemma 4.6 (Approximation in W_h). *Let $k \geq 4$. Let $\sigma_k := k + 2$ for even k and $\sigma_k := k + 1$ for odd k . Then*

$$E_h := \inf_{z \in W_h} \|\tilde{u} - z\| \leq C\varepsilon_h, \quad \varepsilon_h := (1 + (\gamma h_\Gamma)^{1/2}) h^k + h^{\sigma_k} (h_\Gamma^{-1/2} + (\mu/h)^{1/2}).$$

The second term comes from the net flux of the interpolated boundary data. It cannot be removed for the method as posed with g_I . Recall $m_R = \int_{\partial R} g_I \cdot \nu$ (Definition 2.2). Every $z \in W_h$ (including the discrete velocity) has $\int_{\Gamma_h} z \cdot n = -m_R$, while $\int_{\Gamma_h} \tilde{u} \cdot n = 0$. Hence $\|\tilde{u} - z\| \geq (\mu/h)^{1/2} |m_R|/|\Gamma_h|^{1/2}$, and $|m_R| \asymp h^{\sigma_k}$ for generic g .

For bounded γ the term is at most Ch^k whenever $h_\Gamma \geq h^{2(\sigma_k-k)}$: that is, $h_\Gamma \geq h^4$ for even k and $h_\Gamma \geq h^2$ for odd k . If g_I is replaced by the flux-corrected $\tilde{g}_I$ of Definition 2.2 (Lemma 7.21), or for the test data of Section 14, where $g \cdot \nu$ is a polynomial of degree at most 5 on each side and the flux integrals over ∂R are exact, only the contribution of Γ_h to the net flux remains. By step 2 of the proof below it is $O(h_\Gamma^{\sigma_k})$, so the term reduces to $C(1 + \gamma^{1/2})h_\Gamma^{\sigma_k-1/2}$, which is below h^k for bounded γ .

Proof. (1) *Interpolation.* Let $w_h := I_h \tilde{u} \in V_h$ be the Lagrange interpolant of $\tilde{u}$, which is smooth on Ω_h . On ∂R , $w_h = g_I$. Boundary triangles have diameter at most Ch_Γ by shape regularity, so

$$\|\nabla(\tilde{u} - w_h)\| \leq Ch^k, \quad \|\tilde{u} - w_h\|_{L^\infty(\Gamma_h)} \leq Ch_\Gamma^{k+1}, \quad \|\partial_n(\tilde{u} - w_h)\|_{L^\infty(\Gamma_h)} \leq Ch_\Gamma^k.$$

Hence $\|\tilde{u} - w_h\| \leq C(h^k + (\mu/h)^{1/2}h_\Gamma^{k+1} + h_\Gamma^k) \leq C(1 + (\gamma h_\Gamma)^{1/2})h^k$. Moreover $\operatorname{div} w_h = \operatorname{div}(w_h - \tilde{u}) \in \Pi_h$, with $\|\operatorname{div} w_h\| \leq Ch^k$.

(2) *Net flux.* Let ν be the outward normal of R . Since $\tilde{u}|_{\partial R} = g$ and $\operatorname{div} \tilde{u} = 0$, we have $\int_{\Gamma_h} \tilde{u} \cdot n = -\int_{\partial R} g \cdot \nu = 0$. Therefore

$$m := \int_{\Omega_h} \operatorname{div} w_h = \int_{\partial R} (g_I - g) \cdot \nu + \int_{\Gamma_h} (w_h - \tilde{u}) \cdot n, \quad |m| \leq Ch^{\sigma_k}.$$

Here equispaced P_k Lagrange interpolation on an edge integrates like the closed $(k+1)$ -point Newton–Cotes rule. That rule is exact for polynomials of degree $\sigma_k - 1$, so each edge contributes $O(|e|^{\sigma_k+1})$.

(3) *Removing the mean.* Let $b := \sum_{e \in \mathcal{E}_h^\Gamma} \beta_e n_e$, where β_e is the quadratic edge bubble of the boundary triangle on e . Then $b \in V_h^R$, $\int \operatorname{div} b = \int_{\Gamma_h} b \cdot n = |\Gamma_h|/6$, and with $a := 6m/|\Gamma_h|$,

$$\|ab\| \leq C|a|(h_\Gamma^{-1/2} + (\mu/h)^{1/2}) \leq Ch^{\sigma_k}(h_\Gamma^{-1/2} + (\mu/h)^{1/2}) \leq C\varepsilon_h,$$

using $\#\mathcal{E}_h^\Gamma \leq C/h_\Gamma$. Also $\|\operatorname{div}(ab)\| \leq C|a|h_\Gamma^{-1/2} \leq C\varepsilon_h$.

(4) *Divergence correction.* The field $q := \operatorname{div}(w_h - ab) \in \Pi_h$ has mean zero and $\|q\| \leq C\varepsilon_h$. Lemma 4.5 gives $v \in V_h^0$ with $\operatorname{div} v = q$ and $\|\nabla v\| \leq \beta^{-1}\|q\| \leq C\varepsilon_h$. Since $v = 0$ on Γ_h , $\|v\|^2 = \|\nabla v\|^2 + \sum_e |e| \|\partial_n v\|_{L^2(e)}^2 \leq (1 + C_I^2) \|\nabla v\|^2$.

(5) Set $z := w_h - ab - v$. Then $z|_{\partial R} = g_I$ and $\operatorname{div} z = 0$, so $z \in W_h$, and

$$\|\tilde{u} - z\| \leq \|\tilde{u} - w_h\| + \|ab\| + \|v\| \leq C\varepsilon_h. \quad \square$$

Remark 4.7. Version 1 of this preprint proved Lemma 4.6 for $k = 4$ by matching Argyris stream-function data on ∂R . The proof above avoids stream functions and holds for every $k \geq 4$.

5. THE ERROR EQUATION

Lemma 5.1. *For $v \in V_h^R$,*

$$N_h(\tilde{u}, v) - (\tilde{p}, \operatorname{div} v) + \langle \tilde{p}, v \cdot n \rangle = (\tilde{f}, v) + \mathcal{G}(v),$$

where

$$\mathcal{G}(v) := \langle (\nabla \tilde{u})^\top n, v \rangle - \langle \tilde{u}, \partial_n v \rangle + \frac{\mu}{h} \langle \tilde{u}, v \rangle - (F, v).$$

Proof. The argument has four steps.

(1) By symmetry of $D(\tilde{u})$ and since $v = 0$ on ∂R ,

$$a_h(\tilde{u}, v) = (D(\tilde{u}), \nabla v) = -(\operatorname{div} D(\tilde{u}), v) + \langle D(\tilde{u})n, v \rangle.$$

(2) $\operatorname{div} D(\tilde{u}) = \Delta \tilde{u}$.

(3) $-\Delta \tilde{u} = \tilde{f} - F - \nabla \tilde{p}$, and $-(\nabla \tilde{p}, v) = (\tilde{p}, \operatorname{div} v) - \langle \tilde{p}, v \cdot n \rangle$.

(4) $D(\tilde{u})n - \partial_n \tilde{u} = (\nabla \tilde{u})^\top n$. □

Lemma 5.2. For $v \in V_h^R$, $|\mathcal{G}(v)| \leq C(1 + \gamma^{1/2})h_\Gamma^{3/2}\|v\|$. For $v \in V_h^0$, $|\mathcal{G}(v)| \leq Ch_\Gamma^{3/2}\|\nabla v\|$.

Proof. We bound the four terms of $\mathcal{G}(v)$ in turn.

- The transposed term is at most $Ch_\Gamma^{3/2}\|\nabla v\|$ by Lemma 3.5.
- $|\langle \tilde{u}, \partial_n v \rangle| \leq Ch_\Gamma^2 \cdot C_I(\rho/h)^{1/2}\|\nabla v\| \leq Ch_\Gamma^{3/2}\|\nabla v\|$.
- The penalty term is at most $(\mu/h)^{1/2}Ch_\Gamma^2\|v\| = C\gamma^{1/2}h_\Gamma^{3/2}\|v\|$.
- The F term is bounded by Lemma 3.3.

For $v \in V_h^0$, the transposed and penalty terms vanish. □

Corollary 5.3 (GS on Z_h). Let $v \in Z_h$. Then $(\cdot, \operatorname{div} v) = 0$ and $\int_{\Gamma_h} v \cdot n = 0$, so for every constant c ,

$$N_h(\tilde{u} - u_h, v) = \mathcal{R}(v) + \mathcal{G}(v), \quad \mathcal{R}(v) := -\langle \tilde{p} - c, v \cdot n \rangle.$$

6. THE PRINTED METHOD

Throughout this section $\gamma \geq \gamma_0$.

Test field. Let $\varphi_0(\theta) := \int_0^\theta (p - \bar{p})(\theta') d\theta'$, which is periodic, and $\varphi := \chi_1(r)\varphi_0(\theta)$, where $\chi_1 \equiv 1$ near $r = 1$ (so $\chi_1'(1) = 0$) and $\chi_1 \equiv 0$ near ∂R . Then $w := \operatorname{curl} \varphi$ has components $w_r = \partial_\theta \varphi / r$ and $w_\theta = -\partial_r \varphi$, and on Γ

$$w = (p - \bar{p})e_r = -(p - \bar{p})n.$$

Let Π_{k+1} be the C^1 piecewise- P_{k+1} interpolant of Morgan and Scott [34] (for $k = 4$, the Argyris interpolant [1]), which exists because $k+1 \geq 5$; its constants depend on the angles, hence on (M0) and (M2). Since $\varphi \equiv 0$ near ∂R and $h \leq h_*$, $v_\varphi := \operatorname{curl} \Pi_{k+1} \varphi \in Z_h$, and $\|v_\varphi - w\|_{W^{1,\infty}} \leq Ch^k$.

Theorem 6.1 (Energy norm: exact rate 1/2). The error satisfies the upper bound

$$\|\tilde{u} - u_h\| \leq C \left[(h_\Gamma/\gamma)^{1/2}G + (1 + \gamma^{1/2})h_\Gamma^{3/2} + \varepsilon_h \right].$$

If $G > 0$ and $h \leq h_0(G)$, with h_0 independent of $\gamma \geq \gamma_0$, it also satisfies the lower bound

$$\|\tilde{u} - u_h\| \geq (2M)^{-1}(h_\Gamma/\gamma)^{1/2}G - C \left[(1 + \gamma^{1/2})h_\Gamma^{3/2} + \varepsilon_h \right].$$

In Scott's variables, $(h_\Gamma/\gamma)^{1/2} = (h/\mu)^{1/2}$.

Proof of the upper bound. For $z \in W_h$, $e_h := z - u_h \in Z_h$. By Corollary 5.3 and Lemma 4.2,

$$c_1\|e_h\|^2 \leq N_h(z - \tilde{u}, e_h) + \mathcal{R}(e_h) + \mathcal{G}(e_h).$$

Using the Jacobian bound of Lemma 3.1, $|\mathcal{R}(e_h)| \leq (G + Ch_\Gamma^2)(h/\mu)^{1/2}\|e_h\|$. Then apply Lemmas 5.2 and 4.6. □

Proof of the lower bound. (1) We have $v_\varphi \cdot n_e = -(p - \bar{p})(x^*) \cdot n_e + O(h^k)$, $\mathcal{R} = -\langle \tilde{p} - \bar{p}, v \cdot n \rangle$ and $n(x^*) \cdot n_e = 1 + O(s^2)$. Hence

$$\mathcal{R}(v_\varphi) = \langle (\tilde{p} - \bar{p})(p - \bar{p})(x^*), n(x^*) \cdot n_e \rangle + O(h^k) = G^2 + O(h_\Gamma^2 + h^k).$$

$$(2) \|v_\varphi\|^2 \leq (\mu/h)(G + Ch_\Gamma^2 + Ch^k)^2 + C.$$

(3) By continuity, $M\|\tilde{u} - u_h\| \geq (|\mathcal{R}(v_\varphi)| - |\mathcal{G}(v_\varphi)|)/\|v_\varphi\|$, and Lemma 5.2 bounds $\mathcal{G}$.

(4) $|\mathcal{R}|/\|v_\varphi\| \geq (h/\mu)^{1/2}G(1 - C(h_\Gamma^2 + h^k)/G^2 - C(h_\Gamma^2 + h^k)/G - C(h_\Gamma/\gamma_0)^{1/2}/G)$, which is at least $\frac{1}{2}(h/\mu)^{1/2}G$ for $h \leq h_0(G)$. $\square$

The lower bound is carried by the boundary parts of $\|\cdot\|$. By Theorem 6.4, $\|\nabla(\tilde{u} - u_h)\|$ is only $O(h/\mu)$.

Corollary 6.2. *If $G > 0$, $\gamma h_\Gamma \ll G$, $\gamma^{1/2}h_\Gamma \ll G$ and $\varepsilon_h \ll (h_\Gamma/\gamma)^{1/2}G$, then $\|\tilde{u} - u_h\| \asymp (h/\mu)^{1/2}$. The mesh-dependent energy rate is exactly 1/2.*

The two conditions on γ make the terms $\gamma^{1/2}h_\Gamma^{3/2}$ and $h_\Gamma^{3/2}$ of Theorem 6.1 small against $(h_\Gamma/\gamma)^{1/2}G$. The second is implied by the first when $\gamma \geq 1$, which holds for $\gamma \geq \gamma_0$ as soon as $\gamma_0 \geq 1$.

Lemma 6.3 (Normal-load cancellation). *Let S be edgewise $W^{1,\infty}$ on Γ_h , continuous at the vertices of Γ_h , with $|S \cdot \tau_e| \leq Ch_\Gamma$ on every $e \in \mathcal{E}_h^\Gamma$. (For instance, S smooth with $S \cdot \tau = 0$ on Γ .) Then for $v \in Z_h$,*

$$|\langle S, \partial_n v \rangle| \leq C(\|v\|_{\Gamma_h} + h_\Gamma^{1/2}\|\nabla v\|) \leq C\|\nabla v\|.$$

Proof. (1) *Tangential part.* By Lemma 4.1 and (M0), $|\langle (S \cdot \tau)\tau, \partial_n v \rangle| \leq Ch_\Gamma\|\partial_n v\|_{\Gamma_h} \leq Ch_\Gamma^{1/2}\|\nabla v\|$.

(2) *Normal part.* On e , v is a polynomial on T_e with $\operatorname{div} v = 0$, so $\partial_n v \cdot n_e = -\partial_\tau(v \cdot \tau_e)$. With $\sigma_e := S \cdot n_e$,

$$\int_e \sigma_e \partial_n v \cdot n_e = \int_e \partial_\tau \sigma_e (v \cdot \tau_e) - [\sigma_e v \cdot \tau_e]_{\partial e}.$$

(3) At a polygon vertex x shared by e and e' , the endpoint terms combine to $v(x) \cdot (\sigma_{e'}\tau_{e'} - \sigma_e\tau_e)$, of size at most $C(|e| + |e'|)|v(x)|$.

(4) $\sum_e |e|\|v\|_{L^\infty(e)} \leq C \sum_e |e|^{1/2}\|v\|_{L^2(e)} \leq C|\Gamma_h|^{1/2}\|v\|_{\Gamma_h}$.

(5) Finish with Lemma 3.2. $\square$

Numerically (Section 14), the dual norm of $v \mapsto \langle S, \partial_n v \rangle$ stays bounded on Z_h for normal S . It grows like $h_\Gamma^{-1/2}$ on V_h^R (no divergence constraint), and also on Z_h for tangential S . Incompressibility is exactly what the proof uses.

Theorem 6.4 (H^1 seminorm: rate 1). *With $C_p := C(1 + \|p - \bar{p}\|_{W^{1,\infty}(\Gamma)})$,*

$$\|\nabla(\tilde{u} - u_h)\| \leq C_p(h_\Gamma/\gamma) + C\left[(1 + \gamma^{1/2})h_\Gamma^{3/2} + \varepsilon_h\right].$$

Proof. (1) *Split.* For $z \in W_h$, write $z - u_h = \delta_R + \delta_G$, where $\delta_R, \delta_G \in Z_h$ solve

$$N_h(\delta_R, v) = \mathcal{R}(v), \quad N_h(\delta_G, v) = \mathcal{G}(v) - N_h(\tilde{u} - z, v) \quad \forall v \in Z_h.$$

Lax–Milgram on Z_h (Lemma 4.2) gives existence and uniqueness.

(2) *Geometric part.* $\|\nabla \delta_G\| \leq \|\delta_G\| \leq c_1^{-1}(C(1 + \gamma^{1/2})h_\Gamma^{3/2} + ME_h)$.

(3) *Ansatz.* Heuristically $\delta_R \approx -(h/\mu)(p - \bar{p})n = (h/\mu)w$ on Γ_h . Set $S_\varphi := v_\varphi \in Z_h$ and $r_h := \delta_R - (h/\mu)S_\varphi \in Z_h$. Then $N_h(r_h, v) = m(v) + \ell(v)$ with

$$m(v) := \mathcal{R}(v) - \langle S_\varphi, v \rangle = -\langle (\tilde{p} - \bar{p})n + S_\varphi, v \rangle,$$

$$\ell(v) := -(h/\mu)[a_h(S_\varphi, v) - \langle \partial_n S_\varphi, v \rangle - \langle S_\varphi, \partial_n v \rangle].$$

(4) *Bound on m .* On e ,

$$(\tilde{p} - \bar{p})n_e + S_\varphi = (p - \bar{p})(x^*)(n_e - n(x^*)) + O(h_\Gamma^2) + O(h^k) = (p - \bar{p})(m_e^*)s\tau_e + O(h_\Gamma^2) + O(h^k).$$

The leading term is odd in s with a constant coefficient. The argument of Lemma 3.5 handles it, and the $O(h^k)$ remainder contributes $Ch^k\|v\|_{L^1(\Gamma_h)}$. Hence $|m(v)| \leq C_p(h_\Gamma^{3/2} + h^k)\|\nabla v\|$.

(5) *Bound on ℓ .* $|a_h(S_\varphi, v)| + |\langle \partial_n S_\varphi, v \rangle| \leq C_p\|\nabla v\|$. By Lemma 6.3 applied to $S = w$,

$$|\langle S_\varphi, \partial_n v \rangle| \leq |\langle w, \partial_n v \rangle| + |\langle S_\varphi - w, \partial_n v \rangle| \leq C_p\|\nabla v\| + Ch^k h_\Gamma^{-1/2}\|\nabla v\|.$$

So $|\ell(v)| \leq (h/\mu)(C_p + Ch^k h_\Gamma^{-1/2})\|\nabla v\|$.

(6) *Close.* Testing with $v = r_h$,

$$c_1\|r_h\|^2 \leq [C_p(h/\mu + h_\Gamma^{3/2} + h^k) + C(h/\mu)h^k h_\Gamma^{-1/2}]\|\nabla r_h\|.$$

The last two terms are at most $C\varepsilon_h$, since $(h/\mu)h^k h_\Gamma^{-1/2} = \gamma^{-1}h^k h_\Gamma^{1/2} \leq Ch^k$. Hence

$$\|\nabla \delta_R\| \leq (h/\mu)\|\nabla S_\varphi\| + \|\nabla r_h\| \leq C_p(h_\Gamma/\gamma) + C(h_\Gamma^{3/2} + \varepsilon_h). \quad \square$$

Remark 6.5 (Why rate 1 and not 1/2). Without incompressibility, Lemma 6.3 would give only $Ch_\Gamma^{-1/2}$. The H^1 error would then be $(h/\mu)h_\Gamma^{-1/2}$: rate 1/2, with element-scale oscillation in the trace. The inconsistency $-pn$ is purely normal, and $\operatorname{div} v = 0$ converts $\partial_n v \cdot n$ into a tangential derivative. That is why GS still converges at rate 1 in H^1 .

Theorem 6.6 (Boundary slip; H^1 lower bound). *Let $\gamma \geq \gamma_0$ and $G > 0$, and put $X := C_p(h_\Gamma/\gamma) + (1 + \gamma^{1/2})h_\Gamma^{3/2} + \varepsilon_h$, the right-hand side of Theorem 6.4. Suppose that, for a fixed small δ ,*

$$X \leq \delta G \min(G, h_\Gamma^{1/2}), \quad h_\Gamma/\gamma \leq \delta G, \quad h_\Gamma \leq h_0(G).$$

All three hold for fixed γ and bounded ρ as $h_\Gamma \rightarrow 0$. Then

$$(i) \quad c(h_\Gamma/\gamma)G \leq \|\tilde{u} - u_h\|_{L^2(\Gamma_h)} \leq C(h_\Gamma/\gamma)^{1/2}\|\tilde{u} - u_h\|;$$

$$(ii) \quad \|\nabla(\tilde{u} - u_h)\| \geq c(h_\Gamma/\gamma)G - Ch^{k+1/2}.$$

Combined with Theorem 6.4, $\|\nabla(\tilde{u} - u_h)\| \asymp h/\mu$ in this regime.

Proof of (i). Let $e_u := \tilde{u} - u_h$ and $v := v_\varphi$. Corollary 5.3 rearranges to

$$\frac{\mu}{h}\langle e_u, v \rangle = \mathcal{R}(v) + \mathcal{G}(v) - a_h(e_u, v) + \langle \partial_n e_u, v \rangle + \langle e_u, \partial_n v \rangle.$$

We bound the terms, using that v is smooth up to $O(h^k)$.

$$(1) \quad \mathcal{R}(v) = G^2 + O(h_\Gamma^2 + h^k).$$

$$(2) \quad |\mathcal{G}(v)| \leq C(h_\Gamma^{3/2} + \gamma h_\Gamma^3 + \gamma h_\Gamma h^k). \text{ On the chord, } \tilde{u} = \pm d \partial_n u(x^*) + O(d^2) \text{ is tangential while } v \approx -(p - \bar{p})n \text{ is normal. Hence } \tilde{u} \cdot v = O(h_\Gamma^4 + h_\Gamma^2 h^k), \text{ the second term coming from } v_\varphi - w.$$

$$(3) \quad |a_h(e_u, v)| \leq C_p\|\nabla e_u\| \leq C_p X \leq C_p \delta G^2, \text{ by Theorem 6.4.}$$

$$(4) \quad \text{By Theorem 6.1 and } h/\mu = h_\Gamma/\gamma \leq \delta G,$$

$$|\langle e_u, \partial_n v \rangle| \leq C_p\|e_u\|_{\Gamma_h} \leq C_p(h/\mu)^{1/2}\|e_u\| \leq C_p[(h/\mu)G + X] \leq 2C_p \delta G^2.$$

(5) We have

$$|\langle \partial_n e_u, v \rangle| \leq \|\partial_n(\tilde{u} - z)\|_{\Gamma_h}\|v\| + C_I(\rho/h)^{1/2}(\|\nabla(z - \tilde{u})\| + \|\nabla e_u\|)\|v\|_{\Gamma_h}.$$

By Theorem 6.4 and $(\rho/h)^{1/2} \leq (c_0 h_\Gamma)^{-1/2}$, this is at most $Ch_\Gamma^{-1/2} X G \leq C\delta G^2$.

For δ small, $|\langle e_u, v \rangle| \geq \frac{1}{2}(h/\mu)G^2$. Dividing by $\|v\|_{\Gamma_h} \leq G + Ch_\Gamma^2$ gives the lower bound. The upper bound is $\|e_u\|_{\Gamma_h} \leq (h/\mu)^{1/2}\|e_u\|$. $\square$

Proof of (ii). Let $E \in H^1(\Omega_h)$ lift $(g - g_I)|_{\partial R}$, supported near ∂R , with $E|_{\Gamma_h} = 0$ and $\|E\|_{H^1} \leq C\|g - g_I\|_{H^{1/2}(\partial R)} \leq Ch^{k+1/2}$. By Lemma 3.2, $\|e_u\|_{\Gamma_h} = \|e_u - E\|_{\Gamma_h} \leq C_{\text{tr}}(\|\nabla e_u\| + \|\nabla E\|)$. $\square$

Corollary 6.7 (Exact leading term). *Let $G > 0$ and $h_\Gamma \rightarrow 0$ with $h/\mu \rightarrow 0$, $\gamma h_\Gamma \rightarrow 0$ and $\varepsilon_h/(h_\Gamma/\gamma)^{1/2} \rightarrow 0$. Then*

$$\tilde{u} - u_h = (h/\mu)v_\varphi + \vartheta_h, \quad \|\vartheta_h\| = o((h/\mu)^{1/2}), \quad \|\vartheta_h\|_{\Gamma_h} = o(h/\mu),$$

and consequently

$$\frac{\|\tilde{u} - u_h\|_{L^2(\Gamma_h)}}{(h/\mu)G} \rightarrow 1, \quad \frac{\|\tilde{u} - u_h\|}{(h/\mu)^{1/2}G} \rightarrow 1.$$

On Γ_h , $v_\varphi \approx -(p - \bar{p})n$. So to leading order the printed method replaces the no-slip condition by the leak condition $(\mu/h)u_h \cdot n \approx p - \bar{p}$: the penalty, not a traction, balances the pressure.

Proof. Write $\tilde{u} - u_h = (\tilde{u} - z) + \delta_R + \delta_G$ in the notation of Theorem 6.4, with $\delta_R = (h/\mu)S_\varphi + r_h$ and $S_\varphi = v_\varphi$.

(1) From step 6 of Theorem 6.4,

$$c_1\|r_h\|^2 \leq [C_p(h/\mu + h_\Gamma^{3/2}) + C\varepsilon_h]\|r_h\|.$$

So $\|r_h\| \leq C_p(h/\mu + h_\Gamma^{3/2}) + C\varepsilon_h$ and $\|r_h\|_{\Gamma_h} \leq (h/\mu)^{1/2}\|r_h\|$.

(2) $\|\delta_G\| + \|\tilde{u} - z\| \leq C[(1 + \gamma^{1/2})h_\Gamma^{3/2} + \varepsilon_h]$, and the Γ_h norm carries an extra factor $(h/\mu)^{1/2}$.

(3) Relative to $(h/\mu)^{1/2}G$ (energy) and $(h/\mu)G$ (slip), steps 1–2 contribute

$$O((h/\mu)^{1/2}/G + (\gamma^{1/2} + \gamma)h_\Gamma/G + \varepsilon_h(\gamma/h_\Gamma)^{1/2}/G) \rightarrow 0.$$

(4) By Lemma 3.1 (Jacobian and displacement of Γ_h from Γ), $\|v_\varphi\|_{\Gamma_h} = G(1 + O(h_\Gamma^2)) + O(h^k)$ and $\|v_\varphi\|^2 = (\mu/h)\|v_\varphi\|_{\Gamma_h}^2 + O(1)$. $\square$

This gives the sharp answer to Scott's question for the printed method. The estimate $h_\Gamma^{3/2} + h^k$ fails exactly when p is not constant on Γ . The error is then the explicit field $(h/\mu)v_\varphi$, with the universal constant 1 in both the slip norm and the energy norm.

6.1. The H^1 leak constant. Corollary 6.7 identifies the slip and energy constants of GS (both $\rightarrow 1$), but not the H^1 constant $\|\nabla(\tilde{u} - u_h)\| \mu/(hG)$, because the remainder $r_h = \delta_R - (h/\mu)S_\varphi$ of Theorem 6.4 has the same H^1 order as $(h/\mu)S_\varphi$. We identify the limit of $(\mu/h)\delta_R$: it is the Stokes flow in Ω driven by the leak as *Dirichlet* data. The H^1 constant is $K = \|\nabla U\|/G$, a global quantity that depends on the outer box (it tends to $3/\sqrt{7}$ as $L \rightarrow \infty$ for the pressure of test P). For test P and $L = 2.5$ a converged series computation gives $K = 2.7565157$, inside the rigorous bracket $[2.0746, 3.0526]$ of Proposition 6.13.

6.1.1. *The limit problem.*

Definition 6.8 (Leak flow). Let (U, P) solve

$$-\Delta U + \nabla P = 0, \quad \text{div } U = 0 \text{ in } \Omega, \quad U = (p - \bar{p})e_r = -(p - \bar{p})n \text{ on } \Gamma, \quad U = 0 \text{ on } \partial R,$$

and put $K := \|\nabla U\|_{L^2(\Omega)}/G$.

The data are compatible, since $\int_\Gamma U \cdot n = -\int_\Gamma (p - \bar{p}) = 0$. Hence (U, P) exists and is unique (P up to a constant), $U = \text{curl } \Psi$ with a single-valued stream function, and $U|_\Gamma = w|_\Gamma$ with $w = \text{curl } \varphi$ the test field above. U is smooth up to Γ (smooth boundary and data) and $U \in H^2(\Omega)$ (the corners of R are convex). Since $-\Delta U = -\nabla P$, for every divergence-free V with the same boundary values $\int \nabla U : \nabla(V - U) = -\int \nabla P \cdot (V - U) = 0$, so

$$\|\nabla U\|^2 = \min \left\{ \|\nabla V\|_{L^2(\Omega)}^2 : V \in H^1(\Omega)^2, \text{div } V = 0, V|_\Gamma = (p - \bar{p})e_r, V|_{\partial R} = 0 \right\}. \quad (1)$$

In particular K depends on $p|_\Gamma$ and on the outer boundary: it is not a local quantity.

6.1.2. Convergence. Extend Ψ smoothly across Γ to a fixed neighbourhood and put $\tilde{U} := \text{curl } \tilde{\Psi}$ on Ω_h , so $\text{div } \tilde{U} = 0$; extend P smoothly to $\tilde{P}$. Then $F_U := -\Delta \tilde{U} + \nabla \tilde{P}$ vanishes on Ω , is bounded, and is supported in the strip S_φ .

Lemma 6.9. *There is $z_U \in Z_h$ with $\|\tilde{U} - z_U\| \leq E_U := C(h\|U\|_{H^2(\Omega)} + (1 + (\gamma h_\Gamma)^{1/2})h^k + (1 + \gamma^{1/2})h_\Gamma^{\sigma_k - 1/2})$. For bounded γ , $E_U \leq Ch$, since $\sigma_k - \frac{1}{2} \geq k + \frac{1}{2}$.*

Proof. Repeat the proof of Lemma 4.6 with $\tilde{u}$ replaced by $\tilde{U}$ and $g = 0$. Away from Γ only $\tilde{U} \in H^2$ is available, so the interpolation step gives $\|\nabla(\tilde{U} - I_h \tilde{U})\| \leq Ch\|U\|_{H^2(\Omega)}$. Near Γ , $\tilde{U}$ is smooth and the boundary terms are as before. The net flux $m = \int_{\Gamma_h} (I_h \tilde{U} - \tilde{U}) \cdot n$ is $O(h_\Gamma^{\sigma_k})$ because $\int_{\Gamma_h} \tilde{U} \cdot n = \int_{\Omega_h} \text{div } \tilde{U} = 0$. As in step 3 of that proof, the flux bubble then costs $C|m|(h_\Gamma^{-1/2} + (\mu/h)^{1/2}) \leq C(1 + \gamma^{1/2})h_\Gamma^{\sigma_k - 1/2}$; the Bogovskii correction (Lemma 4.5) is unchanged. $\square$

Theorem 6.10 (The leak flow is the H^1 limit). *Let $\gamma \geq \gamma_0$ and let $\delta_R \in Z_h$ solve $N_h(\delta_R, v) = \mathcal{R}(v)$ for all $v \in Z_h$ (Theorem 6.4, step 1). Then*

$$\left\| \nabla \left(\frac{\mu}{h} \delta_R - \tilde{U} \right) \right\|_{\Omega_h} \leq \left\| \frac{\mu}{h} \delta_R - z_U \right\| + E_U \leq C_U \left[\min(\gamma h_\Gamma^{1/2}, (\gamma h_\Gamma)^{1/2}) + h_\Gamma^{1/2} + (h/\mu)^{1/2} + E_U \right],$$

where C_U depends on (M0)–(M2), k , L and on norms of p and U near Γ .

Proof. Put $\epsilon := \delta_R - (h/\mu)z_U \in Z_h$. For $v \in Z_h$,

$$N_h(\epsilon, v) = \mathcal{R}(v) - \frac{h}{\mu} N_h(\tilde{U}, v) + \frac{h}{\mu} N_h(\tilde{U} - z_U, v). \quad (2)$$

(1) *Green's formula for $\tilde{U}$.* As in Lemma 5.1, $\text{div } D(\tilde{U}) = \Delta \tilde{U}$, $\text{div } v = 0$, $v|_{\partial R} = 0$ and $\int_{\Gamma_h} v \cdot n = 0$ give, for any constant c ,

$$N_h(\tilde{U}, v) = (F_U, v) + \langle \sigma_U, v \rangle - \langle \tilde{U}, \partial_n v \rangle + \frac{\mu}{h} \langle \tilde{U}, v \rangle, \quad \sigma_U := (\nabla \tilde{U})^\top n - (\tilde{P} - c)n.$$

(2) *The penalty terms cancel to leading order.* With $\mathcal{R}(v) = -\langle (\tilde{p} - \bar{p})n, v \rangle$, (2) becomes

$$N_h(\epsilon, v) = -\langle \zeta, v \rangle - \frac{h}{\mu} \left[(F_U, v) + \langle \sigma_U, v \rangle - \langle \tilde{U}, \partial_n v \rangle \right] + \frac{h}{\mu} N_h(\tilde{U} - z_U, v), \quad \zeta := (\tilde{p} - \bar{p})n + \tilde{U}.$$

(3) *The mismatch ζ .* On $e \in \mathcal{E}_h^\Gamma$, by Lemma 3.1 and $\tilde{U}(x^*) = -(p - \bar{p})(x^*)n(x^*)$,

$$\zeta = (p - \bar{p})(x^*)(n_e - n(x^*)) + O(h_\Gamma^2) = (p - \bar{p})(m_e^*) s \tau_e + O(h_\Gamma^2),$$

exactly the functional m of Theorem 6.4, step 4 (without the h^k term). Hence $|\langle \zeta, v \rangle| \leq Ch_\Gamma^{3/2} \|\nabla v\|$. Alternatively $\|\zeta\|_{\Gamma_h} \leq Ch_\Gamma$ gives $|\langle \zeta, v \rangle| \leq Ch_\Gamma(h/\mu)^{1/2} \|v\|$.

(4) *The $O(h/\mu)$ terms.* $|(F_U, v)| \leq Ch_\Gamma^2 \|\nabla v\|$ (Lemma 3.3); $|\langle \sigma_U, v \rangle| \leq C\|v\|_{\Gamma_h} \leq C(h/\mu)^{1/2} \|v\|$. On Γ_h , $\tilde{U}$ is edgewise smooth, continuous at the vertices, and $\tilde{U} \cdot \tau_e = -(p - \bar{p})(x^*)n(x^*) \cdot \tau_e + O(h_\Gamma^2) = O(h_\Gamma)$. So Lemma 6.3 applies with $S = \tilde{U}$: $|\langle \tilde{U}, \partial_n v \rangle| \leq C(\|v\|_{\Gamma_h} + h_\Gamma^{1/2} \|\nabla v\|) \leq C((h/\mu)^{1/2} + h_\Gamma^{1/2}) \|v\|$. Finally $|N_h(\tilde{U} - z_U, v)| \leq ME_U \|v\|$ (Lemmas 4.2 and 6.9).

(5) *Close.* Test with $v = \epsilon$ and use Lemma 4.2:

$$c_1 \|\epsilon\| \leq C \left[\min(h_\Gamma^{3/2}, h_\Gamma(h/\mu)^{1/2}) + \frac{h}{\mu} (h_\Gamma^2 + (h/\mu)^{1/2} + h_\Gamma^{1/2} + E_U) \right].$$

Divide by h/μ , and use $h_\Gamma^{3/2} \mu/h = \gamma h_\Gamma^{1/2}$ and $h_\Gamma(\mu/h)^{1/2} = (\gamma h_\Gamma)^{1/2}$. The first inequality of the theorem is the triangle inequality with Lemma 6.9. $\square$

Corollary 6.11 (The H^1 constant). (i) For GS with general data, in the notation of Theorem 6.4,

$$\left\| \nabla \left(\frac{\mu}{h} (\tilde{u} - u_h) - \tilde{U} \right) \right\|_{\Omega_h} \leq (\text{bound of Theorem 6.10}) + C \frac{\mu}{h} \left[(1 + \gamma^{1/2}) h_\Gamma^{3/2} + \varepsilon_h \right].$$

(ii) Consequently, if $h_\Gamma \rightarrow 0$ with $h/\mu \rightarrow 0$, $\gamma h_\Gamma \rightarrow 0$, $\gamma(1 + \gamma^{1/2}) h_\Gamma^{1/2} \rightarrow 0$ and $(\mu/h) \varepsilon_h \rightarrow 0$ (all of which hold for fixed γ and bounded ρ), then

$$\frac{\|\nabla(\tilde{u} - u_h)\|_{\Omega_h}}{(h/\mu)G} \longrightarrow K = \frac{\|\nabla U\|_{L^2(\Omega)}}{G}.$$

(iii) On test P ($u = 0$, $f = \nabla p$) one has $\tilde{u} = 0$, $z = 0$ and $\mathcal{G} \equiv 0$, so $\tilde{u} - u_h = \delta_R$ exactly and Theorem 6.10 applies with no further condition.

(iv) The remainder of Theorem 6.4 converges: $(\mu/h) r_h \rightarrow U - w$ in H^1 . The limit is the Stokes correction of the cut-off field w : it vanishes on $\partial\Omega$ and solves $-\Delta(U - w) + \nabla P = \Delta w$. It depends on the cut-off χ_1 , whereas U does not.

Proof. (i) is $\tilde{u} - u_h = (\tilde{u} - z) + \delta_R + \delta_G$ with steps 2 of Theorem 6.4 and Lemma 4.6. (ii) Multiply out; $\|\nabla \tilde{U}\|_{\Omega_h}^2 - \|\nabla U\|_{\Omega}^2 = \|\nabla \tilde{U}\|_{S_h}^2 = O(h_\Gamma^2)$. (iii) $F = \tilde{f} - \nabla \tilde{p} = 0$ and all terms of $\mathcal{G}$ contain $\tilde{u}$. (iv) $r_h = \delta_R - (h/\mu)v_\varphi$ and $\|\nabla(v_\varphi - w)\| \leq Ch^k$. $\square$

Remark 6.12 (The Dirichlet-dominated regime). The hypothesis $\gamma h_\Gamma = \mu h_\Gamma^2/h \rightarrow 0$ is the one of Corollary 6.7. It fails for the large- μ runs of Table 6 ($\gamma h_\Gamma \approx 240$ at $N = 96$, $\mu = 10^4$). There the discrete problem is close to its $\mu \rightarrow \infty$ limit, a discrete Dirichlet problem whose data $-(p - \bar{p})n_e$ jumps in direction at every vertex of Γ_h . The computations below show that $(\mu/h)\delta_R \rightarrow U$ in that regime too, with relative H^1 distance d_h decaying roughly like $h_\Gamma^{0.6} - h_\Gamma^{0.8}$, the decay slowing as N grows, but Theorem 6.10 does not cover it. Proving convergence uniformly in $\mu \geq \mu_0$ is open.

6.1.3. *Rigorous bracket and exact asymptotics.* For $\varrho > 1$ let $\mathcal{A}_\varrho = \{1 < r < \varrho\}$ and let $\mathcal{E}(\varrho)$ be the minimum (1) over $\mathcal{A}_\varrho$ (zero data on $r = \varrho$). Extending by zero shows that the minimum decreases as the domain grows.

Proposition 6.13. *If $\mathcal{A}_{\varrho_1} \subset \Omega \subset \mathcal{A}_{\varrho_2}$, then $\mathcal{E}(\varrho_2) \leq \|\nabla U\|^2 \leq \mathcal{E}(\varrho_1)$. In the annulus the Fourier modes decouple: if $\Psi|_\Gamma = \sum_n (a_n \cos n\theta + b_n \sin n\theta)$, then $\mathcal{E}(\varrho) = \sum_n (a_n^2 + b_n^2) E_n(\varrho)$, where $E_n(\varrho) = \pi \int_1^\varrho [f''^2 + 2n^2((f/r)')^2 + (f'/r - n^2 f/r^2)^2] r dr$, and f is the combination of $r^n, r^{-n}, r^{n+2}, r^{2-n}$ ($r, r^{-1}, r^3, r \log r$ for $n = 1$) with $f(1) = 1$, $f'(1) = 0$, $f(\varrho) = f'(\varrho) = 0$.*

For test P, $p - \bar{p} = -\frac{3}{4} \sin \theta + \frac{1}{4} \sin 3\theta + \frac{1}{2} \cos \theta$ on Γ , so $\Psi|_\Gamma = \frac{3}{4} \cos \theta + \frac{1}{2} \sin \theta - \frac{1}{12} \cos 3\theta$ and $G^2 = 7\pi/8$. With $\varrho_1 = L = 2.5$ and $\varrho_2 = L\sqrt{2}$ (40-digit quadrature):

$$\boxed{2.0746004 \leq K \leq 3.0525769}$$

The same formulas give the dependence on the box. $E_3(\varrho) \rightarrow 45\pi$ (the exterior clamped solution is $f = \frac{3}{2}r^{-1} - \frac{1}{2}r^{-3}$). $E_1(\varrho) \rightarrow \pi$: the $n = 1$ data split into a potential dipole (exterior energy exactly π) and a rigid translation, whose energy vanishes as $\varrho \rightarrow \infty$ (the Stokes paradox). The annulus values $E_1(\varrho) - \pi = 2.47, 0.87, 0.38$ at $\varrho = 10, 10^2, 10^4$ are consistent with the classical $1/\log \varrho$ rate, which we observe numerically here rather than prove. Hence

$$K(L) \longrightarrow \sqrt{\frac{(\frac{9}{16} + \frac{1}{4})\pi + \frac{45\pi}{144}}{7\pi/8}} = \frac{3}{\sqrt{7}} \approx 1.134 \quad (L \rightarrow \infty),$$

but only logarithmically: the annulus values are $K = 1.420, 1.242, 1.183$ at $\varrho = 10, 10^2, 10^4$. So the constant 2.7–2.8 of Table 6 belongs to the box $L = 2.5$, not to the method.

6.1.4. *The value for test P. Series solution.* Write $\Psi = \sum_{n \text{ odd} \leq n_{\max}} f_n(r) \cos n\theta$ for data $\cos \theta$ and $\cos 3\theta$ separately (the class of $\cos n\theta$, n odd, is invariant under both reflections of the square; the $\sin \theta$ datum is the rotation of the $\cos \theta$ one by $\pi/2$, and cross terms between the two reflection classes vanish). Imposing $\Psi = 0$ on ∂R , rather than an unknown constant, is justified by symmetry: for data $\cos n\theta$ with n odd, the reflection $x \mapsto -x$ maps Ω to itself and changes the sign of the data, so by uniqueness Ψ is odd under it, and its constant value on ∂R is zero; the $\sin \theta$ datum follows by rotation. The conditions on $r = 1$ are imposed exactly, mode by mode, and $\Psi = \partial_\nu \Psi = 0$ on ∂R by least squares at 16 000 Chebyshev-clustered points. The energy is integrated by Gauss quadrature on eight polar sectors. The value is stable to 8 digits under refinement of both the truncation $n_{\max}$ and the quadrature:

$n_{\max}$	max boundary residual	$\ \nabla U\ $	K
15	$4.4 \cdot 10^{-3}$	4.57012143	2.75644148
31	$6.1 \cdot 10^{-5}$	4.57024239	2.75651444
47	$5.1 \cdot 10^{-6}$	4.57024449	2.75651571
63	$1.1 \cdot 10^{-6}$	4.57024450	2.75651571
79	$1.8 \cdot 10^{-7}$	4.57024450	2.75651571

$$K_P = 2.7565157$$

This lies inside the rigorous bracket.

Computation: `code/h1_limit.py annulus`; `code/h1_limit.py mps`; output `logs/h1_limit_series.log`.

Finite element check. Our solver (SuperLU with refinement, as in Section 14) was run on test P, and the discrete field was compared *pointwise* with the series U at the quadrature points. Both the normalised H^1 error K_h and the relative distance $d_h := \|\nabla((\mu/h)(\tilde{u} - u_h) - \tilde{U})\|/\|\nabla U\|$ are shown below. The $\mu = 100$ and $N = 96$ values of K_h reproduce Table 6 to all printed digits.

N	$\mu = 10^2$		$\mu = 10^3$		$\mu = 10^4$		$\mu = 10^8$	
	K_h	d_h	K_h	d_h	K_h	d_h	K_h	d_h
16	2.4642	0.193	2.7844	0.318	3.0562	0.539	3.3254	0.717
24	2.5629	0.147	2.7913	0.259	2.9309	0.398	3.0452	0.497
32	2.6117	0.124	2.7880	0.222	2.8751	0.323	2.9339	0.385
48	2.6601	0.098	2.7808	0.179	2.8283	0.246	2.8514	0.277
64	2.6843	0.084	2.7759	0.154	2.8082	0.206	2.8208	0.226
96	2.7084	0.067	2.7701	0.124	2.7899	0.163	2.7959	0.175

Computation: `code/h1_limit.py fem`; output `logs/h1_limit_fem.log`.

The measured values 2.71–2.79 bracket the limit 2.7565 from both sides. The approach runs in two directions.

- At $\mu = 100$, K_h increases towards K and d_h decays like $h^{0.6}$ ($0.193 \rightarrow 0.067$ as h drops by a factor 5.3). This is consistent with Theorem 6.10: for fixed μ , the terms $(h/\mu)^{1/2}$ and $h_\Gamma^{1/2}$ are both $\propto h^{1/2}$, and $(\gamma h_\Gamma)^{1/2} \propto h^{1/2}$ too.
- At large μ (Remark 6.12), K_h decreases towards K from above. The excess is the energy of an element-scale oscillation that is nearly orthogonal to U : $K_h^2 \approx K^2(1 + d_h^2)$, for example $2.7565\sqrt{1 + 0.175^2} = 2.798$ against 2.796 at $N = 96$.

At fixed $N = 96$ the μ -dependence of Table 6 ($2.708 \rightarrow 2.770 \rightarrow 2.790$) is the crossover between the two regimes, not a sign of a different limit.

7. THE PRESSURE-CONSISTENT METHOD

Proposition 7.1 (The naive consistent form, and its L_0^2 variant). (a) Replace $\langle p_h - \bar{p}_\Gamma(p_h), v \cdot n \rangle$ by $\langle p_h, v \cdot n \rangle$. The resulting linear system is square ($V_h^R \times \Pi_h$ tested by $V_h^R \times \Pi_h$). It has the kernel vector $(0, 1)$ and is therefore singular. Its left null space is therefore nontrivial as well, and the system is solvable only for data satisfying one linear compatibility condition. (b) The L_0^2 variant. Restrict the pressure to $\Pi_h \cap L_0^2(\Omega_h)$ and test the continuity row only with $q \in \Pi_h \cap L_0^2(\Omega_h)$, keeping the momentum row $N_h(u_h, v) - (p_h, \operatorname{div} v) + \langle p_h, v \cdot n \rangle = (\tilde{f}, v)$ for all $v \in V_h^R$. Any solution has $\operatorname{div} u_h \equiv c$, a constant, with

$$c |\Omega_h| = \int_{\partial R} g_I \cdot \nu + \int_{\Gamma_h} u_h \cdot n,$$

n being the outward normal of Ω_h . If $c = 0$, then (u_h, p_h) solves the naive system of (a).

(c) Let $\gamma \geq \gamma_2$ (Theorem 7.2). The naive system of (a) is solvable if and only if the CNS* pressure satisfies $\bar{p}_\Gamma(p_h^{\text{CNS}^*}) = 0$, and then its solutions are $(u_h^{\text{CNS}^*}, p_h^{\text{CNS}^*} + \text{const})$. If moreover the L_0^2 variant is uniquely solvable, then $c = 0$ if and only if the naive system is solvable.

Proof. (a) For $v \in V_h^R$,

$$-(1, \operatorname{div} v) + \langle 1, v \cdot n \rangle = - \int_{\partial \Omega_h} v \cdot n + \int_{\Gamma_h} v \cdot n = 0,$$

and the continuity row vanishes at $u = 0$. A square matrix with a nontrivial kernel has a nontrivial left kernel.

(b) The momentum rows of (a) and (b) coincide. Since $\operatorname{div} u_h \in \Pi_h$ (Lemma 4.3) and $(q, \operatorname{div} u_h) = 0$ for all mean-free $q \in \Pi_h$, $\operatorname{div} u_h$ is constant; integrate over Ω_h , using $u_h = g_I$ on ∂R . If $c = 0$, then $(q, \operatorname{div} u_h) = 0$ for all $q \in \Pi_h$.

(c) Write B^* as in the proof of Theorem 7.2. For every $r \in \Pi_h$ the naive pressure form satisfies $-(r, \operatorname{div} v) + \langle r, v \cdot n \rangle = B^*(r - \bar{p}_\Gamma(r), v)$, and it is invariant under $r \mapsto r + \text{const}$. If (u, p) solves the naive system, then $(u, p - \bar{p}_\Gamma(p))$ solves CNS*, so by uniqueness (Theorem 7.2) $u = u_h^{\text{CNS}^*}$ and $p - \bar{p}_\Gamma(p) = p_h^{\text{CNS}^*}$, which has zero mean on Γ_h . Conversely, if $\bar{p}_\Gamma(p_h^{\text{CNS}^*}) = 0$, the two pressure forms agree at $p_h^{\text{CNS}^*}$. For the last statement: if the naive system is solvable, shift its pressure into L_0^2 ; this gives a solution of the L_0^2 variant with $\operatorname{div} u_h = 0$, which by uniqueness is the solution, so $c = 0$. The converse is (b). $\square$

The singularity is the analogue, for our square Scott–Vogelius system, of the ill-posedness that Liu, Neilan and Otus note for their boundary-multiplier method when the zero-flux constraint is omitted [32, Remark 3.2]. The L_0^2 variant is the standard way to fix the constant; in the fully weak setting of Frachon, Nilsson and Zahedi the same computation shows that $\operatorname{div} u_h$ equals the multiplier of the mean constraint [20, (5.7)–(5.9)]. With weak imposition nothing forces $\int_{\Gamma_h} u_h \cdot n = - \int_{\partial R} g_I \cdot \nu$, even for flux-corrected data, so generically $c \neq 0$ and the L_0^2 variant is *not* exactly divergence-free. Unique solvability of the L_0^2 variant reduces to the nonvanishing of a single scalar (for zero data, write $u_h = u_0 + c w_c$ with $\operatorname{div} w_c \equiv 1$; then $(u_0, p_h - \bar{p}_\Gamma(p_h))$ is c times a fixed CNS* solution, and the constraint $\bar{p}_\Gamma(\cdot) = 0$ on its pressure must force $c = 0$). It is generic and held in every computation below, but we do not prove it.

Numerically ($k = 4$, meshes of Section 14), the kernel of (a) is exactly one-dimensional: in a dense SVD of the naive system for test B with $\mu = 20$ and an element-wise L^2 -orthonormal pressure basis, at $N = 8$ and 12 the smallest singular values are $8 \cdot 10^{-17}$ and $3 \cdot 10^{-16}$, the next ones $2.4 \cdot 10^{-2}$ and $8.1 \cdot 10^{-3}$. For the L_0^2 variant with $\mu = 100$ (SuperLU, relative residual about $6 \cdot 10^{-16}$), $\operatorname{div} u_h$ is constant to round-off. For test A at $N = 16$ it has $c = 9.69 \cdot 10^{-4}$, so $\|\operatorname{div} u_h\|_{L^2(\Omega_h)} = |c| |\Omega_h|^{1/2} = 4.54 \cdot 10^{-3}$, and the CNS* pressure has $\bar{p}_\Gamma(p_h) = 7.5 \cdot 10^{-2} \neq 0$. For

test B at $N = 16$ and 32 , $c = -5.98 \cdot 10^{-5}$ and $-2.94 \cdot 10^{-5}$; there the L_0^2 variant and CNS* have H^1 velocity errors that agree to four digits.

Earlier versions of this paper reported that an iterated-penalty solver and a bordered direct solve of the naive form returned discrete divergence “about $4.5 \cdot 10^{-3}$ ”, and read this as a sign of incompatibility. A bordered solve that fixes the pressure mean computes exactly the L_0^2 variant of (b), and the value coincides with $|c| |\Omega_h|^{1/2} = 4.54 \cdot 10^{-3}$ for test A, $N = 16$, $\mu = 100$. The reported divergence is therefore the constant c of (b), not a symptom of an incompatible system. We have not re-run the original solves, so the identification of the test case rests on this numerical coincidence.

Let β be the uniform inf-sup constant of Lemma 4.5.

Theorem 7.2 (CNS*: well posed, with Scott’s rate). *There is $\gamma_2 \asymp \max(\gamma_0, \beta^{-2})$ such that for $\gamma \geq \gamma_2$ the method CNS* has a unique solution and*

$$\|\tilde{u} - u_h\| \leq C \left[(1 + \gamma^{1/2}) h_\Gamma^{3/2} + \varepsilon_h \right].$$

Consequently, for every fixed $k \geq 4$, for γ in a bounded range $[\gamma_2, \gamma_{\max}]$ and $h_\Gamma \geq h_\Omega^{2(\sigma_k - k)}$ (that is, $h_\Gamma \geq h_\Omega^4$ for even k and $h_\Gamma \geq h_\Omega^2$ for odd k),

$$\|\tilde{u} - u_h\|_{H^1(\Omega_h)} \leq C (h_\Gamma^{3/2} + h_\Omega^k).$$

This condition ensures that the flux term of ε_h is at most Ch^k . Without it, that single extra term is kept, and by the remark after Lemma 4.6 it is then genuinely present for the plain interpolant g_I . With the flux-corrected data $\tilde{g}_I$ of Definition 2.2 no proviso is needed: $\|\tilde{u} - u_h\|_{H^1(\Omega_h)} \leq C(h_\Gamma^{3/2} + h_\Omega^k)$ for every $\gamma \geq \gamma_2$, with C independent of γ and no relation between h_Γ and h_Ω (Theorem 7.22). This is the rate Scott conjectured, for general Stokes data. The L^2 part of the H^1 norm uses the lift E from the proof of Theorem 6.6(ii).

Proof. (1) *Consistency.* Write $B^*(q, v) := -(q, \operatorname{div} v) + \langle q - \bar{p}_\Gamma(q), v \cdot n \rangle$.

- For a constant c , $B^*(q + c, v) = B^*(q, v) - c \int_{\Gamma_h} v \cdot n$.
- $B^*(\tilde{p}, v)$ equals the pressure form of Lemma 5.1 minus $\bar{p}_\Gamma(\tilde{p}) \int_{\Gamma_h} v \cdot n$.
- Hence $p^* := \tilde{p} - \bar{p}_\Gamma(\tilde{p})$ satisfies $N_h(\tilde{u}, v) + B^*(p^*, v) = (\tilde{f}, v) + \mathcal{G}(v)$ for all $v \in V_h^R$.

With $e_u := \tilde{u} - u_h$ and $e_p := p^* - p_h$ this gives

$$N_h(e_u, v) + B^*(e_p, v) = \mathcal{G}(v) \quad \forall v \in V_h^R. \quad (3)$$

(2) *Split.* Write $e_p = \eta + \xi$ with $\eta := p^* - \pi_h p^*$ and $\xi \in \Pi_h$. For $v \in V_h^0$, $(\eta, \operatorname{div} v) = 0$ because $\operatorname{div} V_h^0 \subset \Pi_h$.

(3) *Pressure.* For $v \in V_h^0$, (3) gives

$$(\xi - \bar{\xi}, \operatorname{div} v) = N_h(e_u, v) - \mathcal{G}(v) = a_h(e_u, v) - \langle e_u, \partial_n v \rangle - \mathcal{G}(v).$$

Using $(\rho/\mu)^{1/2} \leq C$ and Lemma 5.2, the right-hand side is at most $C(\|e_u\| + h_\Gamma^{3/2}) \|\nabla v\|$.

Lemma 4.5 then gives $\beta \|\xi - \bar{\xi}\| \leq C(\|e_u\| + h_\Gamma^{3/2})$.

(4) *Velocity.* For $z \in W_h$, $e_h := z - u_h \in Z_h$. Test (3) with $v = e_h$. Since $(e_p, \operatorname{div} e_h) = 0$ and $\langle \bar{p}_\Gamma(e_p), e_h \cdot n \rangle = 0$,

$$c_1 \|e_h\|^2 \leq M E_h \|e_h\| + |\mathcal{G}(e_h)| + |\langle \eta, e_h \cdot n \rangle| + |\langle \xi - \bar{\xi}, e_h \cdot n \rangle|,$$

where

$$|\langle \eta, e_h \cdot n \rangle| \leq C h_\Gamma^k (h/\mu)^{1/2} \|e_h\|,$$

$$|\langle \xi - \bar{\xi}, e_h \cdot n \rangle| \leq C_I (\rho/h)^{1/2} \|\xi - \bar{\xi}\| (h/\mu)^{1/2} \|e_h\| \leq C(c_0 \gamma)^{-1/2} \|\xi - \bar{\xi}\| \|e_h\|.$$

- (5) *Absorb.* Insert step 3 and $\|e_u\| \leq E_h + \|e_h\|$. If $\gamma \geq \gamma_2$ with $C(c_0\gamma_2)^{-1/2}\beta^{-1} \leq c_1/2$, the pressure term is absorbed.
- (6) *Uniqueness.* Apply steps 3–5 to the homogeneous problem ($g_I = 0$, so $z = 0 \in W_h$). This gives $u_h = 0$, and then $p_h = c$ is constant. Now (3) reads $-c \int_{\Gamma_h} v \cdot n = 0$ for all v , and the edge bubble of Lemma 4.3 gives $c = 0$. The system is square, so uniqueness gives existence. $\square$

Remark 7.3 (Relation to the zero-flux formulation of Frachon, Nilsson and Zahedi). Let $V_h^{R,b} := \{v \in V_h^R : \int_{\Gamma_h} v \cdot n = 0\}$, which has codimension one (edge bubble, Lemma 4.3). Transcribed to our setting, the device of [20, Section 5.1] tests the momentum row with the naive form $-(p_h, \operatorname{div} v) + \langle p_h, v \cdot n \rangle$ only for $v \in V_h^{R,b}$, keeps the full continuity row, and fixes the pressure constant separately. On $V_h^{R,b}$ the naive and the CNS* pressure forms coincide, because $\langle \bar{p}_\Gamma(p_h), v \cdot n \rangle = \bar{p}_\Gamma(p_h) \int_{\Gamma_h} v \cdot n = 0$; and $B^*(p + c, w) = B^*(p, w) - c \int_{\Gamma_h} w \cdot n$ for the remaining direction w . Hence, for $\gamma \geq \gamma_2$: the CNS* solution solves the zero-flux formulation; conversely, if (u, p) solves the homogeneous zero-flux formulation, then $(u, p + c)$ solves homogeneous CNS* for the c that satisfies the w -row, so $u = 0$. So the two formulations have the *same velocity*, and pressures that differ by a constant. CNS* is a square packaging of the same device whose constant is fixed automatically, with $p_h \approx \tilde{p} - \bar{p}_\Gamma(\tilde{p})$.

7.1. A matching lower bound: the rate $h_\Gamma^{3/2}$ is sharp. Theorem 7.2 gives $\|\nabla(\tilde{u} - u_h)\| \leq \|\tilde{u} - u_h\| \leq C[(1 + \gamma^{1/2})h_\Gamma^{3/2} + \varepsilon_h]$ for CNS*, and Theorem 6.1 gives the same bound for GS when $G = 0$. Here we prove the reverse inequality in the H^1 seminorm whenever the wall shear is not identically zero.

Status of the hypothesis. The proof needs a local condition (M3) on the boundary stars (Assumption 7.4). For $k \geq 7$ it holds on every mesh satisfying (M0)–(M2) (Lemma 7.5), so the lower bound is unconditional there. For $4 \leq k \leq 6$ we have *not* proved (M3). It is a finite-dimensional, scale-invariant condition that we verified numerically for $k = 4$ at every boundary vertex of the meshes of Section 14 (Table 1); since the space it involves grows with k , this verification also covers $k = 5, 6$ on those meshes (Remark 7.6). For $4 \leq k \leq 6$ on other meshes, the lower bound is conditional on (M3). The proof does *not* go through the transposed-gradient term of Lemma 3.5; see Section 7.1.5 for why that route fails for CNS*.

7.1.1. Notation. On Γ let n be the unit normal pointing into D and τ the unit tangent, oriented so that $\tau(x^*) \cdot \tau_e > 0$ for $x \in e$ (τ_e is n_e rotated in the same sense). By Lemma 3.4, $\partial_n u = W\tau$ on Γ , where

$$W := \partial_n u \cdot \tau \quad (\text{the wall shear}).$$

For $e \in \mathcal{E}_h^\Gamma$, T_e is the triangle with edge e , s is arclength on e from its midpoint m_e , and $W_e := W(m_e^*)$. No triangle has two edges on Γ_h : two such edges would share a vertex z of Γ_h , and the angle of Ω_h at z is $\pi + O(h_\Gamma) > \pi$. Define

$$Y_h := Z_h \cap V_h^0, \quad Y_h^1 := \left\{ v \in Y_h : \int_e \partial_n v = 0 \quad \forall e \in \mathcal{E}_h^\Gamma \right\}.$$

7.1.2. The local assumption. Write $\mathcal{M}(v) := -\frac{1}{2} \sum_{e \in \mathcal{E}_h^\Gamma} \int_e s^2 \partial_n v \cdot \tau_e ds$. For a vertex z of Γ_h , let e_z be the edge of Γ_h that follows z counterclockwise; $z \mapsto e_z$ is a bijection between the vertices and the edges of Γ_h .

Assumption 7.4 (Boundary test fields). **(M3)** There are constants $\kappa_0 > 0$ and K_{ov}, C_s such that every vertex z of Γ_h carries a field $v_z \in Y_h^1$ with the following properties:

- (i) $\|\nabla v_z\| = 1$;
- (ii) v_z is supported in a patch ω_z of triangles within distance $C_s h_\Gamma$ of z , and every triangle lies in at most K_{ov} patches;

(iii) $\mathcal{M}(v_z) \geq \kappa_0 |e_z|^2$, with e_z as above.

Condition (iii) is invariant under scaling, since $\|\nabla v\|$ is scale invariant in two dimensions. Only the sign of v_z is chosen in (iii): $\mathcal{M}(-v) = -\mathcal{M}(v)$.

Lemma 7.5 ((M3) for $k \geq 7$). *If $k \geq 7$, (M3) holds with $\omega_z = T_{e_z}$. Here $K_{\text{ov}} = 1$ and κ_0 depends only on (M0).*

Proof. Let $T = T_{e_z}$, and let $\lambda_1, \lambda_2, \lambda_3$ be its barycentric coordinates with $\lambda_3 = 0$ on $e := e_z$. Put $b := \lambda_1 \lambda_2 \lambda_3$ and $\sigma := 2s/|e| = \lambda_1 - \lambda_2$ on e (up to sign), and set

$$\phi := b^2((\lambda_1 - \lambda_2)^2 - \frac{1}{7}) \in P_8, \quad v := \text{curl } \phi \in P_7^2.$$

- ϕ vanishes to second order on ∂T , so $v = 0$ on ∂T . Extended by zero, v is continuous, lies in $V_h^0 \subset V_h^R$ (because $k \geq 7$), and satisfies $\text{div } v = 0$. Hence $v \in Y_h$.
- On e , $\partial_n v \cdot \tau_e = \pm \partial_{nn} \phi = \pm 2|\nabla \lambda_3|^2(\lambda_1 \lambda_2)^2(\sigma^2 - \frac{1}{7}) = \pm \frac{1}{8}|\nabla \lambda_3|^2(1 - \sigma^2)^2(\sigma^2 - \frac{1}{7})$.
- Since $\int_{-1}^1 (1 - \sigma^2)^2 = \frac{16}{15}$ and $\int_{-1}^1 \sigma^2(1 - \sigma^2)^2 = \frac{16}{105}$, this has mean zero. With $\partial_n v \cdot n_e = 0$ (Lemma 7.7 below), $\int_e \partial_n v = 0$, so $v \in Y_h^1$.
- Its second moment is $\int_{-1}^1 \sigma^2(1 - \sigma^2)^2(\sigma^2 - \frac{1}{7}) d\sigma = \frac{16}{315} - \frac{16}{735} = \frac{64}{2205} \neq 0$.

The other two edges of T are interior edges and do not enter $\mathcal{M}$. Hence $\mathcal{M}(v) \neq 0$. The ratio $|\mathcal{M}(v)|/(|e|^2 \|\nabla v\|)$ is invariant under similarity. It is a continuous, nonvanishing function of the shape of T , and the shapes allowed by (M0) form a compact set, so the ratio is bounded below by some $\kappa_0 > 0$. Normalise and choose the sign. $\square$

Remark 7.6 ($k = 4$). For $k = 4$ no field in Y_h^1 can be supported in a single triangle, or in the two triangles of one first-layer quadrilateral; for the production meshes the latter space is $\{0\}$ (Section 7.1.6). The smallest natural patch is the boundary star ω_z (the triangles at z).

Write $v = \text{curl } \phi$ with ϕ a C^1 piecewise quintic; since $v = 0$ on $\partial \omega_z$ and on Γ_h , we may take $\phi = 0$ and $\nabla \phi = 0$ on $\partial \omega_z$ and on the two edges e, e' of Γ_h at z . On such an edge e , $\partial_n v \cdot \tau_e = \pm \partial_{n_e n_e} \phi|_{T_e}$ is a cubic with zero mean ($v \in Y_h^1$). At the far endpoint of e the gradient of $\phi|_{T_e}$ vanishes along two independent lines, so its Hessian vanishes there. If also $\partial_{n_e n_e} \phi|_{T_e}(z) = 0$, the cubic vanishes at both ends of e and has zero mean, so it is odd about the midpoint and $\int_e s^2 \partial_n v \cdot \tau_e = 0$. Hence (M3) requires $\partial_{n_e n_e} \phi|_{T_e}(z) \neq 0$ for at least one of the two edges e at z , the value being taken from the triangle T_e . This is possible only because C^1 quintics can have Hessians that jump across the interior edges at z : an Argyris (C^2 -vertex) stream function has a single Hessian at z , which vanishes on both boundary directions and hence is zero. Such jumps exist at a boundary vertex with at least three triangles, which is (M1').

We have not proved (M3) for $k = 4$ on general meshes. It is a finite-dimensional, scale-invariant condition on each boundary star, and we verified it numerically:

- for every boundary vertex of every production mesh ($N = 8, 12, 16, 20, 24, 32, 48, 64, 96, 128, 192, 256$) and also for $N = 512$ and 1024 (Table 1): $\dim Y^1(\omega_z) = 1$, $\min_z \kappa_z \geq 0.00288$ in every case, and $\min_z \kappa_z$ converges to about 0.0029 as N grows;
- for randomly generated three- and four-triangle boundary stars: $\dim Y^1(\omega_z) = 2m - 5$ for m triangles, and $\kappa_z > 0$ in every sample.

(M3) is monotone in the degree: a field in Y_h^1 for degree k is continuous piecewise $P_{k'}$ for every $k' \geq k$, and $\mathcal{M}$ and $\|\nabla \cdot\|$ do not depend on k . So (M3) for $k = 4$ on a mesh implies (M3) for $k = 5, 6$ on the same mesh, and the numerical verification above covers $4 \leq k \leq 6$ on the meshes of Section 14. It is a numerical verification in floating point, not a proof.

7.1.3. Three lemmas.

Lemma 7.7. *If $v \in Y_h$ and $e \in \mathcal{E}_h^\Gamma$, then $\partial_n v \cdot n_e = 0$ on e .*

Proof. $v|_{T_e}$ is a polynomial that vanishes on e , so $\partial_\tau v = 0$ on e . Then $0 = \operatorname{div} v = \partial_\tau(v \cdot \tau_e) + \partial_n(v \cdot n_e)$ on e . $\square$

Lemma 7.8 (The error identity on Y_h). *Let u_h be the velocity of CNS* ($\gamma \geq \gamma_2$), or of GS ($\gamma \geq \gamma_0$, any G), and let $e_u := \tilde{u} - u_h$. Then for all $v \in Y_h$,*

$$a_h(e_u, v) - \langle e_u, \partial_n v \rangle = -\langle \tilde{u}, \partial_n v \rangle - (F, v).$$

Proof. For CNS* use (3): $B^*(e_p, v) = -(e_p, \operatorname{div} v) + \langle e_p - \bar{p}_\Gamma(e_p), v \cdot n \rangle = 0$, because $\operatorname{div} v = 0$ and $v = 0$ on Γ_h . For GS use Corollary 5.3: $\mathcal{R}(v) = -\langle \tilde{p} - c, v \cdot n \rangle = 0$. In both cases $N_h(e_u, v) = \mathcal{G}(v)$. Since $v = 0$ on Γ_h , $N_h(e_u, v) = a_h(e_u, v) - \langle e_u, \partial_n v \rangle$, and in $\mathcal{G}(v)$ the transposed and penalty terms vanish, leaving $\mathcal{G}(v) = -\langle \tilde{u}, \partial_n v \rangle - (F, v)$. $\square$

Neither the pressure error nor the penalty enters this identity. It is simply the discrete momentum equation $a_h(u_h, v) - \langle u_h, \partial_n v \rangle = (\tilde{f}, v)$ on Y_h , which $\tilde{u}$ violates by $-\langle \tilde{u}, \partial_n v \rangle$ (Nitsche's symmetric term applied to the nonzero trace of $\tilde{u}$ on the chords) and by (F, v) .

Lemma 7.9 (Trace control on Y_h^1). *Let C_{PT} be the constant in $\|w - \bar{w}_T\|_{L^2(e)} \leq C_{PT}|e|^{1/2}\|\nabla w\|_T$ for $e \subset \partial T$, where $\bar{w}_T$ is the mean of w over T ; it depends only on shape regularity. Then for $w \in H^1(\Omega_h)^2$ and $v \in Y_h^1$,*

$$|\langle w, \partial_n v \rangle| \leq C_{PT} C_I \|\nabla w\| \|\nabla v\|.$$

Proof. On each e , $\int_e \partial_n v = 0$, so $\int_e w \cdot \partial_n v = \int_e (w - \bar{w}_{T_e}) \cdot \partial_n v$. Hence

$$|\langle w, \partial_n v \rangle| \leq \sum_e C_{PT} \|\nabla w\|_{T_e} |e|^{1/2} \|\partial_n v\|_{L^2(e)} \leq C_{PT} C_I \sum_e \|\nabla w\|_{T_e} \|\nabla v\|_{T_e},$$

using the local form of Lemma 4.1. Finish with Cauchy–Schwarz; the T_e are distinct. $\square$

Corollary 7.10 (Abstract lower bound). *With $\Lambda_0 := 2 + C_{PT} C_I$ and $e_u = \tilde{u} - u_h$ as in Lemma 7.8,*

$$\Lambda_0 \|\nabla e_u\| \geq \sup_{0 \neq v \in Y_h^1} \frac{|\langle \tilde{u}, \partial_n v \rangle + (F, v)|}{\|\nabla v\|}.$$

Proof. $|a_h(e_u, v)| \leq 2 \|\nabla e_u\| \|\nabla v\|$; apply Lemmas 7.8 and 7.9. $\square$

Lemma 7.11 (Leading term). *For $v \in Y_h^1$,*

$$\left| \langle \tilde{u}, \partial_n v \rangle - \ell_W(v) \right| \leq Ch_\Gamma^{5/2} \|\nabla v\|, \quad \ell_W(v) := -\frac{1}{2} \sum_{e \in \mathcal{E}_h^\Gamma} W_e \int_e s^2 \partial_n v \cdot \tau_e ds,$$

and $|(F, v)| \leq Ch_\Gamma^2 \|\nabla v\|$.

Proof. (1) *Geometry.* Let $x \in e$ and let $x^* = x/|x|$ be its nearest point on Γ . Then $x = x^* + d n(x^*)$ with $d = 1 - |x|$. Put $\ell = |e|$. Since $|x|^2 = 1 - \ell^2/4 + s^2$,

$$d = \frac{1}{2} \left(\frac{\ell^2}{4} - s^2 \right) + O(\ell^4).$$

(2) *Taylor expansion.* Using $\tilde{u}(x^*) = 0$ and Lemma 3.4,

$$\tilde{u}(x) = d W(x^*) \tau(x^*) + O(d^2).$$

Moreover $\tau(x^*) \cdot \tau_e = 1 + O(s^2)$ and $W(x^*) = W_e + O(|s|)$. Hence

$$\tilde{u} \cdot \tau_e = \frac{1}{2} \left(\frac{\ell^2}{4} - s^2 \right) W_e + O(h_\Gamma^3) \quad \text{on } e.$$

(3) *Pairing on one edge.* By Lemma 7.7, $\tilde{u} \cdot \partial_n v = (\tilde{u} \cdot \tau_e)(\partial_n v \cdot \tau_e)$ on e . The constant part $\frac{\ell^2}{8} W_e$ integrates to zero against $\partial_n v \cdot \tau_e$, whose mean vanishes because $v \in Y_h^1$. Therefore

$$\int_e \tilde{u} \cdot \partial_n v = -\frac{1}{2} W_e \int_e s^2 \partial_n v \cdot \tau_e + O(h_\Gamma^3) \|\partial_n v\|_{L^1(e)}.$$

- (4) *Summation.* $\sum_e \|\partial_n v\|_{L^1(e)} \leq \sum_e |e|^{1/2} \|\partial_n v\|_{L^2(e)} \leq C_I (\#\mathcal{E}_h^\Gamma)^{1/2} \|\nabla v\| \leq Ch_\Gamma^{-1/2} \|\nabla v\|.$
 (5) *The F term.* This is Lemma 3.3. $\square$

Lemma 7.12 (Assembling the test field). *Assume (M3), $W \not\equiv 0$, and $h_\Gamma \leq h_1$, where h_1 depends on $\|W\|_{W^{1,\infty}(\Gamma)}/\|W\|_{L^2(\Gamma)}$, on (M0), on C_I , and on the constants κ_0 , C_s , K_{ov} of (M3). Then $v := \sum_z W(z) v_z \in Y_h^1$ satisfies*

$$\ell_W(v) \geq \frac{1}{2} \kappa_0 c_0^{5/2} K_{\text{ov}}^{-1/2} \|W\|_{L^2(\Gamma)} h_\Gamma^{3/2} \|\nabla v\|.$$

Proof. (1) *Norm.* By bounded overlap, $\|\nabla v\|^2 \leq K_{\text{ov}} \sum_z W(z)^2$.

- (2) *Splitting ℓ_W .* Write $m_{z,e} := -\frac{1}{2} \int_e s^2 \partial_n v_z \cdot \tau_e$, so that $\sum_e m_{z,e} = \mathcal{M}(v_z)$. Then $|m_{z,e}| \leq \frac{h_\Gamma^2}{8} C_I$, and $m_{z,e} \neq 0$ only for the boundedly many edges e in ω_z . For those, $|W_e - W(z)| \leq Ch_\Gamma \|W\|_{W^{1,\infty}}$. Hence

$$\ell_W(v) = \sum_z W(z) \sum_e W_e m_{z,e} \geq \sum_z W(z)^2 \mathcal{M}(v_z) - Ch_\Gamma^3 \|W\|_{W^{1,\infty}} \sum_z |W(z)|.$$

- (3) *Lower bound.* By (M3)(iii) and (M0), $\mathcal{M}(v_z) \geq \kappa_0 c_0^2 h_\Gamma^2$. Moreover $\sum_z |W(z)| \leq \#\mathcal{E}_h^\Gamma \|W\|_\infty \leq Ch_\Gamma^{-1} \|W\|_\infty$.
 (4) *Riemann sum.* Let $\Sigma_W := \sum_z W(z)^2 |e_z|$. Since $z \mapsto e_z$ is a bijection onto the edges, this is a Riemann sum for $\int_\Gamma W^2$, and $|\Sigma_W - \|W\|_{L^2(\Gamma)}^2| \leq Ch_\Gamma \|W\|_{W^{1,\infty}}^2$; and $\sum_z W(z)^2 \in [\Sigma_W/h_\Gamma, \Sigma_W/(c_0 h_\Gamma)]$.
 (5) Combining,

$$\ell_W(v) \geq \kappa_0 c_0^2 h_\Gamma \Sigma_W - Ch_\Gamma^2 \|W\|_{W^{1,\infty}}^2, \quad \|\nabla v\| \leq (K_{\text{ov}} \Sigma_W / (c_0 h_\Gamma))^{1/2}.$$

For $h_\Gamma \leq h_1$ we have $\Sigma_W \geq \frac{3}{4} \|W\|_{L^2(\Gamma)}^2$, and the negative term is at most a quarter of the positive one. $\square$

7.1.4. The theorem.

Theorem 7.13 (H^1 lower bound). *Assume (M0)–(M2) and (D), and assume (M3); for $k \geq 7$ the latter holds automatically (Lemma 7.5), while for $4 \leq k \leq 6$ it is a hypothesis. Let $W = \partial_n u \cdot \tau \not\equiv 0$ on Γ . Let u_h be the velocity of CNS* with $\gamma \geq \gamma_2$, or of GS with $\gamma \geq \gamma_0$ and any G . Then, for $h_\Gamma \leq h_1$,*

$$\|\nabla(\tilde{u} - u_h)\|_{\Omega_h} \geq c_{\text{LB}} \kappa_0 \|W\|_{L^2(\Gamma)} h_\Gamma^{3/2} - Ch_\Gamma^2, \quad c_{\text{LB}} := \frac{c_0^{5/2}}{2\Lambda_0 K_{\text{ov}}^{1/2}}.$$

Here c_{LB} depends on c_0 , on K_{ov} from (M3), and on C_{PT} and C_I (hence on (M0)–(M2)); h_1 is as in Lemma 7.12; and C depends on $\|\tilde{u}\|_{W^{2,\infty}}$, $\|W\|_{W^{1,\infty}}$ and $\|F\|_\infty$. Neither depends on γ , μ , ρ , h_Ω , the pressure, or ε_h .

Proof. Take v from Lemma 7.12 in Corollary 7.10, and use Lemma 7.11:

$$\Lambda_0 \|\nabla e_u\| \geq \frac{\ell_W(v)}{\|\nabla v\|} - Ch_\Gamma^{5/2} - Ch_\Gamma^2.$$

Lemma 7.12 bounds the first term from below. $\square$

Corollary 7.14 (Sharpness of Theorem 7.2 and of Theorem 6.1 at $G = 0$). *Assume (M0)–(M3), $W \not\equiv 0$, $\gamma \in [\gamma_2, \gamma_{\text{max}}]$ and $\varepsilon_h \leq Ch_\Gamma^{3/2}$ (for instance $h_\Omega^k \leq h_\Gamma^{3/2}$ and the flux condition of Theorem 7.2). Then for CNS*, and for GS when p is constant on Γ ,*

$$ch_\Gamma^{3/2} \leq \|\nabla(\tilde{u} - u_h)\| \leq \|\tilde{u} - u_h\| \leq Ch_\Gamma^{3/2}.$$

So the H^1 rate is exactly 3/2. For GS with $G > 0$, Theorems 6.6 and 7.13 together give

$$\|\nabla(\tilde{u} - u_h)\| \geq c \max((h_\Gamma/\gamma)G, h_\Gamma^{3/2})$$

in the regime of Theorem 6.6.

- Remark 7.15.** (1) *Only the chordal bump matters.* The lower bound is carried by the part of $\tilde{u}|_{\Gamma_h}$ that oscillates on each chord, $-\frac{1}{2}s^2W_e$ minus its mean. It has amplitude h_Γ^2 and wavelength h_Γ , hence $H^{1/2}(\Gamma_h)$ -size $h_\Gamma^{3/2}$. No discrete velocity that is Nitsche-consistent on Y_h can reproduce it. The mean part $\frac{\ell^2}{12}W_e$ is smooth along Γ_h and contributes only $O(h_\Gamma^2)$.
- (2) *The terms that do not carry it.* The transposed-gradient term and the penalty term of $\mathcal{G}$ vanish identically on Y_h , as does every pressure term. The lower bound therefore holds for CNS* even though its pressure error is itself $O(h_\Gamma^{3/2})$.
- (3) *A method-independent statement.* Corollary 7.10 holds for *any* discrete velocity with $a_h(u_h, v) - \langle u_h, \partial_n v \rangle = (\tilde{f}, v)$ for all $v \in Y_h$. Every Nitsche variant with this volume form and zero boundary data, tested on Z_h , has this property.

7.1.5. *The route through the transposed-gradient term.* The natural first attempt tests with a field whose trace is odd on each chord, and uses $\mathcal{G}(v) \approx \langle s a_e, v \rangle$. We record what that route gives.

- (1) *The relevant trace is normal, not tangential.* Since $a_e = n(m_e^*)W_e$, we have $\langle s a_e, v \rangle = \sum_e W_e \int_e s v \cdot n_e + O(h_\Gamma^2)\|v\|_{L^1}$. An odd *tangential* trace pairs to zero at leading order.
- (2) *Incompressibility allows an odd normal trace.* For $v = \text{curl } \phi$, $v \cdot n_e = -\partial_\tau \phi$, so $\int_e v \cdot n_e = 0$ just says that ϕ takes equal values at the two ends of e . The identity $\partial_n v \cdot n_e = -\partial_\tau(v \cdot \tau_e)$ constrains the normal *derivative*, not the trace. So there is no obstruction there.
- (3) *The penalty term reinforces, at relative order γh_Γ .* On e ,

$$\tilde{u} \cdot n_e = dW(x^*)\tau(x^*) \cdot n_e + O(h_\Gamma^4) \quad \text{and} \quad (\nabla \tilde{u})^\top n_e \cdot n_e = W(x^*)\tau(x^*) \cdot n_e + O(h_\Gamma^2),$$

where $\tau(x^*) \cdot n_e = \pm s + O(s^3)$. So on the normal trace the penalty term multiplies the transposed term pointwise by $1 + (\mu/h)d(s)$, with $0 \leq (\mu/h)d \leq \gamma h_\Gamma/8$. It is lower order for fixed γ , of the same sign, and becomes leading once $\gamma \gtrsim h_\Gamma^{-1}$. On the tangential trace the penalty pairs $\frac{1}{2}(\frac{\ell^2}{4} - s^2)W_e$ with $v \cdot \tau_e$. This is not lower order unless $v \cdot \tau_e$ is odd or zero; choosing $\partial_n \phi = 0$ on Γ_h removes it.

- (4) *The Nitsche symmetric term is not lower order.* $-\langle \tilde{u}, \partial_n v \rangle$ pairs the chordal bump with $\partial_n v \cdot \tau_e$, which for a first-layer field of amplitude A is of size A/h_Γ . It has the same order $h_\Gamma A$ as the transposed term. This is the term Theorem 7.13 uses.
- (5) *Where the route fails: the pressure.* A test field with $v \cdot n \neq 0$ lies outside Y_h . For CNS* the error equation then contains $\langle \xi - \tilde{\xi}, v \cdot n \rangle$, which by Theorem 7.2 (steps 3–4) is only bounded by

$$C(c_0\gamma)^{-1/2}\beta^{-1}(\|e_u\| + h_\Gamma^{3/2})\|v\|.$$

That is the same order as the term one is trying to bound below, so it can cancel the lower bound unless γ is large compared with constants we do not control.

- (6) *What the route gives when it works.* For GS with $G = 0$, where $|\mathcal{R}(v)| \leq Ch_\Gamma^2\|v\|_{L^1(\Gamma_h)}$ is lower order, the route gives only $\|\tilde{u} - u_h\| \gtrsim h_\Gamma^{3/2}/(1 + \gamma^{1/2})$ in the *energy* norm. That lower bound may be carried entirely by the slip part $(\mu/h)^{1/2}\|e_u\|_{\Gamma_h}$. Numerically this part dominates: for GS on test A at $N = 64$, $\mu = 100$, it is 0.0795 against $\|\nabla e_u\| = 0.0182$. So it says nothing about H^1 .

7.1.6. *Numerical checks.* All computations use $k = 4$ and the meshes of Section 14. The script is `code/lower_bound_tests.py`.

Computation: `code/lower_bound_tests.py stars 8,12,...,1024`; output `logs/lower_bound_stars.log`.

The minimum is attained in the diagonal directions of the square, where the first-layer triangles are most elongated.

TABLE 1. (M3) on the production meshes: star constants $\kappa_z := \mathcal{M}(v_z)/|e_z|^2$, with $\|\nabla v_z\| = 1$ and v_z supported in the three triangles at z . All production levels $N = 8, \dots, 256$ were checked; selected rows are shown.

N	$\dim Y^1(\omega_z)$	$\min_z \kappa_z$	mean	$\max_z \kappa_z$
8	1	0.00288	0.0055	0.0080
16	1	0.00555	0.0109	0.0179
64	1	0.00366	0.0145	0.0252
256	1	0.00303	0.0149	0.0260
1024	1	0.00288	0.0149	0.0261

TABLE 2. The assembled field $v = \sum_z W(z)v_z$ of Lemma 7.12: $\langle \tilde{u}, \partial_n v \rangle / (\|\nabla v\| h_\Gamma^{3/2})$, with the leading term $\ell_W(v)$ in parentheses.

N	test A	rate	test B ₁	rate
32	0.0606 (0.0605)	–	0.0184 (0.0181)	–
64	0.0651 (0.0651)	1.39	0.0204 (0.0203)	1.35
128	0.0665 (0.0665)	1.47	0.0209 (0.0209)	1.46
256	0.0669 (0.0669)	1.49	0.0211 (0.0211)	1.49
512	0.0670 (0.0670)	1.50	0.0212 (0.0212)	1.50

Computation: `code/lower_bound_tests.py` field 16, ..., 512; output `logs/lower_bound_field.log`.

On the full spaces, for test A with $\mu = 100$ and $N = 16 \rightarrow 64$ (`logs/lower_bound_global.log`):

- The exact dual norm of $v \mapsto \langle \tilde{u}, \partial_n v \rangle$ on Y_h^1 has rates 1.12, 1.27, 1.35, 1.40; its ratio to $h_\Gamma^{3/2}$ increases from 0.086 to 0.115.
- It is between 14% and 24% of the actual CNS* error $\|\nabla(\tilde{u} - u_h)\|$.
- On Z_h , the energy-dual norms of $\mathcal{G}$ and of its transposed part decay at rates $1.70 \rightarrow 1.54$ and $1.61 \rightarrow 1.51$. So Lemma 3.5 is sharp on Z_h as well as on V_h^R .
- The $\|\nabla \cdot\|$ -dual norm of $\mathcal{G}$ on Z_h decays only like h_Γ^1 . It is dominated by the penalty part, which is of size γh_Γ in that norm. The $h_\Gamma^{3/2}$ upper bound genuinely needs the mesh-dependent norm.

7.2. Flux-corrected data, a γ -uniform H^1 bound, and a growing penalty. This subsection does three things. It proves an H^1 estimate that is uniform in the penalty (Theorem 7.19). With it, it removes the proviso of Theorem 7.2 for flux-corrected data (Theorem 7.22). And it asks whether a mesh-dependent penalty can rescue the printed method (Corollary 7.23, Theorem 7.24). The computations are in Section 14.10.

Throughout, μ may depend on the mesh. All constants below are independent of $\gamma \geq \gamma_0$ (or $\gamma \geq \gamma_2$ for CNS*) unless displayed; this is the uniformity of Lemma 4.2. When we speak of a *power-law penalty* we assume that h/h_Γ and ρ are bounded (as on the meshes of Section 14) and that

$$\gamma = \frac{\mu h_\Gamma}{h} \asymp h_\Gamma^{-\alpha}, \quad \text{equivalently} \quad \mu = \mu_0 \left(\frac{h_{\text{ref}}}{h} \right)^\alpha, \quad \alpha \geq 0.$$

Scott's normalisation, μ fixed, is $\alpha = 0$; the choice $\mu \propto h^{-1/2}$ is $\alpha = \frac{1}{2}$, i.e. $\mu/h \asymp h^{-3/2}$.

Remark 7.16 (Larger μ only helps stability). If $\mu' \geq \mu \geq \mu_0$ then, with $\|\cdot\|_\mu$ the norm of Section 2 at penalty μ ,

$$N_h^{\mu'}(v, v) = N_h^\mu(v, v) + \frac{\mu' - \mu}{h} \|v\|_{\Gamma_h}^2 \geq c_1 \|v\|_\mu^2 + \frac{\mu' - \mu}{h} \|v\|_{\Gamma_h}^2 \geq \min(c_1, 1) \|v\|_{\mu'}^2,$$

and the continuity constant M of Lemma 4.2 does not depend on μ . The absorption condition $\gamma \geq \gamma_2$ of Theorem 7.2 is also monotone. So coercivity, continuity and the CNS* pressure absorption are uniform for all $\mu \geq \mu_0$; a growing penalty never threatens stability. Its costs are in accuracy in the energy norm (Theorem 7.24) and in conditioning (Proposition 7.26).

7.2.1. A γ -uniform H^1 estimate. Let $Y_h = Z_h \cap V_h^0$ as in Section 7.1. Let $b = \sum_{e \in \mathcal{E}_h^\Gamma} \beta_e n_e$ be the flux bubble of Lemma 4.6, step 3, with $\int_{\Gamma_h} b \cdot n = |\Gamma_h|/6$, and recall that every $z \in W_h$ has $\int_{\Gamma_h} z \cdot n = -m_R$, $m_R = \int_{\partial R} g_I \cdot \nu$, $|m_R| \leq Ch^{\sigma_k}$ (remark after Lemma 4.6).

Lemma 7.17 (Divergence-free lift of a trace). *Let t be continuous and piecewise P_k on Γ_h , with $\int_{\Gamma_h} t \cdot n = 0$. There is $L \in Z_h$ with $L|_{\Gamma_h} = t$ and*

$$\|\nabla L\| \leq C_L h_\Gamma^{-1/2} \|t\|_{L^2(\Gamma_h)}.$$

Proof. (1) Let $L_0 \in V_h$ take the nodal values of t at the Lagrange nodes on Γ_h and vanish at all other nodes. It is supported in the triangles that meet Γ_h ; for $h \leq h_*$ these do not meet ∂R , so $L_0 \in V_h^R$.

(2) Each such triangle T has diameter $\asymp |e|$ for an adjacent $e \in \mathcal{E}_h^\Gamma$ (shape regularity, (M0)). The nodal values on e are bounded by $C|e|^{-1/2} \|t\|_{L^2(e)}$ (equivalence of norms on $P_k(e)$), and $\|\nabla \phi_i\|_{L^2(T)} \leq C$ for each nodal basis function (scale invariance in two dimensions). Summing over the boundedly overlapping triangles, $\|\nabla L_0\| \leq Ch_\Gamma^{-1/2} \|t\|_{\Gamma_h}$ (using $|e| \geq c_0 h_\Gamma$).

(3) $q := \operatorname{div} L_0 \in \Pi_h$, $\|q\| \leq \sqrt{2} \|\nabla L_0\|$, and $\int_{\Omega_h} q = \int_{\Gamma_h} L_0 \cdot n = \int_{\Gamma_h} t \cdot n = 0$. Lemma 4.5 gives $v \in V_h^0$ with $\operatorname{div} v = q$, $\|\nabla v\| \leq \beta^{-1} \|q\|$. Put $L := L_0 - v$. $\square$

Lemma 7.18 (The discrete polygonal Dirichlet problem). *Put $a := -6m_R/|\Gamma_h|$ and $W_h^D := \{z \in W_h : z|_{\Gamma_h} = ab\}$. Then $W_h^D \neq \emptyset$, there is a unique $u^D \in W_h^D$ with*

$$a_h(u^D, v) = (\tilde{f}, v) \quad \forall v \in Y_h,$$

and

$$\|\nabla(\tilde{u} - u^D)\| \leq C(h_\Gamma^{3/2} + h^k + |m_R| h_\Gamma^{-1/2}).$$

Proof. (1) A member of W_h^D close to $\tilde{u}$. Let $t := I_h \tilde{u}|_{\Gamma_h} - ab$ and let L_0 be the nodal extension of t from step 1 of Lemma 7.17. By Lemma 3.3, $\|I_h \tilde{u}\|_{\Gamma_h} \leq \|\tilde{u}\|_{\Gamma_h} + Ch_\Gamma^{k+1} \leq Ch_\Gamma^2$, and $|a||b|_{\Gamma_h} \leq C|m_R|$, so $\|\nabla L_0\| \leq Ch_\Gamma^{-1/2}(h_\Gamma^2 + |m_R|)$. The field $z_0 := I_h \tilde{u} - L_0$ has trace g_I on ∂R and ab on Γ_h , and

$$\int_{\Omega_h} \operatorname{div} z_0 = \int_{\partial R} g_I \cdot \nu + \int_{\Gamma_h} ab \cdot n = m_R + a|\Gamma_h|/6 = 0.$$

Since $\|\operatorname{div} z_0\| \leq \|\operatorname{div}(I_h \tilde{u} - \tilde{u})\| + \sqrt{2} \|\nabla L_0\| \leq C(h^k + h_\Gamma^{3/2} + |m_R| h_\Gamma^{-1/2})$, Lemma 4.5 gives $v_B \in V_h^0$ with $\operatorname{div} v_B = \operatorname{div} z_0$ and the same bound. Then $z := z_0 - v_B \in W_h^D$ and $\|\nabla(\tilde{u} - z)\| \leq C(h^k + h_\Gamma^{3/2} + |m_R| h_\Gamma^{-1/2})$.

(2) *Existence.* W_h^D is an affine space with direction Y_h , and a_h is coercive on Y_h (Lemma 3.2); so u^D exists and is unique.

(3) *Galerkin orthogonality.* For $v \in Y_h$, as in Lemma 5.1 but with $v = 0$ on all of $\partial\Omega_h$ and $\operatorname{div} v = 0$: $a_h(\tilde{u}, v) = (-\Delta \tilde{u}, v) = (\tilde{f} - F - \nabla \tilde{p}, v) = (\tilde{f} - F, v)$. Hence $a_h(\tilde{u} - u^D, v) = -(F, v)$.

- (4) *Céa*. With z from step 1, $y := z - u^D \in Y_h$ and $a_h(y, y) = a_h(z - \tilde{u}, y) - (F, y)$, so by Lemmas 3.2 and 3.3, $\|\nabla y\| \leq \kappa(2\|\nabla(z - \tilde{u})\| + Ch_\Gamma^2)$. $\square$

Theorem 7.19 (H^1 error, uniformly in the penalty). *Let $\gamma \geq \gamma_0$, with no upper bound. Then GS satisfies, with C_p as in Theorem 6.4,*

$$\|\nabla(\tilde{u} - u_h)\| \leq C_p \frac{h_\Gamma}{\gamma} + C(h_\Gamma^{3/2} + h^k + h^{\sigma_k} h_\Gamma^{-1/2}).$$

For CNS* with any $\gamma \geq \gamma_2$,

$$\|\nabla(\tilde{u} - u_h)\| \leq C(h_\Gamma^{3/2} + h^k + h^{\sigma_k} h_\Gamma^{-1/2}).$$

Compared with Theorems 6.4 and 7.2, the terms $\gamma^{1/2} h_\Gamma^{3/2}$, $(\gamma h_\Gamma)^{1/2} h^k$ and $h^{\sigma_k} (\mu/h)^{1/2}$ are gone. In particular the H^1 statement of Theorem 7.2 needs no upper bound $\gamma_{\max}$, and its proviso $h_\Gamma \geq h^{2(\sigma_k - k)}$ is exactly the condition $h^{\sigma_k} h_\Gamma^{-1/2} \leq h^k$, independent of μ .

Proof. GS.

- (1) *Remove the traction defect.* Fix $z \in W_h$ with $\|\tilde{u} - z\| \leq C\varepsilon_h$ (Lemma 4.6) and split $z - u_h = \delta_R + \delta_G$ as in Theorem 6.4. Steps 3–6 of that proof use only coercivity, continuity and Lemma 6.3, all uniform in $\gamma \geq \gamma_0$, and give

$$\|\nabla \delta_R\| \leq C_p(h_\Gamma/\gamma + h_\Gamma^{3/2} + h^k).$$

Put $u'_h := u_h + \delta_R = z - \delta_G \in W_h$. By construction $N_h(\tilde{u} - u'_h, v) = \mathcal{G}(v)$ for all $v \in Z_h$.

- (2) *The identity on Y_h .* For $v \in Y_h$ the penalty and transposed terms of N_h and $\mathcal{G}$ vanish (Lemma 7.8), leaving $a_h(\tilde{u} - u'_h, v) - \langle \tilde{u} - u'_h, \partial_n v \rangle = -\langle \tilde{u}, \partial_n v \rangle - (F, v)$. With step 3 of Lemma 7.18,

$$a_h(u'_h, v) - \langle u'_h, \partial_n v \rangle = (\tilde{f}, v) = a_h(u^D, v) \quad \forall v \in Y_h.$$

- (3) *The trace of u'_h is small.* By Lemma 3.3 and step 2 of Theorem 6.4 ($\|\tilde{u} - u'_h\| \leq \|\tilde{u} - z\| + \|\delta_G\| \leq C[(1 + \gamma^{1/2})h_\Gamma^{3/2} + \varepsilon_h]$),

$$\|u'_h\|_{\Gamma_h} \leq \|\tilde{u}\|_{\Gamma_h} + (h/\mu)^{1/2} \|\tilde{u} - u'_h\| \leq C(h_\Gamma^2 + (h_\Gamma/\gamma)^{1/2} \varepsilon_h),$$

because $(h/\mu)^{1/2} = (h_\Gamma/\gamma)^{1/2}$ and $(h_\Gamma/\gamma)^{1/2} \gamma^{1/2} h_\Gamma^{3/2} = h_\Gamma^2$. Moreover $h_\Gamma^{-1/2} (h_\Gamma/\gamma)^{1/2} \varepsilon_h = \gamma^{-1/2} \varepsilon_h \leq C(h^k + h^{\sigma_k} h_\Gamma^{-1/2})$: the two γ -growing parts of ε_h are exactly compensated, since $\gamma^{-1/2} (\gamma h_\Gamma)^{1/2} = h_\Gamma^{1/2}$ and $\gamma^{-1/2} (\mu/h)^{1/2} = h_\Gamma^{-1/2}$.

- (4) *Compare with u^D .* $w := u'_h - u^D \in Z_h$ (both lie in W_h), with $w|_{\Gamma_h} = u'_h - ab$. Let L be the lift of Lemma 7.17 of this trace (its flux vanishes because $w \in Z_h$), so $\|\nabla L\| \leq Ch_\Gamma^{-1/2}(\|u'_h\|_{\Gamma_h} + |m_R|)$. Then $y := w - L \in Y_h$ and, by step 2, $a_h(y, y) = \langle u'_h, \partial_n y \rangle - a_h(L, y)$. With Lemma 4.1 and $(\rho/h)^{1/2} \leq (c_0 h_\Gamma)^{-1/2}$,

$$\|\nabla w\| \leq \|\nabla L\| + \|\nabla y\| \leq Ch_\Gamma^{-1/2}(\|u'_h\|_{\Gamma_h} + |m_R|) \leq C(h_\Gamma^{3/2} + h^k + h^{\sigma_k} h_\Gamma^{-1/2}).$$

- (5) *Assemble.* $\tilde{u} - u_h = (\tilde{u} - u'_h) + \delta_R = (\tilde{u} - u^D) - w + \delta_R$; apply Lemma 7.18 and steps 1 and 4. CNS*. Lemma 7.8 gives step 2 for u_h itself, and Theorem 7.2 ($\gamma \geq \gamma_2$, any size) gives step 3 for u_h . Steps 4–5 are unchanged, with $\delta_R = 0$. $\square$

The mechanism is the classical one for Nitsche's method as $\mu \rightarrow \infty$: on the fields that vanish on Γ_h the method is exactly the strongly imposed discrete Dirichlet problem, and the only coupling to the penalty is through the trace of the discrete solution, which the energy estimate makes small by the factor $(h/\mu)^{1/2}$. The H^1 seminorm never sees the penalty directly.

7.2.2. Flux-corrected outer data: *Theorem 7.2 without the proviso.* Let $\tilde{g}_I$ and $\tilde{W}_h$ be as in Definition 2.2. Then $\int_{\partial R} \tilde{g}_I \cdot \nu = 0$ exactly (numerically, $|\int_{\partial R} \tilde{g}_I \cdot \nu| \leq 1.3 \cdot 10^{-15}$ in every run of Section 14.10), and $\|\tilde{g}_I - g\|_{L^\infty(\partial R)} \leq Ch^{k+1} + C|m_R| \leq Ch^{k+1}$, since $|m_R| \leq Ch^{\sigma_k}$ and $\sigma_k \geq k+1$; so the accuracy of the boundary data is unchanged. Every $z \in \tilde{W}_h$ has $\int_{\Gamma_h} z \cdot n = 0 = \int_{\Gamma_h} \tilde{u} \cdot n$; the obstruction of the remark after Lemma 4.6 is removed.

Proposition 7.20 (The flux floor). *For GS and CNS* with uncorrected data g_I and any μ ,*

$$\|\tilde{u} - u_h\| \geq \left(\frac{\mu}{h}\right)^{1/2} \|\tilde{u} - u_h\|_{\Gamma_h} \geq \left(\frac{\mu}{h}\right)^{1/2} \frac{|m_R|}{|\Gamma_h|^{1/2}},$$

with constant exactly 1. With flux-corrected data, $\int_{\Gamma_h} (\tilde{u} - u_h) \cdot n = 0$ and the second inequality is void.

Proof. $u_h \in W_h$, so $\int_{\Gamma_h} (\tilde{u} - u_h) \cdot n = 0 + m_R$; apply Cauchy–Schwarz. $\square$

Lemma 7.21 (Approximation with flux-corrected data). *Let $k \geq 4$. Then $\tilde{W}_h \neq \emptyset$ and*

$$\inf_{z \in \tilde{W}_h} \|\tilde{u} - z\| \leq C\tilde{\varepsilon}_h, \quad \tilde{\varepsilon}_h := (1 + (\gamma h_\Gamma)^{1/2})h^k + (1 + \gamma^{1/2})h_\Gamma^{\sigma_k - 1/2} + |m_R|.$$

Proof. Follow Lemma 4.6 with one change in step 1. Let ζ be a smooth cut-off equal to 1 near ∂R and 0 for $r \leq (1 + L)/2$, and put $w_h := I_h \tilde{u} - c I_h(\zeta x)$, $c := m_R/(8L^2)$. Then $w_h|_{\partial R} = \tilde{g}_I$, $w_h = I_h \tilde{u}$ near Γ_h , and $\|c I_h(\zeta x)\| = |c| \|\nabla I_h(\zeta x)\| \leq C|m_R|$ (no boundary terms on Γ_h). In step 2 the net flux becomes

$$\tilde{m} := \int_{\Omega_h} \operatorname{div} w_h = \int_{\partial R} \tilde{g}_I \cdot \nu + \int_{\Gamma_h} I_h \tilde{u} \cdot n = \int_{\Gamma_h} (I_h \tilde{u} - \tilde{u}) \cdot n,$$

using $\int_{\Gamma_h} \tilde{u} \cdot n = -\int_{\partial R} g \cdot \nu = 0$. On each chord $\tilde{u}$ is smooth and $I_h \tilde{u}$ is its equispaced P_k interpolant, so the Newton–Cotes argument of step 2 gives $O(|e|^{\sigma_k+1})$ per chord and $|\tilde{m}| \leq Ch_\Gamma^{\sigma_k}$ (not h^{σ_k} : only the chords of Γ_h contribute). Step 3 then costs $C|\tilde{m}|(h_\Gamma^{-1/2} + (\mu/h)^{1/2}) = C(1 + \gamma^{1/2})h_\Gamma^{\sigma_k - 1/2}$, and $\|\operatorname{div} w_h\| \leq C(h^k + |m_R|)$. Steps 4–5 are unchanged. $\square$

The remaining term $(1 + \gamma^{1/2})h_\Gamma^{\sigma_k - 1/2}$ is the flux of the *interpolation error on Γ_h* , and it is below $(1 + \gamma^{1/2})h_\Gamma^{k+1/2}$ since $\sigma_k \geq k+1$. The method itself has no flux defect left (its velocity and $\tilde{u}$ both carry zero flux through Γ_h), so this term is an artefact of the comparison field, not a floor.

Theorem 7.22 (Theorem 7.2 without the proviso). *Let $\gamma \geq \gamma_2$. CNS* with flux-corrected data satisfies*

$$\|\tilde{u} - u_h\| \leq C[(1 + \gamma^{1/2})h_\Gamma^{3/2} + \tilde{\varepsilon}_h], \quad \|\tilde{u} - u_h\|_{H^1(\Omega_h)} \leq C(h_\Gamma^{3/2} + h^k),$$

the second for every $\gamma \geq \gamma_2$, with C independent of γ , and every $h_\Gamma \leq h$ (no relation between h_Γ and h is needed). For bounded γ also $\|\tilde{u} - u_h\| \leq C(h_\Gamma^{3/2} + h^k)$. The same replacement $\varepsilon_h \rightarrow \tilde{\varepsilon}_h$ holds in Theorems 6.1–6.4 for GS, and in Theorem 7.19 the term $h^{\sigma_k} h_\Gamma^{-1/2}$ becomes $h_\Gamma^{\sigma_k - 1/2} + |m_R|$.

Proof. The consistency identities (Lemma 5.1, (3)) involve only test functions vanishing on ∂R and are unaffected. The proof of Theorem 7.2 uses the data only through E_h , now bounded by Lemma 7.21. For bounded γ , $\tilde{\varepsilon}_h \leq C(h^k + h_\Gamma^{k+1/2} + h^{k+1})$. For the H^1 bound uniform in γ , repeat Theorem 7.19 with $\tilde{g}_I$ in place of g_I . In Lemma 7.18 now $a = 0$ (u^D vanishes on Γ_h), the comparison field of step 1 is $z_0 = I_h \tilde{u} - c I_h(\zeta x) - L_0$ with L_0 the nodal extension of $I_h \tilde{u}|_{\Gamma_h}$, and $\int_{\Omega_h} \operatorname{div} z_0 = \int_{\partial R} \tilde{g}_I \cdot \nu = 0$; so $\|\nabla(\tilde{u} - u^D)\| \leq C(h_\Gamma^{3/2} + h^k + |m_R|)$. In step 3 of Theorem 7.19, $\gamma^{-1/2} \tilde{\varepsilon}_h \leq C(h^k + h_\Gamma^{\sigma_k - 1/2} + |m_R|)$, and in step 4 the term $|m_R| h_\Gamma^{-1/2}$ is absent. Altogether

$\|\nabla(\tilde{u} - u_h)\| \leq C(h_\Gamma^{3/2} + h^k + h_\Gamma^{\sigma_k-1/2} + |m_R|) \leq C(h_\Gamma^{3/2} + h^k)$. The L^2 part uses the lift of $(g - \tilde{g}_I)|_{\partial R}$, of size $C(h^{k+1/2} + |m_R|)$. $\square$

With Theorem 7.13 (which is indifferent to the outer data), Corollary 7.14 now holds with no flux condition: under (M3) and $W \neq 0$, $ch_\Gamma^{3/2} \leq \|\nabla(\tilde{u} - u_h)\| \leq Ch_\Gamma^{3/2}$ whenever $h^k \leq h_\Gamma^{3/2}$, for every $h_\Gamma \leq h$.

7.2.3. A growing penalty for the printed method. What Theorems 6.1 and 6.4 give. Insert $\gamma \asymp h_\Gamma^{-\alpha}$ into Theorem 6.4: $\|\nabla(\tilde{u} - u_h)\| \lesssim h_\Gamma^{1+\alpha} + h_\Gamma^{3/2-\alpha/2} + \varepsilon_h$. The two terms balance at $\alpha = \frac{1}{3}$, so Theorem 6.4 by itself gives at best the H^1 rate $\frac{4}{3}$, and it *loses* rate for $\alpha > \frac{1}{3}$. The culprit is the term $\gamma^{1/2}h_\Gamma^{3/2}$, which Theorem 6.4 inherits from the energy estimate of the geometric part δ_G (Lemma 5.2, penalty term). In the energy norm this term is genuine (Theorem 7.24); in the H^1 seminorm it is not (Theorem 7.19).

Corollary 7.23 (Rescue of the printed method in H^1). *Let $\gamma \geq \gamma_0$ and $\gamma \geq ch_\Gamma^{-1/2}$, that is,*

$$\mu \geq chh_\Gamma^{-3/2} \quad (\text{for } h \asymp h_\Gamma : \mu \gtrsim h^{-1/2}, \mu/h \gtrsim h^{-3/2}).$$

Then GS satisfies $\|\tilde{u} - u_h\|_{H^1(\Omega_h)} \leq C_p(h_\Gamma^{3/2} + h^k + h^{\sigma_k}h_\Gamma^{-1/2})$, and with flux-corrected data $\|\tilde{u} - u_h\|_{H^1(\Omega_h)} \leq C_p(h_\Gamma^{3/2} + h^k)$. Under the hypotheses of Theorem 7.13 ((M3), $W \neq 0$), $\|\nabla(\tilde{u} - u_h)\| \geq ch_\Gamma^{3/2}$ as well, so for $h^k \leq h_\Gamma^{3/2}$ (and, for uncorrected data, $h_\Gamma \geq h^{2(\sigma_k-k)}$) the H^1 rate of GS is exactly $\frac{3}{2}$, for general Stokes data. The L^2 part of the H^1 norm is handled by the lift E of Theorem 6.6(ii).

Proof. Theorems 7.19 and 7.22 with $h_\Gamma/\gamma \leq c^{-1}h_\Gamma^{3/2}$; the lower bound is Theorem 7.13, which holds for GS with any $\gamma \geq \gamma_0$ and any G . $\square$

The energy norm cannot be rescued. Let $W = \partial_n u \cdot \tau$ be the wall shear (Section 7.1).

Theorem 7.24 (The penalty consistency error is real in the energy norm). *Let u_h be the velocity of GS ($\gamma \geq \gamma_0$) or of CNS* ($\gamma \geq \gamma_2$), and $W \neq 0$. There are γ_1 and h_1 , depending on $\|W\|_{L^2(\Gamma)}$ and on the data, such that for $\gamma \geq \gamma_1$, $h_\Gamma \leq h_1$ and $h^k \leq h_\Gamma$,*

$$\|\tilde{u} - u_h\| \geq \left(\frac{\mu}{h}\right)^{1/2} \|\tilde{u} - u_h\|_{\Gamma_h} \geq c_E \|W\|_{L^2(\Gamma)} \gamma^{1/2} h_\Gamma^{3/2} - C(h_\Gamma/\gamma)^{1/2}, \quad c_E := \frac{c_0^2}{72}.$$

Proof. Let $e := \tilde{u} - u_h$.

- (1) *Tangential test field.* Let $\varphi_W := \pm \chi_1(r)(r-1)W(\theta)$, with χ_1 as in Section 6 and the sign chosen so that $w_W := \text{curl } \varphi_W = W\tau$ on Γ . As for v_φ , $v_W := \text{curl } \Pi_{k+1}\varphi_W \in Z_h$ and $\|v_W - w_W\|_{W^{1,\infty}} \leq Ch^k$. On a chord, $|x - x^*| \leq Ch_\Gamma^2$, so $v_W = W(x^*)\tau(x^*) + O(h_\Gamma^2 + h^k)$ and $v_W \cdot n_e = O(h_\Gamma)$; also $\|v_W\|_{\Gamma_h} \leq 2\|W\|_{L^2(\Gamma)}$ for small h .
- (2) *The discrete trace is small along v_W .* Test the discrete momentum equation with v_W ; the pressure term $(p_h, \text{div } v_W)$ vanishes. For GS,

$$\frac{\mu}{h} \langle u_h, v_W \rangle = (\tilde{f}, v_W) - a_h(u_h, v_W) + \langle \partial_n u_h, v_W \rangle + \langle u_h, \partial_n v_W \rangle.$$

Here $\|\nabla u_h\| \leq C + \|e\|$, $\|\partial_n u_h\|_{\Gamma_h} \leq \|\partial_n \tilde{u}\|_{\Gamma_h} + (c_0 h_\Gamma)^{-1/2} \|e\|$ (the last part of $\|\cdot\|$), and $\|u_h\|_{\Gamma_h} \leq Ch_\Gamma^2 + \|e\|_{\Gamma_h}$. Hence $\frac{\mu}{h} |\langle u_h, v_W \rangle| \leq C(1 + h_\Gamma^{-1/2} \|e\|)$. For CNS* the extra term $\langle p_h - \bar{p}_\Gamma(p_h), v_W \cdot n \rangle$ is at most $C_I(\rho/h)^{1/2} \|p_h\| \cdot Ch_\Gamma |\Gamma_h|^{1/2} \leq Ch_\Gamma^{1/2}$, since $\|p_h\|$ is bounded by Theorem 7.2.

(3) *The exact trace along v_W .* By Lemma 7.11, step 2, $\tilde{u} = dW(x^*)\tau(x^*) + O(d^2)$ on e with $d = \frac{1}{2}(\frac{|e|^2}{4} - s^2) + O(|e|^4) \geq 0$, and $\int_e d ds = |e|^3/12 + O(|e|^5)$. Hence

$$\langle \tilde{u}, v_W \rangle = \sum_e W_e^2 \frac{|e|^3}{12} + O(h_\Gamma^3 + h_\Gamma^2 h^k) \geq \frac{c_0^2 h_\Gamma^2}{12} \sum_e W_e^2 |e| - C h_\Gamma^3 \geq \frac{c_0^2 h_\Gamma^2}{12} (\|W\|_{L^2(\Gamma)}^2 - C h_\Gamma),$$

using $|e| \geq c_0 h_\Gamma$, the Riemann sum of Lemma 7.12, step 4, and $h^k \leq h_\Gamma$.

(4) *Close.* $\langle e, v_W \rangle = \langle \tilde{u}, v_W \rangle - \langle u_h, v_W \rangle$, and $\|e\| \geq (\mu/h)^{1/2} \|e\|_{\Gamma_h} \geq (\mu/h)^{1/2} \langle e, v_W \rangle / \|v_W\|_{\Gamma_h}$. With steps 2–3, $(\mu/h)^{1/2} h_\Gamma^2 = \gamma^{1/2} h_\Gamma^{3/2}$ and $(h/\mu)^{1/2} h_\Gamma^{-1/2} = \gamma^{-1/2}$,

$$\|e\| \geq \frac{c_0^2}{24} \|W\| \gamma^{1/2} h_\Gamma^{3/2} - C \gamma^{1/2} h_\Gamma^{5/2} - \frac{C}{\|W\|} \left((h_\Gamma/\gamma)^{1/2} + \gamma^{-1/2} \|e\| \right).$$

For $\gamma \geq \gamma_1 := (2C/\|W\|)^2$ the last term is at most $\frac{1}{2}\|e\|$; move it to the left, and take $h_\Gamma \leq h_1$ to absorb $C\gamma^{1/2} h_\Gamma^{5/2}$ into half of the first term. $\square$

Combined with Theorem 6.1 (upper bound, and lower bound $(2M)^{-1}(h_\Gamma/\gamma)^{1/2}G - C[(1 + \gamma^{1/2})h_\Gamma^{3/2} + \varepsilon_h]$) this pins down the energy error of GS for $G > 0$, $W \neq 0$:

$$\|\tilde{u} - u_h\| \asymp (h_\Gamma/\gamma)^{1/2}G + \gamma^{1/2}h_\Gamma^{3/2}\|W\| \quad \text{whenever } \gamma h_\Gamma \leq \lambda_0 \text{ or } \gamma h_\Gamma \geq \Lambda_0,$$

with λ_0 small and Λ_0 large, both depending on the data. In the window $\lambda_0 < \gamma h_\Gamma < \Lambda_0$ the upper bound is $\asymp h_\Gamma$ and we have no matching lower bound (the leak is normal and the geometric slip tangential, so we expect none is lost, but the constant in step 2 above is not controlled finely enough to exclude a partial cancellation). The same theorem shows that for CNS*, and for GS with $G = 0$, the energy error grows like $\gamma^{1/2}h_\Gamma^{3/2}$: bounded γ is optimal in the energy norm.

Corollary 7.25 (Rates of GS under a power-law penalty). *Let h/h_Γ and ρ be bounded, $\gamma \asymp h_\Gamma^{-\alpha}$ with $\alpha \geq 0$, $G > 0$, $W \neq 0$, (M3) (for $\alpha \geq \frac{1}{2}$), $h^k \leq h_\Gamma^{3/2}$. Then, as $h_\Gamma \rightarrow 0$:*

	H^1 seminorm	energy norm $\ \cdot\ $
$0 \leq \alpha < \frac{1}{2}$	$\asymp h_\Gamma^{1+\alpha}$	$\asymp h_\Gamma^{(1+\alpha)/2}$
$\frac{1}{2} \leq \alpha < 1$	$\asymp h_\Gamma^{3/2}$	$\asymp h_\Gamma^{(1+\alpha)/2}$
$\alpha = 1$	$\asymp h_\Gamma^{3/2}$	$\lesssim h_\Gamma$ ($\asymp h_\Gamma$ unless γh_Γ lies in (λ_0, Λ_0))
$\alpha > 1$	$\asymp h_\Gamma^{3/2}$	$\asymp h_\Gamma^{(3-\alpha)/2}$

So the H^1 rate is $\min(1+\alpha, \frac{3}{2})$, optimal for every $\alpha \geq \frac{1}{2}$, and the energy rate is $\min(\frac{1+\alpha}{2}, \frac{3-\alpha}{2}) \leq 1$, with maximum 1 at $\alpha = 1$. No penalty scaling gives GS the energy rate $\frac{3}{2}$ of CNS*.

Proof. H^1 , upper: Theorem 7.19. H^1 , lower: for $\alpha < \frac{1}{2}$, Theorem 6.6(ii); its hypotheses hold as $h_\Gamma \rightarrow 0$ because $X \asymp h_\Gamma^{1+\alpha} + h_\Gamma^{3/2-\alpha/2} + \varepsilon_h = o(h_\Gamma^{1/2})$ for $\alpha < 2$ and $h_\Gamma/\gamma = o(1)$. For $\alpha \geq \frac{1}{2}$, Theorem 7.13. Energy: Theorem 6.1 (both bounds; its lower bound is effective when $\gamma^{1/2}h_\Gamma^{3/2} = o((h_\Gamma/\gamma)^{1/2})$, i.e. $\alpha < 1$) and Theorem 7.24 (effective for $\alpha > 1$). The flux part of ε_h is $h^{\sigma_k}(h_\Gamma^{-1/2} + (\mu/h)^{1/2}) = o(h_\Gamma^{3/2})$ for $\alpha < 2\sigma_k - 4$. $\square$

The price, in the energy norm, is the product $(h_\Gamma/\gamma)^{1/2} \cdot \gamma^{1/2}h_\Gamma^{3/2} = h_\Gamma^2$: the leak and the penalty consistency error cannot both be below h_Γ . In the slip norm $\|\cdot\|_{L^2(\Gamma_h)}$ the growing penalty is harmless: $\|\tilde{u} - u_h\|_{\Gamma_h} \lesssim (h_\Gamma/\gamma)G + h_\Gamma^2$.

7.2.4. *The price in conditioning.* Work in the Lagrange nodal basis of V_h (Euclidean norm $|\cdot|$ on the free coefficients) and an L^2 -orthonormal basis of Π_h on each element. Let A_μ be the matrix of N_h on V_h^R , B that of $(q, \operatorname{div} v)$, $S_\mu := BA_\mu^{-1}B^\top$ the pressure Schur complement, and

$$K_\mu = \begin{pmatrix} A_\mu & B^\top \\ B & 0 \end{pmatrix}$$

the (symmetric form of the) GS saddle matrix. Then $q^\top S_\mu q = \sup_v (q, \operatorname{div} v)^2 / N_h(v, v)$.

Proposition 7.26 (Conditioning). *Let $\mu \geq \mu_0$ and let $\mathcal{T}_h$ be quasi-uniform, $h_{\min} \asymp h$.*

- (i) $c(1 + \gamma) \leq \lambda_{\max}(A_\mu) \leq C(1 + \gamma)$ and $ch^2 \leq \lambda_{\min}(A_\mu) \leq Ch^2$, uniformly in μ . Hence $\kappa_2(A_\mu) \asymp (1 + \gamma)h^{-2}$.
- (ii) $\lambda_{\max}(S_\mu) \leq 2/c_1$, and

$$\frac{c \min(\beta^2, 1)}{1 + \mu/h} \leq \lambda_{\min}(S_\mu) \leq \frac{|\Gamma_h|}{c_1 |\Omega_h|} \min\left(C_{\text{tr}}^2, \frac{h}{\mu}\right).$$

The small eigenvalue belongs to the constant pressure; on L_0^2 the spectrum stays above $c\beta^2$.

Hence $\kappa_2(S_\mu) \asymp 1 + \mu/h = 1 + \gamma/h_\Gamma$.

- (iii) *Every eigenvalue λ of K_μ satisfies $\min(\lambda_{\min}(A_\mu), \frac{1}{2}\lambda_{\min}(S_\mu)) \leq |\lambda| \leq \lambda_{\max}(A_\mu) + \|B\|$, so*

$$\kappa_2(K_\mu) \leq C(1 + \gamma) \max(h^{-2}, 1 + \mu/h).$$

Proof. (i) $|v|^{-2}\|v\|^2$ is bounded by $C(1 + \gamma)$: $\|\nabla v\|^2 + \sum_e |e| \|\partial_n v\|_e^2 \leq C|v|^2$ by scale invariance in two dimensions, and $(\mu/h)\|v\|_{\Gamma_h}^2 \leq (\mu/h)C \sum_e |e| \|v_e\|^2 \leq C\gamma|v|^2$. With $v^\top A_\mu v \leq M\|v\|^2$ this is the upper bound for $\lambda_{\max}$; the nodal function at a vertex of Γ_h gives the lower bound $v^\top A_\mu v \geq c_1(\mu/h)\|v\|_{\Gamma_h}^2 \geq c\gamma$. For $\lambda_{\min}$: $v^\top A_\mu v \geq c_1\|\nabla v\|^2 \geq c_1 C_F^{-2}\|v\|^2 \geq ch^2|v|^2$; an interior nodal function (vanishing on Γ_h , so the penalty does not see it) gives the upper bound by the standard argument for the stiffness matrix.

(ii) $(q, \operatorname{div} v)^2 \leq 2\|q\|^2\|\nabla v\|^2 \leq (2/c_1)\|q\|^2 N_h(v, v)$. Upper bound for $\lambda_{\min}$: take $q \equiv |\Omega_h|^{-1/2}$; then $(q, \operatorname{div} v) = |\Omega_h|^{-1/2} \int_{\Gamma_h} v \cdot n$ (as $v = 0$ on ∂R), and $(\int_{\Gamma_h} v \cdot n)^2 \leq |\Gamma_h| \|v\|_{\Gamma_h}^2 \leq |\Gamma_h| \min(C_{\text{tr}}^2 \|\nabla v\|^2, (h/\mu)\|v\|^2)$. Lower bound: write $q = q_0 + \bar{q}$ with $\bar{q}$ the mean. Lemma 4.5 gives $v_0 \in V_h^0$ with $\operatorname{div} v_0 = q_0$, $N_h(v_0, v_0) = a_h(v_0, v_0) \leq \beta^{-2}\|q_0\|^2$, and $(\bar{q}, \operatorname{div} v_0) = 0$. Let $v_c \in V_h^R$ be a fixed smooth field (interpolated) with $\int_{\Omega_h} \operatorname{div} v_c = |\Omega_h|$, corrected by Lemma 4.5 so that $\operatorname{div} v_c \equiv 1$; then $N_h(v_c, v_c) \leq C(1 + \mu/h)$. Test with $v = v_0 + tv_c$, $t = \bar{q}/(1 + \mu/h)$: with $X = \|q_0\|^2$ and $Y = \|\bar{q}\|^2/(1 + \mu/h)$ we get $(q, \operatorname{div} v) = X + Y$ and $N_h(v, v) \leq C'(X + Y)$, where C' depends on β^{-2} and $|\Omega_h|$. So $q^\top S_\mu q \geq (X + Y)/C' \geq \|q\|^2/(C'(1 + \mu/h))$ and, on L_0^2 , $\geq \|q_0\|^2/C'$.

(iii) Positive eigenvalues of a symmetric saddle matrix with A_μ positive definite are at least $\lambda_{\min}(A_\mu)$. For $\lambda < 0$ with eigenvector (x, y) , $y \neq 0$ and $B(A_\mu - \lambda)^{-1}B^\top y = -\lambda y$. Since $(A_\mu - \lambda)^{-1} \succeq (1 + |\lambda|/\lambda_{\min}(A_\mu))^{-1}A_\mu^{-1}$, either $|\lambda| \geq \lambda_{\min}(A_\mu)$ or $|\lambda| \geq \frac{1}{2}\lambda_{\min}(S_\mu)$. The upper bound is $\|K_\mu\|$, and $\|B\| \leq C$ in these bases. $\square$

Consequences. Relative to a fixed γ , the penalty $\gamma \asymp h_\Gamma^{-\alpha}$ multiplies $\kappa(A_\mu)$ by $h_\Gamma^{-\alpha}$ (so $h^{-2} \rightarrow h^{-5/2}$ for the H^1 -optimal $\alpha = \frac{1}{2}$) and $\kappa(S_\mu)$ by the same factor ($h^{-1} \rightarrow h^{-3/2}$). For $\alpha \leq 1$ the bound (iii) for the saddle matrix is $\asymp (1 + \gamma)h^{-2}$ asymptotically; for $\alpha > 1$ the constant-pressure mode dominates and $\kappa(K_\mu) \lesssim \gamma^2/h$, quadratic in the penalty. On the meshes of Section 14 the pressure-mode branch is already the active one (Section 14.10), because $\lambda_{\min}(S_\mu) \approx (|\Gamma_h|/|\Omega_h|)h/\mu$ with constant ≈ 1 while $\lambda_{\min}(A_\mu) \approx 0.03h^2$.

8. PRESSURE, BOUNDARY STRESS AND DRAG

All error theorems so far bound the velocity. This section adds pressure, boundary-traction and drag/lift/torque estimates for both methods; drag is what Gjerde and Scott use the method for

[24]. In brief: the CNS* pressure converges at the velocity rate $h_\Gamma^{3/2} + h_\Omega^k$, including its constant, while the GS pressure carries an explicit first-layer boundary layer and converges in L^2 only at the exact rate $\frac{1}{2}$. The variational (Babuška–Miller) drag equals the integral of the Nitsche flux; for GS its error is, to leading order, $-(h/\mu)$ times a reciprocity constant, for CNS* it is $O(h_\Gamma^2)$. The computations are in Section 14.9.

8.1. Setting, notation and summary.

8.1.1. *Setting.* Everything is in the setting of Section 2: $\Omega = R \setminus \overline{D}$, inscribed polygon Γ_h , (M0)–(M2), (D), $k \geq 4$, the forms a_h , N_h , the norm $\|\cdot\|$, the extensions $\tilde{u} = \text{curl } \tilde{\psi}$, $\tilde{p}$, and $F = \tilde{f} + \Delta \tilde{u} - \nabla \tilde{p}$ (supported in $\overline{S_h}$, $S_h = \Omega_h \setminus \overline{\Omega}$). Section 8.6 treats Navier–Stokes and Remark 8.26 general domains. Throughout, $\gamma \geq \gamma_0$ for GS and $\gamma \geq \gamma_2$ for CNS* (Theorem 7.2). We use $\theta = 0$ for GS and $\theta = 1$ for CNS*, so both methods read

$$N_h(u_h, v) - (p_h, \text{div } v) + \theta(p_h - \bar{p}_\Gamma(p_h), v \cdot n) = (\tilde{f}, v) \quad \forall v \in V_h^R, \quad (q, \text{div } u_h) = 0 \quad \forall q \in \Pi_h.$$

Further notation:

- $e_u := \tilde{u} - u_h$; $q := p - \bar{p}$ on Γ ; $G = \|q\|_{L^2(\Gamma)}$; C_p is as in Theorem 6.4; $x^* = x/|x|$ is the nearest point of Γ to $x \in \Gamma_h$.
- T_e is the triangle with edge $e \in \mathcal{E}_h^\Gamma$ (distinct for distinct e , Section 7.1), $H_e := 2|T_e|/|e|$ its height, and $L_1 := \bigcup_e T_e$ the *first layer*. By (M0), $\text{diam } T_e \leq Ch_\Gamma$, $c \leq |e|/H_e \leq C$ and $ch_\Gamma \leq |L_1| \leq Ch_\Gamma$.
- $p^\circ := \tilde{p} - \bar{p}_\Gamma(\tilde{p})$, where $\bar{p}_\Gamma(\cdot)$ is the mean over Γ_h (this is the p^* of Theorem 7.2), and $e_p := p^\circ - p_h$.
- $m_R := \int_{\partial R} g_I \cdot \nu$. Then $\int_{\Gamma_h} u_h \cdot n = -m_R$ and $\int_{\Gamma_h} \tilde{u} \cdot n = 0$, so $\int_{\Gamma_h} e_u \cdot n = m_R$; with flux-corrected data (Definition 2.2) the same holds with m_R replaced by $\int_{\partial R} \tilde{g}_I \cdot \nu = 0$; and $m_R = 0$ for the test data of Section 14.
- $B_h(y, w) := N_h(y, w) - a_h(y, w) = -\langle \partial_n y, w \rangle - \langle y, \partial_n w \rangle + \frac{\mu}{h} \langle y, w \rangle$.
- The error sizes

$$Y := (1 + \gamma^{1/2})h_\Gamma^{3/2} + \varepsilon_h, \quad X_1 := \frac{h}{\mu} + Y.$$

By Theorem 7.2, $\|e_u\| \leq CY$ for CNS*. By Theorems 6.1 and 6.4, $\|e_u\| \leq C[(h/\mu)^{1/2}G + Y]$ and $\|\nabla e_u\| \leq C_p X_1$ for GS.

8.1.2. *Summary.* The rates below are for fixed γ and bounded ρ (so $h \asymp h_\Gamma$ and $h/\mu \asymp h_\Gamma$), $k \geq 4$, and $\varepsilon_h \leq Ch_\Omega^k$ (or flux-corrected data); each entry states the upper bound, and “sharp” means a matching lower bound is proved.

quantity	GS	CNS*
$\ p^\circ - p_h\ _{L^2(\Omega_h)}$	$\asymp (h/\mu)h_\Gamma^{-1/2}G$: rate $\frac{1}{2}$, sharp (Thm 8.11)	$O(h_\Gamma^{3/2})$ (Thm 8.7)
$\ p^\circ - p_h\ _{L^2(\Omega_h \setminus L_1)}$	$O(h/\mu)$	$O(h_\Gamma^{3/2})$
$\ p^\circ - p_h - c\ _{L^\infty(L_1)}$, any c	$\geq cG/\gamma$: no convergence	not proved
drag/lift/torque: Nitsche flux =	$-(h/\mu)K_\psi + O(h^{3/2})$: rate 1 (Thm 8.17)	$-\Theta_f - \Lambda_h + O(\gamma^{-1/2}h_\Gamma^2)$: rate 2 (Thm 8.19)
volume formula with v_ψ		
drag, GS volume formula $\omega_h(\chi_h)$	$-2(h/\mu)K_e + O(h^{3/2})$ (Prop. 8.21)	$J_h^N + O(h_\Gamma^2)$
drag, pointwise traction	$\Xi_h(\psi) = O(1/\gamma)$: no convergence	$O(h_\Gamma)$
$\sigma(u_h, p_h)n$	(Prop. 8.22)	
Nitsche traction in $L^2(\Gamma_h)$	$O(h_\Gamma^{1/2})$	$O(h_\Gamma)$

8.2. Tools.

Lemma 8.1 (Pressure identity on V_h^0). *For both methods and all $v \in V_h^0$,*

$$(p^\circ - p_h, \operatorname{div} v) = a_h(e_u, v) + \langle u_h, \partial_n v \rangle + (F, v).$$

Proof. For $v \in V_h^0$ we have $v = 0$ on Γ_h , so $N_h(w, v) = a_h(w, v) - \langle w, \partial_n v \rangle$ and the θ -term vanishes. The method gives $a_h(u_h, v) - \langle u_h, \partial_n v \rangle - (p_h, \operatorname{div} v) = (\tilde{f}, v)$. On V_h^0 , Lemma 5.1 reads $a_h(\tilde{u}, v) - \langle \tilde{u}, \partial_n v \rangle - (\tilde{p}, \operatorname{div} v) = (\tilde{f}, v) + \mathcal{G}(v)$ with $\mathcal{G}(v) = -\langle \tilde{u}, \partial_n v \rangle - (F, v)$, i.e. $a_h(\tilde{u}, v) - (\tilde{p}, \operatorname{div} v) = (\tilde{f} - F, v)$. Subtract, and note $(c, \operatorname{div} v) = 0$ for constants c . $\square$

The term $\langle u_h, \partial_n v \rangle$ is the *Nitsche coupling*: the symmetric Nitsche term applied to the discrete trace. For $v \in V_h^0$ and $e \in \mathcal{E}_h^\Gamma$, $v|_{T_e}$ vanishes on e , so $\nabla v = \partial_n v \otimes n_e$ there and

$$(\operatorname{div} v)|_{T_e} = \partial_n v \cdot n_e \quad \text{on } e. \quad (4)$$

Hence the *normal* part of the discrete trace acts on the pressure exactly like a boundary source $\langle u_h \cdot n, \operatorname{div} v \rangle_{\Gamma_h}$.

Lemma 8.2 (Edge representers). *For $e \in \mathcal{E}_h^\Gamma$ and $g \in L^\infty(e)$ let $\xi_{e,g} \in P_{k-1}(T_e)$ be defined by $(\xi_{e,g}, r)_{T_e} = \int_e g r$ for all $r \in P_{k-1}(T_e)$.*

(a) *For constant $g \equiv 1$, $\xi_{e,1}(x) = \frac{|e|}{2|T_e|} \hat{r}_k(F_e^{-1}x)$, where F_e is an affine map from the reference triangle $\hat{T} = \{(0,0), (1,0), (0,1)\}$ onto T_e with $F_e(\hat{e}) = e$, $\hat{e} = [0,1] \times \{0\}$, and $\hat{r}_k \in P_{k-1}(\hat{T})$ is the representer of $\hat{r} \mapsto \int_{\hat{e}} \hat{r}$. Consequently*

$$\|\xi_{e,1}\|_{T_e}^2 = \int_e \xi_{e,1} = \hat{c}_k \frac{|e|}{H_e}, \quad \|\xi_{e,1}\|_{L^2(e)}^2 = \hat{d}_k \frac{|e|}{H_e^2},$$

with $\hat{c}_k := \int_{\hat{e}} \hat{r}_k$ and $\hat{d}_k := \|\hat{r}_k\|_{L^2(\hat{e})}^2$ positive and depending only on k .

(b) $\|\xi_{e,g} - g_e \xi_{e,1}\|_{T_e} \leq C \|g - g_e\|_{L^\infty(e)}$ and $\|\xi_{e,g} - g_e \xi_{e,1}\|_{L^2(e)} \leq C |e|^{-1/2} \|g - g_e\|_{L^\infty(e)}$ for every constant g_e .

Exact rational computation (sympy) gives $\hat{c}_k = k(k+1)$ for $1 \leq k \leq 8$; in particular $\hat{c}_4 = 20$.

Proof. (a) With $x = F_e \hat{x}$, $\int_{T_e} \xi r = 2|T_e| \int_{\hat{T}} \hat{\xi} \hat{r}$ and $\int_e r = |e| \int_{\hat{e}} \hat{r}$, which gives the formula. Then $\|\xi_{e,1}\|_{T_e}^2 = \int_e \xi_{e,1}$ (take $r = \xi_{e,1}$), and the scalings follow from $|e|^2/(2|T_e|) = |e|/H_e$. (b) The map $g \mapsto \xi_{e,g}$ is linear, and $\|\xi_{e,g}\|_{T_e} = \sup_r \int_e g r / \|r\|_{T_e} \leq \|g\|_{L^\infty(e)} |e|^{1/2} \sup_r \|r\|_{L^2(e)} / \|r\|_{T_e} \leq C \|g\|_{L^\infty(e)}$ by the inverse trace inequality $\|r\|_{L^2(e)} \leq C |e|^{-1/2} \|r\|_{T_e}$ on P_{k-1} ; the trace bound follows from the same inequality. $\square$

Lemma 8.3 (Normal-load cancellation without incompressibility). *Let S be as in Lemma 6.3: edgewise $W^{1,\infty}$ on Γ_h , continuous at the vertices, with $|S \cdot \tau_e| \leq Ch_\Gamma$. Then for every $\delta \in V_h^R$,*

$$|\langle S, \partial_n \delta \rangle| \leq C \left(\|\nabla \delta\| + \sum_{e \in \mathcal{E}_h^\Gamma} \|(\operatorname{div} \delta)|_{T_e}\|_{L^1(e)} \right).$$

Proof. Repeat Lemma 6.3. The only change is in step 2: on e , $\partial_n \delta \cdot n_e = (\operatorname{div} \delta)|_{T_e} - \partial_\tau(\delta \cdot \tau_e)$. This adds the term $\int_e \sigma_e (\operatorname{div} \delta)|_{T_e}$, which is bounded by $\|S\|_\infty \|(\operatorname{div} \delta)|_{T_e}\|_{L^1(e)}$. $\square$

Lemma 8.4 ($\mathcal{G}$ on non-discrete fields). *For every $w \in H^1(\Omega_h)^2$ that is piecewise H^2 on $\mathcal{T}_h$,*

$$|\mathcal{G}(w)| \leq C(1 + \gamma^{1/2}) h_\Gamma^{3/2} \|w\|.$$

Proof. $|(\nabla \tilde{u})^\top n| \leq Ch_\Gamma$ on Γ_h (step 1 of Lemma 3.5), so $|\langle (\nabla \tilde{u})^\top n, w \rangle| \leq Ch_\Gamma \|w\|_{\Gamma_h} \leq Ch_\Gamma (h/\mu)^{1/2} \|w\| = C\gamma^{-1/2} h_\Gamma^{3/2} \|w\|$. Next, $|\langle \tilde{u}, \partial_n w \rangle| \leq Ch_\Gamma^2 \|\partial_n w\|_{\Gamma_h} \leq Ch_\Gamma^2 (c_0 h_\Gamma)^{-1/2} \|w\|$ and $\frac{\mu}{h} |\langle \tilde{u}, w \rangle| \leq C\gamma^{1/2} h_\Gamma^{3/2} \|w\|$. Finally, the proof of Lemma 11.10 gives $|(F, w)| \leq Ch_\Gamma^2 (\|w\|_{L^1(\Gamma_h)} + h_\Gamma \|\nabla w\|) \leq Ch_\Gamma^2 \|w\|$. $\square$

Lemma 8.5 (B_h is bounded). $|B_h(y, w)| \leq (1 + 2(c_0\gamma)^{-1/2})\|y\|\|w\|$ for all y, w for which the norms are finite.

Proof. $\|\partial_n y\|_{\Gamma_h} \leq (c_0 h_\Gamma)^{-1/2}\|y\|$ and $\|w\|_{\Gamma_h} \leq (h/\mu)^{1/2}\|w\|$, and $(h/(\mu c_0 h_\Gamma))^{1/2} = (c_0\gamma)^{-1/2}$. $\square$

Lemma 8.6 (Normal test fields). Let $\chi_0 \in C^\infty(\mathbb{R}^2)$ with $\chi_0 = 1$ on $\{|x| \leq r_1\}$ and $\chi_0 = 0$ on $\{|x| \geq r_2\}$, $1 < r_1 < r_2 < L$. For $\phi \in C^{k+1}(\Gamma)$ put $\hat{n}(x) := -x/|x|$ and

$$w_\phi := I_h(\chi_0(\phi \circ \pi)\hat{n}), \quad \pi(x) := x/|x|,$$

with I_h the P_k Lagrange interpolant. For $h \leq h_*$: $w_\phi \in V_h^R$, $\|w_\phi\|_{W^{1,\infty}} \leq C_\phi$, and on each $e \in \mathcal{E}_h^\Gamma$

$$|w_\phi - \phi(x^*)n(x^*)| \leq C_\phi h_\Gamma^{k+1}, \quad |w_\phi \cdot \tau_e| \leq C_\phi h_\Gamma, \quad |w_\phi \cdot n_e - \phi(x^*)| \leq C_\phi h_\Gamma^2,$$

and $|\int_{\Gamma_h} w_\phi \cdot n - \int_\Gamma \phi| \leq C_\phi h_\Gamma^2$. For $\phi \equiv 1$ also $(\operatorname{div} w_1)|_{T_e} = -1 + O(h_\Gamma^2)$ on e .

Proof. The triangles meeting Γ_h lie in $\{|x| \leq 1 + Ch_\Gamma\} \subset \{\chi_0 = 1\}$, and those meeting ∂R lie in $\{\chi_0 = 0\}$. On Γ_h , $\hat{n} = n \circ \pi$, and the interpolation error is $O(h_\Gamma^{k+1})$ in L^∞ and $O(h_\Gamma^k)$ in $W^{1,\infty}$ on boundary triangles. By Lemma 3.1, $|n(x^*) \cdot \tau_e| \leq Ch_\Gamma$ and $n(x^*) \cdot n_e = 1 - \frac{1}{2}|n(x^*) - n_e|^2 = 1 + O(h_\Gamma^2)$. The arclength Jacobian of $\pi|_{\Gamma_h}$ is $1 + O(h_\Gamma^2)$. Finally $\operatorname{div} \hat{n} = -1/|x| = -1 + O(h_\Gamma^2)$ on Γ_h . $\square$

8.3. The pressure of CNS*.

Theorem 8.7 (CNS* pressure). Let $\gamma \geq \gamma_2$. The CNS* pressure p_h (unique by Theorem 7.2) satisfies, with $\bar{e}_p$ the mean of $e_p = p^\circ - p_h$ over Ω_h ,

$$\begin{aligned} \|e_p - \bar{e}_p\|_{L^2(\Omega_h)} &\leq CY, \\ |\bar{e}_p| &\leq C[(1 + (\gamma h_\Gamma)^{1/2})Y + (\mu/h)|m_R|]. \end{aligned}$$

Hence $\|p^\circ - p_h\|_{L^2(\Omega_h)} \leq C[(1 + (\gamma h_\Gamma)^{1/2})Y + (\mu/h)|m_R|]$. With flux-corrected data (Definition 2.2) the same bounds hold with the term $(\mu/h)|m_R|$ deleted and ε_h replaced by $\tilde{\varepsilon}_h$ of Lemma 7.21 in Y ; consequently, for $\gamma \in [\gamma_2, \gamma_{\max}]$,

$$\|p^\circ - p_h\|_{L^2(\Omega_h)} \leq C(h_\Gamma^{3/2} + h_\Omega^k),$$

with no relation between h_Γ and h_Ω required. Here C depends on $\gamma_{\max}$: unlike the H^1 bound of Theorem 7.22, this pressure bound is not uniform in γ (Remark 8.8). Moreover $|\bar{p}_\Gamma(p_h)| \leq C[h_\Gamma^{-1/2}Y + (1 + (\gamma h_\Gamma)^{1/2})Y + (\mu/h)|m_R|]$.

Proof. Mean-free part. Write $e_p = \xi + \eta$ with $\eta := p^\circ - \pi_h p^\circ$, so $\|\eta\| \leq Ch^k$ and $\|\eta\|_{L^\infty(\Gamma_h)} \leq Ch_\Gamma^k$. Step 3 of Theorem 7.2 gives $\beta\|\xi - \bar{\xi}\| \leq C(\|e_u\| + h_\Gamma^{3/2}) \leq CY$.

Constant. Let $B^*(r, v) := -(r, \operatorname{div} v) + \langle r - \bar{p}_\Gamma(r), v \cdot n \rangle$. Then $B^*(c, v) = -c \int_{\Gamma_h} v \cdot n$ for constants c , and the error equation (3) gives, for $v := w_1$ of Lemma 8.6,

$$\bar{e}_p \int_{\Gamma_h} w_1 \cdot n = N_h(e_u, w_1) + B^*(e_p - \bar{e}_p, w_1) - \mathcal{G}(w_1). \quad (5)$$

Here $\int_{\Gamma_h} w_1 \cdot n \geq |\Gamma_h|/2$. We bound the three terms on the right.

- $B^*(e_p - \bar{e}_p, w_1) = -(e_p - \bar{e}_p, \operatorname{div} w_1) + \langle e_p - \bar{p}_\Gamma(e_p), w_1 \cdot n - 1 \rangle$, because $\langle e_p - \bar{p}_\Gamma(e_p), 1 \rangle = 0$. The first term is at most $C\|e_p - \bar{e}_p\|$. The second is at most $Ch_\Gamma^2\|e_p - \bar{e}_p\|_{L^1(\Gamma_h)} \leq Ch_\Gamma^{3/2}\|\xi - \bar{\xi}\| + Ch_\Gamma^{k+2}$, by Lemma 8.6 and the inverse trace inequality Lemma 4.1.
- $|\mathcal{G}(w_1)| \leq C(h_\Gamma^2 + \gamma h_\Gamma^3)$. The transposed term is $\sum_e \int_e (sa_e + O(h_\Gamma^2)) \cdot w_1$ with $a_e \parallel n(m_e^*)$; since $sa_e \cdot w_1 = sW_e(1 + O(s^2))$ and $\int_e s ds = 0$, it is $O(h_\Gamma^2)$. Next, $|\langle \tilde{u}, \partial_n w_1 \rangle| \leq Ch_\Gamma^2$. On e , $\tilde{u}(x) = d\partial_n u(x^*) + O(d^2)$ with $\partial_n u(x^*) \perp n(x^*)$, so $|\tilde{u} \cdot w_1| \leq Ch_\Gamma^4$ and $\frac{\mu}{h}|\langle \tilde{u}, w_1 \rangle| \leq C\gamma h_\Gamma^3$. Finally $|(F, w_1)| \leq Ch_\Gamma^2$.

- In $N_h(e_u, w_1) = a_h(e_u, w_1) - \langle \partial_n e_u, w_1 \rangle - \langle e_u, \partial_n w_1 \rangle + \frac{\mu}{h} \langle e_u, w_1 \rangle$:
 - $|a_h(e_u, w_1)| \leq C \|\nabla e_u\|$ and $|\langle e_u, \partial_n w_1 \rangle| \leq C \|e_u\|_{\Gamma_h} \leq C(h/\mu)^{1/2} \|e_u\|$.
 - Write $e_u = (\tilde{u} - I_h \tilde{u}) + \delta$ with $\delta := I_h \tilde{u} - u_h \in V_h^R$. Then $\operatorname{div} \delta = \operatorname{div}(I_h \tilde{u} - \tilde{u}) = O(h_\Gamma^k)$ on L_1 , and $\|\nabla \delta\| \leq \|\nabla e_u\| + Ch^k$. Lemma 8.3 with $S = w_1|_{\Gamma_h}$ gives $|\langle \partial_n e_u, w_1 \rangle| \leq C(\|\nabla e_u\| + h^k)$.
 - $\frac{\mu}{h} \langle e_u, w_1 \rangle = \frac{\mu}{h} [m_R + \langle e_u \cdot n, w_1 \cdot n - 1 \rangle + \langle e_u \cdot \tau, w_1 \cdot \tau \rangle]$, and the last two terms are at most $\frac{\mu}{h} Ch_\Gamma \|e_u\|_{L^1(\Gamma_h)} \leq C(\gamma h_\Gamma)^{1/2} \|e_u\|$.

Collect, using $\|e_u\| \leq CY$, $\gamma h_\Gamma^3 \leq (\gamma h_\Gamma)^{1/2} Y$ and $h^k \leq \varepsilon_h$.

Flux-corrected data. Take $z \in \tilde{W}_h$ from Lemma 7.21, so $\|e_u\| \leq CY$ with $\tilde{\varepsilon}_h$ in Y (Theorem 7.22). In the second bullet replace $I_h \tilde{u}$ by the field $w_h = I_h \tilde{u} - c I_h(\zeta x)$ from the proof of Lemma 7.21: it equals $I_h \tilde{u}$ near Γ_h and has trace $\tilde{g}_I$ on ∂R , so $\delta := w_h - u_h \in V_h^R$, $\operatorname{div} \delta = \operatorname{div}(I_h \tilde{u} - \tilde{u})$ on L_1 , and $\|\nabla \delta\| \leq \|\nabla e_u\| + C(h^k + |m_R|)$ with $|m_R| \leq Ch^{\sigma_k}$. In the third bullet $\int_{\Gamma_h} e_u \cdot n = 0$, so the term $(\mu/h)m_R$ is absent. Finally $h^k \leq \tilde{\varepsilon}_h$, and for bounded γ , $Y \leq C(h_\Gamma^{3/2} + h^k)$. For the last claim, $\bar{p}_\Gamma(p^\circ) = 0$, so $|\bar{p}_\Gamma(p_h)| = |\bar{p}_\Gamma(e_p)| \leq |\bar{e}_p| + |\Gamma_h|^{-1/2} \|e_p - \bar{e}_p\|_{\Gamma_h}$, and $\|\xi - \bar{\xi}\|_{\Gamma_h} \leq Ch_\Gamma^{-1/2} \|\xi - \bar{\xi}\|$. $\square$

Remark 8.8 (Where the pressure error sits). The $h_\Gamma^{3/2}$ is produced by the right-hand side of Lemma 8.1: $a_h(e_u, \cdot)$ and the Nitsche coupling $\langle u_h, \partial_n \cdot \rangle$, where $\|u_h\|_{\Gamma_h} \leq \|\tilde{u}\|_{\Gamma_h} + (h/\mu)^{1/2} \|e_u\| \leq C(1 + \gamma^{-1/2})h_\Gamma^2 + C(h/\mu)^{1/2} \varepsilon_h$ and $\|\partial_n v\|_{\Gamma_h} \leq Ch_\Gamma^{-1/2} \|\nabla v\|$. A $h_\Gamma^{3/2}$ lower bound for the pressure is not proved. The factor $(1 + \gamma^{1/2})$ in Y is real: at fixed N the computed pressure error grows with μ (Section 14.9).

8.4. The pressure of GS.

Proposition 8.9 (The GS pressure is fully determined). *For $\gamma \geq \gamma_0$, GS has a unique solution $(u_h, p_h) \in W_h \times \Pi_h$. No normalisation of p_h is needed or possible: the constant is fixed by the flux-carrying test functions in V_h^R .*

Proof. The velocity is the unique Lax–Milgram solution on W_h , because N_h is coercive on Z_h Lemma 4.2. The functional $v \mapsto N_h(u_h, v) - (f, v)$ vanishes on $Z_h = \ker(\operatorname{div}|_{V_h^R})$. Since $\operatorname{div} V_h^R = \Pi_h$ Lemma 4.3 and Π_h is finite-dimensional, there is exactly one $p_h \in \Pi_h$ with $(p_h, \operatorname{div} v)$ equal to that functional on V_h^R . Adding a constant c to p_h changes the momentum row by $-c \int_{\Gamma_h} v \cdot n \neq 0$ (edge bubble). $\square$

Definition 8.10 (The Nitsche pressure layer). With $g := (h/\mu) q \circ \pi$ on Γ_h , let $\xi_b \in \Pi_h$ be $\xi_{e,g}$ (Lemma 8.2) on each T_e and 0 outside L_1 . Equivalently,

$$(\xi_b, r) = (h/\mu) \langle q \circ \pi, r|_{T_e} \rangle_{\Gamma_h} \quad \forall r \in \Pi_h.$$

Theorem 8.11 (GS pressure). *Let $\gamma \geq \gamma_0$ and $\rho_h := p^\circ - p_h - \xi_b$. Then:*

(a) $\|\rho_h - \bar{\rho}_h\| \leq C_p X_1$ and

$$\|\rho_h\|_{L^2(\Omega_h)} \leq C_p [(1 + (\gamma h_\Gamma)^{1/2}) X_1 + (\mu/h) |m_R|] =: C_p Q.$$

(b) $\|\xi_b\|^2 = \left(\frac{h}{\mu}\right)^2 \left[\hat{c}_k \sum_{e \in \mathcal{E}_h^\Gamma} \frac{|e|}{H_e} q(m_e^*)^2 + O(C_p) \right]$. In particular $c(h/\mu) h_\Gamma^{-1/2} G \leq \|\xi_b\| \leq$

$C(h/\mu) h_\Gamma^{-1/2} (G + C_p h_\Gamma^{1/2})$, the lower bound holding for $h_\Gamma \leq h_0(G)$.

(c) For every constant c ,

$$c_1 \frac{h}{\mu} h_\Gamma^{-1/2} G - C_p Q \leq \|p^\circ - p_h - c\|_{L^2(\Omega_h)}$$

and $\|p^\circ - p_h\|_{L^2(\Omega_h)} \leq C(h/\mu) h_\Gamma^{-1/2} G + C_p Q$.

$$(d) \|p^\circ - p_h\|_{L^2(\Omega_h \setminus L_1)} \leq C_p Q.$$

$$(e) \text{ For every constant } c, \|p^\circ - p_h - c\|_{L^\infty(L_1)} \geq c_2 G/\gamma - C_p h_\Gamma^{-1/2} Q.$$

For fixed γ and bounded ρ with $G > 0$: the L^2 pressure error converges at the exact rate $1/2$. It is carried by ξ_b on the first layer, with $\|p^\circ - p_h\|_{L^2(L_1)}/\|\xi_b\| \rightarrow 1$. It is $O(h/\mu)$ elsewhere, and in the first layer it does not converge pointwise. If $G = 0$, then $\xi_b = 0$ and the CNS* bounds of Theorem 8.7 hold for GS with C in place of C_p .

Proof. Step 1: the equation for ρ_h . By Lemma 8.1, (4) and Definition 8.10, for $v \in V_h^0$,

$$(\rho_h, \operatorname{div} v) = a_h(e_u, v) + \langle u_h - g n, \partial_n v \rangle + (F, v).$$

Step 2: the trace $u_h - g n$. As in the proof of Corollary 6.7, $e_u = (h/\mu)v_\varphi + \vartheta$ with $\vartheta := (\tilde{u} - z) + \delta_G + r_h$ and $\|\vartheta\| \leq C_p X_1$. (By Theorem 6.4, steps 2 and 6: $\|\delta_G\| + \|\tilde{u} - z\| \leq CY$ and $\|r_h\| \leq C_p(h/\mu + h_\Gamma^{3/2}) + C\varepsilon_h$.) On e , $v_\varphi = w + O(h^k)$ and $w(x) = -q(x^*)n(x^*) + O(h_\Gamma^2)$, so

$$v_\varphi + (q \circ \pi)n_e = q(x^*)(n_e - n(x^*)) + O(h_\Gamma^2 + h^k) = O_p(h_\Gamma + h^k).$$

Hence

$$\|u_h - g n\|_{\Gamma_h} \leq \|\tilde{u}\|_{\Gamma_h} + \frac{h}{\mu} C_p(h_\Gamma + h^k) + \|\vartheta\|_{\Gamma_h} \leq C h_\Gamma^2 + C_p \frac{h}{\mu} h_\Gamma + (h/\mu)^{1/2} C_p X_1.$$

Step 3: mean-free part. With $\|\partial_n v\|_{\Gamma_h} \leq C h_\Gamma^{-1/2} \|\nabla v\|$, $\|\nabla e_u\| \leq C_p X_1$ Theorem 6.4 and $|(F, v)| \leq C h_\Gamma^2 \|\nabla v\|$, the right-hand side of Step 1 is at most $C_p[X_1 + h_\Gamma^{3/2} + (h/\mu)h_\Gamma^{1/2} + \gamma^{-1/2}X_1]\|\nabla v\| \leq C_p X_1 \|\nabla v\|$. Now $\rho_h - \eta \in \Pi_h$ with $\eta = p^\circ - \pi_h p^\circ$, and $(\eta, \operatorname{div} v) = 0$. Lemma 8.12 below (Lemma 4.5) gives $\|\rho_h - \bar{\rho}_h\| \leq C_p X_1 + C h^k \leq C_p X_1$.

Step 4: constant. The GS error equation on V_h^R is $N_h(e_u, v) - (p^\circ - p_h, \operatorname{div} v) + \langle p^\circ, v \cdot n \rangle = \mathcal{G}(v)$; the constant in p° cancels because $\int_{\Omega_h} \operatorname{div} v = \int_{\Gamma_h} v \cdot n$. With $v = w_1$,

$$\bar{\rho}_h \int_{\Gamma_h} w_1 \cdot n = N_h(e_u, w_1) + \langle p^\circ, w_1 \cdot n \rangle - \mathcal{G}(w_1) - (\xi_b, \operatorname{div} w_1) - (\rho_h - \bar{\rho}_h, \operatorname{div} w_1).$$

We bound each term.

- $\langle p^\circ, w_1 \cdot n \rangle = \langle p^\circ, w_1 \cdot n - 1 \rangle = O(h_\Gamma^2)$, because $\langle p^\circ, 1 \rangle = 0$.
- $|\mathcal{G}(w_1)| \leq C(h_\Gamma^2 + \gamma h_\Gamma^3)$, as in Theorem 8.7.
- $\operatorname{div} w_1 \in \Pi_h$, so $(\xi_b, \operatorname{div} w_1) = (h/\mu)\langle q \circ \pi, -1 + O(h_\Gamma^2) \rangle = O_p((h/\mu)h_\Gamma^2)$, since $\int_\Gamma q = 0$ and the Jacobian is $1 + O(h_\Gamma^2)$.
- $|\langle \rho_h - \bar{\rho}_h, \operatorname{div} w_1 \rangle| \leq C_p X_1$.
- $N_h(e_u, w_1)$ is treated as in Theorem 8.7, except that $\frac{\mu}{h} \langle e_u \cdot \tau, w_1 \cdot \tau \rangle$ is split along $e_u = (h/\mu)v_\varphi + \vartheta$. On e , $v_\varphi \cdot \tau_e$ and $w_1 \cdot \tau_e$ are both $O_p(h_\Gamma)$, so the first part is $O_p(h_\Gamma^2)$. The second is at most $\frac{\mu}{h} C h_\Gamma (h/\mu)^{1/2} \|\vartheta\| \leq C(\gamma h_\Gamma)^{1/2} C_p X_1$. The term $\frac{\mu}{h} \langle e_u \cdot n, w_1 \cdot n - 1 \rangle$ is at most $C\gamma^{1/2} h_\Gamma^{3/2} \|e_u\| \leq C_p(h_\Gamma^2 + (\gamma h_\Gamma)^{1/2} Y)$, by Theorem 6.1.

Hence $|\bar{\rho}_h| \leq C_p[(1 + (\gamma h_\Gamma)^{1/2})X_1 + (\mu/h)|m_R|]$, which proves (a).

(b) Apply Lemma 8.2 on each T_e with $g_e = (h/\mu)q(m_e^*)$ and $\|g - g_e\|_{L^\infty(e)} \leq C_p(h/\mu)|e|$. Expanding the square, the cross terms are $O_p((h/\mu)^2 \sum_e |e|)$. The two-sided bound uses $c \leq |e|/H_e \leq C$ and the Riemann sum $\sum_e q(m_e^*)^2 |e| = G^2 + O_p(h_\Gamma)$, with $h_\Gamma^{-1} \sum_e q_e^2 |e| \leq \sum_e q_e^2 \leq (c_0 h_\Gamma)^{-1} \sum_e q_e^2 |e|$.

(c) $p^\circ - p_h - c = \xi_b - c + \rho_h$. The L^2 -closest constant to ξ_b is its mean, $|\Omega_h|^{-1}(\xi_b, 1) = |\Omega_h|^{-1}(h/\mu)\langle q \circ \pi, 1 \rangle = O_p((h/\mu)h_\Gamma^2)$. Hence $\|\xi_b - c\| \geq \|\xi_b\| - C_p(h/\mu)h_\Gamma^2$. Use (a) and (b).

(d) $\xi_b = 0$ off L_1 .

(e) $\|\cdot\|_{L^\infty(L_1)} \geq |L_1|^{-1/2} \|\cdot\|_{L^2(L_1)}$ with $|L_1| \leq Ch_\Gamma$. The mean of ξ_b over L_1 is $|L_1|^{-1}(h/\mu)\langle q \circ \pi, 1 \rangle = O_p((h/\mu)h_\Gamma)$, so, as in (c), $\|\xi_b - c\|_{L^2(L_1)} \geq \|\xi_b\| - C_p(h/\mu)h_\Gamma^{3/2}$. Then use (a), (b) and $(h/\mu)h_\Gamma^{-1} = 1/\gamma$.

The regime statement. For fixed γ and bounded ρ , $Q = O(h_\Gamma)$ while $(h/\mu)h_\Gamma^{-1/2} \asymp h_\Gamma^{1/2}$. If $G = 0$, then $g = 0$, and Theorems 6.1 and 6.4 give $\|e_u\| + \|\nabla e_u\| \leq CY$; Steps 1–4 then run with X_1 replaced by Y . $\square$

Lemma 8.12 (Lemma 4.5, restated). *For $r \in \Pi_h$, $\beta\|r - \bar{r}\| \leq \sup_{0 \neq v \in V_h^0}(r, \operatorname{div} v)/\|\nabla v\|$.*

Remark 8.13 (Mechanism). The leak $u_h \cdot n \approx (h/\mu)(p - \bar{p})$ (Corollary 6.7) enters the pressure only through the Nitsche coupling $\langle u_h, \partial_n v \rangle$. By (4), that term is a boundary source $\langle u_h \cdot n, \operatorname{div} v \rangle_{\Gamma_h}$, which the discontinuous pressure resolves exactly, as the first-layer field ξ_b . Its L^2 norm is $(h/\mu)|e|^{-1/2}$ per unit length of boundary, and its pointwise size is $(h/\mu)/H_e \asymp 1/\gamma$. The penalty scaling μ/h is what makes it independent of the mesh at fixed γ . It is a discrete artefact: it has no counterpart in the leak flow (U, P) of Definition 6.8, whose pressure enters only through $\rho_h \approx (h/\mu)P$.

8.5. Boundary stress and drag.

8.5.1. *Functionals.* Let $\operatorname{RM} := \{\psi(x) = a + bx^\perp : a \in \mathbb{R}^2, b \in \mathbb{R}\}$, $x^\perp = (-x_2, x_1)$. Every $\psi \in \operatorname{RM}$ has $\operatorname{div} \psi = 0$, $D(\psi) = 0$ and $\nabla \psi$ skew, and $\int_{\Gamma_h} \psi \cdot n = \int_\Gamma \psi \cdot n = 0$. For $\psi \in \operatorname{RM}$ put

$$J_\psi := \int_\Gamma \sigma(u, p)n \cdot \psi \, ds, \quad \sigma(u, p) = D(u) - pI,$$

with n the outward normal of Ω (pointing into D). J_ψ does not depend on the pressure constant. With $\psi = e_1, e_2, x^\perp$ it is Gjerde–Scott’s $\beta(\chi)$ [24, (3)], $\chi = \psi$ on Γ : drag, lift and torque, up to their sign convention.

Definition 8.14 (Discrete functionals). Let $\phi_\psi(x) := a_1x_2 - a_2x_1 - \frac{b}{2}|x|^2$, so that $\operatorname{curl} \phi_\psi = \psi$. Let χ_0 be as in Lemma 8.6, and put

$$V_\psi := \operatorname{curl}(\chi_0 \phi_\psi), \quad v_\psi := \operatorname{curl} \Pi_{k+1}(\chi_0 \phi_\psi) \in Z_h,$$

where Π_{k+1} is the C^1 interpolant of Section 6. For $w \in V_h^R$ put

$$\omega_h(w) := a_h(u_h, w) - (p_h, \operatorname{div} w) - (\tilde{f}, w)_{\Omega_h}.$$

We consider:

- (i) the *Nitsche-flux (variational) functional* $J_h^N(\psi) := \omega_h(v_\psi)$;
- (ii) the *pointwise traction functional* $J_h^\beta(\psi) := \int_{\Gamma_h} (D(u_h)n - p_h n) \cdot \psi$;
- (iii) for translations $\psi = e$, the *Gjerde–Scott volume functional* $J_h^\chi(e) := \omega_h(\chi_h)$. Here $\chi_h \in Z_h$ is the same method’s discrete Stokes solution with zero body force, zero data on ∂R and Nitsche data e on Γ_h : for GS, $N_h(\chi_h, w) - (\pi_h, \operatorname{div} w) = \ell_e(w)$, with $\ell_e(w) := -\langle e, \partial_n w \rangle + \frac{\mu}{h} \langle e, w \rangle$.

Let $\Theta_f(\psi) := \int_{S_h} \tilde{f} \cdot \psi$ be the body force carried by the strip.

Gjerde–Scott compute (ii), with χ_h in place of ψ on Γ_h , and (iii), with the pressure term dropped because $\operatorname{div} \chi_h = 0$. Their inconsistency parameter is $\varepsilon = (\omega - \beta)/\omega$ [24, (10)].

Lemma 8.15 (Flux identity and error identity). *Let $t_h := \partial_n u_h - \theta(p_h - \bar{p}_\Gamma(p_h))n - \frac{\mu}{h}u_h$ be the Nitsche flux. For both methods:*

- (a) $\omega_h(w) = \langle t_h, w \rangle + \langle u_h, \partial_n w \rangle$ for all $w \in V_h^R$. Consequently $v_\psi = \psi$ on L_1 ,

$$J_h^N(\psi) = \int_{\Gamma_h} t_h \cdot \psi + \langle u_h, \partial_n \psi \rangle,$$

and $J_h^N(\psi) = \omega_h(w)$ for every $w \in V_h^R$ with $w = \psi$ on L_1 . For GS, $J_h^N(\psi)$ contains no pressure at all.

(b) $J_h^N(\psi) - J_\psi = -a_h(e_u, v_\psi) - \Theta_f(\psi)$.

(c) $\Theta_f(\psi) = \sum_e \frac{|e|^3}{12} (f \cdot \psi)(m_e^*) + O(h_\Gamma^3)$, and $\Theta_f = 0$ if $f = 0$ near Γ .

Proof. (a) Subtract the method equation from the definition of ω_h : $\omega_h(w) = a_h(u_h, w) - N_h(u_h, w) - \theta(p_h - \bar{p}_\Gamma(p_h), w \cdot n)$. The triangles of L_1 lie in $\{\chi_0 = 1\}$, where $\chi_0 \phi_\psi = \phi_\psi \in P_2$ is reproduced by Π_{k+1} , so $v_\psi = \psi$ on L_1 and $\partial_n v_\psi = \partial_n \psi$ on Γ_h (traces from T_e). If $w = \psi$ on L_1 , then $w - v_\psi$ has zero trace and zero normal derivative on Γ_h , so $\omega_h(w - v_\psi) = 0$.

(b) For $w \in V_h^R$ let $\tilde{\omega}(w) := a_h(\tilde{u}, w) - (p^\circ, \operatorname{div} w) - (\tilde{f}, w)$. Steps 1–3 of Lemma 5.1, keeping $D(\tilde{u})n$, give $\tilde{\omega}(w) = \langle \sigma(\tilde{u}, p^\circ)n, w \rangle - (F, w)$. The segment between each chord and its arc lies in T_e (the arc leaves the chord at angle $O(h_\Gamma)$, while the base angles of T_e are bounded below), so $v_\psi = \psi$ on S_h . Since $\sigma(\tilde{u}, p^\circ)$ is symmetric and $\nabla \psi$ is skew, $\operatorname{div}(\sigma \psi) = (\operatorname{div} \sigma) \cdot \psi = (F - \tilde{f}) \cdot \psi$ on S_h . The divergence theorem on S_h (outward normal n on Γ_h and $-n$ on Γ ; constants in the pressure drop because $\int \psi \cdot n = 0$) gives $\langle \sigma(\tilde{u}, p^\circ)n, \psi \rangle_{\Gamma_h} = J_\psi + (F - \tilde{f}, \psi)_{S_h}$. Hence $\tilde{\omega}(v_\psi) = J_\psi - \Theta_f(\psi)$. Finally $\omega_h(v_\psi) - \tilde{\omega}(v_\psi) = -a_h(e_u, v_\psi) + (e_p, \operatorname{div} v_\psi) = -a_h(e_u, v_\psi)$.

(c) The segment over e has area $\frac{|e|^3}{12} + O(|e|^5)$ and width $O(|e|^2)$, and $\tilde{f}$ is smooth. $\square$

So the Babuška–Miller volume drag, with any discrete test field equal to ψ on the first layer, is the Nitsche-flux drag, exactly. By (b), its error is the energy error tested against a fixed field that is rigid near the wall. Since $D(v_\psi) = D(\psi) = 0$ on $\{\chi_0 = 1\}$, $a_h(e_u, v_\psi)$ only sees e_u away from Γ .

8.5.2. Dual problems and reciprocity. For $\psi \in \text{RM}$ let (Z_ψ, R_ψ) solve the Stokes problem in Ω with $Z_\psi = \psi$ on Γ and $Z_\psi = 0$ on ∂R . Then $Z_\psi \in H^2(\Omega)$ (the corners of R are convex), Z_ψ is smooth up to Γ , and $Z_\psi = \operatorname{curl} \Phi_\psi$ with Φ_ψ single-valued; extend Φ_ψ, R_ψ smoothly across Γ to $\tilde{Z}_\psi, \tilde{R}_\psi$. Let (U, P) be the leak flow of Definition 6.8: $U = -qn$ on Γ , $U = 0$ on ∂R .

Lemma 8.16 (Reciprocity). $J_\psi(U) := \int_\Gamma \sigma(U, P)n \cdot \psi = \int_\Gamma (p - \bar{p})(R_\psi - \tilde{R}_\psi) ds =: K_\psi$.

Proof. For Stokes pairs with zero body force, $\int_{\partial\Omega} \sigma(U, P)n \cdot Z = \frac{1}{2} \int_\Omega D(U) : D(Z) = \int_{\partial\Omega} \sigma(Z, R)n \cdot U$, using $\operatorname{div} U = \operatorname{div} Z = 0$ and the $H^2 \times H^1$ regularity. Both fields vanish on ∂R . On Γ , $Z_\psi - \psi$ is divergence-free and zero, so $\partial_n(Z_\psi - \psi) \cdot n = 0$ Lemma 3.4. Also $(\nabla \psi)n \cdot n = 0$ by skewness. Hence $\sigma(Z_\psi, R_\psi)n \cdot n = 2\partial_n Z_\psi \cdot n - R_\psi = -R_\psi$. Since $U = -qn$, the right-hand side equals $\int_\Gamma q R_\psi$; and $\int_\Gamma q = 0$. $\square$

8.5.3. The printed method.

Theorem 8.17 (GS drag, lift and torque). *Let $\gamma \geq \gamma_0$ and $\psi \in \text{RM}$. Then*

$$J_h^N(\psi) = J_\psi - \frac{h}{\mu} K_\psi - \Theta_f(\psi) + \mathcal{R}_h, \quad |\mathcal{R}_h| \leq C \frac{h}{\mu} B_U + C[(1 + \gamma^{1/2})h_\Gamma^{3/2} + \varepsilon_h],$$

where $B_U := C_U[\min(\gamma h_\Gamma^{1/2}, (\gamma h_\Gamma)^{1/2}) + h_\Gamma^{1/2} + (h/\mu)^{1/2} + E_U]$ is the bound of Theorem 6.10. Without that theorem, $|J_h^N(\psi) - J_\psi| \leq C_p(h/\mu) + C[(1 + \gamma^{1/2})h_\Gamma^{3/2} + \varepsilon_h]$. For fixed γ and bounded ρ , $J_h^N(\psi) - J_\psi = -(h/\mu)K_\psi + O(h^{3/2})$. So the rate is exactly 1 whenever $K_\psi \neq 0$, and the defect is the force that the leak flow exerts on the body.

Proof. By Lemma 8.15(b), the claim is about $a_h(e_u, v_\psi)$. Split $e_u = \delta_R + [\delta_G + (\tilde{u} - z)]$ as in Theorem 6.4. The bracket contributes at most $2\|\nabla v_\psi\| C[(1 + \gamma^{1/2})h_\Gamma^{3/2} + \varepsilon_h]$. Next, $a_h(\delta_R, v_\psi) = \frac{h}{\mu} a_h(\tilde{U}, v_\psi) + a_h(\delta_R - \frac{h}{\mu} \tilde{U}, v_\psi)$, and the second term is at most $C \frac{h}{\mu} B_U$ by Theorem 6.10. By Green's formula with $F_U := -\Delta \tilde{U} + \nabla \tilde{P}$ (supported in $\overline{S_h}$), $\operatorname{div} v_\psi = 0$ and $v_\psi = \psi$ on $L_1 \supset S_h$,

$$a_h(\tilde{U}, v_\psi) = (F_U, \psi)_{S_h} + \langle \sigma(\tilde{U}, \tilde{P})n, \psi \rangle_{\Gamma_h} = (F_U, \psi)_{S_h} + J_\psi(U) - (F_U, \psi)_{S_h} = J_\psi(U).$$

The middle equality is the divergence theorem on S_h , as in Lemma 8.15(b). Conclude with Lemma 8.16. The unconditional bound is $|a_h(e_u, v_\psi)| \leq C\|\nabla e_u\|$ together with Theorem 6.4. $\square$

Remark 8.18 (The leak drag is the pressure–pressure correlation). The leak $u_h \cdot n \approx (h/\mu)q$ makes the Nitsche flux reproduce the missing traction at leading order: $-\frac{\mu}{h}u_h \approx -qn$. That is why J_h^N converges at all, although the GS flux contains no pressure. The residual $-(h/\mu)K_\psi$ couples the flow’s wall pressure with the wall pressure of the dual problem. In Gjerde–Scott’s cylinder application, $f = 0$ (so $\Theta_f = 0$) and both pressures are dipole-like, so $K_e \neq 0$ generically. With their $\mu = 10^6$ the defect has relative size about $10^{-6}h$ and is invisible.

8.5.4. The pressure-consistent method.

Theorem 8.19 (CNS* drag, lift and torque). *Let $\gamma \geq \max(1, \gamma_2)$, $\gamma h_\Gamma \leq 1$ and $\psi \in \text{RM}$. Define*

$$\Lambda_h(\psi) := \langle \tilde{u}, \partial_n(\tilde{Z}_\psi - \psi) \rangle_{\Gamma_h} = \sum_{e \in \mathcal{E}_h^\Gamma} \frac{|e|^3}{12} (\partial_n u \cdot \partial_n (Z_\psi - \psi))(m_e^*) + O(h_\Gamma^3),$$

and $E_Z := \inf_{z \in Z_h} \|\tilde{\zeta} - z\|$, where $\tilde{\zeta} := V_\psi - \tilde{Z}_\psi$. Then:

- (a) $|J_h^N(\psi) - J_\psi + \Theta_f(\psi)| \leq C\|\nabla e_u\| \leq CY$;
- (b) $|J_h^N(\psi) - J_\psi + \Theta_f(\psi) + \Lambda_h(\psi)| \leq C[\gamma^{-1/2}h_\Gamma^2 + \gamma h_\Gamma^3 + \gamma^{1/2}h_\Gamma^{3/2}E_Z + \varepsilon_h]$.

Consequently $|J_h^N(\psi) - J_\psi| \leq C(h_\Gamma^2 + \varepsilon_h)$: rate 2. As $\gamma \rightarrow \infty$ with $\gamma h_\Gamma \rightarrow 0$, $J_h^N(\psi) - J_\psi = -\Theta_f(\psi) - \Lambda_h(\psi) + o(h_\Gamma^2) + O(\varepsilon_h)$. Since $E_Z \leq Ch$ (argument of Lemma 6.9), the E_Z term is $o(h_\Gamma^2)$ whenever $h \leq Ch_\Gamma$.

The leading term is *geometric* and method-independent. With uniform chords of length ℓ ,

$$\Theta_f + \Lambda_h \approx \frac{\ell^2}{12} \int_\Gamma (f \cdot \psi + \partial_n u \cdot \partial_n (Z_\psi - \psi)) ds.$$

This is the shape derivative of J_ψ under the inward normal displacement $\frac{1}{2}(\ell^2/4 - s^2)$ of the polygon. CNS* is consistent enough that only the polygonal geometry is left.

Proof. (a) is Lemma 8.15(b). For (b), write $e_u = (\tilde{u} - z) + e_h$, with $z \in W_h$ attaining Lemma 4.6 and $e_h := z - u_h \in Z_h$.

Step 1: the auxiliary problem. Let $(\zeta_h, \pi_h) \in Z_h \times \Pi_h$ solve

$$N_h(\zeta_h, w) + B^*(\pi_h, w) = a_h(v_\psi, w) \quad \forall w \in V_h^R. \quad (6)$$

This is the CNS* Stokes system for the smooth-near- Γ solution $(\tilde{\zeta}, \Pi)$ with $\Pi := -\tilde{R}_\psi$, since $-\Delta \tilde{\zeta} + \nabla \Pi = -\Delta V_\psi$ on Ω , $\tilde{\zeta} = 0$ on $\Gamma \cup \partial R$ and $\text{div } \tilde{\zeta} = 0$. Indeed, Lemma 5.1 for $(\tilde{\zeta}, \Pi)$ gives $N_h(\tilde{\zeta}, w) + B^*(\Pi^*, w) = (-\Delta V_\psi, w) + \mathcal{G}_\zeta(w)$, where $\Pi^* := \Pi - \bar{p}_\Gamma(\Pi)$ and $\mathcal{G}_\zeta$ is $\mathcal{G}$ with $\tilde{u} \rightarrow \tilde{\zeta}$ and $F \rightarrow F_\zeta := \Delta \tilde{Z}_\psi - \nabla \tilde{R}_\psi$ (supported in $\bar{S}_h$). Moreover $a_h(v_\psi, w) = (-\Delta V_\psi, w) + a_h(v_\psi - V_\psi, w)$, because $D(V_\psi) = 0$ near Γ_h , and $\|\nabla(v_\psi - V_\psi)\| \leq Ch^k$. Existence and uniqueness, and the bounds

$$\|\tilde{\zeta} - \zeta_h\| \leq CY_Z, \quad \beta\|(\Pi^* - \pi_h) - c_\Pi\| \leq CY_Z, \quad Y_Z := (1 + \gamma^{1/2})h_\Gamma^{3/2} + E_Z + h^k,$$

with c_Π the appropriate mean, follow from the proof of Theorem 7.2. That proof uses the regularity only near Γ (in Lemma 5.2 and in $\|\eta\|_{L^\infty(\Gamma_h)} \leq Ch_\Gamma^k$), and ε_h enters only through E_h , here E_Z . Also $\|\zeta_h\| + \|\nabla \zeta_h\| \leq C$, since $\|\tilde{\zeta}\| \leq C(1 + \gamma^{1/2}h_\Gamma^{3/2})$.

Step 2: the error representation. Test (6) with $w = e_h$ (so $\text{div } e_h = 0$, and $B^*(\pi_h, e_h) = \langle \pi_h - \bar{p}_\Gamma(\pi_h), e_h \cdot n \rangle$), and the error equation (3) with $w = \zeta_h$. By symmetry of N_h ,

$$\begin{aligned} a_h(e_u, v_\psi) &= T_1 + \mathcal{G}(\zeta_h) - T_4 + T_5, & T_1 &:= a_h(v_\psi - \zeta_h, \tilde{u} - z) - B_h(\tilde{u} - z, \zeta_h), \\ T_4 &:= \langle e_p - \bar{p}_\Gamma(e_p), \zeta_h \cdot n \rangle, & T_5 &:= \langle \pi_h - \bar{p}_\Gamma(\pi_h), e_h \cdot n \rangle. \end{aligned}$$

Hence $J_h^N - J_\psi + \Theta_f + \Lambda_h = -T_1 - (\mathcal{G}(\zeta_h) - \Lambda_h) + T_4 - T_5$.

Step 3: T_1 . $|T_1| \leq C\varepsilon_h$, by $\|\nabla(v_\psi - \zeta_h)\| \leq C$, $\|\zeta_h\| \leq C$ and Lemma 8.5.

Step 4: $\mathcal{G}(\zeta_h)$. By Lemma 8.4, $|\mathcal{G}(\zeta_h) - \mathcal{G}(\tilde{\zeta})| \leq C(1 + \gamma^{1/2})h_\Gamma^{3/2}Y_Z$. On Γ_h , $|\tilde{\zeta}| \leq Ch_\Gamma^2$, $|(\nabla\tilde{u})^\top n| \leq Ch_\Gamma$ and $|\tilde{u}| \leq Ch_\Gamma^2$, and near Γ_h , $\partial_n\tilde{\zeta} = \partial_n\psi - \partial_n\tilde{Z}_\psi$. Hence

$$\mathcal{G}(\tilde{\zeta}) = \langle \tilde{u}, \partial_n(\tilde{Z}_\psi - \psi) \rangle + O(h_\Gamma^3 + \gamma h_\Gamma^3 + h_\Gamma^4) = \Lambda_h + O((1 + \gamma)h_\Gamma^3).$$

For the expansion of Λ_h : on e , $\tilde{u}(x) = d\partial_n u(x^*) + O(d^2)$ with $d = 1 - |x| = \frac{1}{2}(\frac{|e|^2}{4} - s^2) + O(|e|^4)$. Also $\nabla\tilde{Z}_\psi(x)n_e = \partial_n(Z_\psi - \psi)(x^*) + \partial_n\psi + O(h_\Gamma^2)$, by Lemma 3.4 applied to $Z_\psi - \psi$ and $n(x^*) \cdot n_e = 1 + O(h_\Gamma^2)$. Finally $\int_e d = \frac{|e|^3}{12} + O(|e|^5)$.

Step 5: T_4 . By step 3 of Theorem 7.2 and Lemma 4.1, $\|e_p - \bar{p}_\Gamma(e_p)\|_{\Gamma_h} \leq \|\xi - \bar{\xi}\|_{\Gamma_h} + \|\eta\|_{\Gamma_h} \leq Ch_\Gamma^{-1/2}Y$. By Lemma 3.4 applied to ζ , $\tilde{\zeta} \cdot n_e = d\partial_n\zeta(x^*) \cdot n_e + O(h_\Gamma^4) = O(h_\Gamma^3)$. So $\|\zeta_h \cdot n\|_{\Gamma_h} \leq Ch_\Gamma^3 + (h/\mu)^{1/2}CY_Z$, and $|T_4| \leq C(h_\Gamma^{5/2}Y + \gamma^{-1/2}YY_Z)$.

Step 6: T_5 . Since $\int_{\Gamma_h} e_h \cdot n = 0$, constants may be subtracted from π_h freely. Then

$$|T_5| \leq |(\Pi - \bar{\Pi}) \circ \pi, e_u \cdot n| + C\|\tilde{u} - z\|_{\Gamma_h} + Ch_\Gamma^2\|e_h\|_{\Gamma_h} + C(h_\Gamma^{-1/2}Y_Z + h_\Gamma^k)\|e_h\|_{\Gamma_h},$$

where $\bar{\Pi}$ is the mean of Π on Γ and $\|e_h\|_{\Gamma_h} \leq (h/\mu)^{1/2}CY$. The first term is bounded by Lemma 8.20 below, with $\phi = \Pi - \bar{\Pi}$.

Step 7: collect. Insert $Y \leq C(\gamma^{1/2}h_\Gamma^{3/2} + \varepsilon_h)$ and $Y_Z \leq C(\gamma^{1/2}h_\Gamma^{3/2} + E_Z + h^k)$, and use $\gamma \geq 1$, $\gamma h_\Gamma \leq 1$ and $|\bar{e}_p| \leq C$ (Theorem 8.7). Every term is bounded by the right-hand side of (b); the $\gamma^{-1/2}h_\Gamma^2$ comes from the term $\frac{h}{\mu}h_\Gamma^{-1/2}Y$ of Lemma 8.20. $\square$

Lemma 8.20 (Weighted normal slip of CNS*). *Let $\gamma \geq \gamma_2$ and $\phi \in C^{k+1}(\Gamma)$ with $\int_\Gamma \phi = 0$. Then*

$$|\langle \phi \circ \pi, e_u \cdot n \rangle_{\Gamma_h}| \leq C_\phi \left[\frac{h}{\mu} (h_\Gamma^{-1/2}Y + h^k + h_\Gamma^2 + \gamma h_\Gamma^3 + h_\Gamma^2|\bar{e}_p|) + h_\Gamma \left(\frac{h}{\mu} \right)^{1/2} Y \right].$$

Proof. Test (3) with w_ϕ of Lemma 8.6:

$$\frac{\mu}{h} \langle e_u, w_\phi \rangle = \mathcal{G}(w_\phi) - a_h(e_u, w_\phi) + \langle \partial_n e_u, w_\phi \rangle + \langle e_u, \partial_n w_\phi \rangle - B^*(e_p, w_\phi).$$

On the left, $w_\phi - (\phi \circ \pi)n = O_\phi(h_\Gamma)$ on Γ_h , so $\frac{\mu}{h} \langle e_u, w_\phi \rangle = \frac{\mu}{h} \langle e_u \cdot n, \phi \circ \pi \rangle + O_\phi((\mu/h)^{1/2}h_\Gamma\|e_u\|)$. On the right:

- $|\mathcal{G}(w_\phi)| \leq C_\phi(h_\Gamma^2 + \gamma h_\Gamma^3)$, exactly as for w_1 in Theorem 8.7.
- $|a_h(e_u, w_\phi)| \leq C_\phi Y$.
- $|\langle \partial_n e_u, w_\phi \rangle| \leq C_\phi(Y + h^k)$, by Lemma 8.3 as in Theorem 8.7.
- $|\langle e_u, \partial_n w_\phi \rangle| \leq C_\phi(h/\mu)^{1/2}Y$.
- $B^*(e_p, w_\phi) = -(e_p - \bar{e}_p, \text{div } w_\phi) - \bar{e}_p \int_{\Gamma_h} w_\phi \cdot n + \langle e_p - \bar{p}_\Gamma(e_p), w_\phi \cdot n \rangle$ is at most $C_\phi(Y + |\bar{e}_p|h_\Gamma^2 + h_\Gamma^{-1/2}Y)$, by Lemma 8.6 and $\int_\Gamma \phi = 0$.

Multiply by h/μ . $\square$

8.5.5. Gjerde–Scott’s volume functional.

Proposition 8.21 (J_h^X). *Let $\psi = e$ be a translation.*

- (CNS*, $\gamma \geq \max(1, \gamma_2)$, $\gamma h_\Gamma \leq 1$.) $|J_h^X(e) - J_h^N(e)| \leq C[h_\Gamma^2 + (h/\mu)^{1/2}(Y + Y_Z)(1 + h_\Gamma^{-1/2}Y_Z)]$, which is $O(h_\Gamma^2 + (h/\mu)^{1/2}E_Z)$.
- (GS, $\gamma \geq \gamma_0$ fixed, bounded ρ , and $E_Z + E_U \leq Ch$.)

$$J_h^X(e) = J_e - 2\frac{h}{\mu}K_e - \Theta_f(e) + O(h^{3/2}).$$

Gjerde-Scott's own drag formula thus has twice the leak defect of the Nitsche flux, because the discrete test field χ_h leaks as well.

Proof. For translations, $\ell_e(w) = N_h(v_e, w) - a_h(v_e, w)$, since $v_e = e$ on L_1 . Hence $\zeta_h := v_e - \chi_h \in Z_h$ satisfies $N_h(\zeta_h, w) - \theta B^*(\pi_h, w) = a_h(v_e, w)$ for all $w \in V_h^R$ (for GS, for all $w \in Z_h$). By Lemma 8.15(a), $J_h^X - J_h^N = -\omega_h(\zeta_h)$.

(a) For CNS*, ζ_h is the solution of (6) (with $\pi_h \rightarrow -\pi_h$). Then $|\omega_h(\zeta_h)| \leq \|t_h\|_{\Gamma_h} \|\zeta_h\|_{\Gamma_h} + \|u_h\|_{\Gamma_h} \|\partial_n \zeta_h\|_{\Gamma_h}$. Here $\|t_h\|_{\Gamma_h} \leq C$ (Proposition 8.22(a) and $\gamma h_\Gamma \leq 1$), $\|\zeta_h\|_{\Gamma_h} \leq Ch_\Gamma^2 + (h/\mu)^{1/2} CY_Z$, $\|u_h\|_{\Gamma_h} \leq Ch_\Gamma^2 + (h/\mu)^{1/2} CY$ and $\|\partial_n \zeta_h\|_{\Gamma_h} \leq C + Ch_\Gamma^{-1/2} Y_Z$.

(b) For GS, write $\omega_h(\chi_h) = \tilde{\omega}(\chi_h) - a_h(e_u, \chi_h)$ (the pressure terms vanish since $\operatorname{div} \chi_h = 0$). With $\tilde{\omega}(\chi_h) = \langle \sigma(\tilde{u}, p^\circ) n, \chi_h \rangle - (F, \chi_h)$, $\chi_h = v_e - \zeta_h$ and Lemma 8.15(b),

$$J_h^X - J_h^N = \langle \sigma(\tilde{u}, p^\circ) n, \chi_h - e \rangle - (F, \chi_h - e) + a_h(e_u, \zeta_h).$$

By Corollary 5.3 with $c = \bar{p}_\Gamma(\tilde{p})$, $a_h(e_u, \zeta_h) = -\langle p^\circ, \zeta_h \cdot n \rangle + \mathcal{G}(\zeta_h) - B_h(e_u, \zeta_h)$, and $-\langle p^\circ, \zeta_h \cdot n \rangle$ cancels the pressure part of the first term. Hence

$$J_h^X - J_h^N = \langle D(\tilde{u}) n, \chi_h - e \rangle - (F, \chi_h - e) + \mathcal{G}(\zeta_h) - B_h(e_u, \zeta_h).$$

The dual error $e_Z := \tilde{Z}_e - \chi_h$ satisfies $N_h(e_Z, w) = -\langle \tilde{R}_e - c, w \cdot n \rangle + \mathcal{G}_Z(w)$ on Z_h , where $\mathcal{G}_Z$ is $\mathcal{G}$ with $\tilde{u} \rightarrow \tilde{Z}_e - e$. This is the structure of Corollary 5.3, so Theorems 6.1, 6.4 and 6.10 and Corollary 6.7 hold for it, with $p \rightarrow R_e$ and $\varepsilon_h \rightarrow E_Z + Ch^k$. In particular $e_Z = (h/\mu)v_\varphi^R + \vartheta^Z$, with $v_\varphi^R \approx -(R_e - \bar{R}_e)n$ on Γ_h and $\|\vartheta^Z\| \leq CX_1^Z := C(h/\mu + Y_Z)$. On Γ_h , $\zeta_h = (e - \tilde{Z}_e) + e_Z$, where $e - \tilde{Z}_e = -d \partial_n Z_e(x^*) + O(h_\Gamma^4)$ is tangential. Then:

- The penalty part of $-B_h(e_u, \zeta_h)$ is $-\frac{\mu}{h} \langle (h/\mu)v_\varphi + \vartheta, (e - \tilde{Z}_e) + (h/\mu)v_\varphi^R + \vartheta^Z \rangle$. Its leading term is $-\frac{h}{\mu} \langle v_\varphi, v_\varphi^R \rangle = -\frac{h}{\mu} K_e + O((h/\mu)(h_\Gamma^2 + h^k))$. The other five products are at most $C(h_\Gamma^4 + h_\Gamma^2 h^k)$, $C(h/\mu)^{1/2} X_1^Z$, $C\gamma^{1/2} h_\Gamma^{3/2} X_1$, $C(h/\mu)^{1/2} X_1$ and $CX_1 X_1^Z$, using $v_\varphi \perp (e - \tilde{Z}_e)$ up to $O(h_\Gamma^2 + h^k)$.
- $|\langle \partial_n e_u, \zeta_h \rangle| \leq Ch_\Gamma^{-1/2} X_1(h_\Gamma^2 + h/\mu + (h/\mu)^{1/2} X_1^Z)$.
- $\langle e_u, \partial_n \zeta_h \rangle = \frac{h}{\mu} \langle v_\varphi, \partial_n \zeta_h \rangle + \langle \vartheta, \partial_n \zeta_h \rangle$. The first part is $O((h/\mu)(\|\zeta_h\|_{\Gamma_h} + h_\Gamma^{1/2} + h^k h_\Gamma^{-1/2}))$, by Lemma 6.3 applied to $w|_{\Gamma_h}$ and $\|v_\varphi - w\|_{W^{1,\infty}} \leq Ch^k$. For the second, Theorem 6.10 gives $r_h = (h/\mu)(z_U - v_\varphi) + \epsilon$ with $\|z_U - v_\varphi\|_{\Gamma_h} \leq (h/\mu)^{1/2} E_U + C(h_\Gamma^2 + h^k)$ and $\|\epsilon\|_{\Gamma_h} \leq (h/\mu)^{3/2} B'_U$ (B'_U the middle bound of that theorem). With $\|\partial_n \zeta_h\|_{\Gamma_h} \leq Ch_\Gamma^{-1/2}$ this part is $O((h/\mu)\gamma^{-1/2}(E_U + B'_U) + (h/\mu)(h_\Gamma^{3/2} + h^k h_\Gamma^{-1/2}) + \gamma^{-1/2} Y)$.
- $|\langle D(\tilde{u}) n, \chi_h - e \rangle| \leq C(h_\Gamma^2 + (h/\mu)h_\Gamma + (h/\mu)^{1/2} X_1^Z)$, because $D(\tilde{u}) n \cdot n_e = 2\partial_n \tilde{u} \cdot n_e = O(h_\Gamma)$ and the normal part of ζ_h is $O(h/\mu)$.
- $|(F, \chi_h - e)| \leq Ch_\Gamma^3$ and $|\mathcal{G}(\zeta_h)| \leq C(1 + \gamma^{1/2})h_\Gamma^{3/2}$.

In the stated regime every remainder is $O(h^{3/2})$. Combine with Theorem 8.17. $\square$

8.5.6. Pointwise traction versus Nitsche flux.

Proposition 8.22 (Traction on Γ_h). *Let $\sigma^\circ := \sigma(\tilde{u}, p^\circ)$ and $T_h^\beta := D(u_h)n - p_h n$.*

(a) CNS* ($\gamma \geq \gamma_2$):

$$\begin{aligned} \|t_h - \sigma^\circ n\|_{\Gamma_h} &\leq C[(1 + \gamma^{1/2})h_\Gamma^{-1/2} Y + (1 + \gamma)h_\Gamma], \\ \|T_h^\beta - \sigma^\circ n\|_{\Gamma_h} &\leq C[h_\Gamma^{-1/2} Y + |\bar{e}_p| + h_\Gamma^k], \\ |J_h^\beta(\psi) - J_\psi| &\leq C(h_\Gamma^{-1/2} Y + h_\Gamma^2). \end{aligned}$$

(b) GS ($\gamma \geq \gamma_0$):

$$\|t_h - \sigma^\circ n\|_{\Gamma_h} \leq C_p[(1 + \gamma^{1/2})h_\Gamma^{-1/2}X_1 + (1 + \gamma)h_\Gamma],$$

$$cG/\gamma - C_p h_\Gamma^{-1/2}(Q + X_1) \leq \|T_h^\beta - \sigma^\circ n\|_{\Gamma_h} \leq C G/\gamma + C_p h_\Gamma^{-1/2}(Q + X_1),$$

and

$$J_h^\beta(\psi) = J_\psi + \Xi_h(\psi) + O_p(h_\Gamma^{-1/2}X_1 + h_\Gamma^2), \quad \Xi_h(\psi) := \langle \xi_b, \psi \cdot n \rangle_{\Gamma_h},$$

where, for translations,

$$\Xi_h(e) = \frac{h}{\mu} \hat{c}_k \sum_{e' \in \mathcal{E}_h^\Gamma} \frac{|e'|}{H_{e'}} q(m_{e'}^*) n_{e'} \cdot e + O_p(h/\mu),$$

so $|\Xi_h| \leq C_p/\gamma$. At fixed γ the pointwise traction of GS does not converge, and J_h^β converges to J_ψ only up to the $O(1/\gamma)$ defect Ξ_h . The Nitsche flux of GS converges in $L^2(\Gamma_h)$ at rate at least $1/2$.

Proof. On Γ_h , $\sigma^\circ n = \partial_n \tilde{u} + (\nabla \tilde{u})^\top n - p^\circ n$ with $|(\nabla \tilde{u})^\top n| \leq Ch_\Gamma$ and $|\tilde{u}| \leq Ch_\Gamma^2$. For the full gradient, $\|\nabla(\tilde{u} - u_h)\|_{\Gamma_h} \leq Ch_\Gamma^{-1/2}(\|\nabla e_u\| + h^k) + Ch_\Gamma^k$ (the inverse trace applied to $I_h \tilde{u} - u_h$).

(a) $t_h - \sigma^\circ n = -\partial_n e_u - (\nabla \tilde{u})^\top n + (e_p - \bar{p}_\Gamma(e_p))n - \frac{\mu}{h}(\tilde{u} - e_u)$. Use $\|e_p - \bar{p}_\Gamma(e_p)\|_{\Gamma_h} \leq Ch_\Gamma^{-1/2}Y$ and $\frac{\mu}{h}\|e_u\|_{\Gamma_h} \leq (\mu/h)^{1/2}\|e_u\| = \gamma^{1/2}h_\Gamma^{-1/2}\|e_u\|$. The bound for T_h^β is similar, and the constant part of e_p is Theorem 8.7. For J_h^β , use $\int \psi \cdot n = 0$ (so only $e_p - \bar{e}_p$ enters) and $\langle \sigma^\circ n, \psi \rangle - J_\psi = -\Theta_f + (F, \psi) = O(h_\Gamma^2)$ (proof of Lemma 8.15).

(b) For t_h : $-\frac{\mu}{h}u_h + p^\circ n = -\frac{\mu}{h}\tilde{u} + (v_\varphi + (q \circ \pi)n) + O(h_\Gamma^2) + \frac{\mu}{h}\vartheta$, and $v_\varphi + (q \circ \pi)n = O_p(h_\Gamma + h^k)$ (Step 2 of Theorem 8.11). Then $\frac{\mu}{h}\|\vartheta\|_{\Gamma_h} \leq (\mu/h)^{1/2}C_p X_1$. For T_h^β : $T_h^\beta - \sigma^\circ n = -D(e_u)n + (\xi_b + \rho_h)n$. Lemma 8.2 gives $\|\xi_b\|_{L^2(e)}^2 = g_e^2 \hat{d}_k |e|/H_e^2 + O_p((h/\mu)^2)$, so $\|\xi_b\|_{\Gamma_h} \asymp (h/\mu)h_\Gamma^{-1}G = G/\gamma$ by the Riemann-sum argument of Theorem 8.11(b). Also $\|\rho_h\|_{\Gamma_h} \leq Ch_\Gamma^{-1/2}\|\rho_h - \eta\| + Ch_\Gamma^k$. For J_h^β : only $\rho_h - \bar{\rho}_h$ enters, $\|\rho_h - \bar{\rho}_h\|_{\Gamma_h} \leq C_p h_\Gamma^{-1/2}X_1$, and $\int_{e'} \xi_b = \int_{e'} g \xi_{e',1} = g_{e'} \hat{c}_k |e'|/H_{e'} + O_p((h/\mu)|e'|)$. $\square$

Remark 8.23 (Gjerde–Scott’s inconsistency parameter). The quantity $\beta_h(\chi_h) = \int_{\Gamma_h} T_h^\beta \cdot \chi_h$ differs from $J_h^\beta(e)$ by $O(h/\mu)$ for GS. So Proposition 8.22(b) and Proposition 8.21(b) give

$$\varepsilon_{\text{GS}} = \frac{\omega - \beta}{\omega} = -\frac{\Xi_h(e)}{J_e} + O(h/\mu + h^{3/2}).$$

At fixed μ , and with h/h_Γ fixed, ε_{GS} tends to a nonzero limit of size $\approx \hat{c}_k(h/\mu h_\Gamma)(|e|/H_e)|J_e^p|/|J_e|$, where $J_e^p = -\int_\Gamma qn \cdot e$ is the pressure drag. It measures the Nitsche pressure layer, not the drag error. With $\mu = 10^6$, $\hat{c}_4 = 20$ and moderate h/h_Γ , this is of order 10^{-5} – 10^{-4} . That is the order of the $\varepsilon \approx 8 \cdot 10^{-5}$ in [24, Table 1b]. We have not checked this against their meshes, which we do not have.

Remark 8.24 (Removing the GS drag defect). To leading order $u_h \cdot n \approx (h/\mu)q$ and $(\chi_h - e) \cdot n \approx (h/\mu)(R_e - \bar{R}_e)$ on Γ_h , so $\frac{\mu}{h}\langle u_h \cdot n, (\chi_h - e) \cdot n \rangle \approx \frac{h}{\mu}K_e$. Moreover J_h^N is affine in $1/\mu$ at leading order, so two solves with different μ allow extrapolation. We have *not* proved rates for either corrected functional. CNS* needs no correction.

8.6. Navier–Stokes. Assume (NS1)–(NS2) of Section 12, with (L) for GS and (L*) for CNS*, and use the convective form c of the scheme also in the drag functional:

$$\omega_h(w) := \nu a_h(u_h, w) + c(u_h; u_h, w) - (p_h, \text{div } w) - (\tilde{f}, w), \quad J_\psi := \int_\Gamma (\nu D(u) - pI)n \cdot \psi.$$

Put $X := h/\mu + (1 + \gamma^{1/2})h_\Gamma^{3/2} + \varepsilon_h$ as in Section 12, and write $e := \tilde{u} - u_h$.

- Proposition 8.25.** (a) (CNS*, pressure.) Theorem 8.7 holds verbatim, with C depending also on ν , N , C_1 and β_* .
- (b) (GS, pressure.) Theorem 8.11 holds with $X_1 \rightarrow X$ and the same ξ_b , i.e. with $g = (h/\mu)q$ and q the Navier–Stokes pressure. In the convention νN_h of Remark 12.2 the layer does not depend on ν . With the unscaled penalty $\hat{\mu} = \nu\mu$ it is $g = \nu(h/\hat{\mu})q$.
- (c) (Drag.) Lemma 8.15(a) holds with t_h multiplied by ν . Lemma 8.15(b) becomes

$$J_h^N(\psi) - J_\psi = -\nu a_h(e, v_\psi) - c(e; \tilde{u}, v_\psi) - c(\tilde{u}; e, v_\psi) + c(e; e, v_\psi) - \Theta(\psi) + O(h_\Gamma^4),$$

where $\Theta(\psi) := (\tilde{f} - (\tilde{u} \cdot \nabla)\tilde{u}, \psi)_{S_h} = \Theta_f(\psi) + O(h_\Gamma^4)$. Hence $|J_h^N - J_\psi| \leq CX$ for GS (rate 1) and $\leq CY$ for CNS* (rate 3/2).

Proof. (a),(b): Lemma 8.1 acquires $c(\tilde{u}; \tilde{u}, v) - c(u_h; u_h, v) + \kappa_c(v)$ on the right. By Lemma 12.4 this is at most $C\|e\|_{H^1}\|\nabla v\|$ plus $Ch_\Gamma^4\|\nabla v\|$. The same convective terms appear when testing with w_1 . The decomposition $e = (h/\nu\mu)v_\varphi + \vartheta$ with $\|\vartheta\| \leq CX$, and $\|\nabla e\| \leq CX$, are Theorem 12.8 and the proof of Corollary 12.9. The layer source is $\nu\langle u_h, \partial_n v \rangle$, whose normal part is $\nu(h/\nu\mu)q$. (c): As in Lemma 8.15, now with $\text{div } \sigma(\tilde{u}, \tilde{p}) = F - \tilde{f} + (\tilde{u} \cdot \nabla)\tilde{u}$; the c_1/c_s difference on v_ψ is $\frac{1}{2}\langle (\tilde{u} \cdot n)\tilde{u}, \psi \rangle = O(h_\Gamma^4)$, and $|\tilde{u}| \leq Ch_\Gamma^2$ on S_h . The bounds use Lemma 12.4. $\square$

Not proved for Navier–Stokes. The GS leading term $-(h/\mu)K_\psi$ would need the Oseen analogue of Theorem 6.10, which is open Remark 12.10. If it holds, the reciprocity argument gives $K_\psi^\dagger = \int_\Gamma (p - \bar{p})(R_\psi^\dagger - \bar{R}_\psi^\dagger)$, with $(Z_\psi^\dagger, R_\psi^\dagger)$ the *adjoint* linearised (Oseen) problem with data ψ on Γ . The CNS* rate 2 and the leading term $-\Theta - \Lambda_h^\dagger$ would need Theorem 8.19 with the adjoint Oseen dual. That requires an adjoint version of (L*), which we have not formulated. Proposition 8.22 transfers verbatim for the Stokes part.

Remark 8.26 (General domains). Theorems 8.7, 8.11 and Proposition 8.22(a) use only lemmas that Section 11 re-proves, together with $|L_1| \asymp h_\Gamma$. They therefore hold on the general domains of Section 11, with the global $\bar{p}$, with $\hat{n} := n \circ \pi$ in Lemma 8.6, and with $\text{div } \hat{n} = \varkappa + O(h_\Gamma^2)$ replacing -1 (only $\int_\Gamma q = 0$ is used). In the drag identities S_h must be replaced by the signed region $\Omega_h \triangle \Omega$: $\Theta_f = (\tilde{f}, \psi)_{\Omega_h \setminus \Omega} - (f, \psi)_{\Omega \setminus \Omega_h}$, and Λ_h carries the factor $\varkappa$ through $d = \frac{1}{2}\varkappa(\ell^2/4 - t^2)$. Theorem 8.17 uses Theorem 6.10, which Section 11 does not cover, so on general domains only its unconditional $O(h/\mu)$ bound is proved.

8.7. What is not proved.

- A lower bound for the CNS* pressure error, or for its pointwise size in L_1 (observed: L^2 rate $\rightarrow 3/2$, $L^\infty(L_1)$ rate ≈ 1).
- Sharpness of the CNS* drag rate 2 at *fixed* γ . Theorem 8.19(b) identifies the leading term only up to $O(\gamma^{-1/2}h_\Gamma^2)$; that term (the weighted normal slip T_5) is not shown to be smaller. Numerically, for lift and torque, the ratio to $-\Theta_f - \Lambda_h$ tends to 1 at $\mu = 100$.
- Lower bounds for the CNS* pointwise traction (observed rate 1 for one component).
- The GS pointwise-traction defect Ξ_h is not bounded *below* in general. Its weights $|e|/H_e$ are mesh dependent, and a sign-indefinite weighted sum can vanish.
- Torque for J_h^χ (the Nitsche data functional with $\psi = x^\perp$ differs from ℓ by $-\langle \partial_n \psi, w \rangle$).
- Everything in Section 8.6 marked as not proved.
- The value $\hat{c}_k = k(k+1)$ for $k > 8$.

9. THE PENALTY THRESHOLD SCALES WITH ρ

Proposition 9.1.

- (a) Sufficiency. $\mu \geq \mu_0 = 4\kappa C_I^2 \rho$ implies coercivity (Lemma 4.2).

- (b) Necessity, with an exact certificate. *Let z be a vertex of a straight part of the Nitsche boundary, and let ℓ_z be the length of its boundary edges. Suppose its star, dilated by the factor $1/\ell_z$, is congruent to one of the reference stars $\mathcal{S}_3$ or $\mathcal{S}'_4$ below. If $\mu/(h/\ell_z) < 9$, then N_h is indefinite on Z_h .*

The same holds, by continuity, at a vertex of Γ_h with interior angle $\pi + O(h_\Gamma)$ whose star is a small perturbation of $\mathcal{S}_3$ or $\mathcal{S}'_4$. This uses that both reference stars are non-singular: every vertex, including the rim vertices, has non-collinear incident edges. The dimension of the divergence-free subspace therefore equals the generic count and is locally constant. This fails at the singular $m = 2$ configuration, which $(M1')$ excludes.

So, for meshes that contain such a boundary star at a vertex with $\rho \asymp h/\ell_z$, the condition $\mu_0 \asymp \rho$ is both necessary and sufficient. Theorem 9.2 and Corollary 9.4 below replace the star hypothesis by a condition on a single boundary triangle. The threshold is local: what is required is $\mu/h \gtrsim c/\ell_z$. The ρ -scaling is the price of Scott's normalisation of the penalty by the global h .

Proof of (b). (1) In two dimensions, $a(v, v)$ and $\langle \partial_n v, v \rangle$ are invariant under dilation, and $\|v\|_{L^2(\Gamma_h)}^2$ scales linearly.

- (2) A witness v on the reference star, extended by zero, lies in Z_h : it is continuous, vanishes on the star's outer chords and on ∂R , and is divergence-free.
- (3) It satisfies $N_h(v, v) = a(v, v) - 2\langle \partial_n v, v \rangle + (\mu\ell_z/h)\|v\|_{\text{ref}}^2$.
- (4) By the certificate below, $2B - A - 9C > 0$ on each reference star, so $N_h(v, v) < 0$ when $\mu\ell_z/h < 9$. $\square$

The exact certificate. The reference stars are fans at $O = (0, 0)$ with rational rim points:

- $\mathcal{S}_3$: rim points $(1, 0), (1, 1), (-1, 1), (-1, 0)$;
- $\mathcal{S}'_4$: rim points $(1, 0), (1, 1), (0, \frac{6}{5}), (-1, 1), (-1, 0)$.

The apex of $\mathcal{S}'_4$ is moved off the line $y = 1$. With apex $(0, 1)$ the rim vertex is singular, the kernel dimension jumps from 16 to 17, and the perturbation argument does not apply.

The Nitsche boundary is $\{y = 0\}$, and the rim chords carry zero Dirichlet data. The velocity is represented on each triangle by P_4 polynomials in global coordinates with rational coefficients. The computation is done in exact rational arithmetic (SymPy):

- Continuity across the spokes, vanishing on the rim, and pointwise $\text{div } v = 0$ are imposed as polynomial identities. This gives 80 and 110 linear constraints and kernel dimensions 14 and 16, the generic counts.
- The quadratic forms $A = \sum_T \int \frac{1}{2} Dv : Dv$, $B = \int (\partial_n v) \cdot v$ and $C = \int |v|^2$ are integrated exactly.
- A rational witness in the kernel satisfies $2B - A - 9C > 0$ *exactly* on each star. Its exact Rayleigh quotients $(2B - A)/C$ are 9.97854... ($\mathcal{S}_3$) and 9.26747... ($\mathcal{S}'_4$); these are lower bounds for the largest Rayleigh quotient λ^* of each star.

Computation: `code/lamstar_exact.py`, `code/lamstar_exact_v2.py`; output `logs/lamstar_exact_*.log`.

The witnesses are continuous piecewise- P_4 and divergence-free, so they lie in Z_h for every $k \geq 4$. The necessity statement therefore holds for all $k \geq 4$; for $k > 4$ the true local threshold is at least as large.

In floating point, the largest Rayleigh quotient λ^* is about 15 on uniform fans with $m = 3, 4$ and ranges over 14.8–38.5 on random 4-fans (`code/lamstar.py`). As a global check, the measured threshold $\gamma^* = \mu^* h_\Gamma / h$ stays in 18.1–20.1 as h/h_Γ varies over 3.3–44 on two mesh levels. Under refinement at fixed $h/h_\Gamma \approx 3.3$ it lies in 14.7–20.8, with increments that roughly halve per doubling (Table 7). Section 9.1 removes the reference-star hypothesis and explains the gap between the certified 9 and the measured 18–21.

9.1. Penalty necessity beyond two reference stars. Proposition 9.1(b) certifies indefiniteness of N_h on Z_h for $\mu\ell_z/h < 9$ only at vertices whose star is congruent to (or a perturbation of) $\mathcal{S}_3$ or $\mathcal{S}'_4$. We replace it by a statement about *every* boundary triangle, with an explicit closed-form constant and an exact certificate over a two-parameter family of shapes. We then explain the gap between the local constant and the global $\gamma^* \approx 18$ –21 of Table 7.

9.1.1. A witness supported in one boundary triangle. Let $T \in \mathcal{T}_h$ have an edge $e \subset \Gamma_h$, and let T not meet ∂R . Rotate, translate and dilate by $1/|e|$ so that $e = [(0, 0), (1, 0)]$, the fluid lies above and the apex is (a, b) , $b > 0$. Let $\lambda_0, \lambda_1, \lambda_2$ be the barycentric coordinates of the apex and of the base vertices $(0, 0)$, $(1, 0)$. For $c \in \mathbb{R}^3$ put

$$\psi_c := \lambda_1^2 \lambda_2^2 (c_0 \lambda_0 + c_1 \lambda_1 + c_2 \lambda_2) \in P_5, \quad v_c := \text{curl } \psi_c \in P_4^2.$$

Since ψ_c vanishes to second order on the two interior edges $\{\lambda_1 = 0\}$ and $\{\lambda_2 = 0\}$, v_c vanishes there. Extended by zero, v_c is continuous, zero on ∂R and pointwise divergence-free. So $v_c \in Z_h$ for every $k \geq 4$ and every mesh containing T . No condition on the vertices (singular or not), on the other triangles, or on the straightness of Γ_h beyond the edge e itself is needed.

Let $A_T = \int_T \frac{1}{2} Dv : Dv$, $B_T = \int_e (\partial_n v) \cdot v$ and $C_T = \int_e |v|^2$, with $n = (0, -1)$, as quadratic forms in c . They are computed exactly (SymPy) and, writing $\varpi := 4a^2 + 4b^2$, are

$$C_T = \frac{1}{1260 b^2} \begin{pmatrix} 2 & a-6 & -(a+5) \\ a-6 & 2(\varpi-3a+9) & \varpi-4a+15 \\ -(a+5) & \varpi-4a+15 & 2(\varpi-5a+10) \end{pmatrix},$$

$$B_T = \frac{1}{840 b^3} \begin{pmatrix} 12 & -(\varpi-6a+27) & -(\varpi-2a+25) \\ -(\varpi-6a+27) & 6(\varpi-3a+9) & 3(\varpi-4a+15) \\ -(\varpi-2a+25) & 3(\varpi-4a+15) & 6(\varpi-5a+10) \end{pmatrix},$$

and $A_T = (\alpha_{ij})/b^3$ with

$$\begin{aligned} 630 \alpha_{00} &= 3a^4 - 6a^3 + 6a^2b^2 + 12a^2 - 6ab^2 - 9a + 3b^4 + 4b^2 + 9, \\ 420 \alpha_{01} &= 3a^4 + 6a^2b^2 - 5a^2 + 6a + 3b^4 - 7b^2 - 9, \\ 420 \alpha_{02} &= 3a^4 - 12a^3 + 6a^2b^2 + 13a^2 - 12ab^2 - 8a + 3b^4 - b^2 - 5, \\ 70 \alpha_{11} &= 3a^4 - 3a^3 + 6a^2b^2 + 5a^2 - 3ab^2 - 3a + 3b^4 + 7b^2 + 3, \\ 140 \alpha_{12} &= a^4 - 2a^3 + 2a^2b^2 + 5a^2 - 2ab^2 - 4a + b^4 + 7b^2 + 5, \\ 70 \alpha_{22} &= 3a^4 - 9a^3 + 6a^2b^2 + 14a^2 - 9ab^2 - 10a + 3b^4 + 10b^2 + 5. \end{aligned}$$

$\det C_T = (3a^4 - 6a^3 + 7a^2b^2 + 3a^2 - 7ab^2 + 4b^4 + 5b^2)/(83\,349\,000 b^6) > 0$. (The forms were checked against an independent quadrature/finite-difference evaluation, relative difference $4.5 \cdot 10^{-9}$, and λ_T below against our penalty-threshold code on single-triangle patches of the Section 14 meshes: 10.48912 vs. 10.48912.)

Theorem 9.2 (Single-triangle necessity). *Let λ_T be the largest eigenvalue of the pencil $(2B_T - A_T, C_T)$; it depends only on the shape of T . If $\mu|e|/h < \lambda_T$, then N_h is indefinite on Z_h for every $k \geq 4$. Consequently every mesh satisfies the necessary condition*

$$\mu \geq h \max_{e \in \mathcal{E}_h^\Gamma} \frac{\lambda_{T_e}}{|e|}.$$

Proof. In two dimensions A and B are dilation invariant and C scales like $|e|$. So for the eigenvector c , $N_h(v_c, v_c) = A_T - 2B_T + (\mu|e|/h)C_T = (\mu|e|/h - \lambda_T) C_T(c) < 0$. $\square$

Proposition 9.3 (Exact certificates over shape families). *For $\varphi \in (0, \pi)$ let Q_φ be the set of apex positions (a, b) for which the apex angle is at least φ and $b \geq s \max(a, 1 - a)$, with $s = 0.087 < \tan 5^\circ$. This contains every triangle with apex angle $\geq \varphi$ and base angles $\geq 5^\circ$. Then*

$$\lambda_T > 9.3 \text{ on } Q_{60^\circ}, \quad \lambda_T > 4 \text{ on } Q_{45^\circ}, \quad \lambda_T > 1 \text{ on } Q_{35^\circ}.$$

The certificate proves only the three strict lower bounds. As numerical observations, not certified (`logs/lamstar_family_levels.log`): the first bound is nearly sharp, since a scan gives the minimum 9.31381 over Q_{60° , at the equilateral triangle; along the level sets of the apex angle the scanned minima are 6.07 (50°), 4.47 (45°), 2.83 (40°), 1.12 (35°) and -0.73 (30°), each at the isosceles shape.

Proof (computer-assisted, exact). Q_φ is the part of the disk through $(0, 0)$, $(1, 0)$ with inscribed angle φ that lies above the two lines. Cover a rational box containing Q_φ by dyadic sub-boxes. A box is discarded only if it provably misses Q_φ , using outward-rounded rational enclosures of the disk. On every other box, take a rational vector q (the eigenvector at the box centre, rounded to denominators ≤ 1000) and verify

$$\Phi_q(a, b) := 2520 b^3 q^\top (2B_T - A_T - cC_T)q > 0$$

on the whole box. Φ_q is a polynomial with rational coefficients (the denominators clear exactly), and the lower bound is evaluated monomial-wise in exact rational interval arithmetic. The boxes are split until this succeeds.

- $c = 9$ and $c = 9.3$ on Q_{60° : 323 and 716 boxes, depth ≤ 11 .
- $c = 4$ on Q_{45° : 259 boxes.
- $c = 1$ on Q_{35° : 297 boxes.

As negative controls, the same procedure fails for $c = 9.4$ on Q_{60° and for $c = 4.5$ on Q_{45° , as it must. $\square$

Computation: `code/lamstar_family.py bubble`; `code/lamstar_family.py certify`; output `logs/lamstar_family_bubble.log`, `logs/lamstar_family_certify.log`.

Corollary 9.4 (ρ -scaling is necessary on a broad class). *Let e_* be a shortest edge of Γ_h and suppose its triangle T_{e_*} has apex angle $\geq 35^\circ$ and base angles $\geq 5^\circ$, and does not meet ∂R (automatic for small h , since $\text{diam } T_{e_*} \leq Ch_\Gamma < L - 1$). Then coercivity of N_h on Z_h requires $\mu > \rho$. If the apex angle is $\geq 45^\circ$ (resp. $\geq 60^\circ$) it requires $\mu > 4\rho$ (resp. $\mu > 9.3\rho$). Together with Lemma 4.2, $\mu_0 \asymp \rho$ is necessary and sufficient on every such mesh, with no star-congruence or perturbation hypothesis.*

Remark 9.5 (What the single triangle cannot do). $\lambda_T < 0$ for tall triangles, for example every right triangle with the right angle at a base vertex and apex angle $\leq 30^\circ$ (numerically $\lambda_T \approx -0.2500$ at apex 30° , decreasing to -15.7 at 10°). A uniform statement under (M0)–(M2) alone would need witnesses on several triangles. In the Section 14 meshes such tall boundary triangles occur near the corner directions; at $N = 64$ the smallest value is $\lambda_T = -3.8$, with scaled apex about $(-0.08, 2.38)$ (`logs/lamstar_family_levels.log`). There the vertex stars and first-layer windows still have positive λ^* . Proving $\lambda_{\text{star}}^* \geq c(\Theta_0) > 0$ for every nonsingular boundary star remains open. The float evidence (random 4-fans, $\lambda^* \in [14.8, 38.5]$) suggests it holds.

9.1.2. *Why the certified constant is below the global threshold.* The global threshold is $\gamma^* = h_\Gamma \sup_{Z_h} (2B - A)/C$, where A is taken over Ω_h , B, C over Γ_h , and $h_\Gamma = \max |e|$. Every patch witness gives a lower bound. For the Section 14 meshes we computed the patch thresholds with the penalty code of Section 14 (bisection on the inertia of $A_\mu + rB^\top M^{-1}B$, fields vanishing outside the patch). We maximised over all boundary triangles and single-vertex stars. *winW* is the first layer restricted to W columns centred at $\theta = 0$, and the column J is the patch made of the first J

full boundary layers, computed for $J \leq 2$ here (the log also has $J = 3, 4$, which change the values by at most 0.002). The *global* column is Table 7(a); for $N \leq 16$ the full mesh is $J = N/4 \leq 4$ layers and the log reproduces it.

N	triangle	star	win2	win4	win8	win16	$J = 1$	$J = 2$	global
8	6.13	13.71	13.91	14.65	–	–	14.68	14.74	14.74
16	8.93	16.58	16.89	17.91	17.94	–	17.95	18.09	18.09
32	10.08	17.48	17.82	19.54	19.69	19.69	19.69	19.85	19.85
64	10.49	17.70	18.03	20.29	20.68	20.69	20.69	20.80	20.80

Computation: `code/lamstar_family.py mesh`; output `logs/lamstar_family_mesh.log`.

The limit geometry. The meshes of Section 14 do not have equal chords. The square points are equispaced and projected radially, so $\max |e| / \min |e|$ grows from 1.43 ($N = 16$) to 1.97 ($N = 256$). The longest chords are at the side midpoints $\theta = 0, \pi/2, \dots$. There the first layer, scaled by the chord, tends to columns $[i, i+1] \times [0, \frac{3}{4}]$ (layer height $(L-1)/2 = \frac{3}{4}$) split on the forward diagonal. The single-vertex star of that geometry is $\mathcal{S}_{3/4}$, with rim $(1, 0), (1, \frac{3}{4}), (0, \frac{3}{4}), (-1, 0)$. The general exact code (continuity, Dirichlet and divergence constraints over $\mathbb{Q}$, exact barycentric moments) gives:

	star	$W = 2$	$W = 4$	$W = 8$	$W = 16$	$W = 32$
1 layer	17.7064	17.9759	20.4872	21.1726	21.3358	21.3758
2 layers	–	18.1738	20.6781	21.2432	21.3675	21.3974
3 layers	–	18.1916	20.6811	21.2462	21.3708	–

Exact certificates in the limit geometry: $2B - A - 17C > 0$ on $\mathcal{S}_{3/4}$ (dimension 14, witness quotient 17.70640; also confirmed by `code/lamstar_exact.py`), and $2B - A - 20C > 0$ on the $W = 4$ window (dimension 41, witness quotient 20.48725).

Computation: `code/lamstar_family.py limit`; `lamstar_exact.run` on $\mathcal{S}_{3/4}$; output `logs/lamstar_family_limit.log`, `logs/lamstar_family_limitstar.log`.

The perturbation argument of Proposition 9.1(b) transfers these strict inequalities to the actual meshes near $\theta = 0$ for all sufficiently large N (non-explicit N_0). All patch vertices keep at least three distinct edge lines, or are Dirichlet corners where v and ∇v vanish, so the kernel dimension is locally constant. This gives $\gamma^*(N) > 20$ for $N \geq N_0$. The finite- N values in the first table agree: win4 is 20.29 at $N = 64$.

The gap between the certified 9 and the measured 18–21 is therefore accounted for, in order of size, by:

- (1) *Shape.* The reference stars $\mathcal{S}_3, \mathcal{S}'_4$ have layer height 1 per unit chord. The meshes have height $\frac{3}{4}$ at the longest chords, which is where γ^* is decided because γ is normalised by $\max |e|$. λ^* grows fast as the layer flattens: the exact witness quotients, which are lower bounds for λ^* , are 9.98 for $\mathcal{S}_3$ against 17.71 for $\mathcal{S}_{3/4}$.
- (2) *Collective boundary modes.* The optimal witness spreads over 4–8 boundary edges and adds about 3.7 ($17.7 \rightarrow 21.4$). Windows wider than about 16 edges add nothing.
- (3) *Depth.* In the limit geometry the second layer adds 0.20 ($W = 2$) and 0.19 ($W = 4$), decreasing to 0.02 at $W = 32$; the third layer adds 0.018 at $W = 2$ and at most 0.004 for $W \geq 4$. On the actual meshes the second layer adds 0.11–0.16 (compare $J = 1$ and $J = 2$). γ^* is a boundary-layer quantity (compare $J = 2$ with the global value).
- (4) *Finite N .* Away from the limit geometry (curvature, the varying chord and height along the layer) $\gamma^*(N)$ is lower. Richardson extrapolation of the window sequences (differences shrink by a factor of about 4 per doubling of W) gives $\gamma_\infty^* \approx 21.39$ (one layer) and 21.41 (two layers), so we expect $\gamma^*(N) \rightarrow \approx 21.4$ on this mesh family.

10. CLOUGH–TOCHER REFINEMENTS: EVERY $k \geq 2$

Sections 2–9 assume continuous P_k velocities with $k \geq 4$ on a mesh satisfying (M0)–(M2). Here we show that the results of Sections 3–9, through Theorem 7.2 and Proposition 9.1, hold for all $k \geq 2$ on Clough–Tocher (barycentric, Alfeld) refinements, and that (M1), (M1') and (M2) can be dropped (the precise scope is Theorem 10.8 and Remark 10.12). Only four ingredients change: the inf-sup constant, the divergence range, the flux exponent in Lemma 4.6, and the C^1 test field v_φ . The penalty-necessity constants of Proposition 9.1 also change, and we recertify them exactly.

10.1. Setting. Let $\mathcal{T}_h^M$ be a conforming triangulation of Ω_h (the *macro mesh*). Its Clough–Tocher refinement $\mathcal{T}_h$ splits every $T \in \mathcal{T}_h^M$ at its barycentre g_T into three triangles. Macro edges are not split, so the edges of $\mathcal{T}_h$ on Γ_h and on ∂R are exactly the macro edges there, and $\mathcal{E}_h^\Gamma$ is unchanged. For $k \geq 2$, V_h is the space of continuous piecewise- P_k fields on $\mathcal{T}_h$ and Π_h the space of discontinuous piecewise- P_{k-1} functions on $\mathcal{T}_h$. V_h^R , V_h^0 , Z_h , W_h , the forms a_h and N_h , the methods GS and CNS* and the norm $\|\cdot\|$ are defined by the formulas of Section 2, applied to these spaces. We put $h := \max_{T \in \mathcal{T}_h^M} \text{diam } T$ and keep h_Γ , ρ , γ as before.

Assumption 10.1.

(CT0) $\mathcal{T}_h^M$ is shape regular: $\text{diam } T \leq \sigma r_T$ for every $T \in \mathcal{T}_h^M$, where r_T is the inradius. Moreover $c_0 h_\Gamma \leq |e| \leq h_\Gamma$ for all $e \in \mathcal{E}_h^\Gamma$.

Assumption (D) of Section 2 is kept. No analogue of (M1), (M1') or (M2) is assumed.

In this section C and c depend on σ , c_0 , k , L and norms of $(u, p, \tilde{\psi})$, and on nothing else.

Lemma 10.2 (Structure of the refined mesh). *Assume (CT0) and $h \leq h_*$.*

- (a) $\mathcal{T}_h$ is shape regular with a constant depending only on σ , and $\text{diam } K \geq c \text{diam } T$ for $K \subset T$.
- (b) Each $e \in \mathcal{E}_h^\Gamma$ is an edge of exactly one $K_e \in \mathcal{T}_h$, namely $K_e = \text{conv}(e \cup \{g_{T_e}\})$, where $T_e \in \mathcal{T}_h^M$ is the macro triangle on e . Its height over e is one third of that of T_e , and $\text{diam } K_e \leq C|e|$. No macro triangle has two edges on Γ_h . Every vertex of Γ_h lies in at least two macro triangles, hence in at least four triangles of $\mathcal{T}_h$.
- (c) $\mathcal{T}_h$ has no singular vertex in the sense of (M1). It satisfies (M2) with a constant $\Theta_0 = \Theta_0(\sigma) > 0$.

In particular, $\mathcal{T}_h$ satisfies (M0)–(M2) of Assumption 2.1, with constants depending only on σ and c_0 .

Proof. (a) For $K \subset T$ we have $|K| = |T|/3$ and $\text{diam } K \leq \text{diam } T$. Hence $\text{diam } K^2/|K| \leq 3 \text{diam } T^2/|T| \leq C(\sigma)$, which bounds the angles of K from below. Also $\text{diam } K \geq$ the length of the macro edge of K , which is $\geq c \text{diam } T$.

- (b) The barycentre lies at one third of each height, so K_e is the only refined triangle with an edge on e , and $\text{diam } K_e \leq \text{diam } T_e \leq C|e|$.

The interior angle of Ω_h at a vertex z of Γ_h is $\pi + O(h_\Gamma)$. Every macro angle is at most $\pi - 2\theta_{\min}(\sigma)$. So for $h \leq h_*$ at least two macro triangles meet at z . A macro triangle with two edges on Γ_h would have both edges at a common vertex z of Γ_h , and its angle at z would be that interior angle, which is impossible.

- (c) Consider a macro vertex z with incident macro triangles $T_1, \dots, T_m$. The refined edges at z are the macro edges and the medians zg_{T_i} , and each median lies strictly inside the angle of T_i at z .

- If z is interior, $m \geq 3$, so z has $2m \geq 6$ refined edges. At most four rays from z lie on two lines, so z is not singular.
- If z is a boundary vertex, it has $2m + 1 \geq 3$ refined edges. For $m \geq 2$ there are at least $5 > 4$ of them. For $m = 1$ the three rays are two macro edges, at an angle $< \pi$, and a median strictly between them, so no two are opposite.

- A barycentre has three edges, and no two are opposite since g_T is interior.
- Every vertex lies in at least two refined triangles.

For (M2), at a macro vertex the two refined angles α'_i, α''_i inside T_i are consecutive (never the wrap-around pair), and they add up to the macro angle $\alpha_i \in [\theta_{\min}, \pi - 2\theta_{\min}]$. Hence $\Theta(z) \geq \sin \theta_{\min}$.

At a barycentre, the angles ϕ_1, ϕ_2, ϕ_3 satisfy $\sum \phi_j = 2\pi$ and, by (a), $\phi_j \leq \pi - 2\theta'_{\min}(\sigma)$. Moreover $|\sin(\phi_j + \phi_{j+1})| = \sin \phi_{j+2}$ and $\max_j \phi_j \geq 2\pi/3$. Hence $\Theta(g_T) \geq \min(\sin \frac{2\pi}{3}, \sin 2\theta'_{\min})$. $\square$

By Lemma 10.2, for $k \geq 4$ the paper applies verbatim to $\mathcal{T}_h$. The new content is $k \in \{2, 3\}$, together with the fact that no angle or singular-vertex hypothesis on the macro mesh is needed for any k . The proofs below cover all $k \geq 2$ uniformly, except the C^1 test field of Lemma 10.7.

10.2. Divergence: local and global. For a triangle T with Clough–Tocher split T^{ct} , let $V_0(T)$ be the space of continuous piecewise- P_k fields on T^{ct} vanishing on ∂T , and $Q_0(T)$ the space of discontinuous piecewise- P_{k-1} functions on T^{ct} with mean zero on T .

Lemma 10.3 (Local surjectivity). *Let $k \geq 2$. Then $\text{div } V_0(T) = Q_0(T)$. For every $q \in Q_0(T)$ there is $v \in V_0(T)$ with $\text{div } v = q$ and $\|\nabla v\|_T \leq C_{\text{loc}} \|q\|_T$, where C_{loc} depends only on k and the shape regularity of T .*

Proof. (1) *Reference element.* Let $\hat{T}$ be the reference triangle, with its Clough–Tocher split. We show that $\text{div} : V_0(\hat{T}) \rightarrow Q_0(\hat{T})$ is onto.

- $k = 2$. Here $\dim V_0(\hat{T}) = 8 = \dim Q_0(\hat{T})$, so it suffices that div is injective. A divergence-free $v \in V_0(\hat{T})$ is $\text{curl } \psi$ with $\psi \in C^1(\hat{T})$ piecewise cubic on the split and $\psi = \nabla \psi = 0$ on $\partial \hat{T}$ (the boundary is connected). All twelve Hsieh–Clough–Tocher degrees of freedom of ψ [15, 14] then vanish, so $\psi = 0$.
- $k \geq 4$. The split is a three-triangle mesh of $\hat{T}$ with no singular vertex (Lemma 10.2(c) applied to one macro triangle), so the Scott–Vogelius theorem [40, 27] applies on the domain $\hat{T}$.
- $k = 3$ (and, as a check, $k = 2, 4, 5, 6$). An exact rational rank computation gives $\text{rank}(\text{div}|_{V_0(\hat{T})}) = \dim Q_0(\hat{T})$. The dimensions are $\dim V_0 = 8, 20, 38, 62, 92$, $\dim Q_0 = 8, 17, 29, 44, 62$ and $\dim \ker \text{div} = 0, 3, 9, 18, 30$.
- The statement for all $k \geq d = 2$ is also the local ingredient of Guzmán–Neilan [26]; for $k = 2$ it is Arnold–Qin [2].

A surjective linear map between finite-dimensional spaces has a bounded right inverse. Let $\hat{C}_k$ be its norm.

- (2) *Piola transfer.* Let $F(\hat{x}) = B\hat{x} + b$ map $\hat{T}$ onto T . Affine maps preserve barycentres, so F maps $\hat{T}^{\text{ct}}$ onto T^{ct} . For $q \in Q_0(T)$, put $\hat{q} := (\det B) q \circ F \in Q_0(\hat{T})$. Take $\hat{v}$ with $\widehat{\text{div}} \hat{v} = \hat{q}$ and $\|\hat{\nabla} \hat{v}\| \leq \hat{C}_k \|\hat{q}\|$, and set $v := (\det B)^{-1} B \hat{v} \circ F^{-1}$. Then $v \in V_0(T)$, $\text{div } v = q$ and $\|\nabla v\|_T \leq \hat{C}_k \|B\| \|B^{-1}\| \|q\|_T$. $\square$

Computation: `code/lamstar_exact_ct.py` (Part 1); output `logs/lamstar_exact_ct.log`.

Lemma 10.4 (Uniform inf-sup constant, Clough–Tocher). *Let $k \geq 2$ and assume (CT0). There is $\beta > 0$, depending only on k, σ and L , such that for every $q \in \Pi_h \cap L_0^2(\Omega_h)$ there exists $v \in V_h^0$ with $\text{div } v = q$ and $\|\nabla v\| \leq \beta^{-1} \|q\|$.*

Proof. (1) *Continuous level.* By the second half of the proof of Lemma 4.5 (the Galdi wedge cover, which involves only Ω_h and not the mesh), there is $\beta_c > 0$ independent of h with the following property: for each $f \in L_0^2(\Omega_h)$ there is $w \in H_0^1(\Omega_h)^2$ with $\text{div } w = f$ and $\|\nabla w\| \leq \beta_c^{-1} \|f\|$.

- (2) *Macro Fortin step.* Let Π^{SZ} be the Scott–Zhang interpolant [41] onto continuous piecewise- P_1 fields on $\mathcal{T}_h^M$ that vanish on $\partial\Omega_h$. For an interior macro edge E , let $b_E = \lambda_a \lambda_b$ be its quadratic macro bubble. Set

$$w_h := \Pi^{\text{SZ}} w + \sum_E c_E b_E, \quad c_E := \frac{6}{|E|} \int_E (w - \Pi^{\text{SZ}} w) \in \mathbb{R}^2.$$

Then w_h is continuous piecewise- P_2 on $\mathcal{T}_h^M$, hence in V_h^0 for $k \geq 2$, and $\int_E w_h = \int_E w$ on every macro edge (both vanish on $\partial\Omega_h$). By the divergence theorem, $\int_T \text{div } w_h = \int_T \text{div } w = \int_T q$ for every $T \in \mathcal{T}_h^M$. The trace inequality and the local estimates of Π^{SZ} give $|c_E| \leq C \|\nabla w\|_{\omega_E}$. Since $\|\nabla b_E\| \leq C$ in two dimensions, $\|\nabla w_h\| \leq C \|\nabla w\| \leq C \beta_c^{-1} \|q\|$.

- (3) *Local step.* On each $T \in \mathcal{T}_h^M$, $r := q - \text{div } w_h$ is piecewise P_{k-1} on T^{ct} (since $\text{div } w_h|_T \in P_1$) and has mean zero on T . Lemma 10.3 gives $v_T \in V_0(T)$ with $\text{div } v_T = r|_T$ and $\|\nabla v_T\|_T \leq C_{\text{loc}} \|r\|_T$. Extended by zero, $v_T \in V_h^0$.
- (4) Put $v := w_h + \sum_T v_T \in V_h^0$. Then $\text{div } v = q$ and $\|\nabla v\| \leq \|\nabla w_h\| + C_{\text{loc}} (\|q\| + \sqrt{2} \|\nabla w_h\|) \leq \beta^{-1} \|q\|$. $\square$

The domain enters only through β_c , in step 1. The transfer of the Galdi argument is therefore immediate: unlike Lemma 4.5, neither a Poincaré constant nor an angle condition is involved. For $k \geq 4$ one may instead apply Lemma 4.5 to $\mathcal{T}_h$, using Lemma 10.2(c).

Lemma 10.5 (Divergence range). *Let $k \geq 2$ and assume (CT0). Then $\text{div } V_h^R = \Pi_h$, and $W_h \neq \emptyset$ for every boundary datum g . The velocities of GS and CNS* are pointwise divergence-free.*

Proof. Follow the proof of Lemma 4.3, using Lemma 10.4 for $\text{div } V_h^0 = \Pi_h \cap L_0^2$, and the quadratic edge bubble $\beta_e n_e$ of the refined triangle K_e for the constants ($\int_e \beta_e = |e|/6$). The corner obstruction in the remark after Lemma 4.3 cannot occur, because by Lemma 10.2(c) no vertex of ∂R is singular in $\mathcal{T}_h$. $\square$

10.3. Approximation in W_h and the test field.

Lemma 10.6 (Approximation in W_h , $k \geq 2$). *Let $k \geq 2$, assume (CT0), and let $\sigma_k := k + 2$ for even k and $\sigma_k := k + 1$ for odd k . In particular $\sigma_2 = \sigma_3 = 4$. Then*

$$E_h = \inf_{z \in W_h} \|\tilde{u} - z\| \leq C \varepsilon_h, \quad \varepsilon_h = (1 + (\gamma h_\Gamma)^{1/2}) h^k + h^{\sigma_k} (h_\Gamma^{-1/2} + (\mu/h)^{1/2}).$$

The lower bound $\|\tilde{u} - z\| \geq (\mu/h)^{1/2} |m_R|/|\Gamma_h|^{1/2}$ for every $z \in W_h$, stated after Lemma 4.6, holds unchanged. The flux term is at most Ch^k for bounded γ whenever $h_\Gamma \geq h^{2(\sigma_k-k)}$: that is, $h_\Gamma \geq h^4$ for $k = 2$ and $h_\Gamma \geq h^2$ for $k = 3$. If $g \cdot \nu$ is a polynomial of degree at most $\sigma_k - 1$ on each side of R , the contribution of ∂R vanishes and the flux term reduces to a term below h^k (as in the remark after Lemma 4.6).

Proof. Steps 1–5 of the proof of Lemma 4.6 go through word for word, with three substitutions.

- (1) The Lagrange interpolant is taken on $\mathcal{T}_h$. Refined triangles meeting Γ_h have diameter at most Ch_Γ by Lemma 10.2(a,b), so the three interpolation bounds of step 1 hold for every $k \geq 1$.
- (2) In step 2, the trace of $I_h \tilde{u}$ on a boundary edge is its equispaced P_k Lagrange interpolant, and the edge is not split. Integrating it is the closed $(k+1)$ -point Newton–Cotes rule, which is exact on P_{σ_k-1} . For $k = 2$ this is Simpson’s rule and for $k = 3$ Simpson’s 3/8 rule, both exact on cubics. Each edge therefore contributes $O(|e|^{\sigma_k+1})$, and $|m| \leq Ch^{\sigma_k}$.
- (3) Step 3 uses the bubbles β_e of the refined triangles K_e , and step 4 uses Lemma 10.4 in place of Lemma 4.5. $\square$

Lemma 10.7 (The C^1 test field). *Let $k \geq 2$, assume (CT0) and $h \leq h_*$, and let φ and $w = \text{curl } \varphi$ be as in Section 6. There is $v_\varphi \in Z_h$ with*

$$\|v_\varphi - w\|_{L^\infty(\Omega_h)} \leq Ch^k, \quad \max_{K \in \mathcal{T}_h} \|\nabla(v_\varphi - w)\|_{L^\infty(K)} \leq Ch. \quad (7)$$

Proof. • $k \in \{2, 3\}$. Let $\Pi^{\text{HCT}}\varphi$ be the Hsieh–Clough–Tocher interpolant on $\mathcal{T}_h^M$ [15]. It is C^1 on Ω_h and piecewise cubic on $\mathcal{T}_h$. Its degrees of freedom are the values and gradients at the macro vertices and the normal derivatives at the macro-edge midpoints, and it reproduces P_3 . The HCT interpolation estimate [14, Section 6.1] gives

$$|\varphi - \Pi^{\text{HCT}}\varphi|_{W^{j,\infty}(T)} \leq Ch_T^{4-j} |\varphi|_{W^{4,\infty}(T)}, \quad j \leq 2,$$

with C depending on σ . Put $v_\varphi := \text{curl } \Pi^{\text{HCT}}\varphi$. It is continuous and piecewise $P_2 \subset P_k$ on $\mathcal{T}_h$, and divergence-free. Since the interpolant is local and $\varphi \equiv 0$ near ∂R , $v_\varphi = 0$ near ∂R . So $v_\varphi \in Z_h$, with L^∞ error $Ch^3 \leq Ch^k$ and gradient error Ch^2 .

- $k \geq 4$. By Lemma 10.2, $\mathcal{T}_h$ satisfies (M0)–(M2), and the construction of Section 6 (the Morgan–Scott interpolant of degree $k+1$ [34] on $\mathcal{T}_h$) gives the errors Ch^{k+1} and Ch^k . $\square$

Section 6 writes $\|v_\varphi - w\|_{W^{1,\infty}} \leq Ch^k$, but its proofs use only the two bounds in (7). The $O(h^k)$ terms in step 1 of the lower bound of Theorem 6.1, in steps 4–5 of Theorem 6.4, in steps 1–2 of Theorem 6.6(i) and in step 4 of Corollary 6.7 are all pointwise values of $v_\varphi - w$ on Γ_h . Everywhere else only boundedness of ∇v_φ is used: in $\|v_\varphi\|$, $a_h(S_h, \cdot)$, $\langle \partial_n S_h, \cdot \rangle$ and $\langle e, \partial_n v \rangle$.

10.4. The theorem.

Theorem 10.8 (Clough–Tocher version of the paper). *Let $k \geq 2$ be fixed, and let $\mathcal{T}_h$ be the Clough–Tocher refinement of a macro mesh satisfying (CT0). Let V_h, Π_h be as in this section, and assume (D).*

- (i) *The following hold as stated, with constants $C, c, C_I, \kappa, c_1, M, \beta, \gamma_0, \gamma_2, h_0, \delta_0$ depending on σ, c_0, k, L (and on the data as in Section 2) instead of on (M0)–(M2), k, L :*

- Lemmas 3.1–3.5, 4.1, 4.2, 5.1, 5.2 and 6.3, and Corollary 5.3;
- Lemma 4.3 in the form of Lemma 10.5, Lemma 4.5 in the form of Lemma 10.4, and Lemma 4.6 in the form of Lemma 10.6;
- Theorems 6.1, 6.4, 6.6 and 7.2, Corollaries 6.2 and 6.7, Proposition 7.1 and Proposition 9.1(a).

In Lemma 4.1 the triangle T_e is replaced by K_e ; no triangle has two edges on Γ_h (Lemma 10.2(b)).

- (ii) *In particular, for $\gamma \in [\gamma_2, \gamma_{\max}]$ the method CNS^* satisfies*

$$\|\tilde{u} - u_h\|_{H^1(\Omega_h)} \leq C(h_\Gamma^{3/2} + h^k) \quad \text{whenever } h_\Gamma \geq h^{2(\sigma_k - k)}.$$

This is $h_\Gamma^{3/2} + h^2$ for $k = 2$ with $h_\Gamma \geq h^4$, and $h_\Gamma^{3/2} + h^3$ for $k = 3$ with $h_\Gamma \geq h^2$. The printed method GS has the leading error $(h/\mu)v_\varphi$ of Corollary 6.7, with constant 1 in the slip and energy norms.

Proof. We go through the paper section by section and record every place where the mesh, the degree or the element enters.

- (1) *Section 3.* Lemmas 3.1–3.4 concern only Γ, Γ_h and Ω_h , which are unchanged. Lemma 3.5 uses, besides these, only the inverse trace inequality $|e| \|\partial_\tau v\|_{L^2(e)}^2 \leq C_I^2 \|\nabla v\|_{K_e}^2$ on the refined triangle K_e . This holds by Lemma 10.2(a,b) and a scaling argument.
- (2) *Section 4.* Lemma 4.1 holds on K_e for the same reason, and the K_e are distinct for distinct e . Lemma 4.2 uses only Lemmas 3.2 and 4.1, so its formulas for μ_0, γ_0, c_1, M are unchanged, with the new C_I . For the remaining three lemmas, see Lemmas 10.5, 10.4 and 10.6.

- (3) *Section 5.* Lemma 5.1 is an identity for smooth $\tilde{u}$ and any $v \in V_h^R$. Lemma 5.2 uses Lemmas 3.3, 3.5 and 4.1. Corollary 5.3 uses $\operatorname{div} v = 0$ pointwise for $v \in Z_h$.
- (4) *Section 6.* Take v_φ from Lemma 10.7; by the discussion after it, (7) is all that Theorems 6.1, 6.4, 6.6 and Corollaries 6.2, 6.7 need. Lemma 6.3 uses only three facts: on e , $v \in Z_h$ is a polynomial on K_e with $\operatorname{div} v = 0$ pointwise, so $\partial_n v \cdot n_e = -\partial_\tau(v \cdot \tau_e)$ on e ; v is continuous at the vertices of Γ_h ; and Lemmas 4.1 and 3.2 hold. All three are available.
- The remaining steps (the split $\delta_R + \delta_G$, Lax–Milgram on Z_h , the ansatz r_h , and the lower-bound arguments) are abstract: they use only Lemma 4.2, Lemma 4.6 through $E_h \leq C\varepsilon_h$, and Lemma 3.2. The lift E in the proof of Theorem 6.6(ii) uses $\|g - g_I\|_{H^{1/2}(\partial R)} \leq Ch^{k+1/2}$, which holds for every $k \geq 1$.
- (5) *Section 7.* Proposition 7.1 is algebraic. In Theorem 7.2, step 2 uses $\operatorname{div} V_h^0 \subset \Pi_h$ with π_h the $L^2(\Omega_h)$ projection onto Π_h (on $\mathcal{T}_h$). Step 3 uses Lemma 10.4. Step 4 uses $\|\eta\|_{L^\infty(K_e)} \leq C(\operatorname{diam} K_e)^k \leq Ch_\Gamma^k$. Step 6 uses the bubble $\beta_e n_e$ on K_e .
- (6) *Section 9(a)* is Lemma 4.2.

The rate in (ii) is Theorem 7.2 combined with Lemma 10.6. $\square$

Remark 10.9. For $k = 2$ the bulk term h^2 dominates $h_\Gamma^{3/2}$ exactly when $h_\Gamma \leq h^{4/3}$. So for quasi-uniform meshes the CNS* error is $O(h^{3/2})$ for every $k \geq 2$, and the degree matters only through ε_h . The constant-pressure benchmark of [24] is the case $G = 0$ of Theorem 6.1. It is also analysed in [13]; the proof here does not use [13].

10.5. The penalty threshold on Clough–Tocher stars. The necessity half of Proposition 9.1 depends on the element, and we recertify it. A *Clough–Tocher star* at a boundary vertex z is the union of the macro triangles containing z , each split at its barycentre. We use the reference stars of Section 9, now split, together with two more. All are fans at $O = (0, 0)$ with Nitsche boundary $\{y = 0\}$:

- $\mathcal{S}_2$: rim points $(1, 0), (0, 1), (-1, 0)$;
- $\mathcal{S}_3$: rim points $(1, 0), (1, 1), (-1, 1), (-1, 0)$;
- $\mathcal{S}_4$: rim points $(1, 0), (1, 1), (0, 1), (-1, 1), (-1, 0)$;
- $\mathcal{S}'_4$: rim points $(1, 0), (1, 1), (0, \frac{6}{5}), (-1, 1), (-1, 0)$.

Without the split, $\mathcal{S}_2$ and $\mathcal{S}_4$ contain singular vertices, and Section 9 had to exclude them. After the split every vertex is non-singular (Lemma 10.2(c)).

Proposition 10.10 (Penalty necessity, Clough–Tocher). *Let z be a vertex of a straight part of the Nitsche boundary, and let ℓ_z be the length of its boundary edges. Suppose its Clough–Tocher star, dilated by the factor $1/\ell_z$, is congruent to the split star $\mathcal{S}$. Let $k \geq 2$, and let $\lambda_k(\mathcal{S})$ be the certified value in Table 3, taking $\lambda_k := \lambda_4$ for $k \geq 5$. If $\mu\ell_z/h < \lambda_k(\mathcal{S})$, then N_h is indefinite on Z_h .*

The same holds, with the same constants and by continuity, at every vertex of Γ_h with interior angle $\pi + O(h_\Gamma)$ whose split star is a small perturbation of a split reference star. No exclusion like $(M1')$ is needed; in particular the case of two macro triangles ($\mathcal{S}_2$), which is typical at vertices of Γ_h , is covered.

Computation: `code/lamstar_exact_ct.py` (Part 2); output `logs/lamstar_exact_ct.log`.

Proof. (1) The witnesses. Steps 1–4 of the proof of Proposition 9.1(b) apply verbatim. The witness is a continuous piecewise- P_k field on the split star, pointwise divergence-free, vanishing on the rim chords, which are macro edges. Extended by zero it therefore lies in Z_h .

- (2) Monotonicity in k . A witness for degree k is also continuous piecewise- $P_{k'}$ for $k' \geq k$, so λ^* is nondecreasing in k ; this justifies $\lambda_k := \lambda_4$ for $k \geq 5$. The witnesses of Section 9 (unsplit

		$\mathcal{S}_2$	$\mathcal{S}_3$	$\mathcal{S}_4$	$\mathcal{S}'_4$
$k = 2$	kernel dimension	6	7	8	8
	certified λ_2	5.71	7.54	7.54	7.52
	exact quotient	5.71715	7.54884	7.54467	7.52180
$k = 3$	kernel dimension	18	24	30	30
	certified λ_3	24.15	24.81	24.80	24.80
	exact quotient	24.15473	24.81227	24.80848	24.80696
$k = 4$	kernel dimension	36	50	64	64
	certified λ_4	48.38	48.78	48.78	48.77
	exact quotient	48.38946	48.78255	48.78181	48.77803

TABLE 3. Clough–Tocher split stars. *Kernel dimension*: the exact dimension of the divergence-free space of continuous piecewise- P_k fields on the split star vanishing on the rim. It equals $\dim V_{\mathcal{S}} - \dim \Pi_{\mathcal{S}}$ in every case, with $\text{rank div} = \dim \Pi_{\mathcal{S}}$. *Certified λ_k* : the rational λ_0 for which $2B - A - \lambda_0 C > 0$ is verified exactly on a rational witness. *Exact quotient*: the exact value of $(2B - A)/C$ at that witness. (Rounded to five decimals; the certified row is the rigorous statement.) It is a lower bound for the largest Rayleigh quotient $\lambda^*(\mathcal{S})$, and it agrees with a floating-point computation of $\lambda^*(\mathcal{S})$ to about 10^{-5} relative.

P_4 fans) are also piecewise P_4 on the split fans. For $k \geq 4$ they certify only 9, which Table 3 supersedes.

- (3) The certificate. The computation follows Section 9, with two changes. First, the per-element polynomials live on the split triangles, whose vertices (including barycentres) are rational. Second, the maximisation is reduced exactly. The kernel $\ker \text{div}$ and the splitting of it into $\ker C$ (fields with zero trace on the Nitsche boundary) and a complement are computed in exact rational arithmetic. On $\ker C$, $B = 0$ and $A > 0$, so the Schur complement of $2B - A$ onto the complement is formed exactly. Only the top eigenvector of this small exact pencil is computed in floating point, then rationalised.

A floating-point splitting misclassifies directions with tiny but non-zero trace. For $k = 3$ it overestimated λ^* (a spurious 25.11), which the exact check then rejected. We report only exactly verified values.

- (4) Continuity. For every Clough–Tocher star, not only the reference ones, $\text{div} : V_{\mathcal{S}} \rightarrow \Pi_{\mathcal{S}}$ is onto. Here $V_{\mathcal{S}}$ is the space of continuous piecewise- P_k fields vanishing on the rim, and $\Pi_{\mathcal{S}}$ the space of discontinuous piecewise- P_{k-1} functions. Indeed, the proof of Lemma 10.4 on the fixed Lipschitz polygon $\mathcal{S}$ (step 1 now being the continuous inf-sup condition on $\mathcal{S}$) gives $\text{div } V_0(\mathcal{S}) = \Pi_{\mathcal{S}} \cap L_0^2$, and the β_e bubble on a Nitsche edge adds the constants.

Hence $\dim \ker \text{div} = \dim V_{\mathcal{S}} - \dim \Pi_{\mathcal{S}}$ does not depend on the vertex positions, for fixed combinatorics. The divergence-free subspace is the kernel of a constraint matrix whose entries are polynomial in the vertex coordinates and whose rank is constant, so it depends continuously on them. The forms A, B, C (with edgewise normals) do as well, so the strict inequality $2B - A - \lambda_0 C > 0$ persists under small perturbations. This includes the rotation by $O(h_{\Gamma})$ of one Nitsche edge at a vertex of Γ_h and unequal edge lengths ℓ, ℓ' with ℓ'/ℓ near 1. $\square$

Remark 10.11 (Size of the threshold). The local thresholds on split stars are much larger than on unsplit ones: the exact witness quotients, which are lower bounds for λ^* , are 48.8 versus 9.98 on $\mathcal{S}_3$ for $k = 4$. They grow with k (7.5, 24.8, 48.8 for $k = 2, 3, 4$). This is consistent with the

inverse-trace constant of the thinner boundary triangle K_e , whose height is one third of that of T_e , and with its growth in k . Sufficiency (Lemma 4.2) and necessity (Proposition 10.10) still both scale with ρ , so the conclusion $\mu_0 \asymp \rho$ of Section 9 holds for Clough–Tocher meshes whose shortest boundary edge carries such a star. The threshold $\gamma^* \approx 18$ –21 measured on unsplit $k = 4$ meshes does not transfer. On the Clough–Tocher refinements of the same meshes we measured $\gamma^* \approx 11.5$ ($k = 2$), 34.6 ($k = 3$) and 65.5 ($k = 4$) at the finest levels (Table 15), each above the corresponding local certificate of Table 3.

Remark 10.12 (Scope). Theorem 10.8 covers Sections 3–7 up to and including Theorem 7.2, and Proposition 9.1. We have not re-checked the identification of the H^1 constant (Section 6.1), the lower bound (Section 7.1), the single-triangle witness of Section 9.1, or Sections 11–13 on Clough–Tocher meshes, and make no claim about them there.

11. GENERAL SMOOTH DOMAINS

Sections 2–9 treat the exterior of the unit disk in a square. Here we extend them to a bounded connected domain whose no-slip boundary Γ consists of finitely many closed C^3 curves, convex or not, each approximated by an inscribed polygon, while the rest of the boundary, $\Sigma \neq \emptyset$, is polygonal and carries strong Dirichlet data. The main new points are:

- a curvature version of the chord lemma;
- a normal-coordinate map Φ_h that works when $\Omega \setminus \Omega_h$ and $\Omega_h \setminus \Omega$ are both nonempty;
- a chart-based star-shaped cover that gives a uniform Bogovskii constant;
- with several curved components, the correct pressure constant is the *global* boundary mean. This forces a new test field (a divergence-free field without a global stream function) and a new discrete-curl interpolant.

The pressure-consistent method needs only one global correction, and its discrete velocity does not depend on how that constant is normalised; a per-component correction is inconsistent. The geometric lemmas use only $\Gamma \in C^3$. All results also assume the regularity (R1) of the exact solution ($u \in C^{k+1}(\bar{\Omega})$, $p \in C^k(\bar{\Omega})$), which for generic f requires $\Gamma \in C^{k+1,\alpha}$. The results that use the leak test field (the lower bound of Theorem 6.1, Theorems 6.4 and 6.6, Corollary 6.7) are proved under the stronger assumption $\Gamma \in C^{k+3}$, $p|_\Gamma \in C^{k+1}$ (assumption (R2) below). The scope is Sections 3–9 up to and including Theorem 7.2 and Proposition 9.1; see Section 11.8. No computations have been run for general domains.

11.1. Setting and assumptions.

11.1.1. Domain.

Assumption 11.1 (Domain). $\Omega \subset \mathbb{R}^2$ is a bounded connected open set with $\partial\Omega = \Gamma \sqcup \Sigma$, where

- $\Gamma = \Gamma_1 \sqcup \cdots \sqcup \Gamma_m$, $m \geq 1$, and each Γ_i is a whole connected component of $\partial\Omega$ and a closed embedded curve of class C^3 ;
- $\Sigma \neq \emptyset$ is a finite disjoint union of closed simple polygons, each a whole component of $\partial\Omega$.

Which components are “inner” or “outer” is not restricted: the outer boundary may be curved (with Σ an inner polygonal hole) or polygonal. Components mixing curved and straight pieces are *not* covered (the junctions are corners of Γ).

For the domain of Section 2, $m = 1$, $\Gamma_1 = \partial D$, $\Sigma = \partial R$.

For each i fix an arclength parametrisation $\gamma_i : \mathbb{R}/|\Gamma_i|\mathbb{Z} \rightarrow \Gamma_i$, let $N_i(s)$ be the unit normal pointing into Ω , and orient γ_i so that (T_i, N_i) , $T_i := \gamma'_i$, is positively oriented. The signed curvature $\varkappa$ is defined by $T'_i = \varkappa N_i$ (so $N'_i = -\varkappa T_i$). Thus $\varkappa > 0$ where Ω is locally convex near Γ and $\varkappa < 0$ where it is locally concave; for the exterior of the unit disk $\varkappa \equiv -1$. The curvature is written $\varkappa$ because κ is the Korn constant of Lemma 3.2. Since $\Gamma \in C^3$, $\varkappa \in C^1(\Gamma)$.

On Γ we write $n := -N$ (the outward normal of Ω); on Γ_h , n is the outward normal of Ω_h , as in Section 2.

Fact 11.2 (Tubular neighbourhood; standard). *There is $r > 0$ with $r\|\varkappa\|_\infty \leq \frac{1}{2}$ such that, with $\mathcal{T}_r := \{x : \text{dist}(x, \Gamma) < r\}$:*

- (i) $\Psi(s, d) := \gamma_i(s) + dN_i(s)$ is a C^2 diffeomorphism from $\bigsqcup_i (\mathbb{R}/|\Gamma_i|\mathbb{Z}) \times (-r, r)$ onto $\mathcal{T}_r$, and d is the signed distance to Γ (positive in Ω), so $\mathcal{T}_r \cap \Omega = \{d > 0\}$;
- (ii) $d \in C^3(\mathcal{T}_r)$, the projection $\pi(x) = x - d(x)\nabla d(x)$ and the arclength coordinate $\sigma := s \circ \Psi^{-1}$ are C^2 , and

$$\nabla d = N \circ \pi, \quad \nabla \sigma = \frac{T \circ \pi}{J}, \quad J(x) := 1 - \varkappa(\pi(x)) d(x) \in [\tfrac{1}{2}, \tfrac{3}{2}], \quad dx = J dd ds;$$

- (iii) $\text{dist}(\mathcal{T}_r, \Sigma) > 0$, and $\text{dist}(\Gamma_i, \Gamma_j) > 2r$ for $i \neq j$.

In the orthogonal coordinates (s, d) (scale factors J and 1), a field $F N \circ \pi + G T \circ \pi$ has

$$\text{div}(F N \circ \pi + G T \circ \pi) = \frac{1}{J}(\partial_d(JF) + \partial_s G). \quad (8)$$

Fact 11.3 (Uniform local graphs; standard). *There is $a_0 \in (0, r/4]$ with $4\|\varkappa\|_\infty a_0 \leq \frac{1}{5}$ such that for every $y = \gamma_i(s_0) \in \Gamma$, in the Cartesian frame $(y; T_i(s_0), N_i(s_0))$ with coordinates (ξ, η) and with $Q'_y := (-2a_0, 2a_0)^2$,*

$$\Gamma \cap Q'_y = \{(\xi, f_y(\xi)) : |\xi| < 2a_0\}, \quad \Omega \cap Q'_y = \{\eta > f_y(\xi)\},$$

where $f_y \in C^3$, $f_y(0) = f'_y(0) = 0$, $f''_y = \varkappa(1 + f'^2_y)^{3/2}$, $|f'_y(\xi)| \leq 2\|\varkappa\|_\infty|\xi| \leq \frac{1}{5}$, $|f_y(\xi)| \leq \|\varkappa\|_\infty \xi^2$ and $\|f'''_y\|_\infty \leq C_\Gamma$ (depending on $\|\varkappa\|_{C^1}$). Moreover $Q'_y \subset \mathcal{T}_r$.

Both facts concern the fixed curve only (tubular neighbourhood theorem, regularity of the distance function for C^3 curves, implicit function theorem and compactness); no h enters.

11.1.2. Inscribed polygons and mesh.

Assumption 11.4 (Inscribed polygons).

- (G1) For each i , $\Gamma_{h,i}$ is the closed polygon whose vertices $z_{i,0}, \dots, z_{i,N_i-1} \in \Gamma_i$ are in cyclic order along γ_i , consecutive vertices being joined by chords. For a chord e , Γ_e denotes the arc of Γ_i between its endpoints containing no other vertex, and $|\Gamma_e| \leq 2|e|$. $\Gamma_h := \bigcup_i \Gamma_{h,i}$.
- (G2) $h_\Gamma \leq h_*$, where $h_* \leq a_0/4$ is small in terms of Γ , Σ and the data (it is decreased finitely often below).

(G1) only excludes a chord that “wraps around” the long way; it is automatic if, e.g., the vertices are images of a quasi-uniform partition of the parameter.

The domain Ω_h is defined in Lemma 11.8. The mesh assumptions (M0)–(M2), (M1'), (D) of Assumption 2.1 are kept verbatim with ∂R replaced by Σ ; $\mathcal{T}_h$ is a conforming triangulation of Ω_h and $\mathcal{E}_h^\Gamma$ its edges on Γ_h . The spaces $V_h, V_h^R, V_h^0, \Pi_h, Z_h, W_h$, the forms a_h, N_h , the methods GS and CNS*, and the norm $\|\cdot\|$ are those of Section 2, with ∂R replaced by Σ . In CNS*,

$$\bar{p}_\Gamma(q) := |\Gamma_h|^{-1} \int_{\Gamma_h} q,$$

the mean over *all* of Γ_h (see Section 11.6 for why this is the right choice).

11.1.3. Exact solution, regularity and the pressure constant.

Assumption 11.5 (Regularity).

- (R1) $u \in C^{k+1}(\bar{\Omega})$, $p \in C^k(\bar{\Omega})$ solve $-\Delta u + \nabla p = f$, $\operatorname{div} u = 0$ in Ω , $u|_{\Gamma} = 0$, $u|_{\Sigma} = g$; $\tilde{f}$ as in (D).
 (R2) (Only for results that use the test field of Section 11.5: the lower bound of Theorem 11.21, Theorems 11.22 and 11.23, Corollary 11.24.) $\Gamma \in C^{k+3}$ and $p|_{\Gamma} \in C^{k+1}(\Gamma)$.

Since $u|_{\Gamma} = 0$ and $\operatorname{div} u = 0$, necessarily $\int_{\Sigma} g \cdot \nu = 0$ (ν the outward normal of Ω).

Definition 11.6 (Pressure constant). $\bar{p} := |\Gamma|^{-1} \int_{\Gamma} p \, ds$ (the mean over all curved components), $q := p - \bar{p}$ on Γ , and

$$G := \|p - \bar{p}\|_{L^2(\Gamma)}, \quad \bar{q}_i := |\Gamma_i|^{-1} \int_{\Gamma_i} q \quad (\text{so } \sum_i |\Gamma_i| \bar{q}_i = 0).$$

$$C_p := C(1 + \|p - \bar{p}\|_{W^{1,\infty}(\Gamma)}).$$

Proposition 11.25 shows that the per-component quantity $G_{\text{per}}^2 := \sum_i \|p - \bar{p}_i\|_{L^2(\Gamma_i)}^2$ ($\bar{p}_i$ the mean over Γ_i) is *not* the right one when $m \geq 2$. Note $G^2 = G_{\text{per}}^2 + \sum_i |\Gamma_i| \bar{q}_i^2$.

Constants. C, c depend on (M0)–(M2), k , the geometry ($\|\kappa\|_{C^1}$, r , a_0 , $|\Gamma|$, Σ , the fixed cover of Lemma 11.14), and norms of $(u, p, \tilde{u}, \tilde{p})$; they are independent of $h, h_{\Gamma}, \mu, \gamma, \rho$.

11.2. Geometry.

Lemma 11.7 (Chords; replaces Lemma 3.1). *Let $e \in \mathcal{E}_h^{\Gamma}$ be a chord of Γ_i of length ℓ , with midpoint m_e and unit tangent τ_e oriented like T_i along Γ_e . For $x \in e$ let $t \in [-\ell/2, \ell/2]$ be its coordinate along e from m_e , $x^* := \pi(x)$, and $\kappa_e := \kappa(m_e^*)$. Let N_e be the unit normal of e with $N_e \cdot N(x^*) > 0$ and $n_e := -N_e$. Then:*

- (a) (sagitta) $d(x) = \frac{1}{2}\kappa(x^*)(\ell^2/4 - t^2) + O(\ell^3)$ and $|d(x)| \leq \|\kappa\|_{\infty}\ell^2/8 + C\ell^4$. If $\kappa \geq 0$ on Γ_e then $e \subset \bar{\Omega}$; if $\kappa \leq 0$ on Γ_e then $e \cap \Omega = \emptyset$.
- (b) (normal rotation) $n(x^*) - n_e = \kappa_e t \tau_e + O(\ell^2)$. In particular $n(x^*) \cdot n_e = 1 + O(\ell^2)$ and $|n(x^*) \cdot \tau_e| + |N(x^*) \cdot \tau_e| \leq C\ell$.
- (c) (Jacobian) $\pi|_e : e \rightarrow \Gamma_e$ is a C^2 bijection with $\frac{d(\sigma \circ \pi)}{dt} = \frac{T(x^*) \cdot \tau_e}{J(x)} = 1 + O(\ell^2)$; hence $|\Gamma_e| = \ell(1 + O(\ell^2))$.
- (d) (turning) If e, e' share a vertex, $|\tau_e - \tau_{e'}| + |n_e - n_{e'}| \leq 2\|\kappa\|_{\infty}(|\Gamma_e| + |\Gamma_{e'}|)$.

The $O(\cdot)$ constants depend only on $\|\kappa\|_{C^1}$. For $\kappa \equiv -1$ this is Lemma 3.1 (there s plays the role of t).

Proof. By (G1), $|\Gamma_e| \leq 2h_{\Gamma} \leq a_0/2$, so Γ_e lies in Q'_z for its first endpoint z and is a graph there. By the mean value theorem there is $y_0 \in \Gamma_e$ with $T(y_0) \parallel e$. Work in the frame of Fact 11.3 at y_0 , with $f := f_{y_0}$ (the arc is still a graph there since $|\Gamma_e| \leq a_0/2$). The chord is the horizontal segment $\{(\xi, c) : \xi_a \leq \xi \leq \xi_b\}$ with $f(\xi_a) = f(\xi_b) = c$, $\xi_b - \xi_a = \ell$, $0 \in [\xi_a, \xi_b]$, and $\tau_e = e_1$, $N_e = e_2$. Put $\xi_m := (\xi_a + \xi_b)/2$, so $t = \xi - \xi_m$.

(a) Linear interpolation error: $c - f(\xi) = \frac{1}{2}f''(\zeta)(\xi - \xi_a)(\xi_b - \xi)$ for some $\zeta \in (\xi_a, \xi_b)$, and $(\xi - \xi_a)(\xi_b - \xi) = \ell^2/4 - t^2$. Since $|f'| \leq 2\|\kappa\|_{\infty}\ell$ on $[\xi_a, \xi_b]$, $f'' = \kappa(1 + f'^2)^{3/2} = \kappa(1 + O(\ell^2))$, and κ at the graph point over ζ differs from $\kappa(x^*)$ by $\|\kappa'\|_{\infty} \cdot O(\ell)$. Next, d is C^2 with $d = 0$ on the graph and $\nabla d = N$, so $d(\xi, c) = (c - f(\xi))N_2 + O((c - f)^2)$ with $N_2 = (1 + f'^2)^{-1/2} = 1 + O(\ell^2)$. Hence $d(x) = \frac{1}{2}\kappa(x^*)(\ell^2/4 - t^2) + O(\ell^3)$, and $|d(x)| \leq \frac{1}{2}\|\kappa\|_{\infty}(1 + C\ell^2)\ell^2/4 + C\ell^4$. The sign statements follow because f'' has the sign of κ and $\Omega \cap Q' = \{\eta > f\}$.

(b) $N(x^*) = (1 + f'(\xi^*)^2)^{-1/2}(-f'(\xi^*), 1)$, so $N(x^*) - N_e = -f'(\xi^*)e_1 + O(\ell^2)$. The projection moves the abscissa by $|d||f'| = O(\ell^3)$, so $f'(\xi^*) = f'(\frac{\xi}{2}) + O(\ell^3) = f'(\xi_m) + f''(\cdot)t + O(\ell^3)$. The midpoint rule for divided differences gives $f'(\xi_m) = \frac{f(\xi_b) - f(\xi_a)}{\ell} + O(\ell^2\|f'''\|_{\infty}) = O(\ell^2)$, and $f''(\cdot) = \kappa(x^*) + O(\ell) = \kappa_e + O(\ell)$. Thus $N(x^*) - N_e = -\kappa_e t \tau_e + O(\ell^2)$; multiply by -1 .

(c) By Fact 11.2(ii), $\frac{d}{dt}\sigma(\pi(x)) = \nabla\sigma \cdot \tau_e = T(x^*) \cdot \tau_e/J(x)$. Here $|T(x^*) - T(y_0)| \leq \|\varkappa\|_\infty |\Gamma_e|$, so $T(x^*) \cdot \tau_e = 1 + O(\ell^2)$, and $J = 1 + O(\ell^2)$ by (a). The derivative is positive, and the endpoints are fixed by π , so $\pi|_e$ is a bijection onto Γ_e .

(d) $\tau_e = T(y_0)$ and $\tau_{e'} = T(y'_0)$ with $y_0 \in \Gamma_e$, $y'_0 \in \Gamma_{e'}$, and $|T(y) - T(y')| \leq \|\varkappa\|_\infty \cdot (\text{arc distance})$; n_e is τ_e rotated by a fixed angle. $\square$

Lemma 11.8 (Γ_h is a normal graph; definition of Ω_h). *For $h_\Gamma \leq h_*$:*

(a) $\Gamma_{h,i} = \{\gamma_i(s) + \eta_h(s)N_i(s)\}$ for a Lipschitz, piecewise C^2 function η_h with

$$\|\eta_h\|_\infty \leq \|\varkappa\|_\infty h_\Gamma^2/8 + Ch_\Gamma^4, \quad \|\eta'_h\|_\infty \leq Ch_\Gamma.$$

In particular the $\Gamma_{h,i}$ are simple closed polygons, pairwise disjoint and disjoint from Σ , and the vertex angle of each is $\pi + O(h_\Gamma)$ (this justifies (M1')).

(b) Define

$$\Omega_h := (\Omega \setminus \mathcal{T}_{r/2}) \cup \{x \in \mathcal{T}_{r/2} : d(x) > \eta_h(\sigma(x))\}.$$

Then Ω_h is a bounded polygonal open set with $\partial\Omega_h = \Sigma \sqcup \Gamma_h$, it lies on the Ω -side of each chord (so the outward normal of Ω_h on e is n_e of Lemma 11.7), and

$$\Omega_h \setminus \Omega = \{\eta_h(\sigma) < d \leq 0\}, \quad \Omega \setminus \Omega_h = \{0 < d \leq \eta_h(\sigma)\}, \quad |\Omega \triangle \Omega_h| \leq Ch_\Gamma^2.$$

Both differences are nonempty as soon as $\varkappa$ takes both signs on Γ (for h_Γ small).

Proof. (a) By Lemma 11.7(c), π maps each chord bijectively onto its arc; the arcs Γ_e of consecutive chords tile Γ_i (G1). So $\pi|_{\Gamma_{h,i}} : \Gamma_{h,i} \rightarrow \Gamma_i$ is a bijection and $\eta_h(s) := d((\pi|_{\Gamma_{h,i}})^{-1}(\gamma_i(s)))$ represents $\Gamma_{h,i}$ as stated. The sup bound is Lemma 11.7(a). On a chord, $\frac{d\eta_h}{ds} = \nabla d \cdot \tau_e \frac{dt}{ds} = N(x^*) \cdot \tau_e (1 + O(\ell^2)) = O(\ell)$ by Lemma 11.7(b),(c). Since Ψ is a diffeomorphism and $|\eta_h| < r/2$, $\Gamma_{h,i}$ is embedded; disjointness follows from Fact 11.2(iii). The vertex angle is π plus the turning angle, which is $O(h_\Gamma)$ by Lemma 11.7(d).

(b) Openness and $\partial\Omega_h = \Sigma \sqcup \Gamma_h$ are immediate from the definition and Fact 11.2. The descriptions of the differences follow from $\Omega \cap \mathcal{T}_r = \{d > 0\}$; their measure is $\int_\Gamma \int |\eta_h| J dd ds \leq Ch_\Gamma^2$. If $\varkappa(y) < 0$ at some y , then for small h_Γ the chord e with $y \in \Gamma_e$ has $\varkappa < 0$ on Γ_e , and Lemma 11.7(a) gives $d < 0$ in its interior; similarly for $\varkappa > 0$. $\square$

Lemma 11.9 (Uniform trace, Friedrichs, Korn; replaces Lemma 3.2). *There are $C_{\text{tr}}, C_F, \kappa, C_P$, independent of h , such that every $v \in H^1(\Omega_h)^2$ with $v|_\Sigma = 0$ satisfies*

$$\|v\|_{L^2(\Gamma_h)} \leq C_{\text{tr}} \|\nabla v\|, \quad \|v\| \leq C_F \|\nabla v\|, \quad \|\nabla v\|^2 \leq \kappa a_h(v, v),$$

and every $v \in H^1(\Omega_h)$ satisfies $\inf_{c \in \mathbb{R}} \|v - c\|_{\Omega_h} \leq C_P \|\nabla v\|_{\Omega_h}$. Moreover Ω_h is connected.

Proof. (1) Let $\chi \in C^\infty(\mathbb{R})$, $0 \leq \chi \leq 1$, $\chi = 1$ on $(-\infty, r/8]$, $\chi = 0$ on $[r/4, \infty)$, $|\chi'| \leq 16/r$. Define Φ_h on $\bar{\Omega}$ by

$$\Phi_h(\Psi(s, d)) := \Psi(s, d + \chi(d)\eta_h(s)) \quad (0 \leq d < r), \quad \Phi_h := \text{id} \text{ on } \bar{\Omega} \setminus \mathcal{T}_r.$$

(2) For fixed s , $d \mapsto d + \chi(d)\eta_h(s)$ has derivative $1 + \chi'\eta_h \geq 1 - Ch_\Gamma^2/r \geq \frac{1}{2}$, equals $\eta_h(s)$ at $d = 0$ and d for $d \geq r/4$. So Φ_h is a homeomorphism of $\bar{\Omega}$ onto $\bar{\Omega}_h$ mapping Γ_i onto $\Gamma_{h,i}$, with $\Phi_h = \text{id}$ outside $\mathcal{T}_{r/4}$, in particular near Σ . Hence Ω_h is connected.

(3) In (s, d) coordinates $\hat{\Phi}_h(s, d) = (s, d + \chi\eta_h)$ has $D\hat{\Phi}_h = I + \begin{pmatrix} 0 & 0 \\ \chi\eta'_h & \chi'\eta_h \end{pmatrix} = I + O(h_\Gamma)$ a.e. Since $D\Psi \in C^1$, $D\Psi(s, d')D\Psi(s, d)^{-1} = I + O(|d' - d|) = I + O(h_\Gamma^2)$. Therefore $D\Phi_h = D\Psi \circ \hat{\Phi}_h \cdot D\hat{\Phi}_h \cdot (D\Psi)^{-1}$ satisfies $\|D\Phi_h - I\|_{L^\infty} + \|D\Phi_h^{-1} - I\|_{L^\infty} \leq Ch_\Gamma$, and $\Phi_h|_\Gamma$ has arclength Jacobian $1 + O(h_\Gamma^2)$ (it is the inverse of $\pi|_{\Gamma_h}$, Lemma 11.7(c)).

- (4) The rest is Lemma 3.2, steps 4–6, verbatim: for $w = v \circ \Phi_h$, $D(w) = D(v) \circ \Phi_h + E$ with $|E| \leq Ch_\Gamma |\nabla v \circ \Phi_h|$; on the fixed connected Lipschitz domain Ω , w vanishes on Σ , which has positive length, so Korn's first inequality, the trace inequality on Γ and the Friedrichs inequality hold; transfer back and absorb $Ch_\Gamma \|\nabla v\|$ for $h_\Gamma \leq h_*$. The Poincaré inequality transfers the same way from Ω . $\square$

Here $\Sigma \neq \emptyset$ is used (Korn and Friedrichs with zero trace on Σ). The map Φ_h works for arbitrary signs of $\varkappa$: it moves points along normals in both directions.

Lemma 11.10 (Extension; replaces Lemma 3.3 and the extension paragraph of Section 2). *There are $\tilde{u} \in C^{k+1}$ and $\tilde{p} \in C^k$ on the open set $\mathcal{O} := \Omega \cup \mathcal{T}_r$ with $\tilde{u} = u$, $\tilde{p} = p$ on Ω and $\operatorname{div} \tilde{u} = 0$ on $\mathcal{O}$. With $F := \tilde{f} + \Delta \tilde{u} - \nabla \tilde{p}$ on Ω_h (so $F = 0$ on $\Omega_h \cap \Omega$ and $\operatorname{supp} F \subset \Omega_h \setminus \Omega$):*

$$\|\tilde{u}\|_{L^\infty(\Gamma_h)} \leq Ch_\Gamma^2, \quad |(F, v)_{\Omega_h}| \leq Ch_\Gamma^2 \|\nabla v\| \quad \forall v \in V_h^R.$$

Proof. Construction. Let $U_i := \Psi(\Gamma_i \times (-r, r))$ and $U_i^+ := U_i \cap \Omega$, an annulus. The flux of u through $\Psi(\Gamma_i \times \{d\})$ is independent of $d \in (0, r)$ since $\operatorname{div} u = 0$, and tends to $\int_{\Gamma_i} u \cdot N = 0$. So $u = \operatorname{curl} \psi_i$ on U_i^+ with ψ_i single-valued and $\psi_i \in C^{k+2}(\overline{U_i^+})$. (A global stream function need not exist: when Σ has several components, g may have nonzero flux through each of them. The single-valued ψ of Section 2 is replaced by these local ones.) By Whitney's extension theorem ($\overline{U_i^+}$ has C^1 boundary, so C^{k+2} functions on it are Whitney jets), extend ψ_i to $\tilde{\psi}_i \in C^{k+2}(U_i)$, and set $\tilde{u} := \operatorname{curl} \tilde{\psi}_i$ on $U_i \setminus \Omega$ and $\tilde{u} := u$ on Ω . On U_i , $\tilde{u} = \operatorname{curl} \tilde{\psi}_i$ on both sides, so $\tilde{u} \in C^{k+1}(\mathcal{O})$ and $\operatorname{div} \tilde{u} = 0$. $\tilde{p}$ is a Whitney extension of p .

Bounds. $\tilde{u} = 0$ on Γ and $|x - \pi(x)| = |d(x)| \leq Ch_\Gamma^2$ on Γ_h (Lemma 11.8) give the first bound. For the second, F is bounded on $\Omega_h \setminus \Omega = \{\eta_h(s) < d \leq 0\}$. For $x = \Psi(s, d)$ there, with $y_h(s) := \Psi(s, \eta_h(s)) \in \Gamma_h$,

$$|v(x)| \leq |v(y_h(s))| + \int_{\eta_h(s)}^0 |\nabla v(\Psi(s, t))| dt.$$

Integrate with $dx = J dd ds$ over the strip of width $|\eta_h(s)| \leq Ch_\Gamma^2$:

$$\int_{\Omega_h \setminus \Omega} |v| \leq Ch_\Gamma^2 \left(\int_{\Gamma} |v(y_h(s))| ds + \int_{\Omega_h \setminus \Omega} |\nabla v| \right) \leq Ch_\Gamma^2 (\|v\|_{L^1(\Gamma_h)} + h_\Gamma \|\nabla v\|),$$

using the Jacobian of Lemma 11.7(c) and $|\Omega_h \setminus \Omega|^{1/2} \leq Ch_\Gamma$. Finish with $\|v\|_{L^1(\Gamma_h)} \leq |\Gamma_h|^{1/2} \|v\|_{L^2(\Gamma_h)}$ and Lemma 11.9. $\square$

Remark 11.11 (The part $\Omega \setminus \Omega_h$). Where Γ is locally convex ($\varkappa > 0$) the chords cut off $\Omega \setminus \Omega_h$. Nothing needs to be extended there: all fields live on Ω_h and all integrations by parts are on Ω_h . The errors are measured on Ω_h as before, so $\Omega \setminus \Omega_h$ (area $O(h_\Gamma^2)$) is simply not approximated. On $\Omega_h \setminus \Omega$ the error $\tilde{u} - u_h$ depends on the extension; two C^2 extensions differ there by $O(h_\Gamma^2)$ in $W^{1,\infty}$ (their difference and its gradient vanish on Γ), hence by $O(h_\Gamma^3)$ in $H^1(\Omega_h)$, which is below every rate in this section.

Lemma 3.4 (wall structure) holds verbatim on any C^1 curve, with $n = -N$.

Lemma 11.12 (Transposed-gradient term; replaces Lemma 3.5). *The statement of Lemma 3.5 holds, with*

$$a_e := -\varkappa_e n(m_e^*) (\partial_n u(m_e^*) \cdot \tau_e).$$

Proof. Only step 1 changes. For $x \in e$, $\tilde{u} \in C^2$ gives $(\nabla \tilde{u}(x))^\top n_e = (\nabla u(x^*))^\top n_e + O(h_\Gamma^2)$. By Lemma 3.4, $(\nabla u)^\top n(x^*) = 0$ and $(\nabla u)^\top b = n(\partial_n u \cdot b)$ at x^* . With Lemma 11.7(b),

$$(\nabla u(x^*))^\top n_e = n(x^*) (\partial_n u(x^*) \cdot (n_e - n(x^*))) = -\varkappa_e t n(x^*) (\partial_n u(x^*) \cdot \tau_e) + O(\ell^2) = t a_e + O(h_\Gamma^2),$$

since x^* and m_e^* are $O(\ell)$ apart. The coefficient a_e is constant on e , and $\int_e t dt = 0$. Steps 2–4 are unchanged; the sum over edges uses $\#\mathcal{E}_h^\Gamma \leq |\Gamma|/(c_0 h_\Gamma)$, since $|\Gamma_h| \leq |\Gamma|$. $\square$

For $\varkappa \equiv -1$ this is the a_e of Lemma 3.5. On straight parts of Γ ($\varkappa = 0$) the chords lie on Γ , and the leading term vanishes.

11.3. Discrete tools. Lemmas 4.1 and 4.2 are mesh-only and hold verbatim, with κ from Lemma 11.9. Lemma 4.3 and Remark 4.4 hold verbatim with $\partial R \rightarrow \Sigma$. The only domain property used in Lemma 4.3, step 1, is that Ω_h is connected (so that $\operatorname{div} V_h^0$ is the mean-zero part of Π_h), and this is Lemma 11.9. The edge bubble of step 2 may sit on any edge of any $\Gamma_{h,i}$.

11.3.1. Uniform continuous inf-sup constant.

Lemma 11.13 (Decomposition of the divergence equation). *Let $U = \bigcup_{j=1}^N U_j$ be a bounded open set, where each U_j is star-shaped with respect to every point of a ball B_j of radius $\geq r_*$ with $\overline{B_j} \subset U_j$, $\operatorname{diam} U_j \leq d_*$, and, for $j < N$, $|A_j| \geq m_*$ where $A_j := U_j \cap D_j$, $D_j := \bigcup_{l>j} U_l$. Then for every $f \in L_0^2(U)$ there is $v \in H_0^1(U)^2$ with $\operatorname{div} v = f$ and $\|\nabla v\| \leq C_B \|f\|$, where C_B depends only on N, r_*, d_*, m_* and $|U|$.*

Proof. Let $E_j := U_j \setminus D_j$ (so $E_N = U_N$ and the E_j partition U). Put $g_1 := f$ and, for $j = 1, \dots, N-1$,

$$f_j := g_j \mathbf{1}_{E_j} - \frac{\int_{E_j} g_j}{|A_j|} \mathbf{1}_{A_j}, \quad g_{j+1} := g_j - f_j, \quad f_N := g_N.$$

By induction, $\operatorname{supp} g_j \subset E_j \cup D_j$ and $\int g_j = 0$. Hence $\operatorname{supp} f_j \subset U_j$, $\int f_j = 0$ and $\operatorname{supp} g_{j+1} \subset D_j$. Also $\|f_j\| \leq (1 + |U|^{1/2} m_*^{-1/2}) \|g_j\|$ and $\|g_{j+1}\| \leq \|g_j\| + \|f_j\|$. So $\|f_j\| \leq C(N, m_*, |U|) \|f\|$, and $f = \sum_j f_j$. By Bogovskii's estimate on star-shaped domains [22, Lemma III.3.1] there is $v_j \in H_0^1(U_j)^2$ with $\operatorname{div} v_j = f_j$ and $\|\nabla v_j\| \leq c_0 (d_*/r_*)^2 (1 + d_*/r_*) \|f_j\|$. Take $v := \sum_j v_j$, each extended by zero. $\square$

Lemma 11.14 (A uniform star-shaped cover of Ω_h). *There are $N, r_*, d_*, m_* > 0$, depending only on Ω , and h_* , such that for $h_\Gamma \leq h_*$, Ω_h admits a cover as in Lemma 11.13.*

Proof. Fix $a := a_0$ (Fact 11.3), so $\|\varkappa\|_\infty a \leq 1/20$.

Chart pieces. Choose $y_1, \dots, y_{N_1} \in \Gamma$ ($N_1 \leq 4|\Gamma|/a + m$) such that the arcs of length $a/2$ centred at the y_j cover Γ . In the frame of Fact 11.3 at y_j , let $Q_j := (-a, a)^2 \subset Q'_j := (-2a, 2a)^2$, $f := f_{y_j}$, and $U_j^h := \Omega_h \cap Q_j$.

- (1) $\Gamma_h \cap Q_j$ is the graph of $f_h := I_1 f$, the piecewise-linear interpolant of f at the abscissae of the vertices of Γ_h in Q'_j . A chord meeting Q_j has length $\leq h_\Gamma < a/4$, so both endpoints lie in $Q'_j \cap \Gamma$, i.e. on the graph of f . Chords of other components, and Σ , do not meet $Q'_j \subset \mathcal{T}_r$ (Fact 11.2(iii)). Moreover $|f'_h| \leq \sup_{|\xi| < 2a} |f'| \leq \frac{1}{5} =: L$ (slopes of $I_1 f$ are values of f'), and $|f_h| \leq \|\varkappa\|_\infty (a + h_\Gamma)^2 \leq a/5 =: b_0$.
- (2) $U_j^h = \{(\xi, \eta) \in Q_j : \eta > f_h(\xi)\}$. $Q_j \setminus \Gamma_h$ has exactly two components, above and below the graph. $\Omega_h \cap Q_j$ is relatively open and closed in $Q_j \setminus \Gamma_h$ (because $\partial\Omega_h \cap Q_j = \Gamma_h \cap Q_j$), so it is a union of components. The point $(0, a/2)$ lies in Ω at distance $> a/4$ from Γ , hence in Ω_h (as $\Omega \setminus \Omega_h \subset \{d \leq Ch_\Gamma^2\}$). The point $(0, -a/2)$ has $d = -a/2 < \eta_h$, so it is not in Ω_h .
- (3) U_j^h is star-shaped with respect to every point of $B_j := B((0, \frac{3}{4}a), \frac{a}{8})$. Let $y \in B_j$, $x \in U_j^h$ and $z_t := (1-t)y + tx$. The constraints $|\xi| < a$, $\eta < a$ are convex. Since $g := f_h$ is L -Lipschitz, $g(\xi_{z_t}) \leq g(\xi_x) + L(1-t)|\xi_x - \xi_y| \leq g(\xi_x) + L(1-t)\frac{9}{8}a$. Hence

$$\eta_{z_t} - g(\xi_{z_t}) \geq t(\eta_x - g(\xi_x)) + (1-t)\left(\frac{5}{8}a - b_0 - \frac{9}{8}La\right) = t(\eta_x - g(\xi_x)) + (1-t)\frac{a}{5} > 0.$$

Also $\overline{B_j} \subset U_j^h$, since $\frac{5}{8}a > b_0$ and $\frac{7}{8}a < a$. Finally $\operatorname{diam} U_j^h \leq 2\sqrt{2}a$.

Interior pieces. Let $\Omega_* := \{x \in \Omega : \text{dist}(x, \Gamma) > a/4\}$. It is independent of h , and $\Omega_* \subset \Omega_h$ for $h_\Gamma \leq h_*$. It is connected: the map $\Psi(s, d) \mapsto \Psi(s, \frac{a}{4} + (1 - \frac{a}{4r})d)$ on $\{0 < d < r\}$, extended by the identity, is a homeomorphism $\Omega \rightarrow \Omega_*$. It is a bounded Lipschitz domain (its boundary is Σ together with the C^2 parallel curves $\{d = a/4\}$). By [22, Lemma II.1.3] it is a finite union $\Omega_* = \bigcup_{l=1}^{N_2} V_l$ of domains, each star-shaped with respect to every point of a ball. Since Ω_* is connected, the intersection graph of $\{V_l\}$ is connected. List the V_l in the reverse of a breadth-first order from a root; then each V_l except the last meets a later one in a nonempty open set. All of this is h -independent.

Covering. If $x \in \Omega_h$ and $\text{dist}(x, \Gamma) \geq a/2$, then $x \in \Omega_*$. If $\text{dist}(x, \Gamma) < a/2$, write $x = \pi(x) + dN$ with $\pi(x)$ within arclength $a/4$ of some y_j . In the frame at y_j , $|\xi_x| \leq \frac{a}{4} + \frac{a}{2} \cdot \frac{1}{5} < a$ and $|\eta_x| \leq \frac{a}{5} + \frac{a}{2} < a$, so $x \in U_j^h$. Hence $\Omega_h = \bigcup_j U_j^h \cup \bigcup_l V_l$.

Overlaps. Order the chart pieces first, then the V_l . Then $U_j^h \cap D_j \supset U_j^h \cap \Omega_* \supset y_j + \{|\xi| < \frac{a}{2}, \frac{a}{2} < \eta < a\}$. These points are above f_h , lie in Ω , and are at distance $\geq (\frac{a}{2} - b_0)/\sqrt{1 + L^2} > \frac{a}{4}$ from the graph part of Γ , and at distance $> \frac{a}{4}$ from $\Gamma \setminus Q'_j$. So $|A_j| \geq a^2/2$. The overlaps among the V_l are fixed positive numbers. Take $N = N_1 + N_2$, $r_* = \min(a/8, \text{radii of the } V_l)$, $d_* = \max(2\sqrt{2}a, \text{diam } V_l)$ and m_* the minimum overlap. $\square$

Lemma 11.15 (Uniform inf-sup constant; replaces Lemma 4.5). *The statement of Lemma 4.5 holds on Ω_h , with β depending only on k , Θ_0 , shape regularity and Ω .*

Proof. As in Lemma 4.5, the Guzmán–Scott argument [27] depends on the domain only through the continuous inf-sup (Bogovskiĭ) constant and a Poincaré constant. The latter is uniform by Lemma 11.9, the former by Lemmas 11.13 and 11.14. The wedge cover and the hypothesis $P_h \subset D$ used in Lemma 4.5 are no longer needed. $\square$

11.3.2. *Approximation.* Lemma 4.6 and the remark after it hold verbatim with $\partial R \rightarrow \Sigma$, ν the outward normal of Ω on Σ , and $m_R := \int_\Sigma g_I \cdot \nu$. The only domain facts used are:

- $\tilde{u} \in W^{k+1, \infty}(\Omega_h)$ (Lemma 11.10);
- boundary triangles have diameter $\leq Ch_\Gamma$;
- $\int_{\Gamma_h} \tilde{u} \cdot n = \int_{\partial\Omega_h} \tilde{u} \cdot n - \int_\Sigma g \cdot \nu = \int_{\Omega_h} \text{div } \tilde{u} - 0 = 0$;
- the edge-bubble field $b = \sum_{e \in \mathcal{E}_h^\Gamma} \beta_e n_e$ over all components;
- Lemma 11.15.

11.4. **Error equation and the flux structure.** Lemma 5.1 holds verbatim. Its proof integrates by parts on the polygon Ω_h , where $\tilde{u} \in C^2(\overline{\Omega_h})$, $\tilde{p} \in C^1(\overline{\Omega_h})$ and $v = 0$ on Σ . Nothing requires $\Omega \subset \Omega_h$. Lemma 5.2 holds verbatim by Lemmas 11.10 and 11.12.

For $v \in V_h^R$ put $\Phi_i(v) := \int_{\Gamma_{h,i}} v \cdot n$.

Lemma 11.16 (Fluxes).

- (a) For $v \in Z_h$, $\sum_{i=1}^m \Phi_i(v) = 0$.
- (b) If $m \geq 2$ and $h_\Gamma, h \leq h_*$ (with (R2) available for the construction, i.e. $\Gamma \in C^{k+3}$), then $v \mapsto (\Phi_i(v))_i$ maps Z_h onto $\{x \in \mathbb{R}^m : \sum_i x_i = 0\}$. In particular individual fluxes of Z_h functions do not vanish.

Proof. (a) $0 = \int_{\Omega_h} \text{div } v = \int_{\Gamma_h} v \cdot n$, since $v = 0$ on Σ . (b) Given $\alpha \in \mathbb{R}^m$ with $\sum_i |\Gamma_i| \alpha_i = 0$, Lemma 11.18 below, applied with $q := \alpha_i$ on Γ_i (which is C^∞ on Γ , the components being disjoint, and has $\int_\Gamma q = 0$), and Lemma 11.19 give $v_\alpha \in Z_h$ with $v_\alpha = \alpha_i N + O(h_\Gamma^2 + h^k)$ on $\Gamma_{h,i}$. By Lemma 11.7(b) and (c), $\Phi_i(v_\alpha) = -|\Gamma_i| \alpha_i + O(h_\Gamma^2 + h^k) |\alpha|$. The map $\alpha \mapsto (\Phi_i(v_\alpha))_i$ is linear from an $(m-1)$ -dimensional space into the hyperplane, and it is a small perturbation of the isomorphism $\alpha \mapsto (-|\Gamma_i| \alpha_i)_i$. So it is onto for small h . $\square$

Corollary 11.17 (GS on Z_h ; replaces Corollary 5.3). *For $v \in Z_h$ and every constant $c \in \mathbb{R}$,*

$$N_h(\tilde{u} - u_h, v) = \mathcal{R}(v) + \mathcal{G}(v), \quad \mathcal{R}(v) := -\langle \tilde{p} - c, v \cdot n \rangle_{\Gamma_h}.$$

The freedom is one global constant. For piecewise constants c_i on $\Gamma_{h,i}$,

$$-\langle \tilde{p} - \sum_i c_i \mathbf{1}_{\Gamma_{h,i}}, v \cdot n \rangle = \mathcal{R}(v) + \sum_i (c_i - c) \Phi_i(v),$$

and the last sum is not zero on Z_h when $m \geq 2$ and the c_i are not all equal (Lemma 11.16(b)).

Proof. As in Corollary 5.3, using only Lemma 11.16(a). $\square$

11.5. The test field and the printed method.

11.5.1. *The test field.* The field $\varphi = \chi_1(r) \int_0^\theta (p - \bar{p})$ of Section 6 generalises, per component, to $\chi(d) \int_0^s (q - \bar{q}_i)$. This is single-valued only with the *per-component* mean. Since the correct constant is global (Corollary 11.17), w must carry the net fluxes $-\lvert \Gamma_i \rvert \bar{q}_i$ through the Γ_i . Such a w has no global stream function, and the construction below replaces φ .

Lemma 11.18 (Test field). *Assume (R2), and let $q \in C^{k+1}(\Gamma)$ with $\int_\Gamma q = 0$. There is $w \in C^{k+1}(\mathcal{O}')^2$ with:*

- (a) $w = 0$ on $\Omega \cap \mathcal{N}_\Sigma$ for a neighbourhood $\mathcal{N}_\Sigma$ of Σ ; extended by 0 to $\mathcal{N}_\Sigma \setminus \Omega$, w is defined on $\mathcal{O}' := \mathcal{O} \cup \mathcal{N}_\Sigma$ (an open neighbourhood of $\bar{\Omega}$) and $\operatorname{div} w = 0$ there;
- (b) $w = q N = -q n$ on Γ ;
- (c) $\|w\|_{W^{1,\infty}(\mathcal{T}_{r/16})} \leq C \|q\|_{W^{1,\infty}(\Gamma)}$, $\|\nabla w\|_{L^2(\Omega)} \leq C \|q\|_{W^{1,\infty}(\Gamma)}$, and $\|w\|_{W^{k+1,\infty}} \leq C \|q\|_{C^{k+1}(\Gamma)}$.

If $m = 1$, or more generally if all $\bar{q}_i = 0$, then $w = -\operatorname{curl}(\chi(d) Q_i(\sigma))$ near Γ_i , which is the construction of Section 6.

Proof. Let χ be as in Lemma 11.9 (so $\chi = 1$ on $(-r, r/8]$, $\chi = 0$ on $[r/4, r)$), and $Q_i(s) := \int_0^s (q - \bar{q}_i)$, which is $|\Gamma_i|$ -periodic. On $\Psi(\Gamma_i \times (-r, r))$ set

$$w_0 := \frac{\chi(d) q(\sigma)}{J} N \circ \pi - \chi'(d) Q_i(\sigma) T \circ \pi,$$

and $w_0 := 0$ elsewhere in $\mathcal{O}$. By (8),

$$\operatorname{div} w_0 = \frac{1}{J} (\chi' q - \chi'(q - \bar{q}_i)) = \frac{\chi'(d) \bar{q}_i}{J} =: g_0 \quad \text{on } U_i.$$

Here g_0 is supported in the bands $B_i := \Psi(\Gamma_i \times [r/8, r/4]) \subset \Omega$, and

$$\int g_0 = \sum_i \bar{q}_i |\Gamma_i| \int_0^r \chi' = - \sum_i |\Gamma_i| \bar{q}_i = - \int_\Gamma q = 0.$$

Write $g_0 = \sum_{i < m} \bar{q}_i g^{(i)}$ with $g^{(i)} := \chi'(d_i)/J - (|\Gamma_i|/|\Gamma_m|) \chi'(d_m)/J$, where d_j is the signed distance to Γ_j . These are $m-1$ fixed, h -independent, compactly supported mean-zero functions in C^{k+1} (since $\Gamma \in C^{k+3}$ gives $J \in C^{k+1}$).

Let $\Omega_{16} := \{x \in \Omega : \operatorname{dist}(x, \Gamma) > r/16\}$; it is connected, by the argument for Ω_* in Lemma 11.14. Let $K \Subset \Omega_{16}$ be a connected finite union of open balls with closures in Ω_{16} containing $\bigcup_i B_i$. It exists because $\bigcup B_i$ is compact and Ω_{16} is path-connected: join the balls covering the bands by finite chains of balls along paths. A tree argument on the intersection graph of the balls writes each $g^{(i)}$ as a finite sum of mean-zero functions in C_c^{k+1} of single balls: use a smooth partition of unity subordinate to the balls, and transfer masses along tree edges with fixed bumps in $C_c^\infty(B \cap B')$. Bogovskii's operator on each ball maps $W_0^{m,p}$ to $W_0^{m+1,p}$ for every integer $m \geq 0$ and $1 < p < \infty$ [22, Thm. III.3.3]; we need only $m = k+1$. This gives $w^{(i)} \in W_0^{k+2,p}(K) \subset C_c^{k+1}(K)$ ($p > 2$) with $\operatorname{div} w^{(i)} = g^{(i)}$.

Set $w := w_0 - \sum_{i < m} \bar{q}_i w^{(i)}$. Then:

- (a) holds, since w_0 vanishes outside the collars and $K \Subset \Omega$.
- (b): at $d = 0$, $\chi = 1$, $\chi' = 0$ and $J = 1$, and $w^{(i)} = 0$ near Γ .
- (c): on $\mathcal{T}_{r/16}$ only the first term of w_0 is present and $w^{(i)} = 0$ there, with $W^{1,\infty}$ norm $\leq C\|q\|_{W^{1,\infty}}$. On the bands $|Q_i| \leq |\Gamma_i| \|q\|_\infty$, and $|\bar{q}_i| \leq \|q\|_\infty$. The regularity count is: $N \circ \pi, T \circ \pi, \sigma \in C^{k+2}$, $J \in C^{k+1}$, $q \circ \sigma \in C^{k+1}$. $\square$

Lemma 11.19 (Discrete curl of a divergence-free field). *Let $\mathcal{O}'$ be an open neighbourhood of $\bar{\Omega}$, and let $w \in W^{k+1,\infty}(\mathcal{O}')^2$ be divergence-free and vanish on a neighbourhood of Σ . For $T \in \mathcal{T}_h$ let $B_T \subset \mathcal{O}'$ be an open disk of radius $2h$ containing $\bar{T}$. It exists for $h \leq h_*$, since $\bar{\Omega}_h \subset \bar{\Omega} \cup \{|d| \leq Ch_\Gamma^2\}$ lies in a fixed compact subset of $\mathcal{O}'$. Let φ_T be a stream function of w on B_T ($w = \text{curl } \varphi_T$), and define*

$$v_w|_T := \text{curl}(\Pi_{k+1}^T \varphi_T),$$

where Π_{k+1}^T is the element restriction of the C^1 piecewise- P_{k+1} nodal interpolant of [34] (Argyris for $k = 4$), whose degrees of freedom on T are functionals of the jet of the function on $\bar{T}$. Then v_w is well defined, $v_w \in Z_h$, and

$$\|v_w - w\|_{W^{1,\infty}(T)} \leq Ch_T^k |w|_{W^{k+1,\infty}(B_T)}.$$

If $w = \text{curl } \varphi$ globally, then $v_w = \text{curl } \Pi_{k+1} \varphi$ is the v_φ of Section 6.

Proof. Well defined. Two stream functions on B_T differ by a constant, and Π_{k+1}^T is linear and reproduces constants.

Continuity. Let T, T' share an edge e , and let $B \subset \mathcal{O}'$ be a disk containing $T \cup T'$ with stream function φ_B . On the connected set $B \cap B_T$, $\varphi_T - \varphi_B$ is constant, so $\text{curl } \Pi_{k+1}^T \varphi_T = \text{curl } \Pi_{k+1}^T \varphi_B$ on T , and likewise on T' . The global interpolant of φ_B on $T \cup T'$ is C^1 across e . So the two curls agree on e .

$v_w \in Z_h$. v_w is a curl elementwise, so $\text{div } v_w = 0$. If $\bar{T}$ meets Σ , then T lies in the neighbourhood where $w = 0$ (for h small), φ_T is constant and $v_w|_T = 0$. So $v_w = 0$ on Σ .

Error. $\|\text{curl}(\Pi_{k+1}^T \varphi_T - \varphi_T)\|_{W^{1,\infty}(T)} \leq Ch_T^k |\varphi_T|_{W^{k+2,\infty}(T)} = Ch_T^k |w|_{W^{k+1,\infty}(T)}$ by the local interpolation estimate for Π_{k+1} (constants depending on (M0), (M2) as in Section 6). $\square$

Definition 11.20 (General test field). Assume (R2). With $q = p - \bar{p}$ (Definition 11.6), let w be the field of Lemma 11.18 and $v_\varphi := v_w \in Z_h$ from Lemma 11.19. Then $\|v_\varphi - w\|_{W^{1,\infty}(\Omega_h)} \leq Ch^k$, and on Γ , $w = (p - \bar{p})N = -(p - \bar{p})n$, exactly as in Section 6.

The following pointwise facts on a chord e replace the uses of “ $n(x^*) = e_r$ ” in Section 6. They follow from Lemma 11.7, $|x - x^*| \leq Ch_\Gamma^2$, and $w \in W^{1,\infty}$:

$$\begin{aligned} v_\varphi \cdot n_e &= -q(x^*) + O(h_\Gamma^2 + h^k), \\ v_\varphi \cdot \tau_e &= O(\|q\|_\infty \|\varkappa\|_\infty h_\Gamma) + O(h_\Gamma^2 + h^k), \\ (\tilde{p} - \bar{p})(x) n_e + v_\varphi(x) &= -q(m_e^*) \varkappa_e t \tau_e + O(h_\Gamma^2 + h^k). \end{aligned} \tag{9}$$

For the third: $(\tilde{p} - \bar{p})(x) = q(x^*) + O(h_\Gamma^2)$, $v_\varphi(x) = q(x^*)N(x^*) + O(h_\Gamma^2 + h^k) = -q(x^*)n(x^*) + \dots$, then Lemma 11.7(b) and $q(x^*)\varkappa(x^*) = q(m_e^*)\varkappa_e + O(\ell)$. Also, by Lemma 11.7(c),

$$\|\tilde{p} - \bar{p}\|_{L^2(\Gamma_h)} \leq G(1 + Ch_\Gamma^2) + Ch_\Gamma^2, \quad \|v_\varphi\|_{L^2(\Gamma_h)} = G(1 + O(h_\Gamma^2)) + O(h^k). \tag{10}$$

11.5.2. The theorems.

Theorem 11.21 (Theorem 6.1 on general domains). *Under Assumptions 11.1, 11.4, (M0)–(M2), (D) and (R1), the upper bound of Theorem 6.1 holds with G of Definition 11.6. If in addition (R2) holds, the lower bound holds as stated. Corollary 6.2 follows as before.*

Proof. Upper bound. Verbatim, with $c = \bar{p}$ in Corollary 11.17 and (10): $|\mathcal{R}(e_h)| \leq (G + Ch_\Gamma^2)(h/\mu)^{1/2}\|e_h\|$.

Lower bound. Use v_φ of Definition 11.20. Step 1 becomes: by (9) and Lemma 11.7(c),

$$\mathcal{R}(v_\varphi) = \langle \tilde{p} - \bar{p}, q \circ \pi \rangle_{\Gamma_h} + O(h_\Gamma^2 + h^k) = \int_\Gamma q^2 + O(h_\Gamma^2 + h^k) = G^2 + O(h_\Gamma^2 + h^k).$$

Step 2 uses (10) and $\|\nabla v_\varphi\| + (\sum_e |e| \|\partial_n v_\varphi\|_{L^2(e)}^2)^{1/2} \leq C$ (Lemma 11.18(c)). Steps 3–4 are verbatim. $\square$

Theorem 11.22 (Theorem 6.4 on general domains). *Under the assumptions of Theorem 11.21 including (R2), with $C_p = C(1 + \|p - \bar{p}\|_{W^{1,\infty}(\Gamma)})$ ($\bar{p}$ global),*

$$\|\nabla(\tilde{u} - u_h)\| \leq C_p(h_\Gamma/\gamma) + C[(1 + \gamma^{1/2})h_\Gamma^{3/2} + \varepsilon_h].$$

Proof. Steps 1, 2 and 6 are verbatim. Step 3: $S_\varphi := v_\varphi$ of Definition 11.20.

Step 4: by the third line of (9), on each e ,

$$(\tilde{p} - \bar{p})n_e + S_\varphi = b_e t + O(h_\Gamma^2 + h^k), \quad b_e := -q(m_e^*)\varkappa_e \tau_e \text{ constant on } e, \quad |b_e| \leq \|q\|_\infty \|\varkappa\|_\infty.$$

The argument of Lemma 11.12 gives $|m(v)| \leq C_p(h_\Gamma^{3/2} + h^k)\|\nabla v\|$.

Step 5: $|a_h(S_\varphi, v)| + |\langle \partial_n S_\varphi, v \rangle| \leq C_p \|\nabla v\|$ by Lemma 11.18(c) and Lemma 11.9. For $\langle S_\varphi, \partial_n v \rangle$ apply Lemma 6.3 to $S = w|_{\Gamma_h}$. Its hypotheses hold:

- w is continuous, and edgewise $W^{1,\infty}$ with $\|\nabla w\|_{L^\infty(\mathcal{T}_{r/16})} \leq C\|q\|_{W^{1,\infty}}$;
- $|w \cdot \tau_e| \leq C\|q\|_{W^{1,\infty}}h_\Gamma$ by the second line of (9) (applied to w);
- in step 3 of Lemma 6.3, $|\sigma_{e'}\tau_{e'} - \sigma_e\tau_e| \leq |w(x)|(|n_{e'} - n_e| + |\tau_{e'} - \tau_e|) \leq C\|q\|_\infty \|\varkappa\|_\infty(|e| + |e'|)$ by Lemma 11.7(d);
- the other steps of Lemma 6.3 are local and use only $\operatorname{div} v = 0$ and (M0).

The remainder $\langle S_\varphi - w, \partial_n v \rangle$ is bounded as in Theorem 6.4. $\square$

Theorem 11.23 (Theorem 6.6 on general domains). *Under the assumptions of Theorem 11.22, Theorem 6.6 holds verbatim with the global G .*

Proof. In (i), item 1 is Theorem 11.21, step 1. For item 2, on e we have $\tilde{u}(x) = \tilde{u}(x^*) + d(x)\partial_N \tilde{u}(x^*) + O(d^2) = -d(x)\partial_n u(x^*) + O(h_\Gamma^4)$. This is tangential up to $O(h_\Gamma^4)$, by $\partial_n u \cdot n = 0$ (Lemma 3.4), while $v_\varphi(x) = q(x^*)N(x^*) + O(h_\Gamma^2 + h^k)$. Hence $\tilde{u} \cdot v_\varphi = O(h_\Gamma^4 + h_\Gamma^2 h^k)$ as in Theorem 6.6. Items 3–5 are verbatim. In (ii), the lift E of $(g - g_I)|_\Sigma$ is supported near Σ , where $\Omega_h = \Omega$. $\square$

Corollary 11.24 (Corollary 6.7 on general domains). *Under the assumptions of Theorem 11.22, Corollary 6.7 holds verbatim, with v_φ of Definition 11.20 and the global G . To leading order the printed method imposes the leak condition $(\mu/h)u_h \cdot n \approx p - \bar{p}$ with $\bar{p}$ the mean of p over all curved components. In particular, with n pointing out of Ω_h , the discrete flux through $\Gamma_{h,i}$ is $\Phi_i(u_h) \approx (h/\mu)|\Gamma_i|(\bar{p}_i - \bar{p})$ (Lemma 11.16).*

Proof. Step 4 of the proof of Corollary 6.7 is (10). The rest is verbatim. $\square$

Proposition 11.25 (The constant is global). *Let $m \geq 2$. No estimate $\|\tilde{u} - u_h\| \leq C[(h/\mu)^{1/2}G_{\text{per}} + (1 + \gamma^{1/2})h_\Gamma^{3/2} + \varepsilon_h]$ can hold. Take $u = 0$, $g = 0$, $\tilde{p}$ any smooth function with $\tilde{p}|_{\Gamma_i} \equiv c_i$ (not all equal), and $\tilde{f} := \nabla \tilde{p}$, so that $f = \nabla p$. Then $G_{\text{per}} = 0$, $F = 0$, $\mathcal{G} \equiv 0$ and $E_h = 0$, but for $h \leq h_0(G)$ (in particular for fixed γ and bounded ρ as $h \rightarrow 0$)*

$$\|u_h\| \geq (2M)^{-1}(h/\mu)^{1/2}G, \quad G^2 = \sum_i |\Gamma_i|(c_i - \bar{c})^2 > 0, \quad \bar{c} = \frac{\sum_i |\Gamma_i|c_i}{|\Gamma|}.$$

Proof. Theorem 11.21, lower bound, with $\tilde{u} = 0$ (so $E_h = 0$: take $z = 0 \in W_h$) and $\mathcal{G} \equiv 0$. Here q is piecewise constant, so (R2) reduces to $\Gamma \in C^{k+3}$. $\square$

11.6. The pressure-consistent method. Proposition 7.1 holds verbatim: for $v \in V_h^R$, $-(1, \operatorname{div} v) + \langle 1, v \cdot n \rangle_{\Gamma_h} = -\int_{\Sigma} v \cdot n = 0$.

More generally, for a linear functional ℓ on $\Pi_h + \mathbb{R}$ with $\ell(1) = 1$, let CNS_ℓ^* be CNS^* with $\bar{p}_\Gamma$ replaced by ℓ :

$$B_\ell(q, v) := -(q, \operatorname{div} v) + \langle q - \ell(q), v \cdot n \rangle_{\Gamma_h}.$$

$\text{CNS}^* = \text{CNS}_\ell^*$ with $\ell = \bar{p}_\Gamma$ (global boundary mean). Extend ℓ to $\tilde{p}$ by the same formula.

Theorem 11.26 (Theorem 7.2 on general domains). *Under Assumptions 11.1, 11.4, (M0)–(M2), (D) and (R1), and for any such ℓ , Theorem 7.2 holds verbatim for CNS_ℓ^* (with β from Lemma 11.15). In particular one global correction suffices for well-posedness and for the rate $h_\Gamma^{3/2} + h_\Omega^k$, for any number m of curved components.*

Proof. We go through the steps of Theorem 7.2.

- (1) *Consistency.* For $c \in \mathbb{R}$, $B_\ell(q + c, v) = B_\ell(q, v) - c \int_{\Gamma_h} v \cdot n$, because $\int_{\Omega_h} \operatorname{div} v = \int_{\Gamma_h} v \cdot n$ for $v \in V_h^R$. With $p^* := \tilde{p} - \ell(\tilde{p})$,

$$B_\ell(p^*, v) = -(\tilde{p}, \operatorname{div} v) + \langle \tilde{p}, v \cdot n \rangle + \ell(\tilde{p}) \left(\int_{\Omega_h} \operatorname{div} v - \int_{\Gamma_h} v \cdot n \right) = -(\tilde{p}, \operatorname{div} v) + \langle \tilde{p}, v \cdot n \rangle,$$

which is the pressure form of Lemma 5.1. So (3) holds for all $v \in V_h^R$.

- (2) *Split.* Verbatim.

- (3) *Pressure.* Verbatim: $v \in V_h^0$ vanishes on all of Γ_h , and Lemma 11.15 applies.

- (4) *Velocity.* $B_\ell(e_p, e_h) = \langle e_p, e_h \cdot n \rangle - \ell(e_p) \sum_i \Phi_i(e_h) = \langle e_p, e_h \cdot n \rangle$ by Lemma 11.16(a). Also $\langle \tilde{\xi}, e_h \cdot n \rangle = \tilde{\xi} \sum_i \Phi_i(e_h) = 0$. Only the *total* flux enters, never the individual Φ_i . The rest is verbatim.

- (5) *Absorb.* Verbatim.

- (6) *Uniqueness* (Proposition 11.27). $\square$

Proposition 11.27 (Kernel analysis). *For $\gamma \geq \gamma_2$, the homogeneous CNS_ℓ^* system ($g_I = 0$, $\tilde{f} = 0$) has only the trivial solution. Hence CNS_ℓ^* is uniquely solvable for every $m \geq 1$.*

Proof. Let (u_h, p_h) solve the homogeneous system. By Lemma 4.3 $u_h \in Z_h$, and steps 3–5 with $z = 0$, $\tilde{u} = 0$, $\tilde{p} = 0$ (so $\mathcal{G} = 0$, $E_h = 0$, $\eta = 0$) give $u_h = 0$. Step 3 then gives $\|p_h - \bar{p}_h\| = 0$, where $\bar{p}_h$ is the mean over Ω_h . Since Ω_h is connected (Lemma 11.9), $p_h \equiv c$. Then for all $v \in V_h^R$,

$$0 = B_\ell(c, v) = -c \int_{\Omega_h} \operatorname{div} v + (c - c\ell(1)) \int_{\Gamma_h} v \cdot n = -c \int_{\Gamma_h} v \cdot n.$$

The edge bubble $b = \beta_e n_e$ on any single $e \in \mathcal{E}_h^\Gamma$ has $\int_{\Gamma_h} b \cdot n \neq 0$, so $c = 0$. The system is square, so uniqueness gives existence. $\square$

Remark 11.28 (The velocity does not depend on ℓ). Let $B_0(q, v) := -(q, \operatorname{div} v) + \langle q, v \cdot n \rangle_{\Gamma_h}$ be the naive form of Proposition 7.1. Then $B_\ell(q, v) = B_0(q, v) - \ell(q) \int_{\Gamma_h} v \cdot n$, so CNS_ℓ^* is the naive system plus the rank-one term $-\ell(p_h) \int_{\Gamma_h} v \cdot n$. For a constant c and $v \in V_h^R$, $B_0(c, v) = -c \int_{\partial\Omega_h} v \cdot n + c \int_{\Gamma_h} v \cdot n = 0$. Hence, if (u_h, p_h) solves CNS_ℓ^* and ℓ' is another normalisation, then $(u_h, p_h + c)$ with $c := \ell(p_h) - \ell'(p_h)$ solves $\text{CNS}_{\ell'}^*$, because $\ell'(p_h + c) = \ell(p_h)$ and the continuity row does not involve ℓ . So the discrete velocity and the scalar $\lambda_h := \ell(p_h)$ do not depend on the normalisation; the pressures differ by constants. The naive system is solvable if and only if $\lambda_h = 0$: if $\lambda_h = 0$, (u_h, p_h) solves it; conversely, if (u, p) solves it, then $(u, p - \ell(p))$ solves CNS_ℓ^* , and uniqueness gives $\lambda_h = \ell(p - \ell(p)) = 0$.

Remark 11.29 (Per-component correction is inconsistent). Let $m \geq 2$ and replace $\langle q - \bar{p}_\Gamma(q), v \cdot n \rangle$ by $\sum_i \langle q - \ell_i(q), v \cdot n \rangle_{\Gamma_{h,i}}$ with $\ell_i(q) := |\Gamma_{h,i}|^{-1} \int_{\Gamma_{h,i}} q$. This is *not* of the form CNS_ℓ^* . For $v \in V_h^R$,

$$B_{\text{per}}(q, v) = B_{\bar{p}_\Gamma}(q, v) - \sum_i (\ell_i(q) - \bar{p}_\Gamma(q)) \Phi_i(v).$$

Adding a constant to q does not change the sum, so no choice of pressure constant makes the exact pair consistent. The residual on Z_h is

$$\mathcal{I}(v) = \sum_i (\ell_i(\tilde{p}) - \bar{p}_\Gamma(\tilde{p})) \Phi_i(v) = \sum_i (\bar{p}_i - \bar{p}) \Phi_i(v) + O(h_\Gamma^2) \|v\|_{L^1(\Gamma_h)}.$$

Test with $v = v_\alpha$ from the proof of Lemma 11.16(b), $\alpha_i := \bar{p}_i - \bar{p}$, $A^2 := \sum_i |\Gamma_i| \alpha_i^2$. Then $\mathcal{I}(v_\alpha) = -A^2 + O(h_\Gamma^2 + h^k) A^2$, and $\|v_\alpha\| \leq (\mu/h)^{1/2} A(1 + o(1)) + CA$ (using $|\alpha| \leq CA$). Hence, if $A > 0$, the dual norm of $\mathcal{I}$ with respect to $\|\cdot\|$ is at least $c(h/\mu)^{1/2} A$ for $h \leq h_0$, because $h/\mu = h_\Gamma/\gamma \leq h_\Gamma/\gamma_0$ is small. We do *not* prove that the discrete error of the per-component method is actually of this size. Unlike GS, its pressure does not drop out on Z_h , so the lower-bound argument of Theorem 11.21 does not apply directly. It is, however, not a consistent method of order $h_\Gamma^{3/2}$ when the component means of p differ. The per-component method is uniquely solvable for γ large: the extra term is absorbed in step 4 like $\langle \xi - \bar{\xi}, e_h \cdot n \rangle$, since $|\ell_i(\xi - \bar{\xi})| \leq |\Gamma_{h,i}|^{-1/2} \|\xi - \bar{\xi}\|_{\Gamma_{h,i}}$, and the kernel argument of Proposition 11.27 goes through.

11.7. The penalty threshold. Proposition 9.1 is local to a boundary star and uses only that the angle of Ω_h at a vertex of Γ_h is $\pi + O(h_\Gamma)$, which is Lemma 11.8(a). It holds verbatim. No computations have been run for general domains.

11.8. Summary of dependencies and scope.

- $\Gamma \in C^3$, $\Sigma \neq \emptyset$, Ω connected: Lemmas 11.7–11.12, Section 11.3, Lemmas 5.1 and 5.2, Theorem 11.21 (upper bound), Theorem 11.26.
- Additionally (R2) ($\Gamma \in C^{k+3}$, $p|_\Gamma \in C^{k+1}$): everything using v_φ , namely the lower bound in Theorem 11.21, Theorems 11.22 and 11.23, Corollary 11.24, Lemma 11.16(b), Proposition 11.25, and the lower bound in Remark 11.29.
- New geometric constants: $\|\varkappa\|_{C^1}$, the tube radius r , the graph scale a_0 , and the h -independent cover of Lemma 11.14. The discrete constants $C_I, c_1, M, \gamma_0, \gamma_2$ depend on these only through κ and β .
- Inherited without re-proof, exactly as in Sections 4 and 6: the reduction of the Guzmán–Scott discrete inf-sup constant to the continuous one plus a Poincaré constant, and the local nodal structure and interpolation estimate of the Morgan–Scott C^1 interpolant.
- Not covered: the identification of the H^1 constant (Section 6.1), the lower bound (Section 7.1), the single-triangle witness (Section 9.1), Clough–Tocher meshes (Section 10), and the case $\Sigma = \emptyset$ or components that mix curved and straight pieces. We make no claim about these on general domains.

12. STEADY NAVIER–STOKES

Gjerde and Scott’s application is cylinder drag, which is a Navier–Stokes computation [24]. This section extends Theorems 6.1–7.2 and Corollary 6.7 to

$$-\nu \Delta u + (u \cdot \nabla)u + \nabla p = f, \quad \text{div } u = 0 \quad \text{in } \Omega, \quad u|_\Gamma = 0, \quad u|_{\partial R} = g.$$

All results of this section assume a discrete stability property of the linearisation at the exact solution, (L) for GS or (L*) for CNS*. We prove these only under a small-data condition (SD) (Lemma 12.7); deriving them from nonsingularity of a large-data solution branch is open (Remark 12.14). The Stokes estimates are reused, in particular steps 4–5 of the proof of Theorem 6.4.

The only new ingredients are a fixed-point argument around the linearisation at the exact solution, and a check that the Stokes estimates are stable under lower-order perturbations that are bounded in the H^1 dual norm.

12.1. What Gjerde–Scott print. The arXiv version of [24] (arXiv:2306.12362) states the continuous problem $-\nu\Delta u + u \cdot \nabla u + \nabla p = 0$. The drag functional (their (7)) contains the standard convective term $(u \cdot \nabla u) \cdot v$. The discrete Nitsche system (27)–(28) is printed for the Stokes operator only. No discrete convective form is printed, and in particular there is no skew-symmetrisation and no inflow/outflow boundary term. The nonlinear solve is described only as the iterated penalty method. We therefore analyse both natural choices below, and show that the conclusions are the same for both.

12.2. Methods. Let $\langle \cdot, \cdot \rangle$, N_h , V_h^R , Z_h , W_h , Π_h and $\|\cdot\|$ be as in Section 2. For $w, u, v \in H^1(\Omega_h)^2$ define

$$c_1(w; u, v) := ((w \cdot \nabla)u, v), \quad c_s(w; u, v) := \frac{1}{2} [((w \cdot \nabla)u, v) - ((w \cdot \nabla)v, u)].$$

Here c denotes either one.

Definition 12.1 (GS and CNS* for Navier–Stokes). Find $u_h \in V_h$ with $u_h|_{\partial R} = g_I$ and $p_h \in \Pi_h$ such that, for all $v \in V_h^R$ and $q \in \Pi_h$,

$$\nu N_h(u_h, v) + c(u_h; u_h, v) - (p_h, \operatorname{div} v) [+ (p_h - \bar{p}_\Gamma(p_h), v \cdot n)] = (\tilde{f}, v), \quad (q, \operatorname{div} u_h) = 0.$$

Without the bracket this is GS; with it, it is CNS*.

Remark 12.2 (Penalty convention). Multiplying N_h by ν makes the wall penalty $\nu\mu/h$. If instead one uses $\nu a_h - \nu \langle \partial_n u, v \rangle - \nu \langle u, \partial_n v \rangle + (\hat{\mu}/h) \langle u, v \rangle$ with an unscaled $\hat{\mu}$, this equals νN_h with $\mu = \hat{\mu}/\nu$. Every statement below holds with $\nu\mu$ replaced by $\hat{\mu}$. In particular the leak velocity below is then $(h/\hat{\mu})(p - \bar{p})$, and coercivity needs $\hat{\mu} \geq \nu\mu_0$.

12.3. Assumptions.

Assumption 12.3.

- (NS1) (u, p) is a smooth solution on $\bar{\Omega}$, $\tilde{f}$ is a smooth extension of f , and (M0)–(M2) and (D) hold.
 (NS2) The extensions are as in Section 2: $\tilde{u} = \operatorname{curl} \tilde{\psi}$ (so $\operatorname{div} \tilde{u} = 0$ on a neighbourhood of $\bar{\Omega}$) and $\tilde{p}$ is smooth. The defect

$$F := \tilde{f} + \nu \Delta \tilde{u} - (\tilde{u} \cdot \nabla) \tilde{u} - \nabla \tilde{p}$$

vanishes on Ω , is supported in $\bar{S}_h$, and $\|F\|_\infty \leq C$.

Stability is assumed in one of two forms, (L) or (L*) below. Lemma 12.7 shows that the standard small-data condition implies both. Let $\mathfrak{A}$ be the linearisation at $\tilde{u}$:

$$\mathfrak{A}(w, v) := \nu N_h(w, v) + c(w; \tilde{u}, v) + c(\tilde{u}; w, v).$$

Set $\|\ell\|_{*,X} := \sup_{0 \neq v \in X} \ell(v) / \|v\|$ for $X = Z_h$ or $X = V_h^R$.

- (L) There is $\beta_L > 0$, independent of h , with $\beta_L \|\delta\| \leq \|\mathfrak{A}(\delta, \cdot)\|_{*,Z_h}$ for all $\delta \in Z_h$.
 (L*) There is $\beta_* > 0$, independent of h , such that for every linear functional ℓ on V_h^R the problem

$$(\delta, \xi) \in Z_h \times \Pi_h : \quad \mathfrak{A}(\delta, v) + B^*(\xi, v) = \ell(v) \quad \forall v \in V_h^R,$$

with $B^*(q, v) = -(q, \operatorname{div} v) + \langle q - \bar{p}_\Gamma(q), v \cdot n \rangle$, has a unique solution, and $\beta_* \|\delta\| \leq \|\ell\|_{*,V_h^R}$.

(L) and (L*) are the discrete forms of “ u lies on a nonsingular branch” for GS and CNS* respectively; see Remark 12.14.

12.4. Tools.

Lemma 12.4 (Uniform Sobolev constants). *There are C_1 and N , independent of h , such that*

- (a) $\|v\|_{H^1(\Omega_h)} \leq C_1 \|\nabla v\|$ for all $v \in H^1(\Omega_h)^2$ with $v|_{\partial R} = 0$;
- (b) $|c(w; u, v)| \leq N \|w\|_{H^1(\Omega_h)} \|u\|_{H^1(\Omega_h)} \|\nabla v\|$ for all $w, u \in H^1(\Omega_h)^2$ and all $v \in H^1(\Omega_h)^2$ with $v|_{\partial R} = 0$, for $c = c_1$ and $c = c_s$.

The same holds in three dimensions.

Proof. (a) is Lemma 3.2. For (b), let Φ_h be the bi-Lipschitz map of Lemma 3.2 with $\|D\Phi_h - I\|_\infty \leq Ch_\Gamma$. The embedding $H^1(\Omega) \subset L^4(\Omega)$ on the fixed Lipschitz domain Ω ($d = 2, 3$) transfers to $\|w\|_{L^4(\Omega_h)} \leq S \|w\|_{H^1(\Omega_h)}$ with S independent of h . Then $|c_1(w; u, v)| \leq \|w\|_{L^4} \|\nabla u\| \|v\|_{L^4}$ and $|((w \cdot \nabla)v, u)| \leq \|w\|_{L^4} \|\nabla v\| \|u\|_{L^4}$. Finish with (a). $\square$

Lemma 12.5 (Convective boundary identities). *Let $w \in H^1(\Omega_h)^2$ with $\operatorname{div} w = 0$, and $u, v \in H^1(\Omega_h)^2$ with $v|_{\partial R} = 0$. Then*

$$c_1(w; u, v) - c_s(w; u, v) = \frac{1}{2} \langle (w \cdot n)u, v \rangle, \quad c_1(w; v, v) = \frac{1}{2} \langle w \cdot n, |v|^2 \rangle, \quad c_s(w; v, v) = 0.$$

Proof. $((w \cdot \nabla)u, v) + ((w \cdot \nabla)v, u) = \int_{\Omega_h} w \cdot \nabla(u \cdot v) = \int_{\partial\Omega_h} (w \cdot n)(u \cdot v) - \int_{\Omega_h} (\operatorname{div} w)(u \cdot v)$. The last integral vanishes, and so does the part of the boundary integral on ∂R , since $v = 0$ there. The outward normal of Ω_h on Γ_h is n . $\square$

So $c_1(u_h; v, v)$ vanishes for all v only if $u_h \cdot n = 0$ on Γ_h . Nitsche's method never enforces this, and for GS the normal trace is the leak $(h/\nu\mu)(p - \bar{p})$ of Corollary 12.9. Section 12.7 shows that this term nevertheless does not affect any result.

Lemma 12.6 (Error equation). *For $v \in V_h^R$,*

$$\nu N_h(\tilde{u}, v) + c(\tilde{u}; \tilde{u}, v) - (\tilde{p}, \operatorname{div} v) + \langle \tilde{p}, v \cdot n \rangle = (\tilde{f}, v) + \mathcal{G}_\nu(v) + \kappa_c(v),$$

where

$$\mathcal{G}_\nu(v) := \nu \left[\langle (\nabla \tilde{u})^\top n, v \rangle - \langle \tilde{u}, \partial_n v \rangle + \frac{\mu}{h} \langle \tilde{u}, v \rangle \right] - (F, v), \quad \kappa_{c_1} = 0, \quad \kappa_{c_s}(v) = -\frac{1}{2} \langle (\tilde{u} \cdot n) \tilde{u}, v \rangle.$$

Moreover $|\mathcal{G}_\nu(v)| \leq C(1 + \gamma^{1/2})h_\Gamma^{3/2} \|v\|$ on V_h^R , $|\mathcal{G}_\nu(v)| \leq Ch_\Gamma^{3/2} \|\nabla v\|$ on V_h^0 , and $|\kappa_{c_s}(v)| \leq Ch_\Gamma^4 \|\nabla v\|$.

Proof. Follow Lemma 5.1: $\nu a_h(\tilde{u}, v) = -\nu(\Delta \tilde{u}, v) + \nu \langle D(\tilde{u})n, v \rangle$ and $-\nu \Delta \tilde{u} = \tilde{f} - F - (\tilde{u} \cdot \nabla)\tilde{u} - \nabla \tilde{p}$. This gives the identity for c_1 . For c_s , apply Lemma 12.5 with $w = u = \tilde{u}$. The bound on $\mathcal{G}_\nu$ is Lemma 5.2, since F enters only through Lemma 3.3, which uses only $\|F\|_\infty \leq C$ and $\operatorname{supp} F \subset \bar{S}_h$. For κ_{c_s} , use $\|\tilde{u}\|_{L^\infty(\Gamma_h)} \leq Ch_\Gamma^2$ (Lemma 3.3) and Lemma 3.2. $\square$

Lemma 12.7 (Small data implies (L) and (L*)). *Let $\gamma \geq \gamma_0$ and*

$$\lambda := \frac{NC_1 \|\tilde{u}\|_{H^1(\Omega_h)}}{\nu c_1} \leq \frac{1}{4}. \tag{SD}$$

Then (L) holds with $\beta_L = \nu c_1/2$. There is $\gamma'_2 \geq \gamma_0$, independent of ν and h , such that for $\gamma \geq \gamma'_2$ (L) holds with $\beta_* = \nu/C_*$, where C_* is independent of ν and h .*

(SD) follows for small h from $NC_1 \|u\|_{H^1(\Omega)} < \nu c_1/4$, since $\|\tilde{u}\|_{H^1(\Omega_h)} \rightarrow \|u\|_{H^1(\Omega)}$. This is the usual small-data (low Reynolds number) condition. It is stated with the *discrete* Nitsche coercivity constant c_1 of Lemma 4.2, so it is more restrictive than the continuous one.

Proof. (1) (L). For $\delta \in V_h^R$, Lemma 12.4 gives $|c(\delta; \tilde{u}, \delta)| + |c(\tilde{u}; \delta, \delta)| \leq 2NC_1 \|\tilde{u}\|_{H^1} \|\nabla \delta\|^2 \leq 2\lambda \nu c_1 \|\delta\|^2$. With Lemma 4.2, $\mathfrak{A}(\delta, \delta) \geq \frac{1}{2} \nu c_1 \|\delta\|^2$.

- (2) *Pressure bound for (L^*) .* Let (δ, ξ) solve the problem in (L^*) and $v \in V_h^0$. Then $B^*(\xi, v) = -(\xi - \bar{\xi}, \operatorname{div} v)$, so $(\xi - \bar{\xi}, \operatorname{div} v) = \mathfrak{A}(\delta, v) - \ell(v)$. On V_h^0 ,

$$N_h(\delta, v) = a_h(\delta, v) - \langle \delta, \partial_n v \rangle \leq C_N \|\delta\| \|\nabla v\|, \quad C_N := 2 + C_I(c_0 \gamma_0)^{-1/2},$$

the convective terms are at most $2\lambda \nu c_1 \|\delta\| \|\nabla v\|$, and $\|v\| \leq (1 + C_I^2)^{1/2} \|\nabla v\|$. Lemma 4.5 gives

$$\beta \|\xi - \bar{\xi}\| \leq \nu(C_N + c_1/2) \|\delta\| + (1 + C_I^2)^{1/2} \|\ell\|_{*, V_h^R}.$$

- (3) *Velocity bound.* Test with $v = \delta \in Z_h$. Since $\operatorname{div} \delta = 0$ and $\int_{\Gamma_h} \delta \cdot n = 0$, we have $B^*(\xi, \delta) = \langle \xi - \bar{\xi}, \delta \cdot n \rangle$. As in step 4 of Theorem 7.2, its modulus is at most $C_I(c_0 \gamma)^{-1/2} \|\xi - \bar{\xi}\| \|\delta\|$. So

$$\frac{1}{2} \nu c_1 \|\delta\| \leq \|\ell\|_{*, V_h^R} + C_I(c_0 \gamma)^{-1/2} \beta^{-1} [\nu(C_N + c_1/2) \|\delta\| + (1 + C_I^2)^{1/2} \|\ell\|_{*, V_h^R}].$$

Choose γ'_2 with $C_I(c_0 \gamma'_2)^{-1/2} \beta^{-1} (C_N + c_1/2) \leq c_1/4$. The factor ν cancels, so γ'_2 does not depend on ν . This gives $\|\delta\| \leq C_* \nu^{-1} \|\ell\|_{*, V_h^R}$.

- (4) *Existence and uniqueness.* If $\ell = 0$, then $\delta = 0$, then ξ is constant by step 2, and $\xi = 0$ by the edge bubble of Lemma 4.3. The system is square, because $\dim Z_h + \dim \Pi_h = \dim V_h^R$ by Lemma 4.3. $\square$

12.5. The printed method. Let $z \in W_h$ attain the bound of Lemma 4.6, so $\|\tilde{u} - z\| \leq C\varepsilon_h$. With the lift E from the proof of Theorem 6.6(ii) and Lemma 12.4(a),

$$\|\tilde{u} - z\|_{H^1(\Omega_h)} \leq C(\varepsilon_h + h^{k+1/2}) \leq C\varepsilon_h. \quad (11)$$

Put $\mathcal{R}(v) := -\langle \tilde{p} - \bar{p}, v \cdot n \rangle$, as in Corollary 5.3, where $\bar{p}$ is the mean over Γ of the Navier–Stokes pressure p . Let $v_\varphi \in Z_h$ be the field of Section 6, built from this p . Write $G = \|p - \bar{p}\|_{L^2(\Gamma)}$ and

$$X := h/\mu + (1 + \gamma^{1/2})h_\Gamma^{3/2} + \varepsilon_h.$$

Theorem 12.8 (GS for Navier–Stokes). *Assume (NS1)–(NS2), (L) and $\gamma \geq \gamma_0$, for either convective form. There are $c_* > 0$ and C , depending on ν , β_L , N , C_1 , the mesh constants and the solution (and on $\|p - \bar{p}\|_{W^{1,\infty}(\Gamma)}$), such that if $X \leq c_*$, then:*

- (a) *Existence and local uniqueness.* GS has a solution (u_h, p_h) . It is the only one with $u_h \in z - (\delta_0 + \{\|\cdot\| \leq r\})$, where $\delta_0 \in Z_h$ is the linearised error defined in (13) and $r = CX$.
- (b) *H^1 rate 1.* $\|\nabla(\tilde{u} - u_h)\| \leq C[h/\mu + (1 + \gamma^{1/2})h_\Gamma^{3/2} + \varepsilon_h]$.
- (c) *Energy rate 1/2.* $\|\tilde{u} - u_h\| \leq \nu^{-1}(h/\mu)^{1/2}G + CX$.
- (d) *Quadratic closeness to the linearisation.* $\|(\tilde{u} - u_h) - (\tilde{u} - z) - \delta_0\| \leq CX^2$.

Proof. (1) *Reformulation.* Subtracting the GS equations from Lemma 12.6, and using $\operatorname{div} v = 0$ and $\int_{\Gamma_h} v \cdot n = 0$ for $v \in Z_h$, the error $e := \tilde{u} - u_h$ satisfies

$$\nu N_h(e, v) + c(\tilde{u}; \tilde{u}, v) - c(u_h; u_h, v) = \mathcal{R}(v) + \mathcal{G}_\nu(v) + \kappa_c(v) \quad \forall v \in Z_h.$$

By bilinearity, $c(\tilde{u}; \tilde{u}, v) - c(\tilde{u} - e; \tilde{u} - e, v) = c(e; \tilde{u}, v) + c(\tilde{u}; e, v) - c(e; e, v)$, so

$$\mathfrak{A}(e, v) = \mathcal{R}(v) + \mathcal{G}_\nu(v) + \kappa_c(v) + c(e; e, v).$$

Write $u_h = z - \delta$ with $\delta \in Z_h$, so $e = e(\delta) := (\tilde{u} - z) + \delta$. Then u_h solves the GS momentum equation on Z_h if and only if

$$\mathfrak{A}(\delta, v) = \ell_0(v) + c(e(\delta); e(\delta), v) \quad \forall v \in Z_h, \quad \ell_0 := \mathcal{R} + \mathcal{G}_\nu + \kappa_c - \mathfrak{A}(\tilde{u} - z, \cdot). \quad (12)$$

Given such $u_h \in W_h$, the functional $v \mapsto \nu N_h(u_h, v) + c(u_h; u_h, v) - (\tilde{f}, v)$ vanishes on $Z_h = \ker(\operatorname{div}|_{V_h^R})$. Since $\operatorname{div} V_h^R = \Pi_h$ (Lemma 4.3), there is a unique $p_h \in \Pi_h$ with $(p_h, \operatorname{div} v)$ equal to it on V_h^R . So solving (12) is equivalent to solving GS.

(2) *Linearised error.* By (L), $\mathfrak{A}$ is invertible on Z_h (a square system). Let

$$\delta_0 := \mathfrak{A}^{-1}\ell_0 = \delta_R + \delta_G, \quad \mathfrak{A}(\delta_R, \cdot) = \mathcal{R}, \quad \mathfrak{A}(\delta_G, \cdot) = \mathcal{G}_\nu + \kappa_c - \mathfrak{A}(\tilde{u} - z, \cdot) \quad \text{on } Z_h. \quad (13)$$

(3) *Bound on δ_G .* Since $|\mathfrak{A}(\tilde{u} - z, v)| \leq \nu M \|\tilde{u} - z\| \|v\| + 2N \|\tilde{u}\|_{H^1} \|\tilde{u} - z\|_{H^1} \|\nabla v\|$, Lemma 12.6 and (11) give $\|\delta_G\| \leq C\beta_L^{-1}[(1 + \gamma^{1/2})h_\Gamma^{3/2} + \varepsilon_h]$.

(4) *Ansatz for δ_R .* Set $\delta_R = (h/\nu\mu)v_\varphi + r$. Since $\nu N_h \cdot (h/\nu\mu) = (h/\mu)N_h$,

$$\mathfrak{A}(r, v) = m(v) + \ell(v) - \frac{h}{\nu\mu} [c(v_\varphi; \tilde{u}, v) + c(\tilde{u}; v_\varphi, v)],$$

where m and ℓ are exactly the functionals of step 3 of Theorem 6.4. Steps 4–5 of that proof give $|m(v)| + |\ell(v)| \leq [C(h/\mu + h_\Gamma^{3/2} + h^k) + C(h/\mu)h^k h_\Gamma^{-1/2}] \|\nabla v\|$. The convective term is at most $C(h/\mu) \|\nabla v\|$ by Lemma 12.4, since $\|v_\varphi\|_{H^1} \leq C$. Using $(h/\mu)h^k h_\Gamma^{-1/2} \leq Ch^k$, $\|\nabla v\| \leq \|v\|$ and (L),

$$\|r\| \leq C\beta_L^{-1}(h/\mu + h_\Gamma^{3/2} + \varepsilon_h).$$

(5) *Consequences for δ_0 .* Since $\|\nabla v_\varphi\| \leq C$ and $\|v_\varphi\| \leq (\mu/h)^{1/2}(G + Ch_\Gamma^2 + Ch^k) + C$ (step 2 of the lower bound in Theorem 6.1),

$$\|\nabla \delta_0\| \leq CX, \quad \|\delta_0\| \leq \nu^{-1}(h/\mu)^{1/2}G + CX. \quad (14)$$

(6) *Contraction.* Let $\Phi(\delta) := \mathfrak{A}^{-1}[\ell_0 + c(e(\delta); e(\delta), \cdot)]$ on Z_h . Set $A := \|\tilde{u} - z\|_{H^1} + C_1 \|\nabla \delta_0\|$, so $A \leq CX$ by (11) and (14). Consider the ball $B := \{\delta \in Z_h : \|\delta - \delta_0\| \leq A/C_1\}$. For $\delta \in B$, Lemma 12.4(a) gives $\|e(\delta)\|_{H^1} \leq A + C_1 \|\delta - \delta_0\| \leq 2A$.

- By (L) and Lemma 12.4(b), $\|\Phi(\delta) - \delta_0\| \leq \beta_L^{-1}N(2A)^2$, which is at most A/C_1 if $A \leq \beta_L/(4NC_1)$.
- For $\delta_1, \delta_2 \in B$ with $d := \delta_1 - \delta_2$, we have $c(e_1; e_1, v) - c(e_2; e_2, v) = c(d; e_1, v) + c(e_2; d, v)$. Hence

$$\|\Phi(\delta_1) - \Phi(\delta_2)\| \leq \beta_L^{-1}N \cdot 4A \cdot C_1 \|d\| \leq \frac{1}{2} \|d\| \quad \text{if } A \leq \beta_L/(8NC_1).$$

Both conditions follow from $X \leq c_*$. Banach's fixed-point theorem gives a unique $\delta \in B$ with $\Phi(\delta) = \delta$, which proves (a), and $\|\delta - \delta_0\| \leq 4\beta_L^{-1}NA^2 \leq CX^2$, which proves (d).

(7) (b) and (c). From $e = (\tilde{u} - z) + \delta_0 + (\delta - \delta_0)$, (14), (d), $\|\tilde{u} - z\| \leq C\varepsilon_h$ and $X^2 \leq X$. $\square$

Corollary 12.9 (Exact leading term; sharpness). *Assume in addition that $G > 0$, $h_\Gamma \rightarrow 0$, $h/\mu \rightarrow 0$, $\gamma h_\Gamma \rightarrow 0$ and $\varepsilon_h/(h_\Gamma/\gamma)^{1/2} \rightarrow 0$ (the hypotheses of Corollary 6.7). Then*

$$\tilde{u} - u_h = \frac{h}{\nu\mu} v_\varphi + \vartheta_h, \quad \|\vartheta_h\| = o((h/\mu)^{1/2}), \quad \|\vartheta_h\|_{\Gamma_h} = o(h/\mu),$$

and consequently

$$\frac{\|\tilde{u} - u_h\|_{L^2(\Gamma_h)}}{(h/\nu\mu)G} \rightarrow 1, \quad \frac{\|\tilde{u} - u_h\|}{\nu^{-1}(h/\mu)^{1/2}G} \rightarrow 1, \quad \|\nabla(\tilde{u} - u_h)\| \geq c \frac{h}{\nu\mu} G - Ch^{k+1/2} \quad \text{for small } h.$$

So $u_h \cdot n \approx (h/\nu\mu)(p - \bar{p})$ on Γ_h , with the Navier–Stokes pressure, and, if moreover μ is bounded (for instance fixed γ and bounded ρ), the H^1 rate 1 of Theorem 12.8(b) is sharp.

Proof. (1) Set $\vartheta_h := (\tilde{u} - z) + \delta_G + r + (\delta - \delta_0)$. Steps 3–4 and 6 of the proof of Theorem 12.8 give $\|\vartheta_h\| \leq C[h/\mu + (1 + \gamma^{1/2})h_\Gamma^{3/2} + \varepsilon_h] + CX^2 \leq CX$.

(2) Under the hypotheses, $X/(h/\mu)^{1/2} \rightarrow 0$. Indeed,

$$\frac{h/\mu}{(h/\mu)^{1/2}} \rightarrow 0, \quad \frac{(1 + \gamma^{1/2})h_\Gamma^{3/2}}{(h_\Gamma/\gamma)^{1/2}} = (\gamma^{1/2} + \gamma)h_\Gamma \rightarrow 0 \quad (\gamma \geq \gamma_0), \quad \frac{\varepsilon_h}{(h/\mu)^{1/2}} \rightarrow 0.$$

Hence $\|\vartheta_h\| = o((h/\mu)^{1/2})$ and $\|\vartheta_h\|_{\Gamma_h} \leq (h/\mu)^{1/2} \|\vartheta_h\| = o(h/\mu)$.

- (3) Step 4 of the proof of Corollary 6.7 gives $\|v_\varphi\|_{\Gamma_h} = G(1 + o(1))$ and $\|v_\varphi\|^2 = (\mu/h)\|v_\varphi\|_{\Gamma_h}^2 + O(1)$. This yields the two limits.
- (4) For the H^1 lower bound, use $\|e\|_{\Gamma_h} \leq C_{\text{tr}}(\|\nabla e\| + \|\nabla E\|)$ with $\|\nabla E\| \leq Ch^{k+1/2}$, where E is the lift from the proof of Theorem 6.6(ii). If μ is bounded, $h^{k+1/2} = o(h/\mu)$ and the term is absorbed. $\square$

Remark 12.10. The slip and energy constants are universal ($= 1$), exactly as for Stokes. The H^1 constant $\lim\|\nabla(\tilde{u} - u_h)\| \nu \mu / (hG)$, if it exists, is not universal. It now depends on the *Oseen* response: r solves the linearised problem, including $c(v_\varphi; \tilde{u}, \cdot) + c(\tilde{u}; v_\varphi, \cdot)$, which is of the same order as ∇v_φ . So the Stokes value $K = \|\nabla U\|/G$ of Corollary 6.11 (2.7565 for test P) is not expected to carry over; we have not identified the Navier–Stokes analogue.

Remark 12.11 (Cylinder drag forces $G > 0$). Under (L) and the hypotheses of Corollary 12.9, the following holds. For flow past the cylinder, the pressure drag is $D_p = \int_\Gamma p n_1$, up to the orientation of n . Since $\int_\Gamma n_1 = 0$, $|D_p| = |\int_\Gamma (p - \bar{p}) n_1| \leq |\Gamma|^{1/2} G$, so $G \geq |D_p|/\sqrt{2\pi}$. In Gjerd–Scott’s own application, the printed method therefore always leaks, at velocity $(h/\nu\mu)(p - \bar{p})$, and its asymptotic H^1 rate at fixed μ is 1, not 3/2. With their $\mu = 10^6$ the leak is small in absolute terms, which is consistent with their accurate drag values. For Stokes, the effect of the leak on the computed drag is quantified in Section 8 (Theorem 8.17, Remark 8.23); for Navier–Stokes only the unconditional $O(h/\mu)$ bound of Proposition 8.25 is proved.

12.6. The pressure-consistent method.

Theorem 12.12 (CNS* for Navier–Stokes: Scott’s rate). *Assume (NS1)–(NS2) and (L*); by Lemma 12.7, (SD) and $\gamma \geq \gamma'_2$ suffice. Let $Y := (1 + \gamma^{1/2})h_\Gamma^{3/2} + \varepsilon_h$. There are c_*, C such that if $Y \leq c_*$, then CNS* has a solution, whose velocity is unique in the $\|\cdot\|$ -ball of radius CY about $z - \delta_0$, where δ_0 is defined in step 2 below. It satisfies*

$$\|\tilde{u} - u_h\| \leq C[(1 + \gamma^{1/2})h_\Gamma^{3/2} + \varepsilon_h].$$

Hence, for fixed $k \geq 4$, $\gamma \in [\gamma'_2, \gamma_{\max}]$ and $h_\Gamma \geq h_\Omega^{2(\sigma_k - k)}$,

$$\|\tilde{u} - u_h\|_{H^1(\Omega_h)} \leq C(h_\Gamma^{3/2} + h_\Omega^k).$$

Proof. (1) *Error equation.* Let $p^* := \tilde{p} - \bar{p}_\Gamma(\tilde{p})$, $\eta := p^* - \pi_h p^*$ (with π_h the L^2 projection onto Π_h), and write $p_h = p^* - \eta - \xi$ and $u_h = z - \delta$. As in step 1 of Theorem 7.2 and step 1 above, (u_h, p_h) solves CNS* if and only if $(\delta, \xi) \in Z_h \times \Pi_h$ solves

$$\mathfrak{A}(\delta, v) + B^*(\xi, v) = \ell_0(v) + c(e(\delta); e(\delta), v) \quad \forall v \in V_h^R, \quad \ell_0 := \mathcal{G}_\nu + \kappa_c - \mathfrak{A}(\tilde{u} - z, \cdot) - B^*(\eta, \cdot).$$

The continuity row holds because $\text{div } u_h = \text{div } z - \text{div } \delta = 0$.

- (2) *Size of ℓ_0 .* We have $(\eta, \text{div } v) = 0$ for all $v \in V_h^R$, because $\text{div } V_h^R \subset \Pi_h$. Hence $|B^*(\eta, v)| \leq 2\|\eta\|_{\Gamma_h} (h/\mu)^{1/2} \|v\| \leq Ch_\Gamma^k \|v\|$. With Lemma 12.6 and (11), $\|\ell_0\|_{*, V_h^R} \leq CY$. Let (δ_0, ξ_0) be the solution of (L*) with data ℓ_0 ; then $\|\delta_0\| \leq \beta_*^{-1} CY$.
- (3) *Fixed point.* Let $\Phi(\delta)$ be the Z_h -component of the (L*) solution with data $\ell_0 + c(e(\delta); e(\delta), \cdot)$. Since $\|c(e; e, \cdot)\|_{*, V_h^R} \leq N\|e\|_{H^1}^2$, step 6 of the proof of Theorem 12.8 applies verbatim, with β_L replaced by β_* and $A := \|\tilde{u} - z\|_{H^1} + C_1\|\delta_0\| \leq CY$.
- (4) *Conclusion.* This gives the unique fixed point and $\|\tilde{u} - u_h\| \leq \|\tilde{u} - z\| + \|\delta_0\| + CY^2 \leq CY$. The H^1 statement follows as in Theorem 7.2, using the lift E for the L^2 part. $\square$

12.7. Is an inflow/outflow correction needed? The next lemma quantifies the convective boundary term.

Lemma 12.13 (Wall Péclet number). *Let $\mu \geq \mu_0$, $w \in V_h$ with $\operatorname{div} w = 0$, and $v \in V_h^R$. Then $\nu N_h(v, v) + c_s(w; v, v) \geq \nu c_1 \|v\|^2$, and*

$$\nu N_h(v, v) + c_1(w; v, v) \geq (\nu c_1 - \tfrac{1}{2} \operatorname{Pe}_\Gamma(w)) \|v\|^2, \quad \operatorname{Pe}_\Gamma(w) := \frac{h}{\mu} \|(w \cdot n)_-\|_{L^\infty(\Gamma_h)},$$

where $(w \cdot n)_- := \max(-w \cdot n, 0)$ is the inflow part. Here n points out of Ω_h (into P_h).

Proof. By Lemma 12.5, $c_1(w; v, v) = \frac{1}{2} \langle w \cdot n, |v|^2 \rangle \geq -\frac{1}{2} \|(w \cdot n)_-\|_\infty \|v\|_{\Gamma_h}^2$, and $\|v\|_{\Gamma_h}^2 \leq (h/\mu) \|v\|^2$. $\square$

Consequences:

- (1) *The results above need no correction, for either form.* (L) and (L*) linearise at $\tilde{u}$, whose normal trace on Γ_h is $O(h_\Gamma^2)$. For c_1 versus c_s , $\mathfrak{A}$ changes by $\frac{1}{2} \langle (\tilde{u} \cdot n) \delta, v \rangle + \frac{1}{2} \langle (\delta \cdot n) \tilde{u}, v \rangle = O(h_\Gamma^2) \|\delta\|_{\Gamma_h} \|v\|_{\Gamma_h}$ (Lemma 12.5). So (L) for one form implies (L) for the other with $\beta_L - Ch_\Gamma^2 h/\mu$. The discrete leak enters only through the quadratic term $c(e; e, v)$, which the contraction controls by continuity alone (Lemma 12.4); no sign is needed.
- (2) *The leak does make $c_1(u_h; u_h, u_h) \neq 0$.* For GS, Corollary 12.9 gives $u_h \cdot n = (h/\nu\mu)(p - \bar{p}) + o(h/\mu)$ in $L^2(\Gamma_h)$, with n pointing into the cylinder. The discrete wall is porous: fluid leaves Ω_h where $p > \bar{p}$ (the front stagnation region) and re-enters where $p < \bar{p}$. The leak law $(\nu\mu/h) u_h \cdot n \approx p - \bar{p}$ is a Darcy-type condition with permeability $h/\nu\mu$. If the leading term also held in $L^\infty(\Gamma_h)$ (Corollary 12.9 is an L^2 statement; an L^∞ version would need an inverse estimate and is not proved here), the inflow part would give $\operatorname{Pe}_\Gamma(u_h) \approx (h/\mu)(h/\nu\mu) \|(p - \bar{p})_-\|_\infty = O((h/\mu)^2)$. The penalty would then dominate the inflow term by a factor of order $(h/\mu)^{-2}$. Formally, the kinetic-energy flux $\frac{1}{2} \int_{\Gamma_h} (u_h \cdot n) |u_h|^2$ is cubic in a boundary trace of size $h_\Gamma^2 + h/\mu$, which is negligible against the penalty dissipation $(\nu\mu/h) \|u_h\|_{\Gamma_h}^2 \approx (h/\nu\mu) G^2$. Point 1 is what the proofs actually use; point 2 is only the physical picture.
- (3) *When a correction would matter.* Lemma 12.13 degrades only when $\operatorname{Pe}_\Gamma(w) \gtrsim \nu c_1$ for iterates w far from $\tilde{u}$. This happens on coarse meshes, at large Reynolds number with fixed μ , or in Picard or iterated-penalty sweeps that start far from the solution. There c_s (for which $c_s(w; v, v) = 0$ identically), or an upwind inflow term $-\langle (w \cdot n)_- u, v \rangle$ in the style of Nitsche methods for Oseen problems (consistent to $O(h_\Gamma^4)$ at $\tilde{u}$), restores the energy inequality. For the asymptotic statements above neither is needed. c_s is also the natural choice for proving existence for arbitrary data, which we do not attempt: that would need a discrete Hopf extension of g_I inside W_h .

Remark 12.14 (What a Brezzi–Rappaz–Raviart version needs). (SD) is sufficient but not necessary for (L) and (L*). To derive (L) from continuous nonsingularity (the linearised Navier–Stokes operator at u is an isomorphism on the divergence-free subspace of H_0^1) one would run a Schatz/Gårding argument. Write $\mathfrak{A} = \nu N_h + K_h$ with K_h of lower order. It would then suffice that the discrete Nitsche–Stokes solution operators S_h (on Z_h , respectively $Z_h \times \Pi_h$) converge to the continuous one S on the moving domains $\Omega_h \rightarrow \Omega$, in operator norm on the compact set $\{K\delta : \|\nabla \delta\| \leq 1\}$. For data that are only in $L^{3/2}$ or H^{-1+s} , this requires Theorems 6.1 and 7.2 without a rate but with rough data. There the pressure is only in $W^{1,3/2}$ near Γ , and Lemma 3.5 must be replaced by an $o(1)$ statement for $\nabla w \in H^s$. We expect this to go through for CNS*, and for GS in the energy norm (whose inconsistency is $(h/\mu)^{1/2} \|p - \bar{p}\|_{L^2(\Gamma)} \rightarrow 0$), but we have not written it out. It is the only gap between the small-data theorems above and a nonsingular-branch theorem.

13. THREE DIMENSIONS

This section asks which of Theorems 6.1–7.2, Corollary 6.7, Proposition 9.1 and Section 12 survive when the curved wall is a surface approximated by an inscribed triangulated polyhedron.

The answer (Theorem 13.13) is that all of them except the necessity half, Proposition 9.1(b), carry over, with the same rates, under one stability hypothesis, (IS3). (IS3) is known only for split meshes (Alfeld/barycentric, $k \geq 3$), modulo the uniformity of the published constants over the perturbed domains (item 3 of Section 13.7). Only a spherical obstacle is treated. Three things change:

- the odd-in- s cancellation of Lemma 3.5 fails on non-equilateral faces (Lemma 13.2(d));
- the vertex terms of Lemma 6.3 become edge terms (Lemma 13.11);
- the Morgan–Scott stream-function construction of v_φ is replaced by an interpolate-and-correct construction (Lemma 13.10).

The first change turns out to be harmless, because the error analysis only needs a penalty-weighted bound (Lemma 13.6); this is also true in 2D. The necessity half of Proposition 9.1 is not proved in 3D.

13.1. Setting. Let $R = (-L, L)^3$ with $L > 1$, let D be the open unit ball, $\Omega = R \setminus \overline{D}$ and $\Gamma = \partial D$. Let P_h be a convex polyhedron with triangular faces whose vertices lie on Γ . Then $P_h \subset D$ and $\Omega \subset \Omega_h := R \setminus P_h$. $\mathcal{T}_h$ is a conforming tetrahedral mesh of Ω_h , and $\mathcal{F}_h^\Gamma$ is its set of faces on $\Gamma_h = \partial P_h$. For simplicity we assume that each $F \in \mathcal{F}_h^\Gamma$ is a whole face of P_h (this holds for an Alfeld refinement of such a mesh; for splits that subdivide boundary faces see Remark 13.12). Put

$$h := \max_T \text{diam } T, \quad h_\Gamma := \max_{F \in \mathcal{F}_h^\Gamma} \text{diam } F, \quad \rho := h / \min_{F \in \mathcal{F}_h^\Gamma} \text{diam } F, \quad \gamma := \mu h_\Gamma / h.$$

The spaces $V_h, V_h^R, V_h^0, \Pi_h, Z_h, W_h$, the forms a_h, N_h , and the methods GS and CNS* are defined exactly as in Section 2, with n the unit normal of Γ_h pointing into P_h . The extension $\tilde{u}$ is built as in two dimensions, with a vector potential in place of the stream function: since $\text{div } u = 0$ and $\int_\Gamma u \cdot n = 0$, on a spherical shell $\{1 < |x| < 1 + r\}$ we have $u = \text{curl } \psi$ for a smooth vector potential ψ ; we extend ψ smoothly across Γ to $\tilde{\psi}$ and set $\tilde{u} := \text{curl } \tilde{\psi}$ there, so that $\text{div } \tilde{u} = 0$ on a neighbourhood of $\overline{\Omega}$. The energy norm is

$$\|v\|^2 := \|\nabla v\|_{\Omega_h}^2 + \frac{\mu}{h} \|v\|_{\Gamma_h}^2 + \sum_{F \in \mathcal{F}_h^\Gamma} \text{diam } F \|\partial_n v\|_{L^2(F)}^2.$$

Assumption 13.1.

(M0³) $\mathcal{T}_h$ is shape regular and $c_0 h_\Gamma \leq \text{diam } F \leq h_\Gamma$ for all $F \in \mathcal{F}_h^\Gamma$. Hence the faces of P_h are shape regular and $\rho/h \leq 1/(c_0 h_\Gamma)$.

(IS3) $k \geq 3$, and there is $\beta > 0$, independent of h , such that for every $q \in \Pi_h \cap L_0^2(\Omega_h)$ there exists $v \in V_h^0$ with $\text{div } v = q$ and $\|\nabla v\| \leq \beta^{-1} \|q\|$.

(D) is as in Assumption 2.1.

Status of (IS3). In 2D, (IS3) is Lemma 4.5 (Guzmán–Scott, $k \geq 4$, general meshes under (M1)–(M2)). In 3D the known results for the Scott–Vogelius pair $P_k - P_{k-1}^{\text{disc}}$ are these.

- *Alfeld (barycentric) refinements* of an arbitrary shape-regular mesh: stable for $k \geq 3$ [44], and for $k \geq d$ in dimension d [26]. Farrell, Mitchell and Scott [19] summarise this as $k \geq d$ for Alfeld splits.
- *Worsey–Farin splits*: lower degrees [25]; [19, Table 2.1] lists Worsey–Farin splits in three dimensions with $k \geq 1$, citing Fabien, Guzmán, Neilan and Zytoon (arXiv:2105.09214). (An earlier version of this paper attached a robustness caveat at nearly singular vertices to this entry; in that table the caveat belongs to the two-dimensional $k \geq 4$ row.)
- *Freudenthal meshes*: $k \geq 6$ [45], conjectured $k \geq 4$ in [19]. This is irrelevant here, because a Freudenthal mesh cannot be fitted to an inscribed polyhedron.

- *General tetrahedral meshes:* we know of no unconditional result for the full pair with P_{k-1}^{disc} pressures. According to Farrell, Mitchell and Scott [19], Neilan [35, Proposition 6.5] extends Zhang's $k \geq 6$ stability result for the Scott–Vogelius pair to more general meshes that satisfy a special condition, which he states as a conjecture; the same paper also constructs smooth Stokes pairs whose pressures carry extra continuity. So a conditional result for $k \geq 6$ exists; it does not give (IS3) unconditionally on meshes of an inscribed polyhedron, and we do not use it.

So the unconditional 3D theory below is for Alfeld-refined meshes with $k \geq 3$. Uniformity of β in h has two parts. The domain part, a Bogovskii constant uniform over the perturbed domains Ω_h , is proved in Lemma 13.5. The discrete part, that the constant of the split-element proof depends only on shape regularity and the continuous inf-sup constant, is how these results are stated. We have not re-audited those proofs line by line for hidden dependence on the domain; see item 3 of Section 13.7.

13.2. Geometry of inscribed triangles.

Lemma 13.2 (Faces). *Let $F \in \mathcal{F}_h^\Gamma$ lie in the plane $\{y \cdot \nu_F = d_F\}$, where $\nu_F = -n_F$ is the unit normal pointing away from the centre. Let r_F be the circumradius of F and $o_F = d_F \nu_F$ its circumcentre; then $d_F = (1 - r_F^2)^{1/2}$ and $r_F \leq Ch_\Gamma$. For $x \in F$ let $x^* = x/|x|$ be its radial projection. Then:*

- (a) $\text{dist}(x, \Gamma) \leq 1 - d_F = r_F^2/2 + O(r_F^4) \leq Ch_\Gamma^2$.
- (b) $n(x^*) - n_F = -(x - o_F) + O(h_\Gamma^2)$, where $n(x^*) = -x^*$. The leading term is tangential to F , linear in x , and vanishes at the circumcentre.
- (c) The area Jacobian of $x \mapsto x^*$ from F to Γ is $d_F/|x|^3 = 1 + O(h_\Gamma^2)$.
- (d) $|F|^{-1} \int_F (n(x^*) - n_F) dA = -(x_c - o_F) + O(h_\Gamma^2)$, where x_c is the centroid. The leading term vanishes if and only if F is equilateral.

Proof. Write $x = o_F + y$ with $y \perp \nu_F$ and $|y| \leq r_F$. Then $|x|^2 = d_F^2 + |y|^2 \in [d_F^2, 1]$, which gives (a). For (b),

$$n(x^*) - n_F = -\frac{o_F + y}{|x|} + \nu_F = -\frac{y}{|x|} + \nu_F \left(1 - \frac{d_F}{|x|}\right),$$

with $|x|^{-1} = 1 + O(r_F^2)$ and $0 \leq 1 - d_F/|x| \leq 1 - d_F$. For (c), central projection onto the unit sphere has area element $(x^* \cdot \nu_F)|x|^{-2}$. (d) follows by integrating (b); the centroid and circumcentre of a triangle coincide exactly when it is equilateral. $\square$

Computation: `notes/v3/face_mean_check.py`. For three faces inscribed in the unit sphere at circumradius $r_F = 0.2, 0.1, 0.05$, the tangential part of the face mean of $n(x^*) - n_F$ is 0 (equilateral face) and $0.105 r_F, 0.122\sqrt{2} r_F$ (two non-equilateral faces), matching $|x_c - o_F|$.

For a general smooth strictly convex D the same holds, with o_F replaced by the point at which the gradient of the quadratic part of Γ , written as a local graph over F , equals the gradient of its linear interpolant. Only (a), (c) and $|n(x^*) - n_F| \leq Ch_\Gamma$ are used below.

In 2D the circumcentre of a chord is its midpoint, so the leading term is odd about the midpoint. That is the cancellation in Lemma 3.5. In 3D it is lost on every non-equilateral face.

Proposition 13.3 (Transposed-gradient term). *For $v \in V_h^R$:*

- (i) On F , $(\nabla \tilde{u})^\top n_F = \sigma n(x^*) + O(h_\Gamma^2)$ with $\sigma := \partial_n u(x^*) \cdot (n_F - n(x^*)) = \partial_n u(x^*) \cdot (x - o_F) + O(h_\Gamma^2)$. So the load is normal, of size $O(h_\Gamma)$, and has nonzero face means in general.
- (ii) $|\langle (\nabla \tilde{u})^\top n, v \rangle| \leq Ch_\Gamma \|v\|_{L^1(\Gamma_h)} \leq Ch_\Gamma (h/\mu)^{1/2} \|v\| = C\gamma^{-1/2} h_\Gamma^{3/2} \|v\|$.

(iii) In the $\|\nabla v\|$ -dual norm on V_h^R the rate is generically only h_Γ . For smooth ψ on Γ , let v be the Lagrange interpolant of $\chi\psi n$, with a fixed cutoff χ . Then

$$\langle (\nabla \tilde{u})^\top n, v \rangle = \sum_{F \in \mathcal{F}_h^\Gamma} |F| \psi(x_F^*) \partial_n u(x_F^*) \cdot (x_c - o_F) + O(h_\Gamma^2),$$

while $\|\nabla v\| \leq C$. This is $\Theta(h_\Gamma)$ whenever $h_\Gamma^{-1} \sum_F |F| \psi \partial_n u \cdot (x_c - o_F) \not\rightarrow 0$. We do not exhibit a specific mesh family for which this holds; the point is only that the two-dimensional cancellation is not available.

Proof. Lemma 3.4 is dimension-free: $\nabla u = \partial_n u \otimes n$ on Γ and $\partial_n u \cdot n = 0$. So $(\nabla u)^\top n_F = n(\partial_n u \cdot n_F) = n(\partial_n u \cdot (n_F - n))$. Taylor-expand $\nabla \tilde{u}$ from x^* to x (distance $O(h_\Gamma^2)$) and apply Lemma 13.2(b); this gives (i). (ii) follows from $|\sigma| \leq Ch_\Gamma$, Cauchy–Schwarz on Γ_h ($|\Gamma_h| \leq C$) and $\|v\|_{\Gamma_h}^2 \leq (h/\mu)\|v\|^2$. For (iii), use $v \cdot n(x^*) = \psi + O(h_\Gamma)$ and Lemma 13.2(d). $\square$

13.3. Tools.

Lemma 13.4 (Dimension-independent tools). *Under $(M0^3)$ and (D) :*

- (a) Uniform Korn, trace and Friedrichs (Lemma 3.2) hold in 3D.
- (b) Extension (Lemma 3.3): $\|\tilde{u}\|_{L^\infty(\Gamma_h)} \leq Ch_\Gamma^2$ and $|(F, v)| \leq Ch_\Gamma^2 \|\nabla v\|$ for $v \in V_h^R$.
- (c) Inverse traces and coercivity (Lemmas 4.1 and 4.2) hold verbatim, with $|e|$ replaced by $\text{diam } F$ and with the same $\mu_0 = 4\kappa\rho C_I^2$, c_1 and M .
- (d) Divergence range (Lemma 4.3): under (IS3), $\text{div } V_h^R = \Pi_h$ and $W_h \neq \emptyset$, and discrete velocities are pointwise divergence-free.

Proof. (a) Let $\rho_h(\omega) = d_F/(\omega \cdot \nu_F)$ be the radial function of Γ_h on the cone over F . Then $\|\rho_h - 1\|_\infty \leq Ch_\Gamma^2$, and ρ_h is Lipschitz with $|\nabla_\omega \rho_h| = d_F |\nu_F - (\omega \cdot \nu_F)\omega|/(\omega \cdot \nu_F)^2 \leq Ch_\Gamma$. The map $\Phi_h(r, \omega) = (r + \chi(r)(\rho_h(\omega) - 1), \omega)$ then satisfies $\|D\Phi_h - I\|_\infty \leq Ch_\Gamma$, and steps 3–6 of Lemma 3.2 apply unchanged.

- (b) The first bound is Lemma 13.2(a). For the second, apply a one-dimensional Poincaré inequality along normal segments across S_h , whose thickness is at most Ch_Γ^2 . This gives $\|v\|_{L^2(S_h)} \leq Ch_\Gamma(\|v\|_{\Gamma_h} + h_\Gamma \|\nabla v\|)$. Then $|(F, v)| \leq \|F\|_\infty |S_h|^{1/2} \|v\|_{L^2(S_h)}$ with $|S_h| \leq Ch_\Gamma^2$, and (a) finishes.
- (c) The proofs use only scaling and shape regularity.
- (d) $\text{div } V_h^0 = \Pi_h \cap L_0^2$ is (IS3). The cubic face bubble $b_F \geq 0$ of the tetrahedron on $F \in \mathcal{F}_h^\Gamma$ gives $b_F n_F \in V_h^R$ (since $k \geq 3$) and $\int \text{div}(b_F n_F) = \int_F b_F \neq 0$. The rest is as in Lemma 4.3. $\square$

Lemma 13.5 (Uniform continuous inf-sup constant). *The Bogovskii (continuous inf-sup) constant of Ω_h is bounded independently of h .*

- Proof.* (1) Fix $\alpha < \pi/2$, $r_b > 1$, $r_B > 0$ with $(r_b - r_B) \cos(\alpha + \arcsin(r_B/r_b) + \alpha_0) > 1$, where $\alpha_0 > 0$ is small. Cover S^2 by finitely many caps of half-angle α about directions ω_j , such that adjacent caps overlap in a cap of fixed size. Let W_j be the corresponding cones, $c_j := r_b \omega_j$ and $B_j := B(c_j, r_B) \subset W_j \cap R$.
- (2) Let $x \in W_j \cap \Omega_h$. Since $0 \in P_h$ (for small h_Γ), the ray from 0 through x leaves P_h through a face F that meets W_j , and x lies on the outer side of the plane of F . Because $|\nu_F - \omega| \leq Ch_\Gamma$ for directions ω hitting F , the angle between ν_F and ω_j is at most $\alpha + Ch_\Gamma$.
 - (3) Hence every $y \in B_j$ satisfies $y \cdot \nu_F \geq (r_b - r_B) \cos(\alpha + \arcsin(r_B/r_b) + Ch_\Gamma) > 1 \geq d_F$. So y is also on the outer side of that plane. The segment $[y, x]$ therefore stays on the outer side of the plane, so it misses the convex set P_h , and it lies in the convex set $W_j \cap R$.
 - (4) Thus $W_j \cap \Omega_h$ is star-shaped with respect to B_j , uniformly in h . Galdi's Lemma III.3.4 [22] gives a Bogovskii operator on $\Omega_h = \bigcup_j (W_j \cap \Omega_h)$ with a bound independent of h . The diameters, ball radii and overlaps are fixed. $\square$

Lemma 13.6 (Geometric consistency). *With $\mathcal{G}$ as in Lemma 5.1 (which is dimension-free), $|\mathcal{G}(v)| \leq C(1 + \gamma^{1/2})h_\Gamma^{3/2}\|v\|$ for $v \in V_h^R$, and $|\mathcal{G}(v)| \leq Ch_\Gamma^{3/2}\|\nabla v\|$ for $v \in V_h^0$.*

Proof. Take the four terms in turn.

- The transposed term is at most $C\gamma^{-1/2}h_\Gamma^{3/2}\|v\| \leq C\gamma_0^{-1/2}h_\Gamma^{3/2}\|v\|$ by Proposition 13.3(ii).
- $|\langle \tilde{u}, \partial_n v \rangle| \leq Ch_\Gamma^2 \cdot C_I(c_0 h_\Gamma)^{-1/2}\|\nabla v\|$.
- $\frac{\mu}{h}|\langle \tilde{u}, v \rangle| \leq (\mu/h)^{1/2}Ch_\Gamma^2\|v\| = C\gamma^{1/2}h_\Gamma^{3/2}\|v\|$.
- $|(F, v)| \leq Ch_\Gamma^2\|\nabla v\|$.

On V_h^0 the first and third terms vanish. $\square$

Remark 13.7 (The 2D cancellation is not needed for any error bound). The proof of Lemma 13.6 works verbatim in 2D. So Lemma 3.5 (the odd-in- s cancellation) is not needed for Lemma 5.2 or Theorem 7.2; it only gives the sharper $\|\nabla v\|$ -dual bound. Likewise, in step 4 of Theorem 6.4 the $O(h_\Gamma)$ tangential load $(p - \bar{p})(n_e - n(x^*))$ can be bounded in the energy dual norm by $C_p h_\Gamma (h/\mu)^{1/2}$. Since step 6 tests with r_h in $\|\cdot\|$, this suffices.

Lemma 13.8 (Approximation in W_h). *Under $(M\theta^3)$ and $(IS3)$, $E_h := \inf_{z \in W_h} \|\tilde{u} - z\| \leq C\varepsilon_h^{(3)}$, where*

$$\varepsilon_h^{(3)} := (1 + (\gamma h_\Gamma)^{1/2})h^k + h^{k+1}(h_\Gamma^{-1/2} + (\mu/h)^{1/2}).$$

For bounded γ the flux term is at most Ch^k if $h_\Gamma \geq h^2$. If g_I is flux-corrected, it reduces to the contribution of Γ_h , which is below h^k for bounded γ . The lower bound $\|\tilde{u} - z\| \geq (\mu/h)^{1/2}|m_R|/|\Gamma_h|^{1/2}$ after Lemma 4.6 holds verbatim.

Proof. As for Lemma 4.6, with three changes.

- *Net flux.* Estimate it crudely by pointwise interpolation error: $|m| \leq \int_{\partial R} |g_I - g| + \int_{\Gamma_h} |w_h - \tilde{u}| \leq Ch^{k+1}$. No extra quadrature exactness of Lagrange interpolation on triangles is used; unlike the 1D Newton–Cotes case, there is none in general (for $k = 2$ the induced rule gives $1/24$ for $\int x^3 = 1/20$ on the reference triangle).
- *Bubble.* $b := \sum_F b_F n_F$ with cubic face bubbles. Then $\int_{\Gamma_h} b \cdot n = c_b |\Gamma_h|$ with $c_b > 0$, $\|\nabla b\| + \|\operatorname{div} b\| \leq Ch_\Gamma^{-1/2}$ (there are $O(h_\Gamma^{-2})$ bubbles, each of H^1 -seminorm squared $O(h_\Gamma)$), $\|b\|_{\Gamma_h} \leq C$, and $\sum_F \operatorname{diam} F \|\partial_n b\|_F^2 \leq Ch_\Gamma^{-1}$.
- *Correction.* Use $(IS3)$ in place of Lemma 4.5. $\square$

13.4. The leak field without stream functions.

Lemma 13.9 (A divergence-free field with normal trace $p - \bar{p}$). *Let $q := (p - \bar{p})|_\Gamma$ and let ϕ_0 solve $\Delta_{S^2} \phi_0 = q$ on the unit sphere; this is solvable since $\int_\Gamma q = 0$. Let $\chi \in C^\infty$ with $\chi \equiv 1$ on $[\frac{1}{2}, 1 + r_1]$ and $\chi \equiv 0$ on $[1 + 2r_1, \infty)$, where $1 + 2r_1 < L$. In spherical coordinates set*

$$w := \chi(r) r^{-2} q(\omega) e_r - \chi'(r) r^{-1} \nabla_{S^2} \phi_0(\omega).$$

Then w is smooth on $\{r > \frac{1}{2}\} \supset \Omega_h$, $\operatorname{div} w = 0$, $w \equiv 0$ near ∂R , and $w|_\Gamma = q e_r = -(p - \bar{p})n$.

Proof. $\operatorname{div}(f(r)q e_r) = r^{-2}(r^2 f)'q$, which equals $r^{-2}\chi'q$ for $f = \chi r^{-2}$. A tangential field $g(r)r^{-1}\nabla_{S^2} \phi_0$ has divergence $g r^{-2}\Delta_{S^2} \phi_0 = g r^{-2}q$. With $g = -\chi'$ the two cancel. At $r = 1$, $\chi = 1$ and $\chi' = 0$. $\square$

Lemma 13.10 (Discrete leak field). *Let I_h be Lagrange interpolation and b as in Lemma 13.8. Set*

$$v_\varphi := I_h w - ab - v_0, \quad a := \frac{\int_{\Gamma_h} I_h w \cdot n}{c_b |\Gamma_h|},$$

where $v_0 \in V_h^0$ is given by (IS3) with $\operatorname{div} v_0 = \operatorname{div}(I_h w - ab)$. Then $v_\varphi \in Z_h$ and

$$\begin{aligned} \|v_\varphi - w\|_{L^\infty(\Gamma_h)} &\leq Ch_\Gamma^{k+1}, \quad \|\nabla(v_\varphi - w)\| \leq Ch^k, \\ \|\partial_n(v_\varphi - w)\|_{\Gamma_h} &\leq Ch^k h_\Gamma^{-1/2}, \quad \sum_F \operatorname{diam} F \|\partial_n(v_\varphi - w)\|_F^2 \leq Ch^{2k}. \end{aligned}$$

These bounds replace $\|v_\varphi - w\|_{W^{1,\infty}} \leq Ch^k$ (Morgan–Scott) in every use in Sections 6 and 12. The same construction works in 2D and removes the dependence on the C^1 interpolant there.

- Proof.* (1) *The mean.* Since $\operatorname{div} w = 0$ on Ω_h and $w = 0$ near ∂R , $\int_{\Gamma_h} w \cdot n = 0$. So $|a| \leq C\|I_h w - w\|_{L^\infty(\Gamma_h)} \leq Ch_\Gamma^{k+1}$, because boundary tetrahedra have diameter at most Ch_Γ . By the choice of a , $\operatorname{div}(I_h w - ab) \in \Pi_h \cap L_0^2$.
- (2) *The correction.* Its norm satisfies $\|\operatorname{div}(I_h w - ab)\| \leq Ch^k + C|a|h_\Gamma^{-1/2} \leq Ch^k$. So $\|\nabla v_0\| \leq Ch^k$. Since $v_0 = 0$ on Γ_h , the inverse trace inequality gives $\|\partial_n v_0\|_{\Gamma_h} \leq Ch_\Gamma^{-1/2} h^k$ and $\sum_F \operatorname{diam} F \|\partial_n v_0\|_F^2 \leq Ch^{2k}$.
- (3) *The remaining terms.* $I_h w - w$ and ab contribute $O(h_\Gamma^{k+1})$ pointwise on Γ_h , $O(h^k)$ in H^1 , and $O(h_\Gamma^k)$ pointwise for ∂_n (for ab this is $|a|h_\Gamma^{-1}$).

The uses are:

- Theorem 6.1 lower bound, steps 1–2 (pointwise trace, and $\|\nabla v_\varphi\| + \sum \operatorname{diam} F \|\partial_n v_\varphi\|^2 \leq C$);
- Theorem 6.4, steps 4–5 ($\|\partial_n v_\varphi\|_{\Gamma_h} \leq C$, and $|\langle v_\varphi - w, \partial_n v \rangle| \leq Ch_\Gamma^{k+1} h_\Gamma^{-1/2} \|\nabla v\|$);
- Theorem 6.6(i), steps 1–5;
- Corollary 6.7, step 4;
- Theorem 12.8, step 5.

Each needs only the listed bounds. $\square$

13.5. Normal-load cancellation with edge terms.

Lemma 13.11 (3D analogue of Lemma 6.3). *Let $S \in W^{1,\infty}$ on a neighbourhood of Γ_h , with $|S - (S \cdot n_F)n_F| \leq Ch_\Gamma$ on every $F \in \mathcal{F}_h^\Gamma$. (For instance $S = w$ of Lemma 13.9, since $w(x) \parallel n(x^*)$ up to $O(h_\Gamma^2)$ and $|n(x^*) - n_F| \leq Ch_\Gamma$.) Then for $v \in Z_h$,*

$$|\langle S, \partial_n v \rangle| \leq C(\|v\|_{\Gamma_h} + h_\Gamma^{1/2} \|\nabla v\|) \leq C\|\nabla v\|,$$

with C depending on $\|S\|_{W^{1,\infty}}$, k and $(M0^3)$.

Proof. (1) *Tangential part.* $|\langle S - (S \cdot n_F)n_F, \partial_n v \rangle| \leq Ch_\Gamma \|\partial_n v\|_{\Gamma_h} \leq Ch_\Gamma^{1/2} \|\nabla v\|$, by Lemma 13.4(c).

- (2) *Normal part.* On F , v is a polynomial on the adjacent tetrahedron with $\operatorname{div} v = 0$ and constant n_F . So $\partial_n v \cdot n_F = -\operatorname{div}_F v_F$, where v_F is the part of v tangential to F and div_F is the in-plane divergence. With $\sigma_F := S \cdot n_F$ and $\nu_{\partial F}$ the in-plane outward conormal,

$$\int_F \sigma_F \partial_n v \cdot n_F = \int_F \nabla_F \sigma_F \cdot v_F - \int_{\partial F} \sigma_F v \cdot \nu_{\partial F}.$$

The first term is at most $C\|v\|_{L^1(F)}$.

- (3) *Edge terms.* Each edge E of the closed triangulated surface Γ_h is shared by exactly two faces F, F' . Since v is continuous across E , the two contributions combine to $\int_E v \cdot (\sigma_F \nu_{E,F} + \sigma_{F'} \nu_{E,F'})$. With t_E a unit tangent oriented so that $\nu_{E,F} = t_E \times n_F$, we have $\nu_{E,F'} = -t_E \times n_{F'}$. Hence $\nu_{E,F} + \nu_{E,F'} = t_E \times (n_F - n_{F'}) = O(h_\Gamma)$ and $\sigma_F - \sigma_{F'} = S \cdot (n_F - n_{F'}) = O(h_\Gamma)$. So the coefficient is $O(h_\Gamma)$.

- (4) *Summing.* The inverse inequality $\|v\|_{L^1(\partial F)} \leq C(\text{diam } F)^{-1}\|v\|_{L^1(F)}$ for $v|_F \in P_k$ (scaling and shape regularity) gives

$$\sum_E h_\Gamma \|v\|_{L^1(E)} \leq C \sum_F \|v\|_{L^1(F)} = C\|v\|_{L^1(\Gamma_h)} \leq C\|v\|_{\Gamma_h}.$$

- (5) Finish with the uniform trace inequality. $\square$

Remark 13.12. If boundary faces of $\mathcal{T}_h$ subdivide the faces of P_h (Worsey–Farin splits, or Alfeld refinements of a mesh whose boundary faces already subdivide the faces of P_h), the extra edges lie inside a plane. There $n_F = n_{F'}$ and the conormals are opposite, so their contributions in step 3 cancel exactly. Lemma 13.2 is applied to the faces of P_h , which must then be shape regular of size $\asymp h_\Gamma$.

13.6. Main results in three dimensions.

Theorem 13.13. *Assume $(M0^3)$, $(IS3)$ and (D) , and let (u, p) be a smooth Stokes solution on $\bar{\Omega}$. Then Theorems 6.1, 6.4, 6.6 and 7.2, Corollaries 6.2 and 6.7, Proposition 7.1 and Proposition 9.1(a) hold verbatim in 3D, with ε_h replaced by $\varepsilon_h^{(3)}$ and v_φ by the field of Lemma 13.10. In particular:*

- (i) GS converges at rate h/μ in H^1 and $(h/\mu)^{1/2}$ in energy. To leading order its error is $(h/\mu)v_\varphi$ with $v_\varphi|_{\Gamma_h} \approx -(p - \bar{p})n$, with universal constant 1 in the slip and energy norms.
- (ii) CNS* is well posed for $\gamma \geq \gamma_2$ and satisfies $\|\tilde{u} - u_h\| \leq C[(1 + \gamma^{1/2})h_\Gamma^{3/2} + \varepsilon_h^{(3)}]$. For bounded γ and $h_\Gamma \geq h^2$ this gives $\|\tilde{u} - u_h\|_{H^1(\Omega_h)} \leq C(h_\Gamma^{3/2} + h^k)$.

On Alfeld-refined meshes this holds for every $k \geq 3$, subject to the uniformity caveat in item 3 of Section 13.7.

Proof. The 2D proofs use the following facts, each replaced as indicated.

- Lemma 3.1 is replaced by Lemma 13.2: sagitta $O(h_\Gamma^2)$, $|n(x^*) - n_F| \leq Ch_\Gamma$, Jacobian $1 + O(h_\Gamma^2)$. Hence $|\mathcal{R}(v)| \leq (G + Ch_\Gamma^2)(h/\mu)^{1/2}\|v\|$ and $\mathcal{R}(v_\varphi) = G^2 + O(h_\Gamma^2 + h^k)$. Note $v_\varphi \cdot n_F = -(p - \bar{p})(x^*) \cdot n \cdot n_F + O(h_\Gamma^{k+1})$ with $n \cdot n_F = 1 + O(h_\Gamma^2)$.
- Lemmas 3.2, 3.3, 4.1, 4.2 and 4.3 are replaced by Lemma 13.4; Lemmas 4.5 and 4.6 by (IS3), Lemma 13.5 and Lemma 13.8; Lemma 5.2 by Lemma 13.6; Lemma 6.3 by Lemma 13.11; and the Morgan–Scott v_φ by Lemma 13.10.
- Corollary 5.3, Proposition 7.1 and the consistency and uniqueness steps of Theorem 7.2 are dimension-free. They use $\int_{\partial\Omega_h} v \cdot n = \int \text{div } v$ and a boundary bubble.
- Step 4 of Theorem 6.4. On F , $(\bar{p} - \bar{p})n_F + v_\varphi = (p - \bar{p})(x^*)(n_F - n(x^*)) + O(h_\Gamma^2) + O(h_\Gamma^{k+1})$. This is $O(h_\Gamma)$ with nonzero face means, so we use the energy-dual bound (Remark 13.7): $|m(v)| \leq C_p(h_\Gamma(h/\mu)^{1/2} + h_\Gamma^2 + h^k)\|v\|$. Step 6 then gives $\|r_h\| \leq C_p(h/\mu + h_\Gamma^2) + C\varepsilon_h^{(3)}$, using $h_\Gamma(h/\mu)^{1/2} \leq \frac{1}{2}(h_\Gamma^2 + h/\mu)$. The conclusion of Theorem 6.4, and the bound on $\|r_h\|$ in Corollary 6.7, are unchanged or better.
- Step 2 of Theorem 6.6(i). Here $|\mathcal{G}(v_\varphi)| \leq C(h_\Gamma + \gamma h_\Gamma^3 + \gamma h_\Gamma h^k)$, with h_Γ in place of $h_\Gamma^{3/2}$ by Proposition 13.3(i). The proof only needs this to be at most δG^2 , which follows from the hypothesis $h_\Gamma \leq h_0(G)$ and the X -condition. The penalty part $\tilde{u} \cdot v_\varphi = O(h_\Gamma^4 + h_\Gamma^2 h^k)$ is unchanged: $\tilde{u} \approx \pm d \partial_n u(x^*)$ is tangential and v_φ is normal to Γ up to $O(d)$.
- Flux proviso. The condition $h_\Gamma \geq h_\Omega^{2(\sigma_k - k)}$ of Theorem 7.2 becomes $h_\Gamma \geq h^2$, since the 3D flux exponent is $k + 1$. $\square$

Corollary 13.14 (Navier–Stokes in 3D). *Lemma 12.7, Theorems 12.8 and 12.12, Corollary 12.9 and Lemma 12.13 hold in 3D under the assumptions of Theorem 13.13 together with (NS1)–(NS2)*

(with $\tilde{u} = \text{curl } \tilde{\psi}$ for a vector potential $\tilde{\psi}$, which exists because $\int_{\Gamma} u \cdot n = 0$), with $\varepsilon_h^{(3)}$ in place of ε_h .

Proof. Section 12 reuses the Stokes estimates, in particular steps 4–5 of the proof of Theorem 6.4, together with Lemma 12.4, which holds for $d = 3$ because $H^1 \subset L^4$. Replace the Stokes lemmas by Theorem 13.13 and its ingredients. In step 5 of Theorem 12.8 use $\|v_{\varphi}\|_{H^1} \leq C$ from Lemma 13.10. In 3D the functional m of step 4 of Theorem 6.4 is bounded only in the energy-dual norm (proof of Theorem 13.13); this suffices, because (L) and (L*) are stated in the energy-dual norms $\|\cdot\|_{*,Z_h}$ and $\|\cdot\|_{*,V_h^R}$, so step 4 of Theorem 12.8 needs only an energy-dual bound on m . $\square$

13.7. What is not proved in 3D.

- (1) *Penalty necessity* (Proposition 9.1(b)). Scaling gives the same threshold form in 3D: under dilation by ℓ , a and $\langle \partial_n v, v \rangle$ scale like ℓ and $\|v\|_{\Gamma_h}^2$ like ℓ^2 , so $N_h(v, v) = \ell[A - 2B + (\mu\ell/h)C]$. But no 3D witness (a divergence-free $v \in Z_h$ supported in a boundary vertex or face star with $2B > A$) has been computed or certified. Sufficiency $\mu \geq 4\kappa C_I^2 \rho$ holds.
- (2) *(IS3) on general tetrahedral meshes.* It is unknown in 3D for unsplit meshes, as far as we know. The theory is unconditional only on split meshes.
- (3) *Uniformity of the split-mesh inf-sup constant over Ω_h .* The domain part is Lemma 13.5. We rely on the published statements for the claim that the discrete part depends only on shape regularity; we have not re-audited Zhang’s or Guzmán–Neilan’s proofs for hidden domain dependence.
- (4) *Sharpness of $h_{\Gamma}^{3/2}$ for CNS*.* Not proved in 3D. In 2D it is Theorem 7.13, unconditionally for $k \geq 7$ and under (M3) for $4 \leq k \leq 6$; that proof uses the 2D chord geometry and 2D boundary stars.
- (5) *Non-convex D and other domains.* Not treated in 3D, because $\Omega \subset \Omega_h$ is used. In 2D, Section 11 treats general smooth, possibly nonconvex, multiply connected domains.

14. COMPUTATIONS

14.1. Code and meshes. Except in Section 14.11, we use P_4 Scott–Vogelius elements with Nitsche’s method on the inscribed polygon and strong Dirichlet data on the square $R = (-2.5, 2.5)^2$. Divergence is enforced exactly: all ten P_3 moments per triangle.

Saddle-point systems are solved by SuperLU [30] with partial pivoting and three steps of iterative refinement. Over the 148 production solves (74 cases, two methods each), the relative residual is at most $3.8 \cdot 10^{-15}$ and the discrete divergence at most $7 \cdot 10^{-11}$ (median $9 \cdot 10^{-12}$).

For the zero-pressure-gradient benchmark of [13], the code gives the coercivity threshold $\mu^* = 60.343$ at $N = 20$ (results/pod/thresh/N20.L2.5.m5.json), against the transition $60.33821 < \mu^* < 60.33831$ reported there.

The vertices of Γ_h are the radial projections of N equispaced points on the square, so the boundary chords are *not* equal (except for $N = 8$): $\max |e| / \min |e|$ is 1.43, 1.72, 1.87 and 1.97 at $N = 16, 32, 64$ and 256. The ratio $h/h_{\Gamma} = h_{\Omega} / \max |e|$ stays in 3.28–3.48, while $\rho = h / \min |e|$ of Section 2 is 4.70, 5.73 and 6.35 at $N = 16, 32$ and 64. h_{Ω} decreases from 1.505 to 0.109 for $N = 16 \rightarrow 256$; since h/h_{Γ} is nearly constant, rates in h_{Ω} and h_{Γ} coincide. Penalty thresholds are reported as $\gamma^* = \mu^* h_{\Gamma} / h$, which is the γ of Section 2. Rates in the tables are computed between consecutive runs of the full sequence $N = 16, 24, 32, 48, 64, 96, 128, 192, 256$, some of which are omitted from the tables.

14.2. Test problems.

A Scott’s benchmark: the shear flow $u = (1 - r^{-2})(-y, x)$ with $p \equiv 0$, so $G = 0$.

B_a $\psi = (r^2 - 1)^2(x + y^2/2)/10$ and $p = a(x^2y - y + x/2)$, so $G = a(7\pi/8)^{1/2}$: $G = 1.658$ for $a = 1$ and 165.8 for $a = 100$.

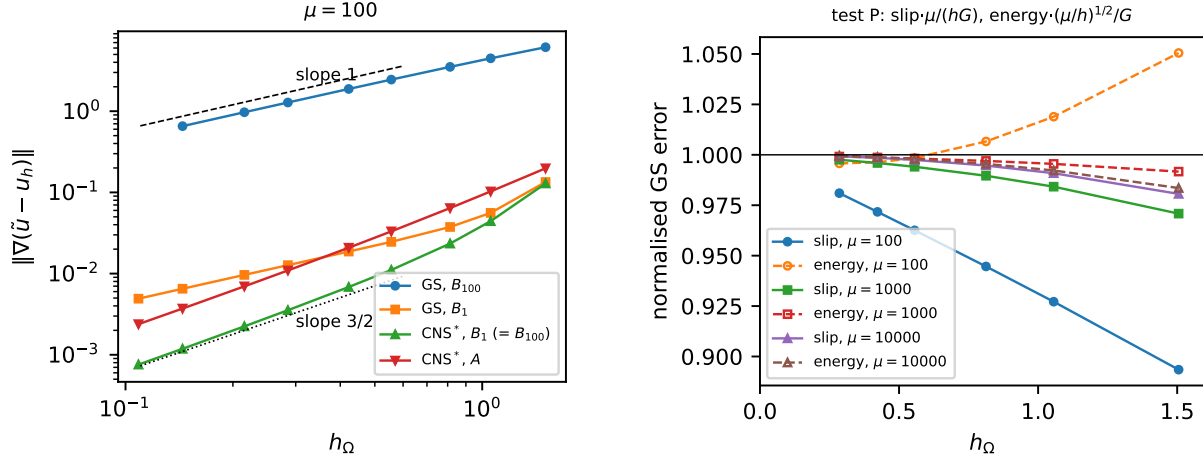


FIGURE 1. Left: H^1 errors at $\mu = 100$. Once $G > 0$, GS eventually has slope 1, while CNS* tends to slope $3/2$ independently of the pressure. Right: the normalised slip and energy errors of GS on test P approach 1 (Corollary 6.7); at $N = 96$ the energy ratios are within 0.5% of 1 and the slip ratios within 2%.

TABLE 4. CNS* attains Scott's rate $h_\Gamma^{3/2}$ (Theorem 7.2). H^1 error, $\mu = 100$.

N	h_Ω	A	rate	B_1	rate
32	0.813	$6.41 \cdot 10^{-2}$	1.80	$2.34 \cdot 10^{-2}$	2.43
64	0.422	$2.08 \cdot 10^{-2}$	1.69	$6.83 \cdot 10^{-3}$	1.76
96	0.285	$1.09 \cdot 10^{-2}$	1.64	$3.54 \cdot 10^{-3}$	1.68
128	0.215	$6.96 \cdot 10^{-3}$	1.60	$2.24 \cdot 10^{-3}$	1.63
192	0.144	$3.71 \cdot 10^{-3}$	1.58	$1.19 \cdot 10^{-3}$	1.60
256	0.109	$2.39 \cdot 10^{-3}$	1.56	$7.58 \cdot 10^{-4}$	1.57

P Pure pressure: $u \equiv 0$, $f = \nabla p$, with p as in B_1 . CNS* reproduces this exactly ($u_h = 0$ and $p_h = p - \bar{p}_\Gamma(p)$, since $p \in \Pi_h$); its computed velocity is below 10^{-13} in H^1 . The GS output is therefore *exactly* the response to the missing traction.

C ψ as in B, with the non-polynomial pressure $p = e^{x/2} \cos y$. This exercises the term $p^* - \pi_h p^*$ of Theorem 7.2.

Tests A and B were run with $\mu \in \{100, 1000, 10000\}$ and $N \leq 192$, plus $N = 256$ for $(A, 100)$ and $(B_1, 100)$, on a 32-core machine (36 minutes in total). Test P (all three μ) and test C ($\mu = 100$) were run with $N \leq 96$ on a 2-core machine.

14.3. Theorem 7.2: CNS*. Table 4 gives the CNS* errors. At $N = 256$ the energy-norm rate is 1.52 and the slip rate is 2.02; the slip carries an extra factor $(h/\mu)^{1/2}$. The H^1 rates decrease monotonically toward $3/2$, the geometric term of Lemma 3.5. The h_Ω^4 term is invisible at these levels because the geometric term dominates.

The error is independent of the pressure.

- For polynomial pressures this is structural ($p \in \Pi_h$). B_1 and B_{100} agree to all printed digits wherever both were run ($N \leq 192$, every μ).

TABLE 5. GS eventually loses Scott’s rate once $G > 0$. H^1 rate between $N = 128$ and $N = 192$, test B_{100} .

μ	GS	CNS*
100	0.99	1.60
1000	1.00	1.56
10000	1.06	1.58

- With the non-polynomial pressure of test C, the CNS* H^1 error differs from that of B_1 by a relative $7.9 \cdot 10^{-7}$, $3.9 \cdot 10^{-8}$ and $9.7 \cdot 10^{-9}$ at $N = 16, 64$ and 96 . This is the size of the contribution $\langle p^* - \pi_h p^*, v \cdot n \rangle$, which is of higher order, as Theorem 7.2 predicts.

14.4. **Theorems 6.1–6.4: the printed method.** Table 5 compares the H^1 rates of GS and CNS* on B_{100} .

For the non-polynomial test C at $\mu = 100$, the GS H^1 rate from $N = 64$ to 96 is already 1.12 , against 1.68 for CNS*. For the small pressure B_1 at $\mu = 1000$, GS still shows 1.51 at $N = 192$. The $(h/\mu)G$ term dominates only once $(h/\mu)G \gtrsim h_\Gamma^{3/2}$, which always happens eventually because $h_\Gamma^{3/2}$ decays faster. Larger μ or smaller G delays the crossover but does not remove it.

14.5. **Corollary 6.7: the leading term.** Table 6 and Figure 1 (right) show the normalised GS error on test P, and Figure 2 shows the leak field itself.

- At $N = 32$, $\mu = 100$, the normal trace of u_h on Γ_h matches the prediction $(h/\mu)(p - \bar{p})$ with relative $L^2(\Gamma_h)$ misfit 0.055 . The maximum of the tangential trace is about 12% of that of the normal one, and the tangential trace oscillates at the element scale.
- The slip and energy ratios approach 1: at $N = 96$ the energy ratios are within 0.5% of 1 for every μ . The defect in the slip ratio shrinks with h/μ : at $N = 96$ it is $0.019, 0.0025$ and 0.0007 for $\mu = 10^2, 10^3$ and 10^4 .
- The H^1 constant is not covered by the corollary. For $\mu = 100$ it rises from 2.46 to 2.71 ; for $\mu = 10^3$ and 10^4 it falls, from 2.78 to 2.77 and from 3.06 to 2.79 . Section 6.1 identifies the limit as $K = \|\nabla U\|/G = 2.7565$ for this box, proves convergence to it as $h \rightarrow 0$ at fixed μ (Corollary 6.11; on these meshes h/h_Γ is nearly constant, so fixed μ means fixed γ and bounded ρ), and compares it pointwise with these runs. The $\mu = 100$ values approach it from below; the larger- μ values approach it from above, in a regime ($\gamma h_\Gamma \gg 1$) that the proof does not cover (Remark 6.12).
- On test B, GS itself (B_1 , $\mu = 100$, $N = 256$) gives slip and energy ratios 0.995 and 1.001 .
- For $\mu = 10^4$ with the small pressure B_1 , $\gamma h_\Gamma \approx 120 \gg G$ at $N = 192$, so the hypothesis of the corollary fails. The GS energy ratio there is about 7.8 , dominated by the geometric error rather than the traction response.
- The difference $u_{\text{GS}} - u_{\text{CNS}^*}$ is not a clean proxy for the traction response on test B. It also contains the difference between the two methods on the velocity part, which is $O(h_\Gamma^{3/2})$ and non-zero even for test A. Test P removes this.

14.6. **Proposition 9.1: the penalty study.** The coercivity threshold μ^* on Z_h is the smallest μ for which the unpivoted symmetric factorisation of $A_\mu + rB^\top M^{-1}B$ has only positive pivots; we locate it by bisection. Table 7 and Figure 3 show the results.

Under refinement at fixed h/h_Γ , γ^* increases by 3.4 ($N = 8 \rightarrow 16$), 1.8 ($16 \rightarrow 32$) and 0.95 ($32 \rightarrow 64$). The increments roughly halve. The patch computations of Section 9.1 (limit geometry, windows of up to 32 boundary edges) predict a limit $\gamma_\infty^* \approx 21.4$; this is an extrapolation, not a proof.

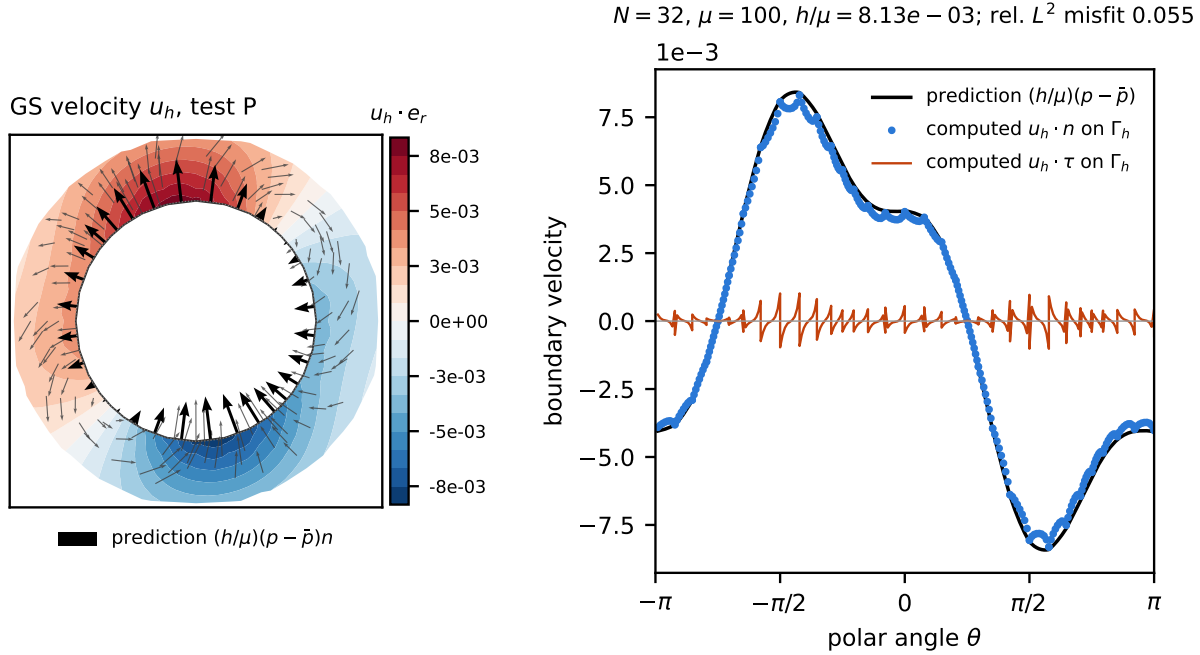


FIGURE 2. The leak of the printed method on test P ($u = 0$, so $u_h = -(\tilde{u} - u_h)$), $N = 32$, $\mu = 100$. Left: radial velocity $u_h \cdot e_r$ near the wall (colour) and u_h (thin arrows); thick arrows show the prediction $(h/\mu)(p - \bar{p})n$ of Corollary 6.7 at the chord midpoints. Right: normal and tangential traces of u_h along Γ_h against $(h/\mu)(p - \bar{p})$.

TABLE 6. The leading term of Corollary 6.7. Normalised GS error on test P ($G = 1.658$), where $e = \tilde{u} - u_h$.

N	μ	$\ \nabla e\ \mu / (hG)$	$\ e\ _{\Gamma_h} \mu / (hG)$	$\ e\ (\mu/h)^{1/2} / G$
16	100	2.464	0.893	1.050
32	100	2.612	0.945	1.007
64	100	2.684	0.972	0.996
96	100	2.708	0.981	0.996
96	1000	2.770	0.998	0.999
96	10000	2.790	0.9993	0.9994

In part (b) of Table 7, h/h_Γ is varied by coarsening the far field while the vertices of Γ_h are held fixed and the boundary layer approximately fixed. Then μ^* is linear in h/h_Γ over more than a decade, with a constant slope of about 18–20. The vertices of Γ_h do not depend on L , so $\max |e| / \min |e|$ is fixed at fixed N , and μ^* is equally linear in ρ , with the slope divided by that ratio. The threshold is thus set locally at Γ_h ($\mu/h \gtrsim c/h_\Gamma$), and the scaling with h/h_Γ (hence with ρ) is the cost of normalising the penalty by the global h .

In practical terms:

- for $k = 4$ a safe choice is $\gamma = \mu h_\Gamma / h \geq 25$, that is, $\mu \geq 25 h/h_\Gamma$; since $\rho \geq h/h_\Gamma$, $\mu \geq 25\rho$ is also safe;
- a fixed μ fails once the far field is coarsened beyond $h/h_\Gamma \approx \mu/20$;

TABLE 7. Coercivity threshold μ^* and $\gamma^* = \mu^* h_\Gamma / h$ ($k = 4$). The row h/h_Γ is $h_\Omega / \max |e|$; the $\rho = h / \min |e|$ of Section 2 is larger by the factor $\max |e| / \min |e|$ (for example $\rho = 4.70, 5.73, 6.35$ at $N = 16, 32, 64$ in part (a)).

(a) Refinement at fixed $h/h_\Gamma \approx 3.3$.								
N	8	12	16	20	24	32	48	64
h/h_Γ	3.41	3.02	3.28	3.16	3.30	3.33	3.37	3.40
ρ	3.41	4.16	4.70	5.08	5.36	5.73	6.13	6.35
μ^*	50.3	51.8	59.3	60.3	63.3	66.0	69.1	70.7
γ^*	14.7	17.1	18.1	19.1	19.2	19.9	20.5	20.8

(b) Varying h/h_Γ through the outer box $(-L, L)^2$.							
	L	2.5	5	10	20	40	
$N = 16$	h/h_Γ	3.28	5.90	11.3	22.2	43.9	
	μ^*	59.3	112.1	208.0	411.5	809.3	
	γ^*	18.1	19.0	18.4	18.5	18.4	
$N = 32$	h/h_Γ	3.33	5.79	10.8	21.1	41.5	
	μ^*	66.0	113.9	218.0	422.2	835.7	
	γ^*	19.9	19.7	20.1	20.0	20.1	

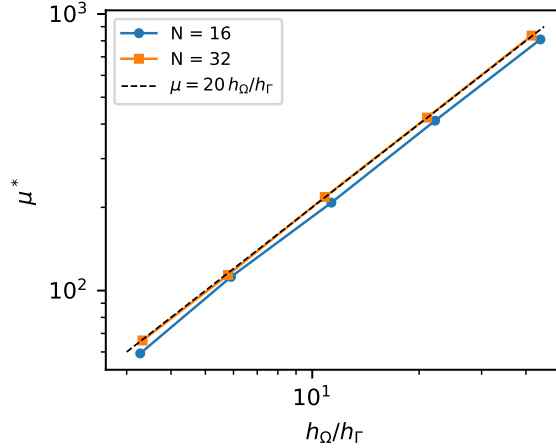


FIGURE 3. Coercivity threshold μ^* against h/h_Γ (Table 7(b)).

- the threshold grows quickly with k , and is larger on Clough–Tocher refinements (Section 14.11), so these values do not transfer to other elements.

The exact certificate $\lambda \geq 9$ on $\mathcal{S}_3$ and $\mathcal{S}'_4$ is a local lower bound, consistent with the global $\gamma^* \approx 18$ –21; Section 9.1 accounts for the gap.

14.7. Lemma tests. We compute exact discrete dual norms, $\sup_v \ell(v) / \|\nabla v\|$ over Z_h or V_h^R , by a saddle-point solve with the pointwise divergence constraint. Table 8 gives the rates for $N = 48 \rightarrow 64$.

Computation: `code/lemma_tests.py 16,24,32,48,64`; output `logs/lemma_tests.log`.

TABLE 8. Exact discrete dual norms; rate in h_Γ for $N = 48 \rightarrow 64$ (positive means decay).

Functional	Space	Rate
$\langle S, \partial_n v \rangle, S$ normal	Z_h	+0.05 (bounded, ≈ 1.75)
$\langle S, \partial_n v \rangle, S$ normal	V_h^R	-0.52
$\langle S, \partial_n v \rangle, S$ tangential	Z_h	-0.52
$\langle (\nabla \tilde{u})^\top n, v \rangle$, tests A / B	V_h^R	1.57 / 1.56 (Lemma 3.5: 3/2)

14.8. Strong imposition and the $h^{1/2}$ phenomenon. Table 9 compares strongly imposed no-slip, $u_h = 0$ at every node of Γ_h with no Nitsche terms and the same P_4 - P_3^{disc} pair, with GS and CNS* on test A.

Computation: `code/strong.bc.py`; `code/report_v5.py`; output `results/v5/strong/`, `results/v5/letter_tables.txt` (T1, T1').

- *Mesh (a): the rate $h^{1/2}$ is reproduced.* On mesh (a) the first-layer diagonals alternate, so every other polygon vertex lies in only two triangles. The H^1 rate of strong imposition falls from 0.75 to 0.56 at $N = 128$, towards $\frac{1}{2}$. The largest vertex-gradient error $\max_{z, T \ni z} |\nabla u_h|_T(z) - \nabla u(z)|$ tends to 1.99, which is $|\partial_n u \cdot \tau| = 2$ at $r = 1$ for this flow: the discrete gradient vanishes at those vertices. The pressure norm grows like $h^{-1/2}$ ($\|p_h\| = 14.0 \rightarrow 35.7$, although $p \equiv 0$).
- *Mesh (s): no locking in the range computed.* On the mesh of the rest of this section every polygon vertex lies in three triangles. Strong imposition then converges with H^1 rates $1.93 \rightarrow 1.61$ ($N = 24 \rightarrow 128$), like the Nitsche methods and with an error about 2.5 times larger; the vertex-gradient error decays like h ($1.28 \rightarrow 0.16$), and $\|p_h\|$ decays. Imposing $u_h = \tilde{u}$ at the nodes of Γ_h instead of $u_h = 0$ gives rate ≈ 4 , so the defect of strong imposition is entirely the transfer of the no-slip value $u = 0$ from Γ to Γ_h .
- *GS and CNS* do not see the change of mesh.* On mesh (a) their H^1 errors at $N = 128$ are $6.00 \cdot 10^{-3}$ and $7.17 \cdot 10^{-3}$, against $6.11 \cdot 10^{-3}$ and $6.96 \cdot 10^{-3}$ on mesh (s); at every $N \leq 128$ the two meshes differ by at most 6%.

Remark 14.1 (How to read Table 9). These are observations for one flow and two mesh families, not theorems; we have not analysed strong imposition. They are consistent with the local count of Section 1: at a wall vertex in two triangles, a conforming piecewise-polynomial divergence-free field vanishing on both boundary edges has zero gradient, while at a vertex in three triangles a one-parameter family of vertex gradients remains. We did not test three-triangle stars that are nearly singular, nor other flows, so we do not claim that three triangles per wall vertex prevent locking in general. Mesh (a) violates (M1') and (M2) at its two-triangle vertices ($\Theta(z) = O(h_\Gamma)$ there), so it lies outside the hypotheses of our theorems for GS and CNS* as well; that these methods are unaffected on it is an observation. The comparison does support the conclusion of Gjerde and Scott that weak imposition removes the zero-gradient defect, and it shows that the size of that defect for strong imposition depends on the mesh at the wall.

14.9. Pressure errors, stress and drag. All runs use $k = 4$ and $\mu = 100$, and measure $\|p^\circ - p_h\|_{L^2(\Omega_h)}$ with $p^\circ = \tilde{p} - \bar{p}_\Gamma(\tilde{p})$, the target of Theorems 8.7 and 8.11, for both methods. Removing the optimal constant instead changes these errors in the third digit at most; the discrete pressures are thus $\bar{p}_\Gamma$ -normalised up to a vanishing constant, even though $\bar{p}_\Gamma(p_h) \neq 0$ ($\bar{p}_\Gamma(p_h) = -4.6 \cdot 10^{-3} \rightarrow -9.7 \cdot 10^{-4}$ for CNS* on B₁, $N = 16 \rightarrow 96$). The velocity H^1 errors of these runs reproduce the stored ones to a relative $2.1 \cdot 10^{-7}$. The results are in Tables 10–12.

Computation: `code/pressure_err.py`; `code/report_v5.py`; output `results/v5/pressure/`, `results/v5/letter_tables.txt` (T2, T4).

TABLE 9. Strong imposition against Nitsche. Test A, $k = 4$, $\mu = 100$: H^1 semi-norm error $|\tilde{u} - u_h|_{1,\Omega_h}$. *Strong*: $u_h = 0$ at all nodes of Γ_h , no Nitsche terms, on mesh (s), the mesh of Section 14 (every polygon vertex in three triangles), and on mesh (a), the same mesh with alternating first-layer diagonals (polygon vertices in two or four triangles). GS and CNS* on mesh (s).

N	h	strong, mesh (a)		strong, mesh (s)		GS		CNS*	
		error	rate	error	rate	error	rate	error	rate
16	1.5052	$9.03 \cdot 10^{-1}$	–	$4.62 \cdot 10^{-1}$	–	$1.72 \cdot 10^{-1}$	–	$1.98 \cdot 10^{-1}$	–
24	1.0559	$6.91 \cdot 10^{-1}$	0.75	$2.33 \cdot 10^{-1}$	1.93	$8.87 \cdot 10^{-2}$	1.86	$1.03 \cdot 10^{-1}$	1.85
32	0.8127	$5.72 \cdot 10^{-1}$	0.72	$1.43 \cdot 10^{-1}$	1.86	$5.55 \cdot 10^{-2}$	1.79	$6.41 \cdot 10^{-2}$	1.80
48	0.5560	$4.44 \cdot 10^{-1}$	0.67	$7.29 \cdot 10^{-2}$	1.78	$2.88 \cdot 10^{-2}$	1.73	$3.31 \cdot 10^{-2}$	1.74
64	0.4224	$3.75 \cdot 10^{-1}$	0.62	$4.56 \cdot 10^{-2}$	1.71	$1.81 \cdot 10^{-2}$	1.68	$2.08 \cdot 10^{-2}$	1.69
96	0.2853	$2.98 \cdot 10^{-1}$	0.59	$2.38 \cdot 10^{-2}$	1.65	$9.57 \cdot 10^{-3}$	1.63	$1.09 \cdot 10^{-2}$	1.64
128	0.2153	$2.54 \cdot 10^{-1}$	0.56	$1.52 \cdot 10^{-2}$	1.61	$6.11 \cdot 10^{-3}$	1.59	$6.96 \cdot 10^{-3}$	1.60
192	0.1445	–	–	–	–	$3.27 \cdot 10^{-3}$	1.57	$3.71 \cdot 10^{-3}$	1.58
256	0.1087	–	–	–	–	$2.10 \cdot 10^{-3}$	1.55	$2.38 \cdot 10^{-3}$	1.56

- CNS*. The pressure error does not depend on the pressure data: for B_1 , B_{100} and C it agrees to 10^{-6} at $N = 16$ and $3 \cdot 10^{-11}$ at $N = 96$. Its L^2 rates 3.0, 3.0, 2.6, 2.1, 1.8 decrease towards $\frac{3}{2}$, as Theorem 8.7 predicts; the first-layer part already has rate about 1.6, and the L^∞ error on the first layer has rate about 1 (0.98 at $N = 96$), for which we have no proof.
- GS with $G > 0$. The L^2 rate decreases towards $\frac{1}{2}$ (0.60, 0.62, 0.63 at $N = 96$ for B_1 , B_{100} , C). The first-layer part has rate 0.48–0.50 and the rest rate 0.9–1.0; the L^∞ error on the first layer does not converge (0.78 for B_1 , 77 for B_{100}). This is Theorem 8.11. For B_{100} the GS pressure error is 9.2 at $N = 96$, against $8 \cdot 10^{-3}$ for CNS*.
- GS with $G = 0$ (test A) behaves like CNS*: L^2 rate 1.68 at $N = 96$ and still decreasing, and the first layer converges.

Layer, drag and traction. On test P with GS at $N = 64$ and 96, the first-layer pressure error is 0.0958 and 0.0793, against $\|\xi_b\| = 0.0961$ and 0.0796 from Definition 8.10; the rest is 0.069 and 0.047. The Nitsche-flux drag defect $J_h^N - J_{e_1} + \Theta_f$ is 0.970 and 0.981 times the prediction $-(h/\mu)K_{e_1}$ of Theorem 8.17 ($K_{e_1} \approx 17.16$, $K_{e_2} \approx -25.36$); for Gjerde–Scott’s volume formula the ratio to $-2(h/\mu)K_{e_1}$ is 0.905 and 0.952 at $N = 32$ and 64 (Proposition 8.21). The pointwise-traction drag error does not converge: $J_h^\beta - J_{e_1} = -1.211$ and -1.225 at $N = 64$ and 96, against $\Xi_h = -1.192$ at $N = 64$ (Proposition 8.22). For CNS* on B_1 , the lift defect divided by $-\Theta_f - \Lambda_h$ is 0.788, 0.895, 0.941, 0.955 for $N = 16$ –96, and on test A the torque ratio is 0.84, 0.92, 0.95 (Theorem 8.19). At fixed $N = 64$ the CNS* pressure error on B_1 grows with the penalty (0.016, 0.039, 0.059 for $\mu = 10^2, 10^3, 10^4$), which is the non-uniformity in γ of Remark 8.8.

Computation: notes/v4/pressure_drag_check.py; output
notes/v4/pressure_drag_report.md, Section 3 (no log file).

14.10. A growing penalty and flux-corrected data. All runs use $k = 4$ on the meshes of Section 14 ($h/h_\Gamma \in [3.28, 3.43]$, so $\gamma \approx 0.29\mu$ and rates in h and h_Γ agree), SuperLU for $N \leq 64$ and PARDISO (Section 14.11) for $N = 96$; relative residuals are at most $1.8 \cdot 10^{-12}$ and the discrete divergence at most $1.6 \cdot 10^{-9}$. The $\mu = 100$ values for test P reproduce Table 6 ($\|\nabla e\| = 1.2810 \cdot 10^{-2}$ at $N = 96$).

TABLE 10. Test B₁₀₀ (large pressure variation, $p = 100(x^2y - y + x/2)$), $k = 4$, $\mu = 100$: velocity error $|\tilde{u} - u_h|_{1,\Omega_h}$ and pressure error $\|p^\circ - p_h\|_{L^2(\Omega_h)}$, $p^\circ = \tilde{p} - \bar{p}_\Gamma(\tilde{p})$.

N	h	GS, $ \cdot _1$		CNS*, $ \cdot _1$		GS, L^2 pressure		CNS*, L^2 pressure	
		error	rate	error	rate	error	rate	error	rate
16	1.5052	6.14	–	$1.29 \cdot 10^{-1}$	–	$2.75 \cdot 10^1$	–	$4.97 \cdot 10^{-1}$	–
24	1.0559	4.48	0.89	$4.42 \cdot 10^{-2}$	3.02	$2.17 \cdot 10^1$	0.67	$1.69 \cdot 10^{-1}$	3.04
32	0.8127	3.51	0.93	$2.34 \cdot 10^{-2}$	2.43	$1.82 \cdot 10^1$	0.68	$7.67 \cdot 10^{-2}$	3.03
48	0.5560	2.45	0.95	$1.11 \cdot 10^{-2}$	1.97	$1.41 \cdot 10^1$	0.67	$2.87 \cdot 10^{-2}$	2.59
64	0.4224	1.88	0.97	$6.83 \cdot 10^{-3}$	1.76	$1.18 \cdot 10^1$	0.65	$1.62 \cdot 10^{-2}$	2.08
96	0.2853	1.28	0.98	$3.54 \cdot 10^{-3}$	1.68	9.23	0.62	$8.01 \cdot 10^{-3}$	1.80
128	0.2153	$9.71 \cdot 10^{-1}$	0.98	$2.24 \cdot 10^{-3}$	1.63	–	–	–	–
192	0.1445	$6.54 \cdot 10^{-1}$	0.99	$1.18 \cdot 10^{-3}$	1.60	–	–	–	–

TABLE 11. Test B₁, $\mu = 100$: pressure error $\|p^\circ - p_h\|_{L^2(\Omega_h)}$ ($p^\circ = \tilde{p} - \bar{p}_\Gamma(\tilde{p})$); the off-layer column is the error on $\Omega_h \setminus L_1$ with the optimal constant removed.

N	h	GS		GS, off first layer		CNS*	
		error	rate	error	rate	error	rate
16	1.5052	$5.30 \cdot 10^{-1}$	–	$5.00 \cdot 10^{-1}$	–	$4.97 \cdot 10^{-1}$	–
24	1.0559	$2.47 \cdot 10^{-1}$	2.15	$1.96 \cdot 10^{-1}$	2.63	$1.69 \cdot 10^{-1}$	3.04
32	0.8127	$1.80 \cdot 10^{-1}$	1.22	$1.21 \cdot 10^{-1}$	1.84	$7.67 \cdot 10^{-2}$	3.03
48	0.5560	$1.37 \cdot 10^{-1}$	0.72	$8.05 \cdot 10^{-2}$	1.08	$2.87 \cdot 10^{-2}$	2.59
64	0.4224	$1.15 \cdot 10^{-1}$	0.61	$6.32 \cdot 10^{-2}$	0.88	$1.62 \cdot 10^{-2}$	2.08
96	0.2853	$9.13 \cdot 10^{-2}$	0.60	$4.45 \cdot 10^{-2}$	0.89	$8.01 \cdot 10^{-3}$	1.80

TABLE 12. Test A ($p \equiv 0$), $\mu = 100$: pressure error $\|p^\circ - p_h\|_{L^2(\Omega_h)}$ ($p^\circ = \tilde{p} - \bar{p}_\Gamma(\tilde{p})$); the off-layer column is the error on $\Omega_h \setminus L_1$ with the optimal constant removed. Here $G = 0$, and GS behaves like CNS*.

N	h	GS		GS, off first layer		CNS*	
		error	rate	error	rate	error	rate
16	1.5052	$2.05 \cdot 10^{-1}$	–	$1.36 \cdot 10^{-1}$	–	$3.78 \cdot 10^{-1}$	–
24	1.0559	$1.05 \cdot 10^{-1}$	1.88	$7.03 \cdot 10^{-2}$	1.86	$1.95 \cdot 10^{-1}$	1.87
32	0.8127	$6.48 \cdot 10^{-2}$	1.85	$4.34 \cdot 10^{-2}$	1.84	$1.21 \cdot 10^{-1}$	1.81
48	0.5560	$3.29 \cdot 10^{-2}$	1.79	$2.20 \cdot 10^{-2}$	1.79	$6.23 \cdot 10^{-2}$	1.75
64	0.4224	$2.04 \cdot 10^{-2}$	1.73	$1.36 \cdot 10^{-2}$	1.74	$3.92 \cdot 10^{-2}$	1.69
96	0.2853	$1.06 \cdot 10^{-2}$	1.68	$6.99 \cdot 10^{-3}$	1.70	$2.05 \cdot 10^{-2}$	1.65

Computation: `code/rescue.py`; `code/rescue_run.sh`; `code/rescue_report.py`; output `logs/rescue_study.log`, `rescue_musweep.log`, `rescue_flux.log`, `rescue_cond.log`; `results/v4/rescue/`.

Power-law penalty. We take $\mu = c(h_{16}/h)^\alpha$, $h_{16} = 1.505$. Table 13 gives the rates between $N = 64$ and 96. On the leak-only test P they are exactly those of Corollary 7.25 (H^1 : $1+\alpha$; energy: $(1+\alpha)/2$), with the normalised constants of Corollary 6.7 unchanged ($\|\nabla e\|_\mu/(hG) = 2.71\text{--}2.78$ and $\|e\|_\mu(\mu/h)^{1/2}/G = 0.996\text{--}0.999$ at $N = 96$). On B₁ the GS H^1 rate moves from 0.96 (fixed μ) to

TABLE 13. GS (and CNS*) with $\mu = c(h_{16}/h)^\alpha$: rates between $N = 64$ and $N = 96$, and the predicted asymptotic rates (Corollary 7.25; for test P, where $\tilde{u} = 0$, only the leak is present and the rates are $1 + \alpha$ and $(1 + \alpha)/2$ for every α).

test	method	(α, c)	μ at $N = 96$	$\ \nabla e\ $ at $N = 96$	H^1 rate	energy rate	predicted
P	GS	(0, 100)	100	$1.28 \cdot 10^{-2}$	0.95	0.49	1, 0.5
P	GS	$(\frac{1}{2}, 100)$	230	$5.64 \cdot 10^{-3}$	1.45	0.73	1.5, 0.75
P	GS	$(\frac{1}{2}, 1000)$	2297	$5.72 \cdot 10^{-4}$	1.47	0.73	1.5, 0.75
P	GS	(1, 100)	528	$2.47 \cdot 10^{-3}$	1.94	0.97	2, 1
B ₁	GS	(0, 100)	100	$1.27 \cdot 10^{-2}$	0.96	0.56	1, 0.5
B ₁	GS	$(\frac{1}{2}, 100)$	230	$6.42 \cdot 10^{-3}$	1.42	0.86	1.5, 0.75
B ₁	GS	$(\frac{1}{2}, 1000)$	2297	$5.81 \cdot 10^{-3}$	1.52	1.28	1.5, 0.75
B ₁	GS	(1, 100)	528	$5.03 \cdot 10^{-3}$	1.47	1.06	1.5, 1
B ₁	CNS*	(0, 100)	100	$3.54 \cdot 10^{-3}$	1.64	1.53	1.5, 1.5
B ₁	CNS*	(1, 100)	528	$5.21 \cdot 10^{-3}$	1.45	1.16	1.5, 1
A	GS	(0, 100)	100	$9.57 \cdot 10^{-3}$	1.59	1.54	1.5, 1.5
A	GS	$(\frac{1}{2}, 100)$	230	$1.21 \cdot 10^{-2}$	1.43	1.35	1.5, 1.25
A	GS	$(\frac{1}{2}, 1000)$	2297	$1.86 \cdot 10^{-2}$	1.52	1.32	1.5, 1.25
A	GS	(1, 100)	528	$1.48 \cdot 10^{-2}$	1.34	1.16	1.5, 1

1.42–1.52, and the H^1 error at $N = 96$ halves ($1.27 \cdot 10^{-2} \rightarrow 6.4 \cdot 10^{-3}$, against $3.5 \cdot 10^{-3}$ for CNS* at $\mu = 100$); on B₁₀₀ it drops by a factor 22 at $c = 1000$. The energy rates follow Corollary 7.25: on B₁ with $\alpha = \frac{1}{2}$, $c = 1000$ the measured 1.28 is the geometric branch $(3 - \alpha)/2 = 1.25$ (the leak branch 0.75 takes over only later), and with $\alpha = 1$ it is 1.06.

The price when G is small. On test A ($G = 0$) a growing penalty only costs. Theorem 7.19 bounds the H^1 error uniformly, but at $N = 64$ the constant moves from its value at $\gamma \approx 30$ ($\|\nabla e\| = 1.82 \cdot 10^{-2}$) to the value of the Dirichlet limit $\mu \rightarrow \infty$ ($4.56 \cdot 10^{-2}$), 2.5 times larger. Along $\alpha = 1$ the ratio of the GS error to the Dirichlet-limit error climbs from 0.37 to 0.56 between $N = 16$ and 64; this transient is why the measured H^1 rate on A with $\alpha = 1$ is 1.34 rather than 1.5. The Dirichlet-limit error itself has rates 1.90, 1.80, 1.72, 1.66, decreasing towards $\frac{3}{2}$ as for CNS*. The energy error, by contrast, grows without bound: at $N = 64$, $\|e\|/(\gamma^{1/2} h_\Gamma^{3/2}) \rightarrow 0.339$ (A) and 0.105 (B₁) as $\mu \rightarrow \infty$, and these limits are exactly $\|\tilde{u}\|_{\Gamma_h}/h_\Gamma^2$: the discrete trace tends to 0 and the energy error becomes $(\mu/h)^{1/2} \|\tilde{u}\|_{\Gamma_h}$, the mechanism of Theorem 7.24. CNS* agrees with GS to four digits for $\mu \geq 10^6$.

Conditioning (Proposition 7.26; $N = 32, 48$, $\mu = 10^2$ – 10^5). $\lambda_{\min}(A_\mu) \approx 0.03h^2$ does not depend on μ , and $\lambda_{\max}(A_\mu) \approx \max(102$ – $118, 0.47\gamma)$, so for $\gamma \lesssim 200$ the P_4 stiffness part dominates and a moderate growth of μ is free. The smallest Schur eigenvalue is the constant-pressure mode and equals $(|\Gamma_h|/|\Omega_h|) h/\mu$ to within 2.5% at every N and μ ; the next one (0.007–0.010, the squared inf-sup constant on L_0^2) does not depend on μ . So $\kappa(K_\mu) \approx \lambda_{\max}(A_\mu) (|\Omega_h|/|\Gamma_h|) \mu/h$, from $4.9 \cdot 10^4$ ($N = 32$, $\mu = 10^2$) to $6.0 \cdot 10^9$ ($\mu = 10^5$): linear in μ while the stiffness part dominates $\lambda_{\max}$, quadratic after. For the H^1 -optimal $\alpha = \frac{1}{2}$ with $c = 100$ the whole study ($\mu \leq 230$) stays in the linear regime.

Flux correction. The test data above have $m_R = 0$ up to round-off ($|m_R| \leq 2 \cdot 10^{-13}$), either because $g \cdot \nu$ is a polynomial of low degree (B, P) or by symmetry (A). Test F uses $\psi = (r^2 - 1)^3 \sin(2x + 1) \cos(\frac{3}{2}y + \frac{7}{10})/400$ and $p = x^2y - y + x/2$: the cube makes $W \equiv 0$, so the geometric slip $\|\tilde{u}\|_{\Gamma_h} = O(h_\Gamma^4)$ is small, and the phase shift breaks the symmetries, so $m_R \neq 0$; it decays like $h^{6.6}$ ($\sigma_4 = 6$). Table 14 shows the floor of Proposition 7.20 directly. Without correction the CNS* slip error at large μ is $(|m_R|^2/|\Gamma_h| + \|\tilde{u}\|_{\Gamma_h}^2)^{1/2}$ to three digits: the uniform normal flux defect

TABLE 14. Test F, CNS*, uncorrected (g_I) against flux-corrected ($\tilde{g}_I$) outer data. Slip errors $\|\tilde{u} - u_h\|_{\Gamma_h}$ at $\mu = 10^6$; energy errors at $\mu = 10^8$, with the floor $(\mu/h)^{1/2}|m_R|/|\Gamma_h|^{1/2}$ of Proposition 7.20.

N	m_R	$\frac{ m_R }{ \Gamma_h ^{1/2}}$	$\ \tilde{u}\ _{\Gamma_h}$	slip, g_I	slip, $\tilde{g}_I$	$\ e\ $, g_I	$\ e\ $, $\tilde{g}_I$	floor
16	$2.04 \cdot 10^{-3}$	$8.19 \cdot 10^{-4}$	$2.40 \cdot 10^{-5}$	$8.204 \cdot 10^{-4}$	$2.396 \cdot 10^{-5}$	6.752	0.955	6.672
24	$1.54 \cdot 10^{-4}$	$6.14 \cdot 10^{-5}$	$5.21 \cdot 10^{-6}$	$6.172 \cdot 10^{-5}$	$5.210 \cdot 10^{-6}$	0.701	0.366	0.598
32	$2.56 \cdot 10^{-5}$	$1.02 \cdot 10^{-5}$	$1.70 \cdot 10^{-6}$	$1.038 \cdot 10^{-5}$	$1.697 \cdot 10^{-6}$	0.187	0.149	0.114
48	$2.14 \cdot 10^{-6}$	$8.55 \cdot 10^{-7}$	$3.42 \cdot 10^{-7}$	$9.213 \cdot 10^{-7}$	$3.422 \cdot 10^{-7}$	0.0372	0.0354	0.0115
64	$3.75 \cdot 10^{-7}$	$1.50 \cdot 10^{-7}$	$1.09 \cdot 10^{-7}$	$1.851 \cdot 10^{-7}$	$1.091 \cdot 10^{-7}$	0.0123	0.0121	0.0023

and the tangential geometric slip, orthogonal to each other. With the correction the slip error is $\|\tilde{u}\|_{\Gamma_h}$ to five digits, the flux through Γ_h is at most $2.5 \cdot 10^{-14}$, and the energy error loses the term $(\mu/h)^{1/2}|m_R|/|\Gamma_h|^{1/2}$ (a factor 7 at $N = 16$, $\mu = 10^8$). The H^1 errors with and without correction agree to a relative $3 \cdot 10^{-6}$ ($N = 16$) down to 10^{-9} ($N = 64$), as Theorem 7.19 predicts: the flux enters the H^1 error only as $|m_R|h_\Gamma^{-1/2}$, far below the interpolation error of this steep solution. For the companion test F2 ($(r^2 - 1)^2$ in place of $(r^2 - 1)^3$, so $W \neq 0$) the geometric slip exceeds the floor by a factor 23 ($N = 16$) to $2 \cdot 10^4$ ($N = 64$), and the correction changes nothing visible.

14.11. Higher degree and Clough–Tocher refinements. We repeated the study with $k = 5, 6$ on the same meshes, and with $k = 2, 3$ on their Clough–Tocher refinements (Section 10), using a degree-general version of the solver (`code/svn.k.py`). The refinement leaves Γ_h , the square and $h = h_\Omega$ unchanged, so the h values are those of the standard meshes. At $k = 4$ with SuperLU the new code is bit-identical to `svn.py`, and agrees with the stored production records to $2.6 \cdot 10^{-10}$ relative. SuperLU runs out of memory at the target sizes, so these runs use MKL PARDISO. The saddle-point matrix is ill-conditioned, and for robustness of PARDISO we regularise the pressure block by 10^{-12} times the pressure mass matrix. The result agrees with SuperLU to $\leq 10^{-9}$ relative ($k = 4$ and $k = 6$, $N \leq 32$). Over the 260 solves (390 records, including 130 derived records for D) the relative residual of the unregularised system is at most $2.2 \cdot 10^{-12}$ and the discrete divergence at most 10^{-9} . One side effect: CNS* on test P returns $\|\nabla u_h\| \approx 4 \cdot 10^{-11}$, the regularisation floor, instead of round-off.

Thresholds. Table 15 gives γ^* . It grows quickly with k and is larger on the Clough–Tocher refinements, whose boundary triangles are three times thinner while h is the macro size. In every case the measured value exceeds the corresponding certified local constant (Proposition 9.1 for $k \geq 4$, Proposition 10.10 on the refinements), which is consistent with those constants being local lower bounds. γ^* still drifts upward with N (+7% to +24% from $N = 16$ to 64). In particular $\mu = 100$ is *below* the threshold for $k = 5$ ($N \geq 32$), for $k = 6$ and for Clough–Tocher $k = 3$ (all N). The $\mu = 100$ runs in those cases are erratic, as they should be (for instance the CNS* H^1 error for $k = 5$ on B_1 jumps between 0.05 and 3.7), and they are not covered by the theory. We therefore report $\mu = 300$ and $\mu = 1000$ for them.

The leak (test P). Table 16 shows that the slip and energy constants of Corollary 6.7 tend to 1 for every element, with rates 1 (slip), $1/2$ (energy) and 1 (H^1). The H^1 constants are close to the $K = 2.7565$ of Section 6.1, which does not depend on the element; this agreement is observed, not proved, for these elements. On test B_1 the leak part $D := u_{GS} - u_{CNS^*}$ has the same normalised slip and energy ratios (0.986–1.003 and 0.991–1.006 at the finest levels), H^1 rate 1.01–1.15 and energy rate 0.49–0.52. Its normalised H^1 constant is still drifting down at the finest levels (3.03 for $k = 5$, $\mu = 1000$).

Rates. Table 17 gives H^1 rates at the finest pair of levels.

TABLE 15. Coercivity threshold $\gamma^* = \mu^* h_\Gamma / h$ for higher degrees and on Clough-Tocher (CT) refinements. The $k = 4$ row is Table 7(a).

	$N = 8$	16	32	64	128
$k = 4$	14.7	18.1	19.9	20.8	–
$k = 5$	25.7	30.3	32.1	33.0	–
$k = 6$	–	45.2	47.4	48.3	–
CT, $k = 2$	5.7	9.1	10.5	11.2	11.5
CT, $k = 3$	–	31.6	33.9	34.6	–
CT, $k = 4$	–	61.9	65.5	–	–

TABLE 16. Normalised GS error on test P at the finest level run, $e = \tilde{u} - u_h$.

Element	μ	N	$\ \nabla e\ \mu / (hG)$	$\ e\ _{\Gamma_h} \mu / (hG)$	$\ e\ (\mu/h)^{1/2} / G$
$k = 4$	1000	96	2.770	0.998	0.999
$k = 5$	1000	128	2.768	0.9983	0.9994
$k = 6$	1000	96	2.769	0.9976	0.9993
CT, $k = 2$	100	256	2.737	0.9928	0.9974
CT, $k = 2$	1000	128	2.751	0.9982	0.9992
CT, $k = 3$	1000	128	2.763	0.9983	0.9996

TABLE 17. H^1 rates at the finest pair of levels (energy-norm rate of CNS* in parentheses). $D = u_{\text{GS}} - u_{\text{CNS}^*}$ is the leak part; it is not reported for test A, where $G = 0$.

Element	Test	μ	N	GS	CNS*	D
$k = 5$	B ₁	300	96 $\rightarrow$ 128	1.20	1.59 (1.53)	1.03
$k = 5$	B ₁	1000	96 $\rightarrow$ 128	1.53	1.58 (1.54)	1.13
$k = 6$	B ₁	300	64 $\rightarrow$ 96	1.27	1.61 (1.54)	1.04
$k = 6$	B ₁	1000	64 $\rightarrow$ 96	1.57	1.61 (1.54)	1.15
CT, $k = 3$	A	1000	96 $\rightarrow$ 128	1.56	1.56 (1.55)	–
CT, $k = 2$	A	100	192 $\rightarrow$ 256	1.78	1.78 (1.55)	–
CT, $k = 2$	B ₁	100	192 $\rightarrow$ 256	1.91	1.91 (1.91)	1.01
CT, $k = 3$	B ₁	1000	96 $\rightarrow$ 128	2.83	2.83 (1.64)	1.05

- CNS* tends to rate $3/2$ for $k = 5, 6$ on B₁ (1.58–1.61, energy rate 1.53–1.54), non-increasing from $N = 32$ on. The CNS* errors for $k = 4, 5, 6$ are almost equal (for example $1.05 \cdot 10^{-2}$, $9.02 \cdot 10^{-3}$, $8.47 \cdot 10^{-3}$ at $N = 64$, $\mu = 1000$): the geometric term dominates. On Clough-Tocher $k = 3$, test A, the H^1 rate is 1.56 at $N = 128$.
- GS tends to rate 1 when $G > 0$. For $k = 5$, $\mu = 300$, the GS H^1 rate on B₁ decreases monotonically, 1.63, 1.58, 1.48, 1.38, 1.28, 1.20 for $N = 24, \dots, 128$, while its leak part D already has rate 1.03. At $\mu = 1000$ the crossover comes later (1.53 at $N = 128$), exactly as for $k = 4$.
- CNS* is pressure-independent for $k = 5, 6$: on test P it returns $u_h = 0$ to the regularisation floor at every N and every $\mu \geq \mu^*$. By linearity this is the statement that $u_h(B_a) = u_h(B_1)$ for every a .

Caveats for Clough-Tocher $k = 2, 3$.

- (1) *CNS* on test P is not exactly zero.* The pressure $p = x^2y - y + x/2$ is cubic, so $p \notin \Pi_h$ for $k \leq 3$, and the term $\eta = p^* - \pi_h p^*$ of Theorem 7.2 is present. The spurious velocity is small and converges at a rate close to $k + \frac{1}{2}$: $\|\nabla u_h\| = 1.4 \cdot 10^{-6}$ at $N = 256$ for $k = 2$ (rate 2.56), and $4.7 \cdot 10^{-9}$ at $N = 128$ for $k = 3$ (rate 3.53). So for $k \leq 3$ the pressure-independence of CNS^* holds only up to a term of order $h^{k+1/2}$ that scales with the amplitude of p . It is below $h_\Gamma^{3/2}$ asymptotically, but for B_{100} it would be 100 times larger.
- (2) *On test B_1 the rate 3/2 is not yet visible.* The H^1 error is dominated by interpolation of the steep velocity (u grows like r^5 , since ψ grows like r^6 , on the square of side 5): for $k = 2$ it is 0.41 at $N = 256$, against $7.6 \cdot 10^{-4}$ for $k = 4$, with rate 1.91 still rising towards 2. For $k = 3$ the rate is 2.83, close to 3. The runs are pre-asymptotic and the crossover to $h_\Gamma^{3/2}$ is out of reach. The leak part D is on theory (rate 1.01, ratios 1.0009 and 1.0056 at $N = 256$ for $k = 2$).
- (3) *On test A (bounded velocity) the energy-norm rate is already 1.55 for $k = 2$, while the H^1 rate (1.78 at $N = 256$) is still decreasing.*

Memory limits (a shared 5.8 GB budget) prevented $k = 6$ at $N = 128$ and Clough–Tocher $k = 3$ at $N = 192$.

APPENDIX A. REPRODUCIBILITY

All code, raw outputs and logs are in the repository that accompanies this paper. Paths below are relative to its root; commands are run from `code/`. `code/regen_logs.sh` regenerates every log in `logs/` with the commands below, and `make report` checks that all five reports (`report.txt`, `report_local.txt`, `report_v3.txt`, `letter_tables.txt` and `results/v4/rescue/summary.md`) are reproduced byte-for-byte from the shipped raw records.

Code (code/).

- `svn.py`: the solver. It contains the mesh generator, the P_4 space, assembly of N_h , B and the CNS^* coupling, the SuperLU saddle-point solver (`solve.lu`), error norms, and the exact solutions for tests A, B, P and C.
- `job.py`: one study or threshold job per process.
- `run_all.py`: memory- and CPU-aware scheduler for the full campaign.
- `report.py` and `report_local.py`: produce the tables.
- `plots.py`: produces the figures.
- `penalty.py`: computes the coercivity threshold μ^* .
- `lemma_tests.py`: computes the dual norms of Table 8.
- `lamstar_exact.py` and `lamstar_exact_v2.py`: the exact certificates of Proposition 9.1.
- `lamstar.py`: the floating-point λ^* on other fans.

Added in this version:

- `svn_k.py`: degree-general solver (P_k velocities, $k \geq 2$; standard or Clough–Tocher mesh; SuperLU or PARDISO). `job_k.py` runs one study or threshold job, `penalty_k.py` computes μ^* , `report_v3.py` writes the tables, and `validate_k.py` checks $k = 4$ against `svn.py` and PARDISO against SuperLU. The scheduler is `results/v3/sched.py`, with job list `results/v3/jobs.txt`.
- `lamstar_exact_ct.py`: the exact local-surjectivity ranks and penalty certificates on Clough–Tocher stars (Section 10).
- `lamstar_family.py`: the single-triangle forms, the shape-family certificates and the patch thresholds (Section 9.1).
- `h1_limit.py`: the annulus bracket, the series solution and the finite element comparison for the H^1 constant (Section 6.1).

- `lower_bound_tests.py`: the (M3) star constants, the assembled test field and the dual norms (Section 7.1).
- `plot_leak.py`: Figure 2.
- `notes/v3/face_mean_check.py`: the face-mean check of Lemma 13.2(d).

Added in this version (v4):

- `rescue.py`: the growing-penalty study, the μ -sweeps, the flux-corrected data $\tilde{g}_I$ (tests F and F2) and the spectra of Proposition 7.26 (Section 14.10); it reuses `svn.k.py` unchanged. `rescue_run.sh` is its batch driver; it writes the four logs `logs/rescue_*.log` and the records in `results/v4/rescue/`. From these records `rescue_report.py` writes the summary `summary.md` in the same directory.
- `pressure_err.py`: pressure errors of GS and CNS* against p° , with the first-layer split (Section 14.9); writes `results/v5/pressure/`.
- `strong_bc.py`: strongly imposed no-slip on Γ_h , and GS/CNS* on the alternating mesh (a) (Section 14.8); writes `results/v5/strong/`.
- `report_v5.py`: writes `results/v5/letter_tables.txt` from the JSONL records (and, with argument `thresh`, recomputes h/h_Γ and ρ for the threshold runs into `results/v5/thresh_rho.jsonl`). The job lists are `results/v5/jobs*.txt`; see `results/v5/README.md`.
- `notes/v4/pressure_drag_check.py`: the layer, drag and traction checks of Section 14.9; its output is recorded in `notes/v4/pressure_drag_report.md`, Section 3.

Where each number comes from.

Result	Command	Output
Tables 4, 5, 7	SVN_OUT=../results/pod python3 run_all.py, then python3 report.py ../results/pod	results/pod/report.txt
Table 6, test C	SVN_OUT=../results/local python3 job.py study N P1 MU and SVN_OUT=../results/local python3 job.py study N C 100, then python3 report_local.py	results/local/report_local.txt
Certificate for S_3	python3 lamstar_exact.py	logs/lamstar_exact_ S3_S4singular.log
Certificate for S'_4	python3 lamstar_exact_v2.py	logs/lamstar_exact_S4prime.log
Table 8	python3 lemma_tests.py 16,24,32,48,64	logs/lemma_tests.log
Floating-point λ^*	python3 -W ignore::DeprecationWarning lamstar.py	logs/lamstar_float.log
Figures 1, 3	python3 plots.py	paper/figures/h1_rates.pdf etc.
Figure 2	python3 plot_leak.py	paper/figures/leak_field.pdf
Tables 15–17	SVN_SOLVER=pardiso python3 ../results/v3/sched.py ../results/v3/jobs.txt LOG (runs job_k.py; output root results/v3 by default, or SVN_OUT), then python3 report_v3.py	results/v3/report_v3.txt
Validation of <code>svn.k.py</code>	python3 validate_k.py	results/v3/validation.txt
Lemma 10.3 ($k = 3$), Table 3	python3 lamstar_exact_ct.py	logs/lamstar_exact_ct.log
Theorem 9.2, Proposition 9.3	python3 lamstar_family.py bubble; ... certify	logs/lamstar_family_bubble.log, ..._certify.log

Result	Command	Output
Patch thresholds (Section 9.1)	<code>python3 lamstar_family.py mesh; ... limit; ... limitstar</code>	<code>logs/lamstar_family_mesh.log, ..._limit.log, ..._limitstar.log</code>
Level-set and tall-triangle values of λ_T (Section 9.1)	<code>python3 lamstar_family.py levels</code>	<code>logs/lamstar_family_levels.log</code>
H^1 constant (Section 6.1) (M3) and Tables 1, 2	<code>python3 h1_limit.py annulus and ... mps (series log); ... fem python3 lower_bound_tests.py with stars 8,12,...,1024, field 16,...,512, global 16,...,64</code>	<code>logs/h1_limit_series.log, ..._fem.log logs/lower_bound_stars.log, ..._field.log, ..._global.log</code>
Lemma 13.2(d)	<code>python3 notes/v3/ face_mean_check.py (from the root)</code>	standard output
Tables 13, 14; Section 14.10 Tables 9–12; Section 14.9	<code>bash rescue_run.sh all, then python3 rescue_report.py the commands in results/v5/jobs*.txt (pressure_err.py, strong_bc.py), then python3 report_v5.py</code>	<code>logs/rescue_*.log, results/v4/ rescue/, .../summary.md results/v5/letter_tables.txt</code>
Layer, drag and traction checks (Section 14.9)	<code>python3 notes/v4/ pressure_drag_check.py main xi K lam chi torque ... (from the root)</code>	<code>notes/v4/ pressure_drag_report.md</code>

Resources. The full A/B campaign (74 study jobs up to $N = 256$ and 16 threshold jobs) took 36 minutes on 32 cores with 125 GB of memory. The largest single job ($N = 256$, both methods) takes about 23 minutes; the scheduler reserved up to 55 GB for it. Every job with $N \leq 96$ runs on a 2-core, 7 GB machine.

The runs of Section 14.11 used a 2-core machine with a 5.8 GB memory limit shared with other jobs; the largest single job used 3.6 GB (peak RSS is recorded in each record). Each job took at most about 2.5 minutes.

The v4 and v5 runs (Sections 14.8–14.10) were sequential single-thread jobs on the same 2-core machine; the largest used 2.2 GB (SuperLU, $N = 64$), and the $N \geq 64$ strong-imposition, mesh-(a) and $N = 96$ runs used PARDISO as in Section 14.11. Over the 83 v5 records the relative residual is at most $1.3 \cdot 10^{-13}$ and the discrete divergence at most $9.6 \cdot 10^{-7}$ (the nearly singular strong system on mesh (a) at $N = 128$; SuperLU at $N = 64$ agrees to five digits with divergence $1.5 \cdot 10^{-10}$).

The environment was Python 3.11 with NumPy 2.4, SciPy 1.17 (SuperLU) and SymPy 1.14. The runs of Section 14.11 also used MKL PARDISO through `pypardiso`.

USE OF GENERATIVE-AI TOOLS DECLARATION

The author used Anthropic’s Claude Opus 5.5 as a research-support tool. It assisted with the following:

- developing and checking the proofs;
- implementing, running and validating the finite element code and the exact-arithmetic certificates;
- independently reviewing the arguments as a referee;
- preparing the manuscript and the accompanying repository.

The author directed the project and retains authorship of the work. Responsibility for all mathematical statements, computations, references and conclusions rests with the author.

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